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# MAX86178

# Ultra-Low-Power, Clinical-Grade Vital-Sign AFE

## General Description

The MAX86178 is a highly integrated, multiple vital-sign monitoring device with a complete photoplethysmogram (PPG), electrocardiogram (ECG) and biopotential (BioZ) analog front end (AFE) for wearable applications. The MAX86178 offers high performance for wellness and clinical applications with low power for long battery life.

The PPG data acquisition system supports up to 6 LEDs and 4 photodiode inputs. The LEDs are programmable from two high-current, 8-bit LED drivers. The receive path has two low-noise, high-resolution readout channels that each include independent 20-bit ADCs and industry-leading ambient light cancellation (ALC) circuits, producing the highest performing integrated optical data acquisition system on the market today.

The ECG channel has EMI filtering, internal lead biasing, right-leg drive, and extensive calibration voltages for built-in self-test. The ECG channel also has high-input impedance, low noise, high CMRR, programmable gain, an anti-aliasing low-pass filter, and a high-resolution ADC. It is designed to meet IEC 60601-2-47 Ambulatory ECG Systems monitoring compliance requirements.

The BioZ receive channel has EMI filtering and extensive calibration features. The BioZ receive channel also has high input impedance, low noise, programmable gain, low-pass and high-pass filter options, and a high-resolution ADC. There are several modes for generating input stimulus: balanced square-wave source/sink current, sine-wave current, and both sine-wave and square-wave voltage stimuli. A wide range of stimulus magnitudes and frequencies is available.

The MAX86178 has DC and AC lead-off detection, a flexible timing system, and a PLL. All three sensor channels are synchronized. The MAX86178 is available in a 7 x 7 49-bump wafer-level package (WLP) with package dimensions of 2.77mm x 2.57mm, and operates over -40ºC to +85ºC temperature range.

## Applications

- Wearable Vital-Sign Monitors
- Ambulatory Heart Rate Monitors
- Pulse-Oximetry Devices
- Smart-Clothing Applications
- Impedance Cardiography/Hemodynamic Monitors
- Single- and Multi-Frequency Bioimpedance Analysis
- Pulse Arrival Time (PAT), Pulse Transit Time (PTT), Pulse Wave Velocity (PWV) Assessments

## Benefits and Features

PPG

- Dual-Channel Optical Data-Acquisition System
- 113dB $SpO_2$ System SNR
- Excellent Ambient Range and Rejection Capability
  - > 200µA Ambient Photodetector Current
  - > 90dB Ambient Rejection at 120Hz with Averaging

ECG

- Clinical Grade ECG Acquisition System
  - 15.3 ENOB with 0.72$\mu V_{RMS}$ (0.05Hz to 40Hz)
  - 230f$A_{RMS}$ Input-Current Noise (0.05Hz to 40Hz, +25°C)
- Fully Differential Signal with CMRR > 110dB at 50Hz and 60Hz
- High Input Impedance > 1GΩ for Extremely Low Common-to-Differential Mode Conversion
- High DC-Differential Input Range of ±1000mV (typ 1.8V) Allows a Wide Variety of Electrodes to be Used
- High AC-Dynamic Range of 200m$V_{P-P}$ Prevents Saturation Due to Motion, or Varying Half-Cell Potentials

BIOZ

- High-Performance BioZ Data Acquisition System
- Low-Noise BioZ Receive Channel (17 ENOB, 0.16µ$V_{RMS}$)
- Sine-Wave, Square-Wave, Current and Voltage Stimulus (16Hz to 500kHz)
- BIA/BIS Measurement with High Absolute Resistance and Reactance Accuracy

SYSTEM

- Ultra-Low-Power Systems for Wearable Devices
- DC and AC Lead-Off Detect Capability
- Shutdown Current of 0.5µA (typ)
- 256-Word FIFO for ECG, PPG, and BioZ
- Flexible Interrupt and Multi-AFE Synchronization
- FIFO Timing Data Allows Synchronized PPG, ECG, and BioZ

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***Ordering Information** appears at end of data sheet.*

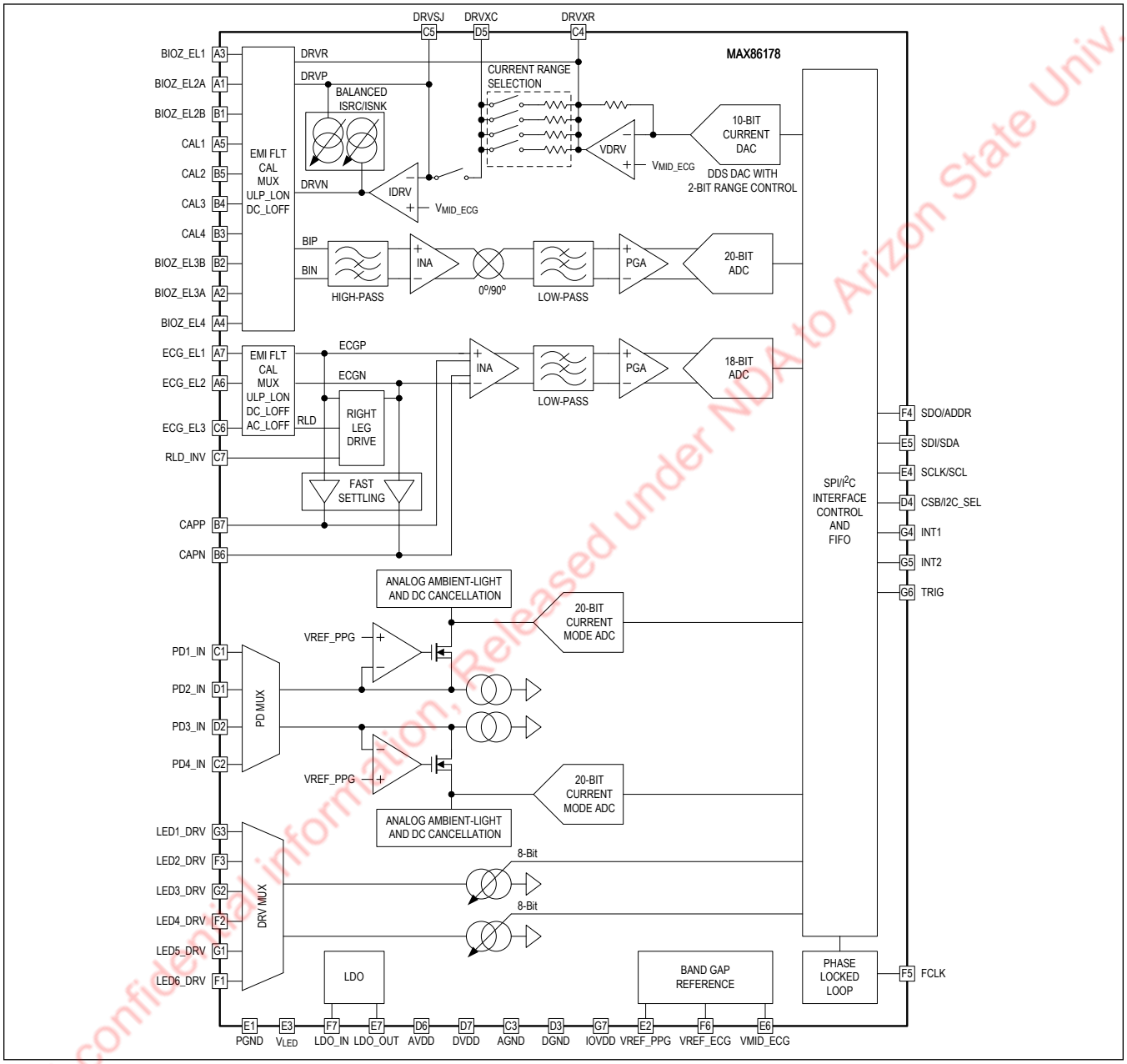
*19-100991; Rev 4; 8/23*

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## Simplified Block Diagram




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## Absolute Maximum Ratings

LDO_IN to AGND ..... -0.3V to +6.0V
IOVDD to DGND..... -0.3V to +6.0V
AVDD to AGND ..... -0.3V to +2.2V
DVDD to DGND..... -0.3V to +2.2V
$V_{LED}$ to PGND..... -0.3V to +6.0V
AVDD to DVDD ..... -0.3V to +0.3V
DGND, PGND to AGND ..... -0.3V to +0.3V
PDm_IN to AGND..... -0.3V to +2.2V
LEDn_DRV to PGND ..... -0.3V to ($V_{LED}$ + 0.3V)
CSB/I2C_SEL, SCLK/SCL, SDI/SDA to DGND .... -0.3V to +6.0V
TRIG, FCLK, INT1, INT2 to DGND ..... -0.3V to +6.0V
SDO/ADDR to DGND ..... -0.3V to (IOVDD + 0.3V)
Output Short-Circuit Duration..... Continuous
All Other Pins to AGND..... -0.3V to +2.2V
Continuous Current into Any Pin..... ±50mA
Operating Temperature Range ..... -40ºC to +85ºC
Junction Temperature ..... +150ºC
Storage Temperature Range ..... -65ºC to +150ºC
Lead Temperature (Soldering, 10sec) ..... +300ºC
Soldering Temperature (Reflow) ..... +260ºC

*Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.*

## Package Information

### 49 WLP

| | |
|---|---|
| Package Code | N492A2+1 |
| Outline Number | 21-100520 |
| Land Pattern Number | Refer to *Application Note 1891* |
| **THERMAL RESISTANCE, FOUR-LAYER BOARD** | |
| Junction-to-Ambient ($\theta_{JA}$) | 48.76°C/W |

For the latest package outline information and land patterns (footprints), go to *www.maximintegrated.com/packages*. Note that a "+", "#", or "-" in the package code indicates RoHS status only. Package drawings may show a different suffix character, but the drawing pertains to the package regardless of RoHS status.

Package thermal resistances were obtained using the method described in JEDEC specification JESD51-7, using a four-layer board. For detailed information on package thermal considerations, refer to *www.maximintegrated.com/thermal-tutorial*.

## Electrical Characteristics

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|
| **POWER SUPPLY** | | | | | | |
| LDO_IN Supply Voltage | $V_{LDO_IN}$ | Guaranteed by DC PSRR | 2.3 | | 5.5 | V |
| LDO_OUT Voltage | $V_{LDO_OUT}$ | $V_{LDO_IN}$ > 2.3V, LDO_OUT Load Current = 10mA. | 1.70 | 1.8 | 1.95 | V |
| IOVDD Supply Voltage | $V_{IOVDD}$ | Guaranteed by Digital I/O $V_{IL}$ and $V_{IH}$ Thresholds | 1.70 | | 5.5 | V |
| $V_{LED}$ Supply Voltage (Note 3) | $V_{LED}$ | Guaranteed by DC PSRR | 3.1 | | 5.5 | V |
| AVDD Supply Voltage | $V_{AVDD}$ | Guaranteed by DC PSRR | 1.7 | 1.8 | 2.0 | V |
| DVDD Supply Voltage | $V_{DVDD}$ | Guaranteed by Scan Test | 1.7 | 1.8 | 2.0 | V |

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## Electrical Characteristics (continued)

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|---|
| **POWER SUPPLY / PPG SUPPLY CURRENT** | | | | | | | |
| PPG Supply Current ($I_{AVDD}$ + $I_{DVDD}$) | $I_{PPG}$ | Single Channel, One Meas/Frame (Note 4a) | FR = 512fps | | 91 | 140 | µA |
| | | | FR = 64fps | | 16 | | |
| | | | FR = 8fps | | 7.5 | | |
| | | Single Channel, Four Meas/Frame (Note 4a) | FR = 512fps | | 325 | 450 | |
| | | | FR = 64fps | | 52 | | |
| | | | FR = 8fps | | 20 | | µA |
| | | Dual Channel, One Meas/Frame (Note 4b) | FR = 512fps | | 140 | 210 | µA |
| | | | FR = 64fps | | 23 | | |
| | | | FR = 8fps | | 9.5 | | |
| | | Dual Channel, Four Meas/Frame (Note 4b) | FR = 512fps | | 530 | 672 | |
| | | | FR = 64fps | | 87 | | |
| | | | FR = 8fps | | 30 | | |
| $V_{LED}$ Supply Current | $I_{LED}$ | Single Channel, One Meas/Frame (Note 4a) | FR = 512fps | | 840 | 970 | µA |
| | | | FR = 64fps | | 135 | | |
| | | | FR = 8fps | | 40 | | |
| **POWER SUPPLY / ECG SUPPLY CURRENT** | | | | | | | |
| Biopotential Channel Supply Current ($I_{AVDD}$+$I_{DVDD}$) | $I_{BPC}$ | ECG Mode (Note 5) | RLD Amplifier Disabled, ECG Lead Bias = 100MΩ, M = 128, ECG_FDIV = 16, ECG_NDIV = 8, SR_ECG = 128sps | | 105 | 150 | µA |
| | | | RLD Enabled, M = 128, ECG_FDIV = 16, ECG_NDIV = 8, SR_ECG = 128sps | | 107 | 150 | |
| | | | RLD Amplifier Disabled, ECG Lead Bias = 100MΩ, M = 125, ECG_FDIV = 16, ECG_NDIV = 8, SR_ECG = 250sps | | 105 | 150 | |

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## Electrical Characteristics (continued)

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|---|
| **POWER SUPPLY / BIOZ SUPPLY CURRENT** | | | | | | | |
| BIA Supply Current ($I_{AVDD}$+$I_{DVDD}$) | $I_{BIOZ}$ | BIA/BIS Conditions (Note 6a) | F_BIOZ = 1kHz, SR_BIOZ = 62.5sps, PLL_CLK = 16.384MHz | | 880 | | µA |
| | | | F_BIOZ = 50.048kHz, SR_BIOZ = 48.875sps, PLL_CLK = 12.812MHz | | 1030 | 2250 | µA |
| | | | F_BIOZ = 50.048kHz, SR_BIOZ = 48.875sps, PLL_CLK = 12.812MHz, REF_CLK_SEL = 0 | | 1020 | | |
| | | | F_BIOZ = 100.096kHz, SR_BIOZ = 48.875sps, PLL_CLK = 12.812MHz | | 1025 | | |
| | | | F_BIOZ = 249.856kHz, SR_BIOZ = 61sps, PLL_CLK = 15.991MHz | | 1100 | 2350 | µA |
| | | | F_BIOZ = 499.712kHz, SR_BIOZ = 61sps, PLL_CLK = 15.991MHz | | 1130 | | |
| Respiration Supply Current ($I_{AVDD}$ + $I_{DVDD}$) | $I_{RESP}$ | Respiration Conditions (Note 6b) | F_RESP = 32kHz, SR_RESP = 62.5sps, PLL_CLK = 8.192MHz , RESP_EN = 1 | | 325 | 425 | µA |
| | | | F_BIOZ = 32kHz, SR_BIOZ = 31.25sps, PLL_CLK = 4.096MHz, BIOZ_DRV_MODE = H-Bridge, DRVP and DRVN unconnected | | 150 | | |

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## Electrical Characteristics (continued)

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|---|
| ICG Supply Current ($I_{AVDD}$ + $I_{DVDD}$) | $I_{ICG}$ | ICG Conditions (Note 6c) | F_BIOZ = 100.096kHz, SR_BIOZ = 97.75sps, PLL_CLK = 12.812MHz | | 1925 | | µA |
| GSR/EDA Supply Current ($I_{AVDD}$ + $I_{DVDD}$) | $I_{GSR_EDA}$ | GSR/EDA Conditions (Note 6d) | F_BIOZ = 64Hz , SR_BIOZ = 32sps, PLL_CLK = 4.194MHz, $I_{DRV}$ = 320n$A_{RMS}$, BIOZ_GAIN = 1V/V | | 220 | | µA |
| **POWER SUPPLY / ECG + BIOZ SUPPLY CURRENT** | | | | | | | |
| ECG + BIOZ Supply Current ($I_{AVDD}$+$I_{DVDD}$) | | Respiration and ECG Conditions (Note 7) | F_RESP = 32kHz, SR_RESP = 31.25sps, SR_ECG = 250sps, PLL_CLK = 4.096MHz | | 285 | 450 | µA |
| **POWER SUPPLY / PPG + ECG + BIOZ SUPPLY CURRENT** | | | | | | | |
| PPG + ECG + BIOZ Supply Current ($I_{AVDD}$+$I_{DVDD}$) | | Respiration, ECG, and PPG Conditions (Note 8) | | | 435 | | µA |
| **POWER SUPPLY / IOVDD SUPPLY CURRENT** | | | | | | | |
| IOVDD Supply Current | $I_{IOVDD}$ | SCLK = 24MHz | | | 11.5 | 17 | µA |
| **POWER SUPPLY / SHUTDOWN CURRENT** | | | | | | | |
| Shutdown Current ($I_{AVDD}$ + $I_{DVDD}$) | $I_{SHDN}$ | $T_A$ = +25°C | | | 1.2 | 3 | µA |
| | | | | | | 25 | |
| $V_{LED}$ Shutdown Current | $I_{VLED_SHDN}$ | $T_A$ = +25°C | | | | 0.25 | µA |
| | | | | | | 0.5 | |
| IOVDD Shutdown Current | $I_{IOVDD_SHDN}$ | | | | | 0.5 | µA |
| Lead-On Detect Current | | EN_ECG_LON = 1 or EN_BIOZ_LON = 1 | | | 3.6 | 28 | µA |
| **PPG CHARACTERISTICS / READOUT CHANNEL** | | | | | | | |
| ADC Resolution | | | | | 20 | | bits |
| INL | $INL_{RX}$ | MEASx_TINT = 117.0µs | | | ±10 | | LSB |
| | | MEASx_TINT = 14.6µs | | | ±40 | | |
| DNL | $DNL_{RX}$ | MEASx_TINT = 117.0µs | | | ±3 | | LSB |
| | | MEASx_TINT = 14.6µs | | | ±10 | | |
| ADC Full-Scale Input Current | $I_{FS}$ | MEASx_PPGy_ADC_RGE = 0x0 | | | 4.0 | | µA |
| | | MEASx_PPGy_ADC_RGE = 0x1 | | | 8.0 | | |
| | | MEASx_PPGy_ADC_RGE = 0x2 | | | 16.0 | | |
| | | MEASx_PPGy_ADC_RGE = 0x3 | | | 32.0 | | |

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## Electrical Characteristics (continued)

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|---|
| Offset DAC Full-Scale Current | | MEASx_PPGy_DACOFF = 0xE | | | 30.0 | | µA |
| ADC Integration Time | $t_{INT}$ | MEASx_TINT = 0x0 | | | 14.6 | | µs |
| | | MEASx_TINT = 0x1 | | | 29.2 | | |
| | | MEASx_TINT = 0x2 | | | 58.6 | | |
| | | MEASx_TINT = 0x3 | | | 117.1 | | |
| Minimum Free-Running Frame Rate | | CLK_FREQ_SEL = 32.768kHz | | | 1 | | fps |
| Maximum Free-Running Frame Rate | | CLK_FREQ_SEL = 32.768kHz | | | 1927.53 | | fps |
| Internal Power-Up Time | | | | | 200 | | µs |
| ADC Clock Frequency | | | | 9.8 | 10 | 10.2 | MHz |
| DC Ambient-Light Rejection Range | ALR | ALC_OVF_ENx = 1 | | 0 | | 200 | µA |
| Dynamic Ambient-Light Rejection | | $I_{EXPOSURE}$ = 1µA, $I_{AMBIENT}$ = 1µA DC with 0.4µ$A_{P-P}$ 120Hz sinewave | | | 80 | | dB |
| DC Ambient-Light Rejection | | $I_{EXPOSURE}$ = 1µA, $I_{AMBIENT}$ = 1µA and 30µA | | | 0.5 | | nA |
| Dark Current Offset | DC_O | ALC = ON, PDm_BIAS = 0x1, MEASx_TINT = 117.0µs | | | ±1 | | Counts |
| Dark Current-Input Referred Noise | | MEASx_TINT = 14.6µs | | | 212 | | $pA_{RMS}$ |
| | | MEASx_TINT = 29.2µs | | | 150 | | |
| | | MEASx_TINT = 58.6µs | | | 106 | | |
| | | MEASx_TINT = 117.1µs | | | 75 | | |
| $V_{DD}$ DC PSR | | $V_{AVDD}$, $V_{DVDD}$ = 1.7V to 2.0V (Internal Loop Back mode) | MEASx_TINT = 14.6µs | | 0.9 | | %FS/V |
| | | | MEASx_TINT = 29.2µs | | 0.8 | | |
| | | | MEASx_TINT = 58.6µs | | 0.7 | | |
| | | | MEASx_TINT = 117.1µs | | 0.7 | 1.25 | |
| **PPG CHARACTERISTICS / LED DRIVER** | | | | | | | |
| LED Current Resolution | | | | | 8 | | bits |
| Driver DNL | $DNL_{TX}$ | MEASx_LED_RGE = 0x3 | | -1 | | +1 | LSB |
| Driver INL | $INL_{TX}$ | MEASx_LED_RGE = 0x3 | | | ±1 | | LSB |

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## Electrical Characteristics (continued)

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|---|
| Full-Scale LED Current | $I_{LED}$ | MEASx_DRVy_PA = 0xFF | MEASx_LED_RGE = 0x0 | | 32 | | mA |
| | | | MEASx_LED_RGE = 0x1 | | 64 | | |
| | | | MEASx_LED_RGE = 0x2 | | 96 | | |
| | | | MEASx_LED_RGE = 0x3 | 116 | 128 | 140 | |
| LED Driver Rise Time | | MEASx_DRVy_PA = 0xFF, 10% to 90%, All LED Range Settings | | | | 3 | µs |
| LED Driver Fall Time | | MEASx_DRVy_PA = 0xFF, 10% to 90%, All LED Range Settings | | | | 3 | µs |
| Minimum Output Voltage | $V_{OL}$ | MEASx_DRVy_PA = 0xFF, 95% of the Desired LED Current | MEASx_LED_RGE = 0x0 | | 170 | | mV |
| | | | MEASx_LED_RGE = 0x1 | | 260 | | |
| | | | MEASx_LED_RGE = 0x2 | | 380 | | |
| | | | MEASx_LED_RGE = 0x3 | | 500 | 800 | |
| LED Driver DC $V_{LED}$ PSR | | MEASx_DRVy_PA = 0xFF, $V_{LEDn_DRV}$ = 1.2V, $V_{LED}$ = 3.1V to 5.5V | MEASx_LED_RGE = 32mA | | ±16 | | µA/V |
| | | | MEASx_LED_RGE = 64mA | | ±13 | | µA/V |
| | | | MEASx_LED_RGE = 96mA | | ±14 | | |
| | | | MEASx_LED_RGE = 128mA | -200 | ±18 | +200 | |
| LED Driver Compliance Interrupt Threshold | | MEASx_LED_RGE = 0x0 | | 115 | 148 | 180 | mV |
| | | MEASx_LED_RGE = 0x1 | | 255 | 287 | 320 | |
| | | MEASx_LED_RGE = 0x2 | | 395 | 425 | 460 | |
| | | MEASx_LED_RGE = 0x3 | | 530 | 560 | 600 | |
| **ECG CHARACTERISTICS / READOUT CHANNEL** | | | | | | | |
| Input-Voltage Range | $V_{ECGP}$, $V_{ECGN}$ | $f_{IN}$ = 64Hz, $V_{IN}$ = 12m$V_{P-P}$, shift from gain at 0.76V < 2.5% | | 0.3 | | AVDD-0.6 | V |
| | | $f_{IN}$ = 64Hz, $V_{IN}$ = 12m$V_{P-P}$, shift from gain at 0.76V < 2.5% | | | 0.15 to 1.35 | | |
| DC Differential-Input Range | $V_{ECGP-ECGN}$ | $f_{IN}$ = 64Hz, $V_{IN}$ = 12m$V_{P-P}$, shift from gain at 0V < 2.5% | | -1000 | | +1000 | mV |
| AC Differential-Input Range | $V_{IN}$ | $f_{IN}$ = 64Hz, INA_GAIN = 10V/V, THD < 0.3% | | 170 | 195 | | m$V_{P-P}$ |

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## Electrical Characteristics (continued)

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|---|
| Common-Mode Rejection Ratio (Note 9) | CMRR | $f_{IN}$ = 64Hz, Electrode Offset = ±300mV, $V_{CM}$ = 0.76V, $T_A$ = +25ºC | 0Ω Source Impedance | 100 | 114 | | dB |
| | | | ETI Impedance Mismatch as Described in Note 10 | | 78 | | |
| | | | ETI Impedance Mismatch as Described in Note 10. RLD Closed Loop Gain = 97V/V. | 100 | 107 | | |
| Input-Referred Noise (Note 9) | | $T_A$ = 25°C | BW = 0.05Hz to 150Hz | | 0.99 | 1.6 | $\mu V_{RMS}$ |
| | | | BW = 0.05Hz to 150Hz | | 6.00 | 12 | $\mu V_{P-P}$ |
| | | | BW = 0.05Hz to 40Hz | | 0.72 | 1.3 | $\mu V_{RMS}$ |
| | | | BW = 0.05Hz to 40Hz | | 3.99 | 7.9 | $\mu V_{P-P}$ |
| | | | BW = 0.05Hz to 40Hz, ECG_INA_GAIN = 10V/V | | 0.95 | 1.9 | $\mu V_{RMS}$ |
| | | | BW = 0.05Hz to 40Hz, ECG_INA_GAIN = 10V/V | | 5.26 | 10.6 | $\mu V_{P-P}$ |
| Input-Current Noise Density | | 100MΩ external bias resistor to $V_{MID_ECG}$ | | | 37 | | $fA/\sqrt{Hz}$ |
| ECGP and ECGN Input-Leakage Current | | $V_{ECGP/N}$ = AVDD - 0.2V or AGND + 0.2V, $T_A$ = +25°C | | -500 | ±50 | +500 | pA |
| | | $V_{ECGP/N}$= AVDD - 0.2V or AGND + 0.2V | | -2500 | | +2500 | |
| Input Impedance (Note 9) | INA | Differential, $f_{IN}$ = 8Hz | | | 1.0 | | GΩ |
| | | Differential, $f_{IN}$ = 64Hz | | | 0.74 | | |
| | | Common-Mode, $f_{IN}$ = 8Hz | | | 14 | | |
| | | Common-Mode, $f_{IN}$ = 64Hz | | | 5 | | |
| Total Harmonic Distortion | THD | $f_{IN}$ = 64Hz, Electrode Offset = ±300mV | $V_{IN}$ = 65m$V_{P-P}$ | | 0.035 | | % |
| | | | $V_{IN}$ = 30m$V_{P-P}$ | | | 0.3 | |
| Biopotential-Channel Gain Setting | GCH | Selected by ECG_INA_GAIN and ECG_PGA_GAIN | | | 2.5 to 960 | | V/V |

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## Electrical Characteristics (continued)

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|---|
| Biopotential-Channel Gain Error | | $f_{IN}$ = 64Hz, $V_{IN}$ = 12m$V_{P-P}$, ECG_PGA_GAIN = X | ECG_INA_RGE = 0x0, ECG_INA_GAIN = 0x1 | -2.5 | | +2.5 | % |
| | | | ECG_INA_RGE = 0x0, 0x1, ECG_INA_GAIN = 0x0, 0x1, 0x2 | | ±1.8 | | |
| | | | ECG_INA_RGE = 0x2, 0x3, ECG_INA_GAIN = 0x0, 0x1, 0x2 | | ±2.8 | | |
| | | | ECG_INA_RGE = X, ECG_INA_GAIN = 0x3 | | ±5.0 | | |
| Biopotential-Channel Input-Referred Offset Voltage | | ECG_INA_GAIN = 20V/V | | | ±400 | | µV |
| ADC Resolution | | | | | 18 | | bits |
| ADC Sample Rate | | | | | 64 to 2048 | | sps |
| CAPP to CAPN Impedance | $R_{HPF}$ | $f_{HPF_3dB}$ = 1/(2π x $R_{HPF}$ x $C_{HPF}$), $C_{HPF}$ = Capacitance between CAPP and CAPN, ECG_IMP_HI = 0 | | 240 | 400 | 650 | kΩ |
| Analog High-Pass Filter Slew Current | | Fast Recovery Enabled | | | 100 | | µA |
| | | Fast Recovery Disabled | | | 0.3 | | |
| Biopotential-Channel Power-Supply Rejection | PSRR | Lead Bias Disabled, $V_{ECGP}$ = $V_{ECGN}$ = $V_{MID_ECG}$, Change in Input Referred Offset | $V_{LDO_IN}$ = 2.3V to 5.5V, | 94 | 114 | | dB |
| | | | $V_{LDO_IN}$ = AGND, $V_{AVDD}$ = $V_{DVDD}$ = 1.7V to 2.0V | 84 | 97 | | |
| **ECG CHARACTERISTICS / ECG MUX / DC/AC LEAD-OFF DETECTION** | | | | | | | |
| Full-Scale Current | | Selected by ECG_LOFF_IMAG | | | 5 to 400 | | nA |
| Full-Scale Current Accuracy | | ECG_LOFF_IMAG = 400nA | $T_A$ = 25°C | | ±5 | | % |
| | | | | -30 | | +30 | |
| Comparator Threshold | $V_{ECG_TH_H}$, $V_{ECG_TH_L}$ | Selected by ECG_LOFF_THRESH | | $V_{MID_ECG}$ ±25 | | $V_{MID_ECG}$ ±400 | mV |
| Comparator-Threshold Accuracy | | ECG_LOFF_THRESH = ±400mV | | -12.5 | | +12.5 | % |
| Full-Scale Electrode Resistance | | ECG_LOFF_IMAG = 400nA, ECG_LOFF_THRESH = ±25mV | | | 62.5 | | kΩ |
| | | ECG_LOFF_IMAG = 5nA, ECG_LOFF_THRESH = ±400mV | | | 80 | | MΩ |
| AC Lead-Off Frequency | | Selected by ECG_LOFF_FREQ and ECG_ADC_CLK (see *Timing Subsystem*) | | | 74 to 8192 | | Hz |

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## Electrical Characteristics (continued)

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|---|
| **ECG CHARACTERISTICS / ECG MUX / LEAD BIAS** | | | | | | | |
| Lead-Bias Impedance | | ECG Lead-Bias Enabled to $V_{MID_ECG}$ or $V_{RLD}$ (see the RLD_RBIAS bit) | ECG_RBIAS_VALUE = 0x0 | | 50 | | MΩ |
| | | | ECG_RBIAS_VALUE = 0x1 | | 100 | | |
| | | | ECG_RBIAS_VALUE = 0x2 | | 200 | | |
| Lead-Bias Voltage | | ECG Lead-Bias Enabled, RLD_RBIAS = 0 | | 0.66 | 0.76 | 0.86 | V |
| **ECG CHARACTERISTICS / ECG MUX / CALIBRATION PULSE** | | | | | | | |
| Calibration-Voltage Magnitude | | Single-Ended | ECG_CAL_MAG = 0x0 | | 0.25 | | mV |
| | | | ECG_CAL_MAG = 0x1 | | 0.5 | | |
| Calibration-Voltage Magnitude Error | | ECG_CAL_MAG = 0x1, Bipolar Mode (Note 11) | $T_A$ = +25°C | -3 | | +3 | % |
| | | | | -6 | | +6 | |
| Calibration-Voltage Frequency | | Selected by ECG_CAL_FREQ | | | 0.0156 to 256 | | Hz |
| Calibration-Voltage Pulse-Time | | ECG_CAL_DUTY = 0x0, Selected by ECG_CAL_HIGH | | | 0.03052 to 62.474 | | µs |
| | | ECG_CAL_DUTY = 0x1 | | | 50 | | % |
| **ECG CHARACTERISTICS / RIGHT LEG / BODY BIAS DRIVE** | | | | | | | |
| Output-Voltage Range | | RLD_GAIN = 24V/V, Shift from Small Signal Gain < 6.5% | | 0.45 | | $V_{AVDD}$ - 0.45 | V |
| RLD Out-of-Range Comparator Limits | | High Limit, RLD_OOR asserted | | | 0.870 x $V_{AVDD}$ | | V |
| | | Low Limit, RLD_OOR asserted | | | 0.127 x $V_{AVDD}$ | | |
| Input-Referred Noise of RLD Amplifier | | BW = 0.05Hz to 150Hz | | | 8 | | $µV_{P-P}$ |
| DC Gain Accuracy | | RLD_GAIN = 24V/V | | -5 | | +5 | % |
| | | External Gain Resistor: RLD_EXT = 3.65MΩ, Nominal $R_{INT}$ = 150kΩ, RLD_EXT_RES = 0x1 | | -14 | ±9 | +14 | |
| **BIOZ CHARACTERISTICS / RECEIVE PATH** | | | | | | | |
| ADC Resolution | | | | | 20 | | bits |
| ENOB | | BIOZ_ADC_OSR = 128 | | | 16.3 | | bits |
| | | BIOZ_ADC_OSR = 8 | | | 9.6 | | |
| ADC Sample Rate | | Programmable, see *Timing Subsystem* | | | 16 to 4546 | | sps |
| Phase-Measurement Accuracy at 16Hz | | 453kΩ in series with 22nF load at 16Hz after calibration (Note 12), $T_A$ = +25°C | | | ±1 | | deg |

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## Electrical Characteristics (continued)

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|---|
| Phase-Measurement Accuracy at 50kHz | | Cole Impedance (800Ω \|\| (2500Ω + 1nF)) load at 50kHz after calibration (Note 12), $T_A$ = +25°C. See TOC87 for Histogram Distribution (Note 9) | | -0.1 | | +0.1 | deg |
| Phase-Measurement Accuracy at 500kHz | | 316Ω in series with 1nF load at 500kHz after calibration (Note 12), $T_A$ = +25°C | | | ±1 | | deg |
| Phase-Measurement Drift | | Drift of impedance phase at 50kHz after calibration (Note 12), drift relative to $T_A$ = +25°C (Note 9) | | | ±0.01 | | deg/°C |
| Magnitude-Measurement Accuracy at 16Hz | | 453kΩ in series with 22nF load at 16Hz after calibration (Note 12), $T_A$ = +25°C | | | ±1 | | % |
| Magnitude-Measurement Accuracy at 50kHz | | Cole Impedance (800Ω \|\| (2500Ω + 1nF)) load at 50kHz after calibration (Note 12), $T_A$ = +25°C. See TOC86 for Histogram Distribution (Note 9) | | -0.1 | | +0.1 | % |
| Magnitude-Measurement Accuracy at 500kHz | | 316Ω in series with 1nF load at 500kHz after calibration (Note 12), $T_A$ = +25°C | | | ±1 | | % |
| Magnitude-Measurement Drift | | Drift of impedance magnitude at 50kHz after calibration (Note 12), drift relative to $T_A$ = +25°C | | | ±0.01 | | %/°C |
| Impedance Repeatability | | $R_{BODY}$ = 100kΩ, conditions for GSR (Note 13a) | | | 4.4 | | $Ω_{RMS}$ |
| | | $R_{BODY}$ = 680Ω, conditions for respiration (Note 13b) | | | 8.3 | | $mΩ_{RMS}$ |
| | | $R_{BODY}$ = 25Ω, conditions for ICG/AED body impedance (Note 13c) | | | 1.0 | | |
| DC Power-Supply Rejection | PSRR | $I_{DRV}$ = 64μ$A_{RMS}$, F_BIOZ = 1kHz, BIOZ_GAIN = 10V/V, $R_{BODY}$ = 1kΩ | $V_{LDO_IN}$ = 2.3V to 5.5V | | 5 | 125 | LSB/V |
| | | | LDO_IN = AGND, $V_{AVDD}$ = $V_{DVDD}$ = 1.7V to 2.0V | | 600 | 4500 | |
| | | BIOZ_DRV_MODE = 0x1, BIOZ_VDRV_MAG = 50m$V_{PK}$, F_BIOZ = 64Hz, BIOZ_GAIN = 10V/V | $V_{LDO_IN}$ = 2.3V to 5.5V | | 10 | 80 | |
| | | | LDO_IN = AGND, $V_{AVDD}$ = $V_{DVDD}$ = 1.7V to 2.0V | | 425 | 3500 | |
| Channel Gain | | Selected by BIOZ_GAIN | | | 1 to 10 | | V/V |
| Channel-Gain Accuracy | | $f_{IN}$=1kHz, BIOZ_GAIN = 10V/V, $V_{IN}$ =100m$V_{P-P}$ | | -5 | | +5 | % |
| Input-Voltage Range | | $f_{IN}$ = 1kHz, $V_{IN}$ = 100m$V_{P-P}$, BIOZ_GAIN = 10V/V, shift from gain at 0.76V < 2% | | 0.5 | | $V_{AVDD}$ - 0.75 | V |

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## Electrical Characteristics (continued)

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|---|
| Maximum AC-Differential Input Range | | $f_{IN}$=1kHz, BIOZ_GAIN = 1V/V, shift from small-signal gain at 100m$V_{P-P}$ < 2% | | | 1000 | | $mV_{P-P}$ |
| Input-Referred Voltage Noise | | $R_{BODY}$ = 0Ω, DRVP and DRVN Disconnected, BW = 0.05Hz to 67Hz (Note 14) | | | 1.6 | | $\mu V_{RMS}$ |
| Input-Referred Current Noise | | DRVP and DRVN disconnected, BW = 0.05Hz to 67Hz (Note 14) | | | 300 | | $fA/\sqrt{Hz}$ |
| Input-Referred Offset Voltage | | $R_{BODY}$ = 0Ω, BIOZ_GAIN = 10V/V, BIOZ_AHPF = bypassed, BIOZ_DM_DIS = 1 | | -3.0 | | +3.0 | mV |
| Differential Input Impedance | | Lead bias disabled, BIOZ_AHPF = bypassed, impedance at 1kHz | | | 3.0 // 2.4 | | GΩ // pF |
| Common-Mode Input Impedance | | Lead bias disabled, BIOZ_AHPF = bypassed, impedance at 1kHz | | | 2.2 // 1.5 | | GΩ // pF |
| Input-Analog High-Pass Filter | | Programmable from the Register Map | | | 100 to 10,000 | | Hz |
| Input-Analog High-Pass Filter Variation | | BIOZ_AHPF = 200Hz | | -50 | | +100 | % |
| Input-Analog High-Pass Filter Resistor (Differential) | | Differential Filter Resistance | BIOZ_AHPF = 0x8 | 17.0 | 44.8 | 84.8 | MΩ |
| | | | BIOZ_AHPF = 0x9 | 10.6 | 23.1 | 40.3 | |
| | | | BIOZ_AHPF = 0xA | 4.2 | 9.3 | 16.0 | |
| | | | BIOZ_AHPF = 0xB | 2.3 | 4.6 | 7.1 | |
| | | | BIOZ_AHPF = 0xC | 1.1 | 2.2 | 3.6 | |
| | | | BIOZ_AHPF = 0xD or 0xE | 0.46 | 0.92 | 1.40 | |
| Input-Analog High-Pass Filter Bias Voltage | $V_{MID_RX}$ | | | 700 | 760 | 820 | mV |
| Input-Leakage Current | | BIP, BIN = AVDD - 0.2V or AGND + 0.2V, $T_A$ = +25°C | BIOZ_ELx, CAL1 and CAL2 | -2 | 0.1 | +2 | nA |
| | | | CAL3 and CAL4 (Note 7) | -2 | 0.1 | +2 | |
| **BIOZ CHARACTERISTICS / TRANSMIT PATH** | | | | | | | |
| DDS DAC Sine Wave Resolution | | | | | 10 | | bits |
| Current-Drive THD | | $I_{DRV}$ = 64$\mu A_{RMS}$, $f_{STIM}$ = 50kHz, $R_{BODY}$ = 1kΩ. Includes odd harmonics h3, h5, and h7. | | | 0.05 | 0.25 | % |
| Current-Drive Amplitude Resolution | | See BIOZ_VDRV_MAG and BIOZ_IDRV_RGE | | | 4 | | bits |
| Current Drive (Range 1) | | BIOZ_VDRV_MAG = 0x0, 0x1, 0x2, 0x3 | BIOZ_IDRV_RGE = 0x0 | | 16, 32, 80, 160 | | $nA_{RMS}$ |
| Current Drive (Range 2) | | BIOZ_VDRV_MAG = 0x0, 0x1, 0x2, 0x3 | BIOZ_IDRV_RGE = 0x1 | | 0.32, 0.64, 1.6, 3.2 | | $\mu A_{RMS}$ |

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## Electrical Characteristics (continued)

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|---|
| Current Drive (Range 3) | | BIOZ_VDRV_MAG = 0x0, 0x1, 0x2, 0x3 | BIOZ_IDRV_RGE = 0x2 | | 6.4, 12.8, 32, 64 | | $\mu A_{RMS}$ |
| Drive Current (Range 4) | | BIOZ_VDRV_MAG = 0x0, 0x1, 0x2, 0x3 | BIOZ_IDRV_RGE = 0x3 | | 128, 256, 640, 1280 | | $\mu A_{RMS}$ |
| Current-Drive Accuracy | | $I_{DRV}$ ≥ 32nA | | -6 | | +6 | % |
| | | $I_{DRV}$ = 16nA | | -10 | | +10 | |
| Current-Drive Temperature Coefficient | | | | | 250 | | ppm/ºC |
| Short-Circuit Current | | BIOZ_DRV_MODE = Voltage Mode, BIOZ_VDRV_MAG = 0x11, BIOZ_AMP_BW = 0x3, BIOZ_AMP_RGE = 0x3, DRVP, DRVN shorted to AGND, AVDD | | 120 | | 650 | µA |
| | | BIOZ_DRV_MODE = H-Bridge Mode, DRVP, DRVN shorted to AGND, AVDD | | 0.69 | | 1.65 | mA |
| Drive Frequency Range | | | | | 0.016 to 500 | | kHz |
| Current-Drive Compliance Voltage | $V_{DRVP}$, $V_{DRVN}$ | BIOZ_DRV_MODE = 0x0, shift from current at 0.76V < ±4% | | 0.4 | | $V_{AVDD}$ - 0.4 | V |
| Current-Drive Out-of-Range Comparator Limits | | EN_BIOZ_DRV_OOR = 1 | | 0.27 | | $V_{AVDD}$ - 0.52 | V |
| Drive CM Voltage | $V_{MID_TX}$ | Voltage at DRVSJ in Current Mode | | 700 | 760 | 820 | mV |
| **BIOZ CHARACTERISTICS / DIGITAL FILTER** | | | | | | | |
| Output-Digital Low-Pass Filter | | BIOZ_DLPF = 0x1 | | | 0.005 x SR_BIOZ | | Hz |
| | | BIOZ_DLPF = 0x2 | | | 0.02 x SR_BIOZ | | |
| | | BIOZ_DLPF = 0x3 | | | 0.08 x SR_BIOZ | | |
| | | BIOZ_DLPF ≥ 0x4 | | | 0.25 x SR_BIOZ | | |
| | | BIOZ_DHPF = 0x1 | | | 0.00025 x SR_BIOZ | | |
| | | BIOZ_DHPF ≥ 0x2 | | | 0.002 x SR_BIOZ | | |

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## Electrical Characteristics (continued)

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|---|
| **BIOZ CHARACTERISTICS / RESPIRATION** | | | | | | | |
| Current Amplitude | | Programmable, see RESP_CGMAG | RESP_CG_MAG_4X = 0 | | 8 to 96 | | µA |
| | | | RESP_CG_MAG_4X = 1 | | 32 to 384 | | |
| Current-Amplitude Accuracy | | RESP_CG_MAG = 0x7 | RESP_CG_MAG_4X = 0 | -20 | | +20 | % |
| | | | RESP_CG_MAG_4X = 1 | -25 | | +25 | |
| Current-Compliance Voltage | | RESP_CG_MAG_4X = 1, RESP_CG_MAG = 384µA, shift from current at 0.76V < 5% | | 0.7 | | $V_{AVDD}$ - 0.7 | V |
| Frequency Range | | See *Timing Subsystem* | | | 10 to 131 | | kHz |
| Impedance Repeatability | | $R_{BODY}$ = 680Ω, conditions for respiration (Note 13d) | | | 6.3 | | $mΩ_{RMS}$ |
| **BIOZ I/O MUX / DC LEAD-OFF DETECTION** | | | | | | | |
| Full-Scale Current | | Selected by BIOZ_LOFF_IMAG | | | 5,10,20, 50,100 | | nA |
| Full-Scale Current Accuracy | | BIOZ_LOFF_IMAG = 100nA | $T_A$ = 25°C | | ±5 | | % |
| | | | | -30 | | +30 | |
| Comparator Threshold | $V_{BIOZ_TH_H}$, $V_{BIOZ_TH_L}$ | Selected by BIOZ_LOFF_THRESH | | $V_{MID_ECG}$ ± 200 | | $V_{MID_ECG}$ ± 575 | mV |
| Comparator-Threshold Accuracy | | BIOZ_LOFF_THRESH = ±575mV | | -12.5 | | +12.5 | % |
| Full-Scale Electrode Resistance | | BIOZ_LOFF_IMAG = 100nA, BIOZ_LOFF_THRESH = ±200mV | | | 4 | | MΩ |
| | | BIOZ_LOFF_IMAG = 10nA, BIOZ_LOFF_THRESH = ±500mV | | | 100 | | |
| **BIOZ I/O MUX / LEAD BIAS** | | | | | | | |
| Lead-Bias Impedance | | BioZ lead-bias enabled to $V_{MID_ECG}$ or $V_{RLD}$ (see the RLD_RBIAS bit) | BIOZ_RBIAS_VALUE = 0x0 | | 50 | | MΩ |
| | | | BIOZ_RBIAS_VALUE = 0x1 | | 100 | | |
| | | | BIOZ_RBIAS_VALUE = 0x2 | | 200 | | |
| Lead-Bias Voltage | | BioZ lead-bias enabled, RLD_RBIAS = 0 | | 660 | 760 | 860 | mV |
| **BIOZ I/O MUX / INTERNAL RESISTOR LOADS** | | | | | | | |
| Internal BIA Resistive-Load Nominal Value | $R_{VAL}$ | Selected by BMUX_RSEL | | | 200, 500, 800, 5000 | | Ω |
| Internal GSR Resistive-Load Nominal Value | $R_{GSR}$ | Selectable by BMUX_GSR_RSEL | | | 25, 100, 500, 1000 | | kΩ |

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## Electrical Characteristics (continued)

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|
| **TIMING SUBSYSTEM / PLL** | | | | | | |
| PLL Frequency Range | | FCLK = 32.768kHz, FREQ_LOCK asserted < 3.5ms | 4 | | 28 | MHz |
| PLL Lock Time | | Change in FCLK to FREQ_LOCK asserted, MDIV = 0x356 | | 1.2 | 3.5 | ms |
| | | Change in FCLK to PHASE_LOCK asserted, MDIV = 0x356 | | 3.5 | 8.5 | |
| FCLK Input Frequency | | Must match CLK_FREQ_SEL | | 32.0 or 32.768 | | kHz |
| Maximum FCLK Rise Time (10% to 90%) | tRISE | $T_A$ = +25°C, $V_{IN}$ = 0V to 1.8V | | 100 | | ns |
| Maximum FCLK Fall Time (90% to 10%) | tFALL | $T_A$ = +25°C, $V_{IN}$ = 1.8V to 0V | | 100 | | ns |
| PLL External-Reference Jitter | | Cycle-to-Cycle Period, PLL_LOCK_WNDW = 0 | | 3 | | $ns_{RMS}$ |
| FCLK Input Capacitance | $C_{FCLK}$ | | | 15 | | pF |
| **TIMING SUBSYSTEM / INTERNAL CLOCK** | | | | | | |
| Internal Oscillator | $f_{CLKINT}$ | REF_CLK_SEL = 32kHz | -2 | ±0.4 | +2 | % |
| | | REF_CLK_SEL = 32.768kHz | -2 | ±0.4 | +2 | |
| **INTERNAL REFERENCE** | | | | | | |
| ECG Common-Mode Reference Output Voltage | $V_{MID_ECG}$ | | 740 | 760 | 780 | mV |
| ECG Common-Mode Reference Temperature Coefficient | $TC_{MID_ECG}$ | | | 10 | | ppm/°C |
| ECG Reference Output Voltage | $V_{REF_ECG}$ | | 0.990 | 1.000 | 1.010 | V |
| ECG Reference Temperature Coefficient | $TC_{REF_ECG}$ | | | 10 | | ppm/°C |
| PPG Reference Output Voltage | $V_{REF_PPG}$ | | 1.15 | 1.205 | 1.215 | V |
| PPG Reference Temperature Coefficient | $TC_{REF_PPG}$ | | | 10 | | ppm/°C |
| **DIGITAL I/O CHARACTERISTICS** | | | | | | |
| Input-Voltage Low | $V_{IL}$ | CSB/I2C_SEL, SDO/ADDR, SDI/SDA, SCLK/SCL, TRIG, FCLK | | | 0.2 x $V_{IOVDD}$ | V |
| Input-Voltage High | $V_{IH}$ | CSB/I2C_SEL, SDO/ADDR, SDI/SDA, SCLK/SCL, TRIG, FCLK | 0.8 x $V_{IOVDD}$ | | | V |
| Input Hysteresis | $V_{HYS}$ | CSB/I2C_SEL, SDO/ADDR, SDI/SDA, SCLK/SCL, TRIG, FCLK | | 525 | | mV |
| Input Capacitance | $C_{IN}$ | CSB/I2C_SEL, SDO/ADDR, SDI/SDA, SCLK/SCL, TRIG, FCLK | | 10 | | pF |

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## Electrical Characteristics (continued)

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|
| Input-Leakage Current | $I_{IN}$ | CSB/I2C_SEL, SDO/ADDR, SDI/SDA, SCLK/SCL, TRIG, FCLK, $T_A$ = +25°C, $V_{IN}$ = $V_{DGND}$ or $V_{DVDD}$ | -1 | +0.01 | +1 | µA |
| Output Low Voltage | $V_{OL}$ | SDO, INT1, INT2, $I_{SINK}$ = 4mA | | | 0.4 | V |
| Output High Voltage | $V_{OH}$ | SDO, INT1, INT2, $I_{SOURCE}$ = 4mA | $V_{IOVDD}$ - 0.4 | | | V |
| Open-Drain Output Low Voltage | $V_{OL_OD}$ | INT1_OCFG = INT2_OCFG = 0, $I_{SINK}$ = 4mA | | | 0.4 | V |
| **I²C TIMING CHARACTERISTICS (Note 9)** | | | | | | |
| I²C Write Address | | SDO/ADDR is low | | D8 | | Hex |
| | | SDO/ADDR is high | | DA | | |
| I²C Read Address | | SDO/ADDR is low | | D9 | | Hex |
| | | SDO/ADDR is high | | DB | | |
| Serial Clock Frequency | $f_{SCL}$ | | 100 | | 400 | kHz |
| Bus Free-Time Between STOP and START Conditions | $t_{BUF}$ | | 1.3 | | | µs |
| Hold-Time START and Repeat START Condition | $t_{HD,STA}$ | | 0.6 | | | µs |
| SCL Pulse-Width Low | $t_{LOW}$ | | 1.3 | | | µs |
| SCL Pulse-Width High | $t_{HIGH}$ | | 0.6 | | | µs |
| Setup Time for a Repeated START Condition | $t_{SU,STA}$ | | 0.6 | | | µs |
| Data Hold Time | $t_{HD,DAT}$ | | 0 | | 900 | ns |
| Data Setup Time | $t_{SU,DAT}$ | | 100 | | | ns |
| Setup Time for STOP Condition | $t_{SU,STO}$ | | 0.6 | | | µs |
| Pulse-Width of Suppressed Spike | $t_{SP}$ | | 0 | | 50 | ns |
| Bus Capacitance | $C_B$ | | | | 400 | pF |
| SDA and SCL Receiving Rise Time | $t_R$ | | 20 + 0.1$C_B$ | | 300 | ns |
| SDA and SCL Receiving Fall Time | $t_F$ | | 20 + 0.1$C_B$ | | 300 | ns |
| SDA Transmitting Fall Time | $t_{TF}$ | | 20 + 0.1$C_B$ | | 300 | ns |
| **SPI TIMING CHARACTERISTICS (Note 9)** | | | | | | |
| SCLK Frequency | $f_{SCLK}$ | | 0.1 | | 24 | MHz |
| SCLK Period | $t_{CP}$ | | 41.7 | | | ns |
| SCLK Pulse-Width High | $t_{CH}$ | | 18 | | | ns |
| SCLK Pulse-Width Low | $t_{CL}$ | | 18 | | | ns |

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## Electrical Characteristics (continued)

($V_{LDO_IN}$ = 3.7V, $V_{LED}$ = 3.7V, $V_{AVDD}$ = 1.8V, $V_{DVDD}$ = 1.8V, $V_{IOVDD}$ = 1.8V, $T_A$ = +25°C, MIN/MAX are from $T_A$ = -40°C to +85°C, unless otherwise noted.) (Notes 1, 2, )

| PARAMETER | SYMBOL | CONDITIONS | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|
| CSB-Fall to SCLK-Rise Setup Time | $t_{CSS0}$ | Applies to the 1st SCLK rising edge after CSB/I2C_SEL goes low | 20 | | | ns |
| CSB-Fall to SCLK-Rise Hold Time | $t_{CSH0}$ | Applies to the inactive rising edge preceding the 1st rising edge | 10 | | | ns |
| Last SCLK Rise to CSB Rise | $t_{CSH1}$ | Applies to the last SCLK rising edge in a transaction | 20 | | | ns |
| Last SCLK Rise to Next CSB Fall | $t_{CSF}$ | Applies to the last SCLK rising edge to the next CSB falling edge (new transaction) | 60 | | | ns |
| CSB Pulse-Width High | $t_{CSPW}$ | | 40 | | | ns |
| SDI to SCLK-Rise Setup Time | $t_{DS}$ | | 10 | | | ns |
| SDI to SCLK-Rise Hold Time | $t_{DH}$ | | 5 | | | ns |
| SCLK Fall to SDO Transition | $t_{DOT}$ | $C_{LOAD}$ = 30pF | | | 15 | ns |
| CSB Fall to SDO Fall | $t_{DOE}$ | $C_{LOAD}$ = 0pF | 10 | | | ns |
| CSB Rise to SDO Hi-Z | $t_{DOZ}$ | Disable time | | | 30 | ns |
| TRIG Pulse-Width | $t_{TRIG}$ | | 1 | | | µs |
| **DIGITAL I/O TIMING CHARACTERISTICS** | | | | | | |
| TIMING_SYS_RESET Delay on INT2 (Note 9) | | Relative to rising edge of FCLK | | | 10 | ns |
| TIMING_SYS_RESET Setup Time on TRIG | | Relative to falling edge of FCLK | | ≥ 10 | | ns |
| TIMING_SYS_RESET Hold Time on TRIG | | Relative to falling edge of FCLK | | ≥ 10 | | ns |

**Note 1:** Limits are 100% tested at TA = +25°C. Limits over the operating temperature range and relevant supply voltage range are guaranteed by design and characterization

**Note 2:** All register settings use default values, unless otherwise noted below or in specific EC conditions.

REF_CLK_SEL = 0, CLK_FREQ_SEL = 1, FR_CLK_DIV = 32 ($f_{FRAME}$ = 1kfps),

CHPF = 10µF

**Note 3:** $V_{LED}$ should be set greater than the minimum output voltage + LED forward voltage.

**Note 4:** FR = PPG Frame Rate
a. MEASx_DRVy_PA = 0x7F, REF_CLK_SEL = 0, PPG1_PWRDN = 0, PPG2_PWRDN = 1, MEASx_TINT = 0x0
b. MEASx_DRVy_PA = 0x7F, REF_CLK_SEL = 0, PPG1_PWRDN = 0, PPG2_PWRDN = 0, MEASx_TINT = 0x0

**Note 5:** PLL Enabled, FCLK = 32768Hz, ECG_INA_GAIN = 20V/V

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**Note 6:** FCLK = 32768Hz, REF_CLK_SEL = 1, $I_{DRV}$ = 32µ$A_{RMS}$, BIOZ_GAIN = 10V/V, BIOZ_DRV_MODE = Current Drive, BIOZ_INA_MODE = low power mode, DRVP shorted to DRVN (unless otherwise noted).
a. Digital filters disabled, BIOZ_AMP_RGE = medium-high, BIOZ_AMP_BW = medium-high, BIOZ_INA_MODE = high-power.
b. BIOZ_DLPF = 0.08 x SR_BIOZ, BIOZ_DHPF = bypass.
c. BIOZ_DLPF = 0.25 x SR_BIOZ, BIOZ_DHPF = 0.002 x SR_BIOZ, IDRV = 1.28m$A_{RMS}$, BIOZ_AMP_RGE = medium-high, BIOZ_AMP_BW = medium-high, BIOZ_INA_MODE = high-power.
d. Digital filters disabled, BIOZ_AMP_RGE = low, BIOZ_AMP_BW = low.

Register settings for F_BIOZ:

| F_BIOZ [Hz] | SR [Hz] | PLL [Hz] | M | BIOZ_KDIV | BIOZ_DAC_OSR | BIOZ_NDIV | BIOZ_ADC_OSR |
|---|---|---|---|---|---|---|---|
| 1000 | 62.5 | 16,384,000 | 500 | 64 | 256 | 512 | 512 |
| 50,048 | 48.875 | 12,812,288 | 391 | 1 | 256 | 512 | 512 |
| 100,096 | 48.875 | 12,812,288 | 391 | 1 | 128 | 512 | 512 |
| 100,096 | 97.75 | 12,812,289 | 391 | 1 | 128 | 512 | 256 |
| 249,856 | 61 | 15,990,784 | 488 | 1 | 64 | 512 | 512 |
| 499,712 | 61 | 15,990,784 | 488 | 1 | 32 | 512 | 512 |
| 32,000 | 62.5 | 8,192,000 | 250 | 1 | 256 | 256 | 512 |
| 64 | 32 | 4,194,304 | 128 | 256 | 256 | 256 | 512 |

**Note 7:** M = 125, PLL_CLK = 4.096MHz, BIOZ_KDIV = 1, BIOZ_DAC_OSR = 128, F_RESP = 32kHz, BIOZ_NDIV = 256, BIOZ_ADC_OSR = 512, SR_RESP = 31.25sps, ECG_FDIV = 1, ECG_NDIV = 128, ECG_DEC_RATE = 128, SR_ECG = 250sps, INA_MODE = low power, RESP_EN = 1, $I_{DRV}$ = 32µ$A_{PK}$, CG_MODE = 0x2, CG_LPF_DUTY = 0x3

**Note 8:** M = 125, PLL_CLK = 4.096MHz, BIOZ_KDIV = 1, BIOZ_DAC_OSR = 128, F_RESP = 32kHz, BIOZ_NDIV = 256, BIOZ_ADC_OSR = 512, SR_RESP = 31.25sps, ECG_FDIV = 1, ECG_NDIV = 128, ECG_DEC_RATE = 128, SR_ECG = 250sps, INA_MODE = low power, RESP_EN = 1, $I_{DRV}$ = 32µ$A_{PK}$, CG_MODE = 0x2, CG_LPF_DUTY = 0x3

FR_PPG = 128fps, PPG1_PWRDN = 0, PPG2_PWRDN = 1, MEASx_TINT = 14.6µs, MEASx_PPG1_ADC_RGE = 32µA, MEASx_PD_SETLNG = 11.8µs, MEASx_LED_SETLNG = 11.7µs, MEAS1_EN = 1, MEAS2_EN = 1, MEASx_DRVA_PA = MEASx_DRVB_PA = 25mA

**Note 9:** Guaranteed by design and characterization. Not tested in production.

**Note 10:** Electrode tissue interface impedance mismatch such that one electrode has < 10Ω source impedance and the other has an impedance of 51kΩ || 47nF per IEC60601-2-47.

**Note 11:** This specification defines the accuracy of the internal calibration voltage source as measured through the ADC channel.

**Note 12:** Overall accuracy must include calibration resistor accuracy and the overall calibration accuracy. Calibration uses an external 32.768kHz, 2.5ns$_{RMS}$ jitter, ±5ppm TC oscillator with a 680Ω external calibration resistor. The calibration resistor and Cole impedance were measured to within 0.05% magnitude and 0.1° phase accuracy using a calibrated Zurich Instruments MFIA.

**Note 13:** a. BIOZ_DRV_MODE = 0x0, $I_{DRV}$ = 160n$A_{RMS}$, F_BIOZ = 16Hz, SR_BIOZ = 16sps, BIOZ_GAIN = 10V/V, BIOZ_DLPF = 0x4, BIOZ_AHPF = 42.4MΩ with external 47nF BIP and BIN capacitors. Effective signal band = DC to 4Hz.
b. BIOZ_DRV_MODE = 0x0, $I_{DRV}$ = 32µ$A_{RMS}$, F_BIOZ = 32kHz, SR_BIOZ = 31.25sps, BIOZ_GAIN = 10V/V, BIOZ_DLPF = 0x3, BIOZ_AHPF = 2kHz. Effective signal band = DC to 2.5Hz.
c. BIOZ_DRV_MODE = 0x0, $I_{DRV}$ = 1.28m$A_{RMS}$, F_BIOZ = 64kHz, SR_BIOZ = 250sps, BIOZ_GAIN = 10V/V, BIOZ_AHPF = 5kHz. Effective signal band = DC to 65Hz.
d. RESP_EN = 1, RESP_CG_MAG_4x = 0, RESP_CG_MAG = 32µ$A_{PK}$, F_RESP = 32kHz, SR_RESP = 31.25sps, BIOZ_GAIN = 10V/V, BIOZ_DLPF = 0x3, BIOZ_AHPF = 2kHz. Effective signal band = DC to 2.5Hz.

**Note 14:** BIOZ_GAIN = 10V/V, F_BIOZ = 65.5kHz, SR_BIOZ = 256sps, BioZ Lead Bias = 200MΩ, BIOZ_AHPF = 5kHz

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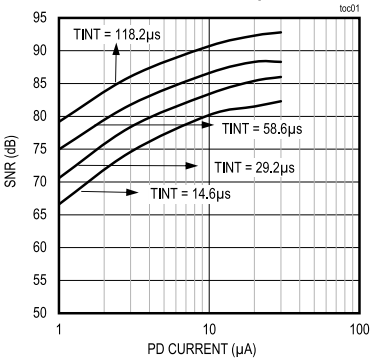
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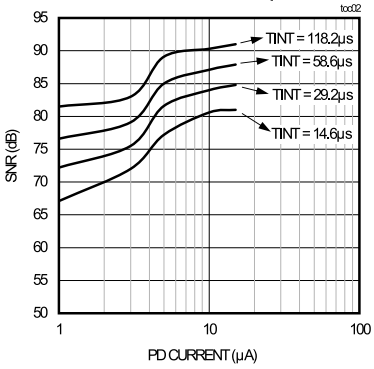
## Typical Operating Characteristics

($V_{DD_ANA}$ = $V_{DD_DIG}$ = 1.8V, $V_{LED}$ = 5.0V, GND = PGND = 0V, $T_A$ = +25°C, unless otherwise noted.)

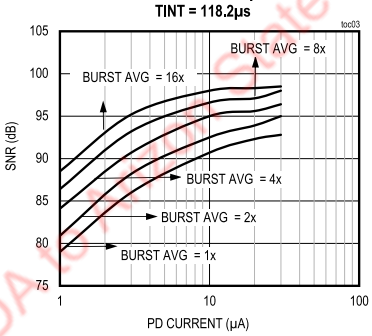
SNR vs. INPUT CURRENT
ADC RANGE = 32µA



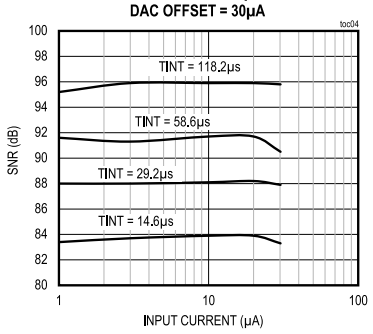

SNR vs. INPUT CURRENT
ADC RANGE = 16µA



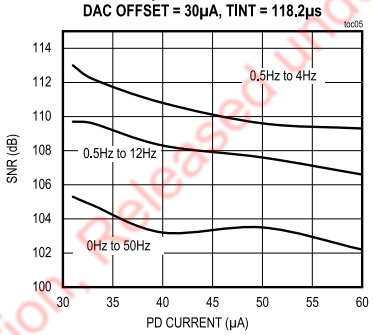

SNR vs. INPUT CURRENT
ADC RANGE = 32µA
TINT = 118.2µs



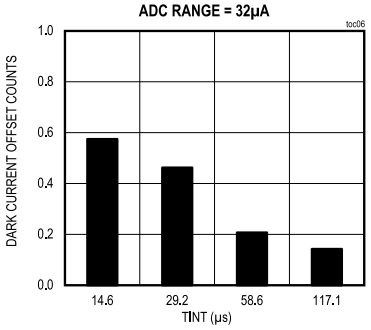

SNR vs. INPUT CURRENT
ADC RANGE = 32µA
DAC OFFSET = 30µA



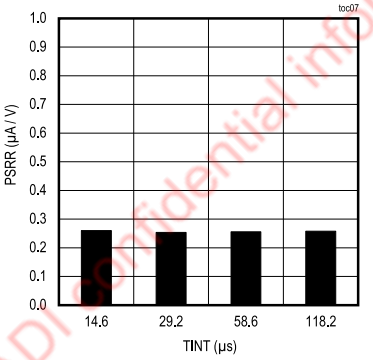

SNR vs. INPUT CURRENT
ADC RANGE = 32µA, ON-CHIP AVERAGING = 16x,
DAC OFFSET = 30µA, TINT = 118.2µs



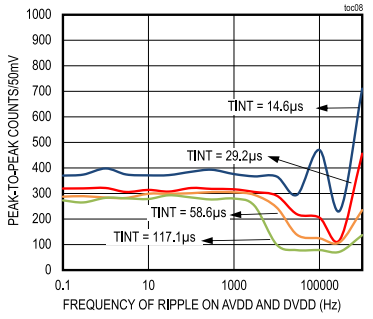

DARK CURRENT OFFSET COUNTS
ADC RANGE = 32µA



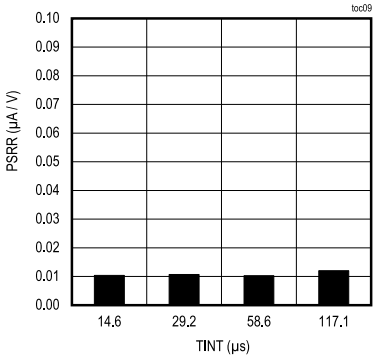

AVDD AND DVDD PPG DC PSRR
PD CURRENT = 31µA




AVDD AND DVDD PPG AC PSRR
PD CURRENT = 31µA




$V_{LED}$ PPG DC PSRR
PD CURRENT = 31µA




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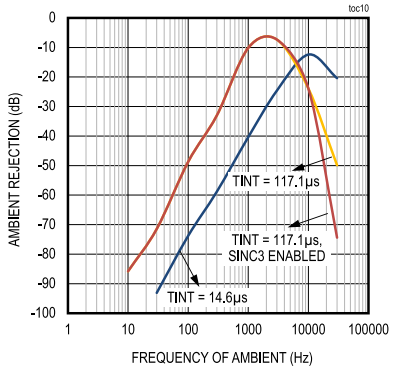
ANALOG DEVICES CONFIDENTIAL/DISTRIBUTE ONLY UNDER NDA



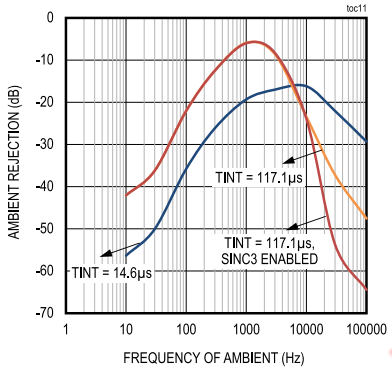
## Typical Operating Characteristics (continued)

($V_{DD_ANA}$ = $V_{DD_DIG}$ = 1.8V, $V_{LED}$ = 5.0V, GND = PGND = 0V, $T_A$ = +25°C, unless otherwise noted.)

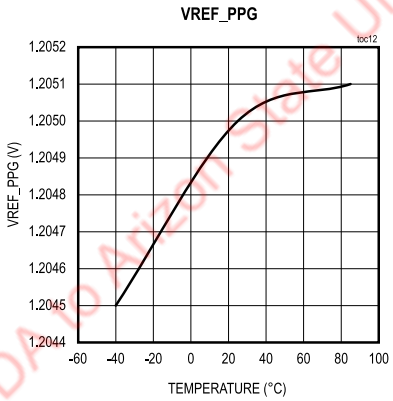
AC AMBIENT REJECTION WITH CDM



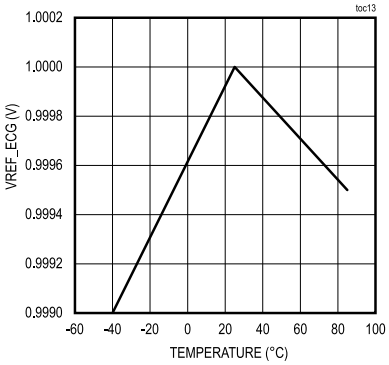

AC AMBIENT REJECTION WITH FDM



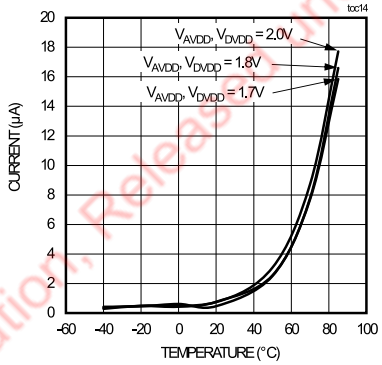

VREF_PPG



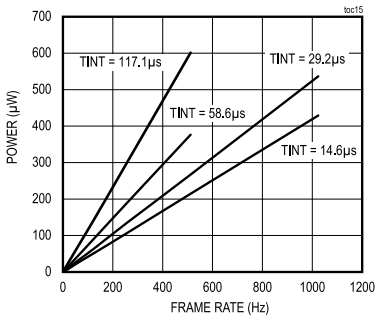

VREF_ECG



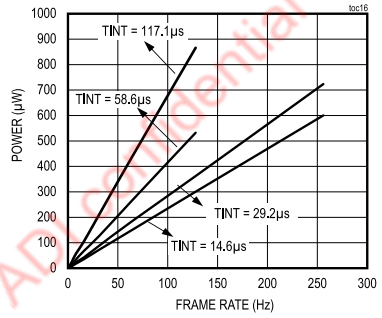

AVERAGE ($I_{AVDD}$ + $I_{DVDD}$) SHUTDOWN CURRENT



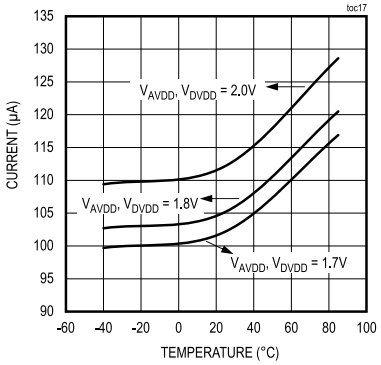

PPG POWER DISSIPATION
1 MEASUREMENT/FRAME



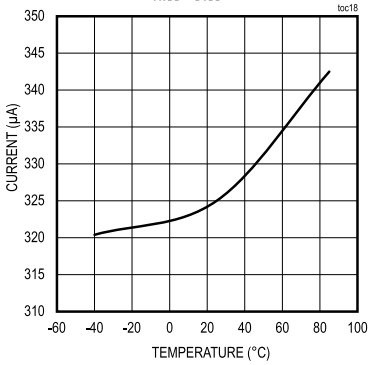

PPG POWER DISSIPATION
6 MEASUREMENTS/FRAME




AVERAGE ($I_{AVDD}$ + $I_{DVDD}$) SUPPLY CURRENT
ECG MODE, RLD DISABLED, RBIAS ENABLED
SAMPLE RATE = 128sps




AVERAGE ($I_{AVDD}$ + $I_{DVDD}$) SUPPLY CURRENT
BIOZ, RESPIRATION MODE
$V_{AVDD}$,$V_{DVDD}$ = 1.8V




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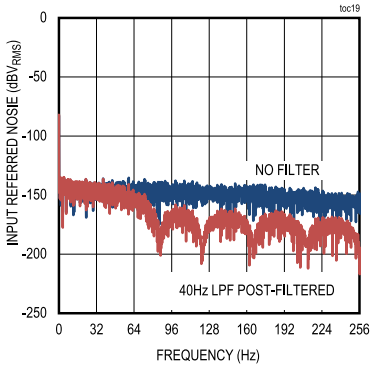
ANALOG DEVICES CONFIDENTIAL/DISTRIBUTE ONLY UNDER NDA



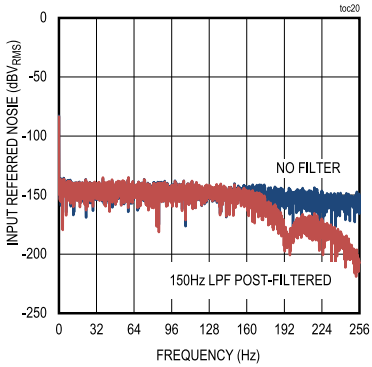
## Typical Operating Characteristics (continued)

($V_{DD_ANA}$ = $V_{DD_DIG}$ = 1.8V, $V_{LED}$ = 5.0V, GND = PGND = 0V, $T_A$ = +25°C, unless otherwise noted.)

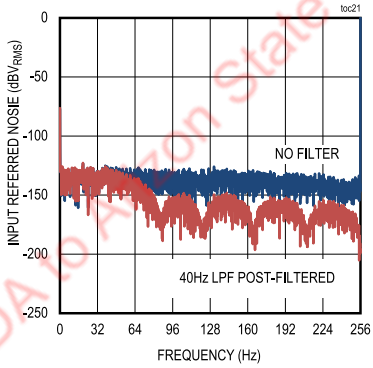
ECG NOISE SPECTRUM vs. FREQUENCY
INPUTS SHORTED TO $V_{MID_ECG}$
GAIN = 20x, LPF = 40Hz



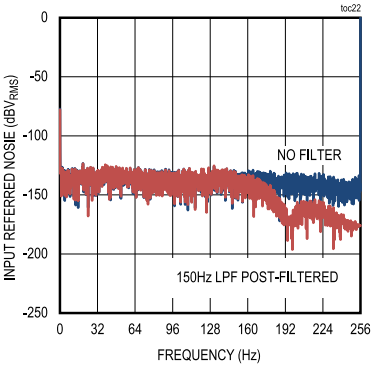

ECG NOISE SPECTRUM vs. FREQUENCY
INPUTS SHORTED TO $V_{MID_ECG}$
GAIN = 20x, LPF = 150Hz



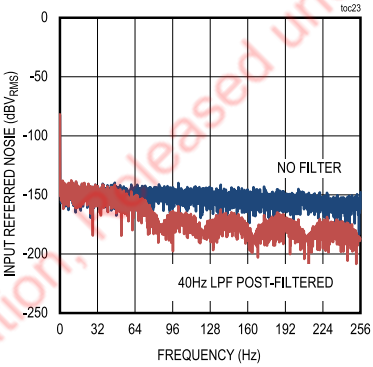

ECG NOISE SPECTRUM vs. FREQUENCY
INPUTS SHORTED TO $V_{MID_ECG}$
GAIN = 160x, LPF = 40Hz



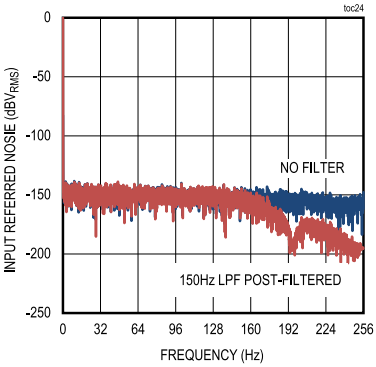

ECG NOISE SPECTRUM vs. FREQUENCY
INPUTS SHORTED TO $V_{MID_ECG}$
GAIN = 160x, LPF = 150Hz



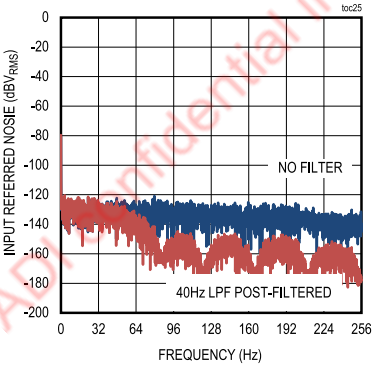

ECG NOISE SPECTRUM vs. FREQUENCY
INPUTS SHORTED THROUGH 100MΩ TO $V_{MID_ECG}$
GAIN = 20x, LPF = 40Hz



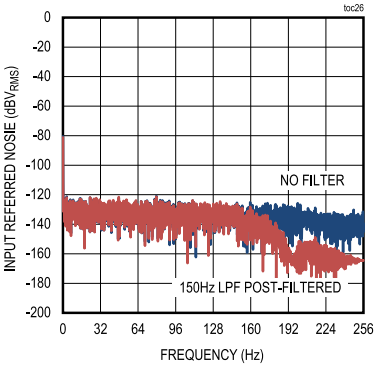

ECG NOISE SPECTRUM vs. FREQUENCY
INPUTS SHORTED THROUGH 100MΩ TO $V_{MID_ECG}$
GAIN = 20x, LPF = 150Hz



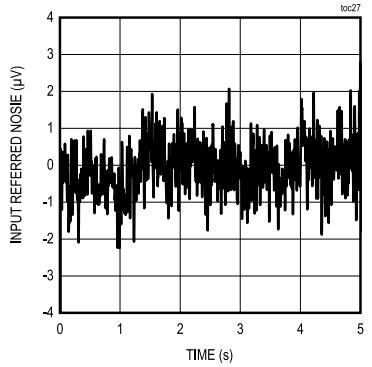

ECG NOISE SPECTRUM vs. FREQUENCY
INPUTS SHORTED THROUGH 100Ω TO $V_{MID_ECG}$
GAIN = 160x, LPF = 40Hz




ECG NOISE SPECTRUM vs. FREQUENCY
INPUTS SHORTED THROUGH 100MΩ TO $V_{MID_ECG}$
GAIN = 160x, LPF = 150Hz




ECG NOISE vs. TIME
INPUTS SHORTED TO $V_{MID_ECG}$
GAIN = 20x, LPF = 40Hz




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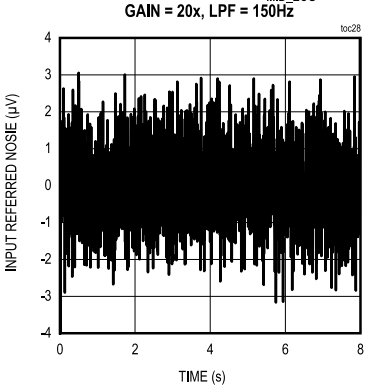
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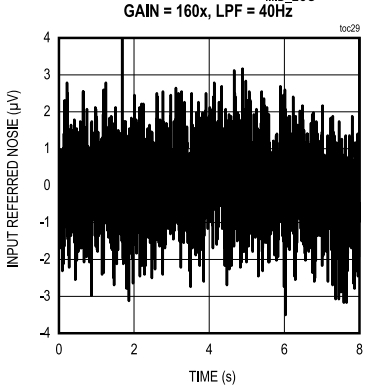
## Typical Operating Characteristics (continued)

($V_{DD_ANA}$ = $V_{DD_DIG}$ = 1.8V, $V_{LED}$ = 5.0V, GND = PGND = 0V, $T_A$ = +25°C, unless otherwise noted.)

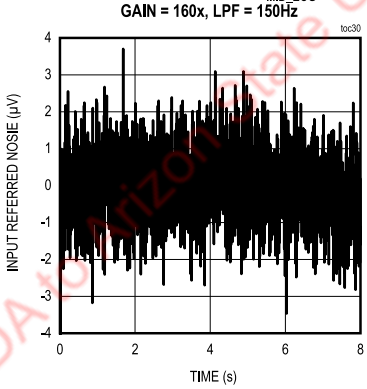
ECG NOISE vs. TIME
INPUTS SHORTED TO $V_{MID_ECG}$
GAIN = 20x, LPF = 150Hz



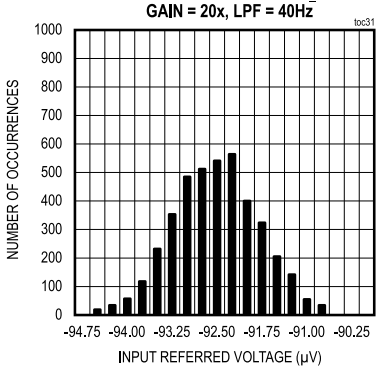

ECG NOISE vs. TIME
INPUTS SHORTED TO $V_{MID_ECG}$
GAIN = 160x, LPF = 40Hz



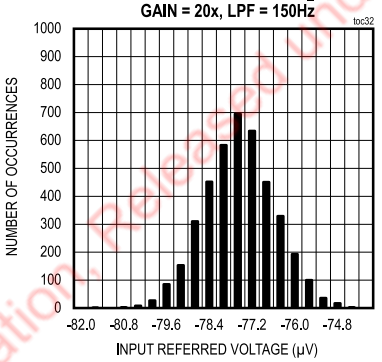

ECG NOISE vs. TIME
INPUTS SHORTED TO $V_{MID_ECG}$
GAIN = 160x, LPF = 150Hz



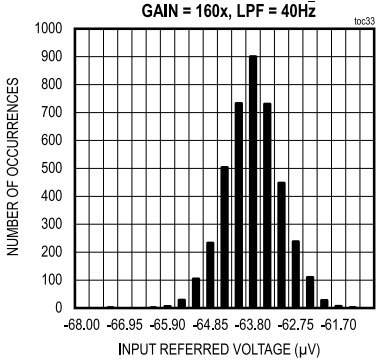

ECG NOISE HISTOGRAM
INPUTS SHORTED TO $V_{MID_ECG}$
GAIN = 20x, LPF = 40Hz



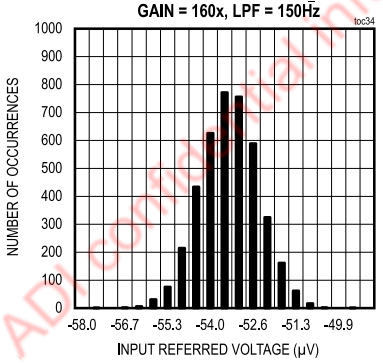

ECG NOISE HISTOGRAM
INPUTS SHORTED TO $V_{MID_ECG}$
GAIN = 20x, LPF = 150Hz



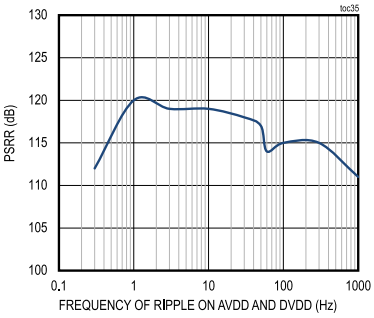

ECG NOISE HISTOGRAM
INPUTS SHORTED TO $V_{MID_ECG}$
GAIN = 160x, LPF = 40Hz



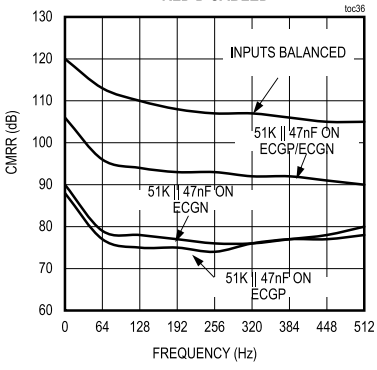

ECG NOISE HISTOGRAM
INPUTS SHORTED TO $V_{MID_ECG}$
GAIN = 160x, LPF = 150Hz




ECG AC PSRR vs. FREQUENCY
GAIN = 20x, SAMPLE RATE = 2048sps




ECG CMRR vs. FREQUENCY
GAIN = 20x, SAMPLE RATE = 2048sps
RLD DISABLED




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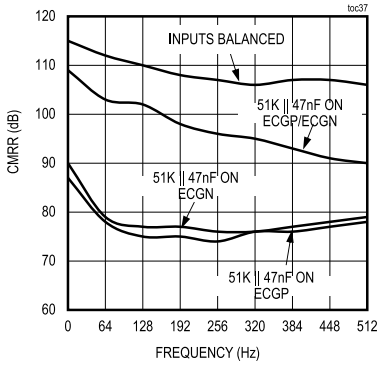
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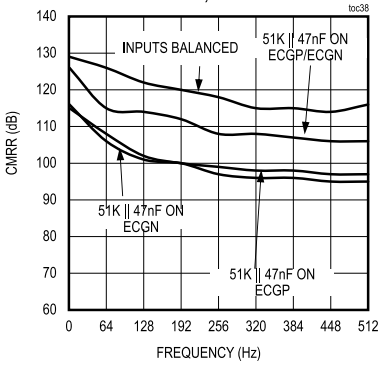
## Typical Operating Characteristics (continued)

($V_{DD_ANA}$ = $V_{DD_DIG}$ = 1.8V, $V_{LED}$ = 5.0V, GND = PGND = 0V, $T_A$ = +25°C, unless otherwise noted.)

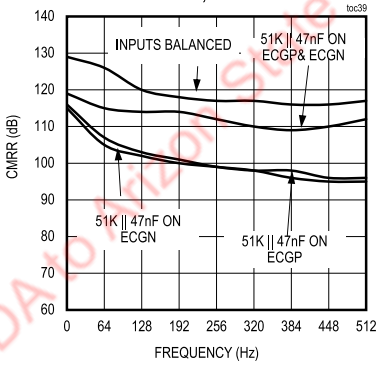
ECG CMRR vs. FREQUENCY
GAIN = 160x, SAMPLE RATE = 2048sps
RLD DISABLED



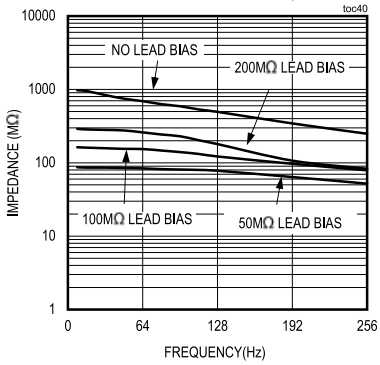

ECG CMRR vs. FREQUENCY
GAIN = 20x, SAMPLE RATE = 2048sps
RLD ENABLED, RLD GAIN = 98V/V



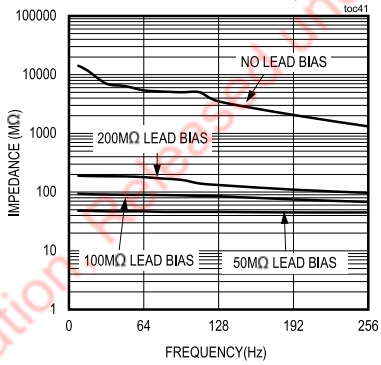

ECG CMRR vs. FREQUENCY
GAIN = 160x, SAMPLE RATE = 2048sps
RLD ENABLED, RLD GAIN = 98V/V



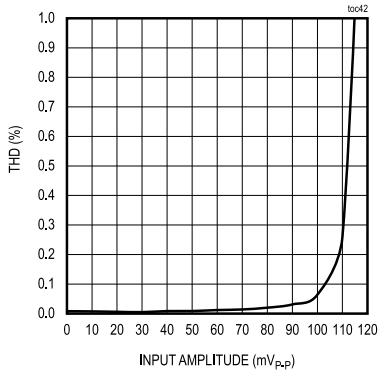

DIFFERENTIAL MODE
INPUT IMPEDANCE vs. FREQUENCY



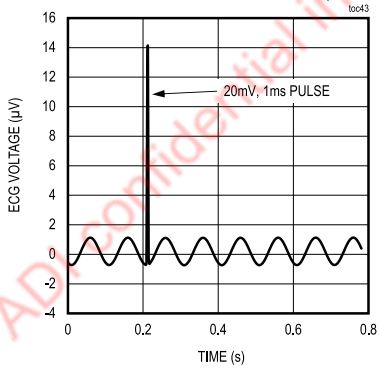

COMMON MODE
INPUT IMPEDANCE vs. FREQUENCY



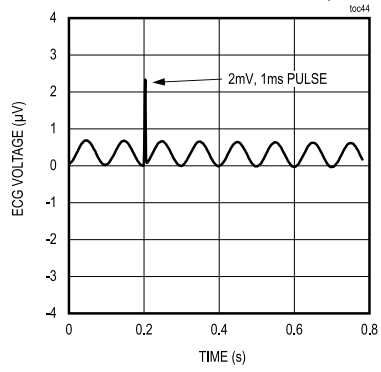

ECG THD vs. INPUT AMPLITUDE
GAIN = 20x, INPUT FREQUENCY = 8Hz



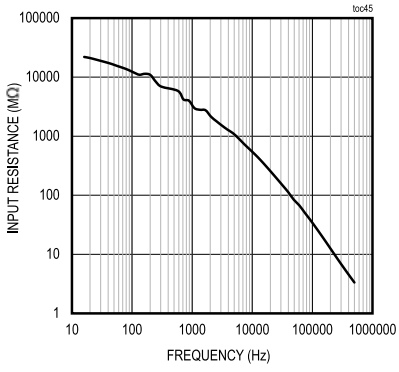

ECG PACEMAKER PULSE TOLERANCE
GAIN = 20x, SAMPLE RATE = 512sps




ECG PACEMAKER PULSE TOLERANCE
GAIN = 20x, SAMPLE RATE = 512sps




BIOZ CHANNEL DIFFERENTIAL INPUT RESISTANCE
16Hz TO 500kHz, GAIN = 1, AHPF BYPASSED




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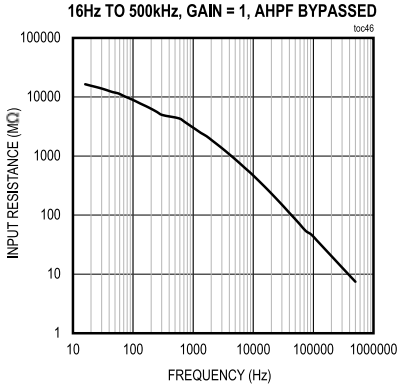
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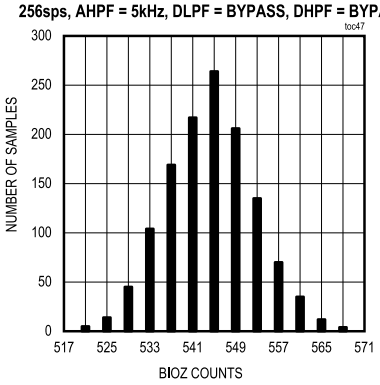
## Typical Operating Characteristics (continued)

($V_{DD_ANA}$ = $V_{DD_DIG}$ = 1.8V, $V_{LED}$ = 5.0V, GND = PGND = 0V, $T_A$ = +25°C, unless otherwise noted.)

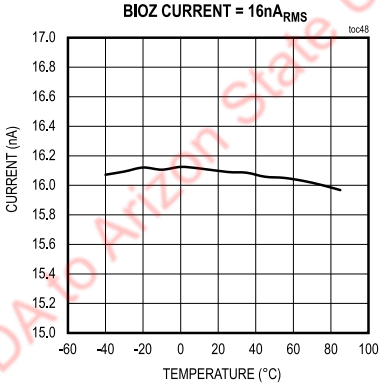
BIOZ CHANNEL COMMON MODE INPUT RESISTANCE
16Hz TO 500kHz, GAIN = 1, AHPF BYPASSED



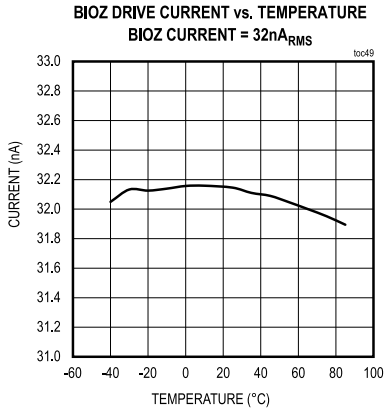

BIOZ NOISE HISTOGRAM
INPUTS SHORTED, GAIN = 10V/V, F_BIOZ = 65.536kHz
256sps, AHPF = 5kHz, DLPF = BYPASS, DHPF = BYPASS



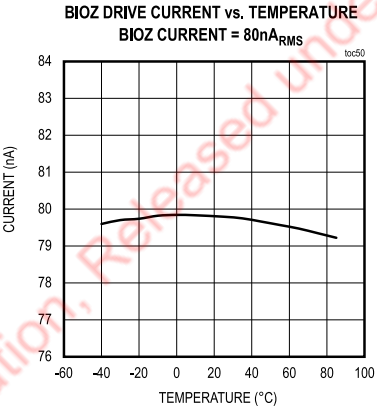

BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 16nA$_{RMS}$



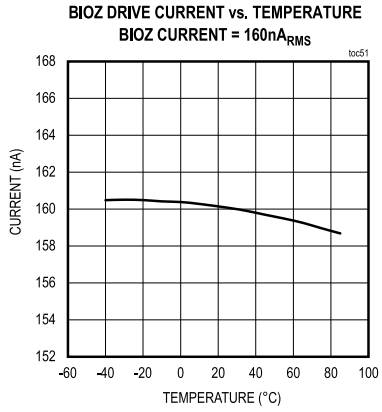

BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 32nA$_{RMS}$



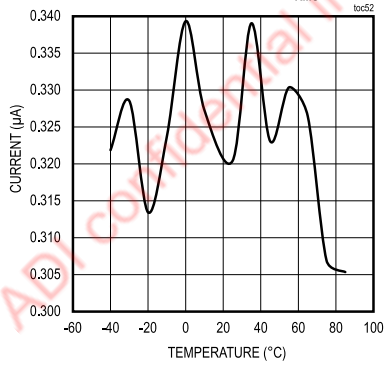

BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 80nA$_{RMS}$



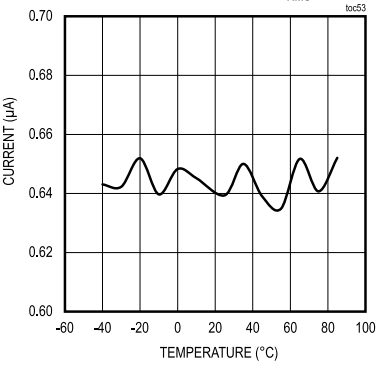

BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 160nA$_{RMS}$



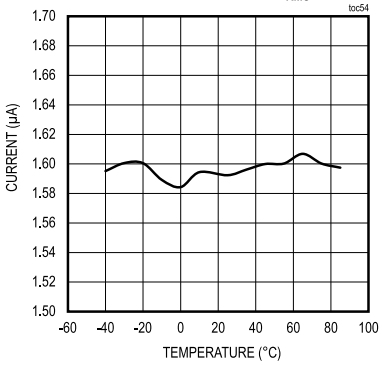

BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 0.32µA$_{RMS}$




BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 0.64µA$_{RMS}$




BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 1.6µA$_{RMS}$




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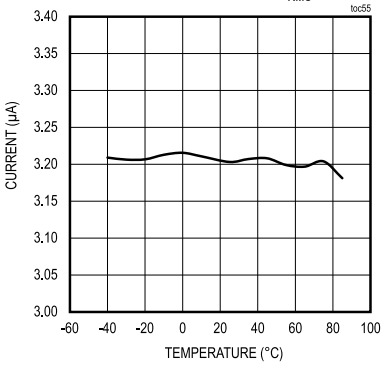
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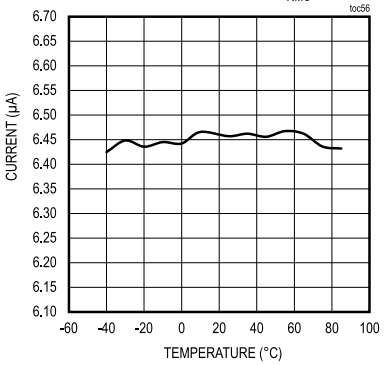
## Typical Operating Characteristics (continued)

($V_{DD_ANA}$ = $V_{DD_DIG}$ = 1.8V, $V_{LED}$ = 5.0V, GND = PGND = 0V, $T_A$ = +25°C, unless otherwise noted.)

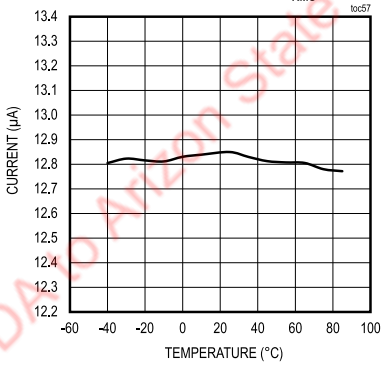
BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 3.2μA$_{RMS}$



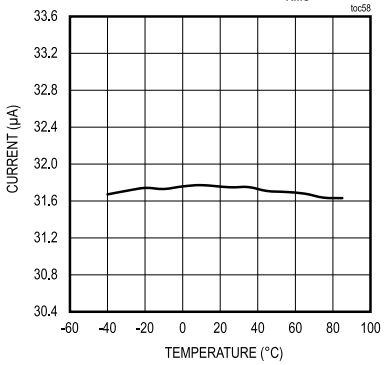

BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 6.4μA$_{RMS}$



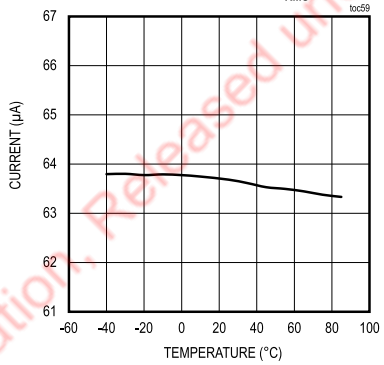

BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 12.8μA$_{RMS}$



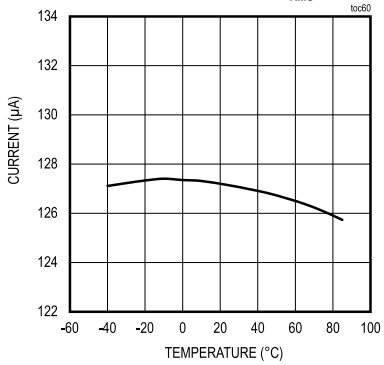

BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 32μA$_{RMS}$



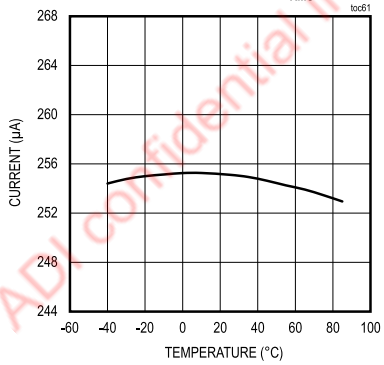

BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 64μA$_{RMS}$



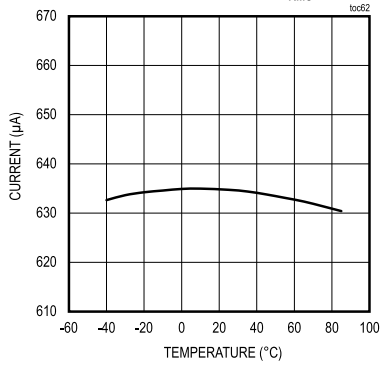

BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 128μA$_{RMS}$



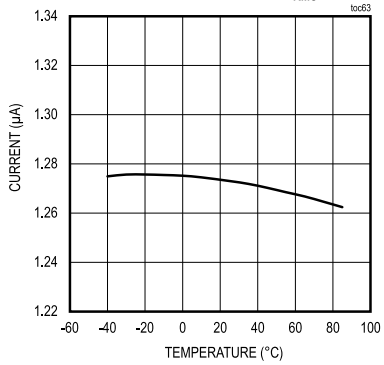

BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 256μA$_{RMS}$




BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 640μA$_{RMS}$




BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ CURRENT = 1.28mA$_{RMS}$




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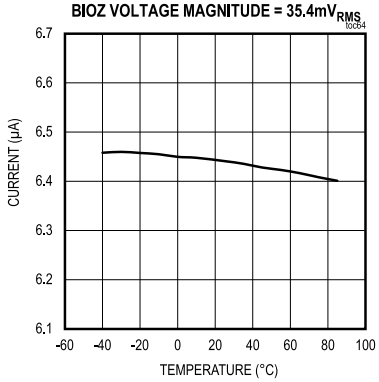
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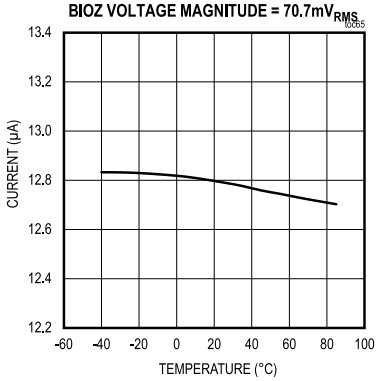
## Typical Operating Characteristics (continued)

($V_{DD_ANA}$ = $V_{DD_DIG}$ = 1.8V, $V_{LED}$ = 5.0V, GND = PGND = 0V, $T_A$ = +25°C, unless otherwise noted.)

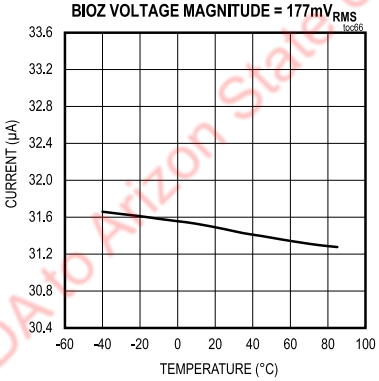
BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ EXT RES = 1, $R_{EXT}$ = 5.525kΩ
BIOZ VOLTAGE MAGNITUDE = 35.4m$V_{RMS}$



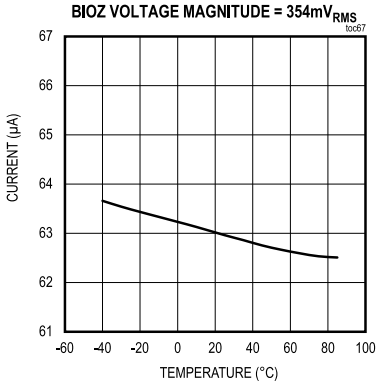

BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ EXT RES = 1, $R_{EXT}$ = 5.525kΩ
BIOZ VOLTAGE MAGNITUDE = 70.7m$V_{RMS}$



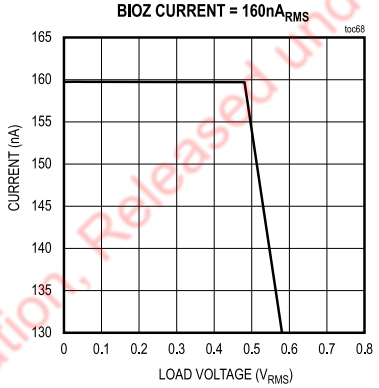

BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ EXT RES = 1, $R_{EXT}$ = 5.525kΩ
BIOZ VOLTAGE MAGNITUDE = 177m$V_{RMS}$



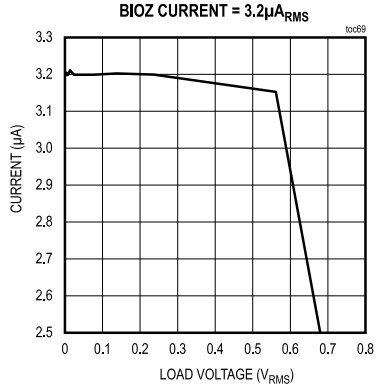

BIOZ DRIVE CURRENT vs. TEMPERATURE
BIOZ EXT RES = 1, $R_{EXT}$ = 5.525kΩ
BIOZ VOLTAGE MAGNITUDE = 354m$V_{RMS}$



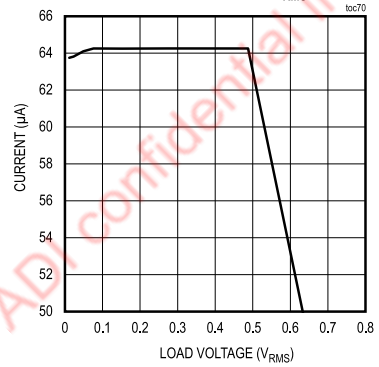

BIOZ DRIVE CURRENT vs. LOAD VOLTAGE
BIOZ CURRENT = 160n$A_{RMS}$



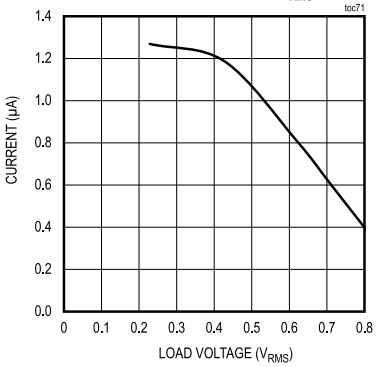

BIOZ DRIVE CURRENT vs. LOAD VOLTAGE
BIOZ CURRENT = 3.2µ$A_{RMS}$



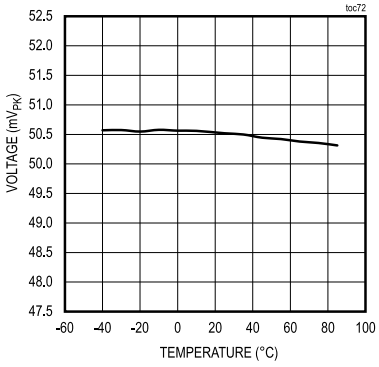

BIOZ DRIVE CURRENT vs. LOAD VOLTAGE
BIOZ CURRENT = 64µ$A_{RMS}$




BIOZ DRIVE CURRENT vs. LOAD VOLTAGE
BIOZ CURRENT = 1.28m$A_{RMS}$




BIOZ DRIVE VOLTAGE vs. TEMPERATURE
BIOZ VOLTAGE = 50m$V_{PK}$




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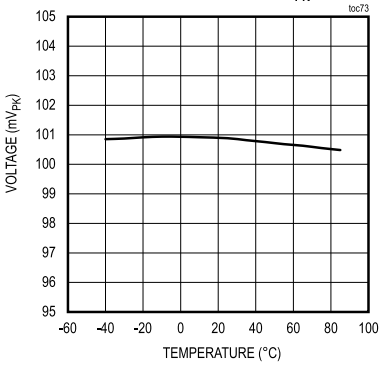
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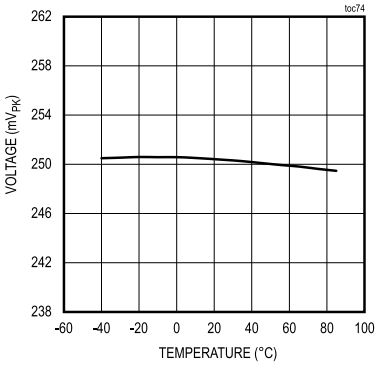
## Typical Operating Characteristics (continued)

($V_{DD_ANA}$ = $V_{DD_DIG}$ = 1.8V, $V_{LED}$ = 5.0V, GND = PGND = 0V, $T_A$ = +25°C, unless otherwise noted.)

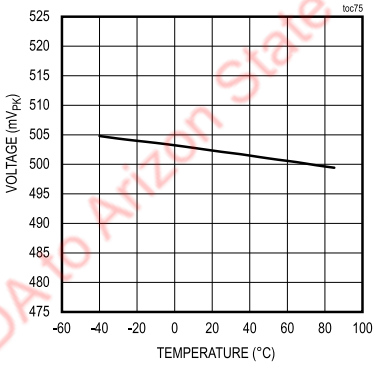
BIOZ DRIVE VOLTAGE vs. TEMPERATURE
BIOZ VOLTAGE = 100mV$_{PK}$



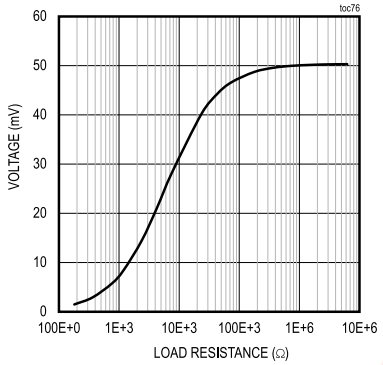

BIOZ DRIVE VOLTAGE vs. TEMPERATURE
BIOZ VOLTAGE = 250mV$_{PK}$



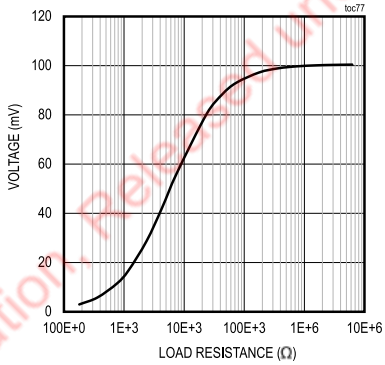

BIOZ DRIVE VOLTAGE vs. TEMPERATURE
BIOZ VOLTAGE = 500mV$_{PK}$



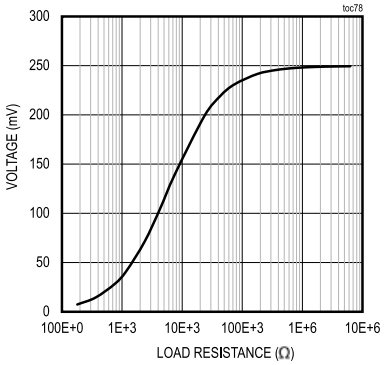

BIOZ DRIVE VOLTAGE vs. LOAD RESISTANCE
BIOZ VOLTAGE = 50mV$_{DC}$
BIOZ_AMP_BW = BIOZ_AMP_RGE = 0x0



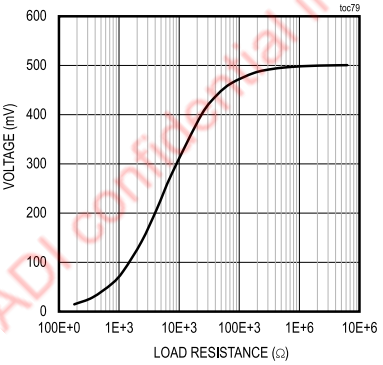

BIOZ DRIVE VOLTAGE vs. LOAD RESISTANCE
BIOZ VOLTAGE = 100mV$_{DC}$
BIOZ_AMP_BW = BIOZ_AMP_RGE = 0x0



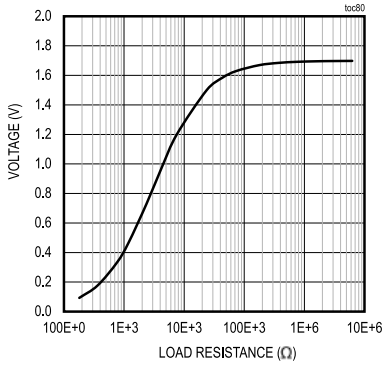

BIOZ DRIVE VOLTAGE vs. LOAD RESISTANCE
BIOZ VOLTAGE = 250mV$_{DC}$
BIOZ_AMP_BW = BIOZ_AMP_RGE = 0x0



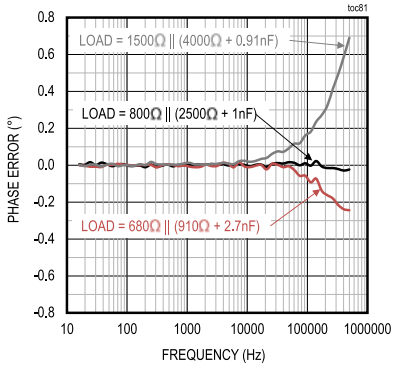

BIOZ DRIVE VOLTAGE vs. LOAD RESISTANCE
BIOZ VOLTAGE = 500mV$_{DC}$
BIOZ_AMP_BW = BIOZ_AMP_RGE = 0x0




H-BRIDGE DRIVE VOLTAGE vs. LOAD RESISTANCE




BIOZ PHASE ERROR vs. FREQUENCY
16Hz TO 500kHz, AFTER CALIBRATION
GAIN = 10V/V, $R_{CAL}$ = 680Ω




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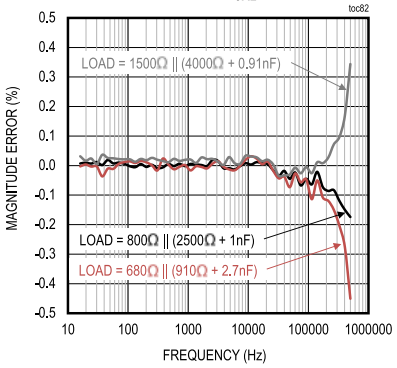
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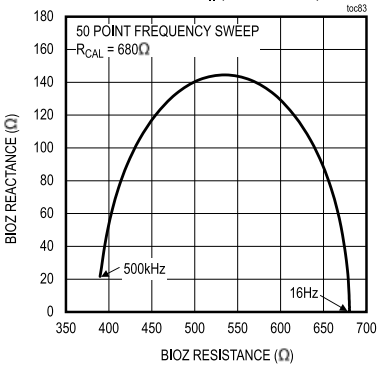
## Typical Operating Characteristics (continued)

($V_{DD_ANA}$ = $V_{DD_DIG}$ = 1.8V, $V_{LED}$ = 5.0V, GND = PGND = 0V, $T_A$ = +25°C, unless otherwise noted.)

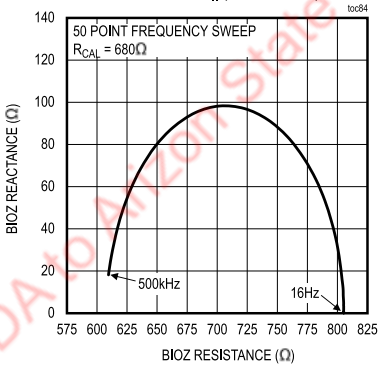
BIOZ MAGNITUDE ERROR vs. FREQUENCY
16Hz TO 500kHz, AFTER CALIBRATION
GAIN = 10V/V, $R_{CAL}$ = 680Ω



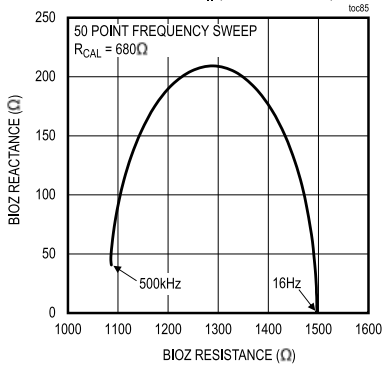

BIOZ COLE-COLE PLOT
16Hz TO 500kHz, AFTER CALIBRATION
LOAD = 680Ω || (910Ω + 2.7nF)



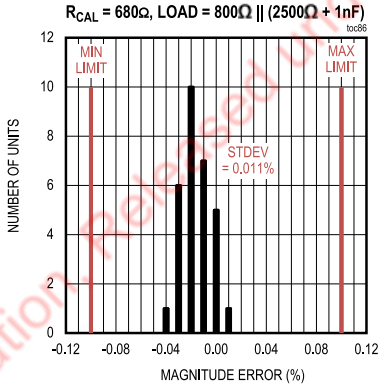

BIOZ COLE-COLE PLOT
16Hz TO 500kHz, AFTER CALIBRATION
LOAD = 800Ω || (2500Ω + 1nF)



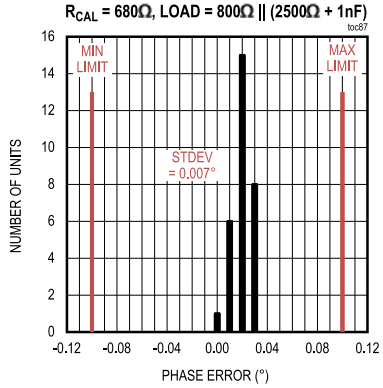

BIOZ COLE-COLE PLOT
16Hz TO 500kHz, AFTER CALIBRATION
LOAD = 1500Ω || (4000Ω + 0.91nF)




BIOZ CALIBRATION MAGNITUDE ERROR HISTOGRAM
50kHz, 32μ$A_{RMS}$ SINE WAVE STIMULUS
$R_{CAL}$ = 680Ω, LOAD = 800Ω || (2500Ω + 1nF)




BIOZ CALIBRATION PHASE ERROR HISTOGRAM
50kHz, 32μ$A_{RMS}$ SINE WAVE STIMULUS
$R_{CAL}$ = 680Ω, LOAD = 800Ω || (2500Ω + 1nF)



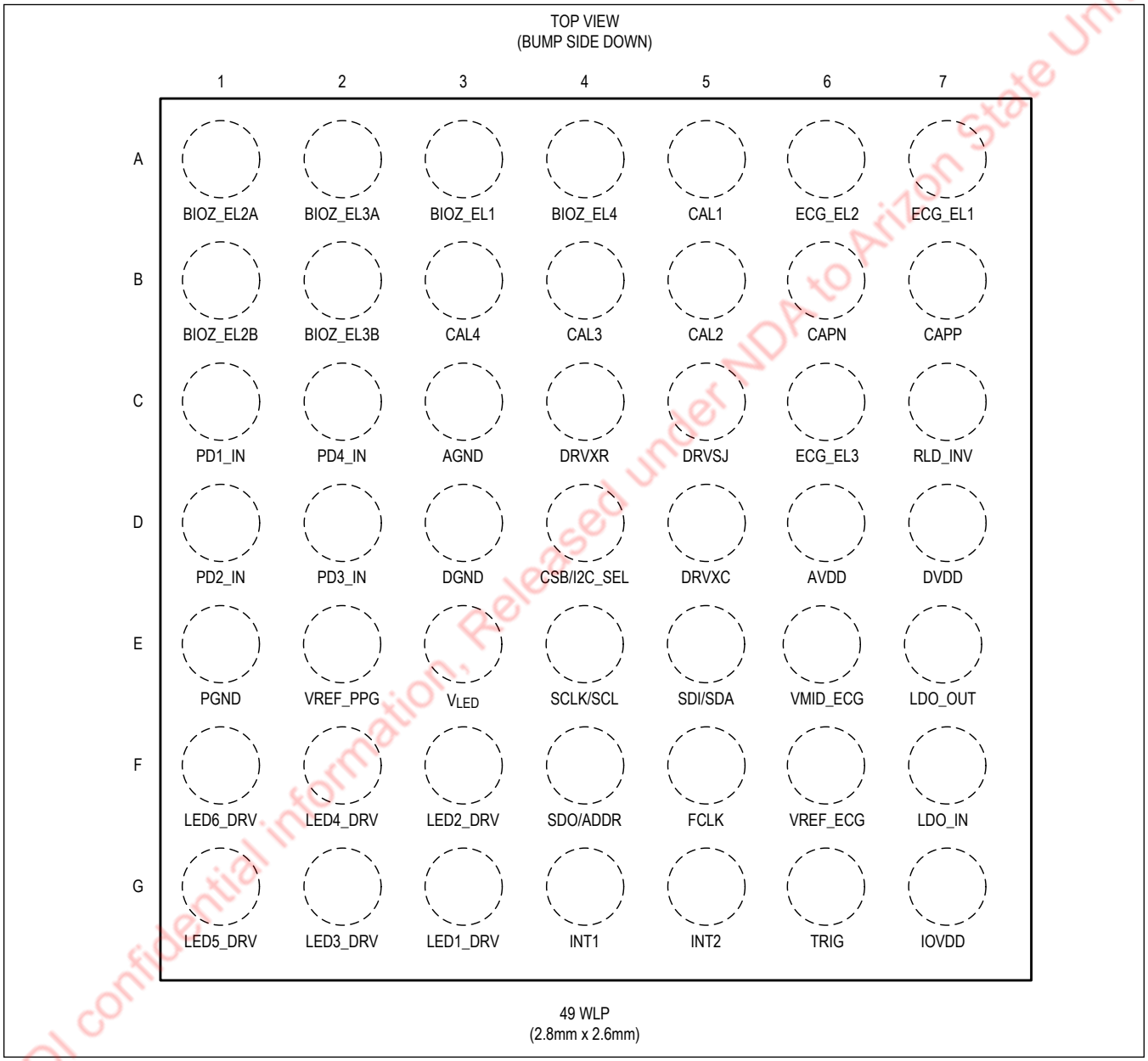

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## Pin Configuration

### MAX86178 WLP




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## Pin Description

| PIN | NAME | FUNCTION |
|---|---|---|
| **POWER** | | |
| D6 | AVDD | Analog Core Supply Voltage. Bypass to AGND with a 0.1µF and a 10µF X5R 0603 capacitor or equivalent effective capacitance. Connect to an externally regulated supply or LDO_OUT if LDO_IN ≥ 2.3V |

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## Pin Description (continued)

| PIN | NAME | FUNCTION |
|---|---|---|
| D7 | DVDD | Digital Core Supply Voltage. Bypass to DGND with a 0.1µF and a 10µF X5R 0603 capacitor or equivalent effective capacitance. Connect to an externally regulated supply or LDO_OUT if LDO_IN ≥ 2.3V. |
| C3 | AGND | Analog Power and Reference Ground. Connect to the PCB ground plane. |
| D3 | DGND | Digital Ground for both Digital Core and I/O Pad Drivers. Recommended to connect to the AGND plane. |
| E1 | PGND | LED Power Return. Connect to PCB ground. |
| E3 | $V_{LED}$ | $V_{LED}$ Power Supply Input. In a configuration with more than one LED supply, connect $V_{LED}$ to the highest LED supply voltage. Bypass with a 10µF X5R 16V 0603 capacitor or minimum equivalent effective capacitance to PGND. |
| F7 | LDO_IN | LDO Input. Connect to AGND or DGND when not in use. |
| E7 | LDO_OUT | LDO Output. Connect to AVDD, DVDD, and IOVDD only. Bypass to AGND. Not for external use. |
| G7 | IOVDD | Digital Input Output Supply Voltage. Bypass to DGND |
| **PPG** | | |
| C1 | PD1_IN | Photodiode 1 Cathode Input. This is connected to either PPG1 or PPG2 depending on the MEASx_PD1_SEL (x = 1 to 6) setting. Tie to AGND when this pin is not used. |
| D1 | PD2_IN | Photodiode 2 Cathode Input. This is connected to either PPG1 or PPG2 depending on the MEASx_PD2_SEL (x = 1 to 6) setting. Tie to AGND when this pin is not used. |
| D2 | PD3_IN | Photodiode 3 Cathode Input. This is connected to either PPG1 or PPG2 depending on the MEASx_PD3_SEL (x = 1 to 6) setting. Tie to AGND when this pin is not used. |
| C2 | PD4_IN | Photodiode 4 Cathode Input. This is connected to either PPG1 or PPG2 depending on the MEASx_PD4_SEL (x = 1 to 6) setting. Tie to AGND when this pin is not used. |
| G3 | LED1_DRV | LED Output 1. Driven when MEASx_DRVy = 0 (x = 1 to 6, y = A, B). Connect the LED cathode to LED1_DRV and its anode to the $V_{LED}$ supply. This pin can be left unconnected when not used. |
| F3 | LED2_DRV | LED Output 2. Driven when MEASx_DRVy = 1 (x = 1 to 6, y = A, B). Connect the LED cathode to LED2_DRV and its anode to the $V_{LED}$ supply. This pin can be left unconnected when not used. |
| G2 | LED3_DRV | LED Output 3. Driven when MEASx_DRVy = 2 (x = 1 to 6, y = A, B). Connect the LED cathode to LED3_DRV and its anode to the $V_{LED}$ supply. This pin can be left unconnected when not used. |
| F2 | LED4_DRV | LED Output 4. Driven when MEASx_DRVy = 3 (x = 1 to 6, y = A, B). Connect the LED cathode to LED4_DRV and its anode to the $V_{LED}$ supply. This pin can be left unconnected when not used. |
| G1 | LED5_DRV | LED Output 5. Driven when MEASx_DRVy = 4(x = 1 to 6, y = A, B). Connect the LED cathode to LED5_DRV and its anode to the $V_{LED}$ supply. This pin can be left unconnected when not used. |
| F1 | LED6_DRV | LED Output 6. Driven when MEASx_DRVy = 5 (x = 1 to 6, y = A, B). Connect the LED cathode to LED6_DRV and its anode to the $V_{LED}$ supply. This pin can be left unconnected when not used. |
| **ECG** | | |
| A7 | ECG_EL1 | ECG Electrode 1 Connection. Configured as an ECG positive input, ECG negative input, or ECG right leg drive output based on the ECG_MUX_SEL[1:0](0x82) register setting. This pin can be left unconnected when not used. |
| A6 | ECG_EL2 | ECG Electrode 2 Connection. Configured as an ECG positive input, ECG negative input, or ECG right leg drive output based on the ECG_MUX_SEL[1:0](0x82) register setting. This pin can be left unconnected when not used. |
| C6 | ECG_EL3 | ECG Electrode 3 Connection. Configured as an ECG positive input, ECG negative input, or ECG right leg drive output based on the ECG_MUX_SEL[1:0](0x82) register setting. This pin can be left unconnected when not used. |
| B7 | CAPP | Analog High-Pass Filter Input. Connect a 1µF X7R capacitor ($C_{HPF}$) between CAPP and CAPN to form a 0.5Hz high-pass response in the ECG channel. For more options of CHPF and corner frequencies, see Table 12. This pin can be left unconnected when not used. |

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## Pin Description (continued)

| PIN | NAME | FUNCTION |
|---|---|---|
| B6 | CAPN | Analog High-Pass Filter Input. Connect a 1µF X7R capacitor ($C_{HPF}$) between CAPP and CAPN to form a 0.5Hz high-pass response in the ECG channel. For more options of $C_{HPF}$ and corner frequencies, see Table 12. This pin can be left unconnected when not used. |
| C7 | RLD_INV | Inverting Input of RLD Amplifier. Connect a gain-setting resistor between RLD_INV and the ECG_ELx pin assigned to RLD, or leave disconnected to use internal gain settings. This pin can be left unconnected when not used. |
| **BIOZ** | | |
| **BIOZ / ELECTRODE CONNECTIONS** | | |
| A1 | BIOZ_EL2A | BIOZ Electrode 2A Connection. BIOZ_EL2A or BIOZ_EL2B are normally connected to the BIOZ receive channel BIP input, but can be switched to connect to the DRVP current generator output under program control. Two BIOZ_EL2 inputs are provided so that one can use the device for both GSR/EDA applications that require an external AC-coupling capacitor and BIA/BIS applications that use the internal AHPF, and thus, do not require the use of an external AC-coupling capacitor. |
| B1 | BIOZ_EL2B | BIOZ Electrode 2B Connection. See the description for BIOZ_EL2A. |
| A2 | BIOZ_EL3A | BIOZ Electrode 3A Connection. BIOZ_EL3A or BIOZ_EL3B are normally connected to the receive channel BIN input, but can be switched to connect to the DRVN current generator output under program control. Two BIOZ_EL3 inputs are provided so that one can use the device for both GSR/EDA applications that require an external AC-coupling capacitor and BIA/BIS applications that use the internal AHPF and thus do not require the use of an external AC-coupling capacitor. |
| B2 | BIOZ_EL3B | BIOZ Electrode 3B Connection. See the description for BIOZ_EL3A. |
| A3 | BIOZ_EL1 | BIOZ Electrode 1 Connection. BIOZ_EL1 is normally connected to the DRVP current generator output, but can be switched to the receive channel BIP input under program control. |
| A4 | BIOZ_EL4 | BIOZ Electrode 4 Connection. BIOZ_EL4 is normally connected to the DRVN current generator output, but can be switched to the receive channel BIN input under program control. |
| **BIOZ / CALIBRATION SUPPORT** | | |
| A5 | CAL1 | Calibration Port 1. Connects the internal DRVP node to an external reference resistor when calibration is enabled. This pin can be left unconnected when not used. |
| B5 | CAL2 | Calibration Port 2. Connects the internal BIP node to an external reference resistor when calibration is enabled. This pin can be left unconnected when not used. |
| B4 | CAL3 | Calibration Port 3. Connects the internal BIN node to an external reference resistor when calibration is enabled. This pin can be left unconnected when not used. |
| B3 | CAL4 | Calibration Port 4. Connects the internal DRVN node to an external reference resistor when calibration is enabled. This pin can be left unconnected when not used. |
| **BIOZ / REFERENCE** | | |
| C5 | DRVSJ | Drive Amplifier Summing Junction. Virtual ground in sine-wave current mode. This pin can be left unconnected when not used. |
| C4 | DRVXR | Drive Amplifier External Resistor. Connect a precision resistor between DRVXR and DRVXC if setting the BioZ drive current externally. Leave unconnected if using internal current settings. This pin can be left unconnected when not used. |
| D5 | DRVXC | Drive Amplifier External Capacitor. Connect a 47nF capacitor between DRVXC and DRVSJ to AC couple the VDRV and IDRV amplifiers in sine-wave current drive applications. Otherwise, short DRVXC to DRVSJ. This pin can be left unconnected when not used. |
| **REFERENCE** | | |
| E6 | VMID_ECG | ECG Common Mode Reference Output. Bypass to AGND with a 10µF X5R ceramic capacitor. This pin can be left unconnected when ECG and BioZ subsystems are not being used. |
| E2 | VREF_PPG | PPG ADC Reference Buffer Output. Bypass to AGND with a 1µF X5R ceramic capacitor. This pin can be left unconnected when PPG subsystem is not being used. |

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## Pin Description (continued)

| PIN | NAME | FUNCTION |
|---|---|---|
| F6 | VREF_ECG | ECG ADC Reference Buffer Output. Bypass to AGND with a 10µF X5R ceramic capacitor. This pin can be left unconnected when ECG and BioZ subsystems are not being used. |
| **CONTROL INTERFACE** | | |
| E4 | SCLK/SCL | SPI Clock in SPI Mode or $I^2C$ Clock in $I^2C$ Mode |
| E5 | SDI/SDA | SPI Data Input in SPI Mode or $I^2C$ Data Input and Output in $I^2C$ Mode |
| F4 | SDO/ADDR | SPI Data Output in SPI Mode or $I^2C$ Address Select in $I^2C$ Mode. Do not leave unconnected. |
| D4 | CSB/ I2C_SEL | Active-Low Chip Select Input in SPI Mode. Pull high or connect to IOVDD to select $I^2C$ Mode. Do not leave unconnected. |
| G4 | INT1 | Interrupt 1 Output. INT1 is a programmable active-high or active-low or open-drain status output. It can be used to interrupt an external device. When not used, it can be left unconnected. |
| G5 | INT2 | Interrupt 2 Output. INT2 is a programmable active-high or active-low or open-drain status output. It can be used to interrupt an external device. When not used, it can be left unconnected. |
| G6 | TRIG | Configurable Synchronization Trigger Input/Output. PPG frame trigger input, PLL sync input/ output, ECG sync output, or LED pulse sync output. When not used, connect to DGND. |
| F5 | FCLK | External Clock Input. Connect to a 32.0kHz or 32.768kHz external clock source (optional). When not used, it can be left unconnected. |

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## Detailed Description

The MAX86178 is a complete, integrated data acquisition systems, ideal for various applications including optical pulse oximetry, heart-rate detection, ECG, respiration, bioimpedance analysis/bioimpedance spectroscopy (BIA/BIS), and numerous other applications. It is designed for the demanding requirements of medical, mobile and wearable devices and requires minimal external hardware components for integration.

The PPG data acquisition system in the MAX86178 supports up to 6 LEDs and 4 photodiode inputs and includes two high-resolution optical-readout signal processing channels with robust ambient light cancellation and high-current LED-driver DACs to form a complete dual-optical readout signal chain.

The ECG channel in the MAX86178 has high input impedance, low noise, high CMRR, programmable gain, right-leg drive, lead-on and lead-off detection, and a high-resolution analog-to-digital converter.

The BioZ transmit channel has an independent current stimulus circuit for providing injected body currents. The stimulus current generation circuit can be supplied in a four-electrode (tetrapolar) manner as well as a two-electrode (bipolar) manner. This injected current is programmable and available over a wide frequency range (16Hz to 500kHz) and a wide range of stimulus current magnitudes ($16nA_{RMS}$, up to $1.28mA_{RMS}$ max). These ranges support GSR (galvanic-skin response) and electrodermal activity (EDA) measurements, BIA/BIS applications, and impedance cardiography (ICG) measurements such as cardiac output and stroke volume, or for impedance plethysmography (IPG) measurements. The transmit channel also includes an independent high-impedance, low-noise current-source driver for measuring respiration simultaneously with ECG.

The BioZ receive channel also has high input impedance, low noise, high CMRR, programmable gain, various low-pass and high-pass filter options, a high resolution analog to digital converters, and I and Q measurement capability to provide resistance and reactance measurements for BIA/BIS applications.

The BioZ channel also includes DC lead-off detection, DRVN lead-off detection, ultra-low-power lead-on detection during standby mode, and extensive calibration features and programmable resistive loads for built-in self-test. Soft power-up sequencing ensures that no large transients are injected into the electrodes.

The MAX86178 provides a calibration port for a four-wire external precision reference resistance to use during calibration. This calibration is required when using the MAX86178 for bioimpedance measurements needing absolute accuracy such as BIA/BIS or AED body-impedance. The four-wire calibration port can also be used to support multiple calibration resistances. Alternatively, there are trimmed load resistors internal to the device that can be used for calibration, but they are not as accurate as using an external reference resistor.

The MAX86178 is fully adjustable through software registers and the digital output data is stored in a 256-word FIFO. The FIFO allows the MAX86178 to be connected to a microcontroller or processor on a shared $I^2C$ or SPI bus. Both operate in fully autonomous mode for low-power battery applications.

The MAX86178 operates on a 1.8V main supply voltage or directly from a battery when using the internal LDO regulator, plus an additional 2.7V to 5.5V LED driver power supply. Both have flexible timing and shutdown configurations as well as control of individual blocks so that an optimized measurement can be made with minimum power consumption and a high level of accuracy.

### Timing Subsystem

The MAX86178 timing subsystem is shown in Figure 1.

Figure 1 shows all the registers bits and formulas needed for setting the PPG frame rate, ECG and BioZ sample rates, as well as the stimulus frequencies for BioZ, respiration and ECG AC lead-off detect.

The sample rates for the three data paths, PPG, ECG and BIOZ are all referenced to a common clock called REF_CLK, which is sourced either from an external oscillator on the FCLK or from the internal slow oscillator clock INT_FCLK depending on the REF_CLK_SEL[6](0x1D) setting. A separate 10MHz oscillator provides PPG_ADC_CLK, which is the clock for the PPG ADC.

The MAX86178 timing system offers a great deal of flexibility. However, certain considerations must be taken into account when configuring the timing system. The following sections describe these considerations in detail.

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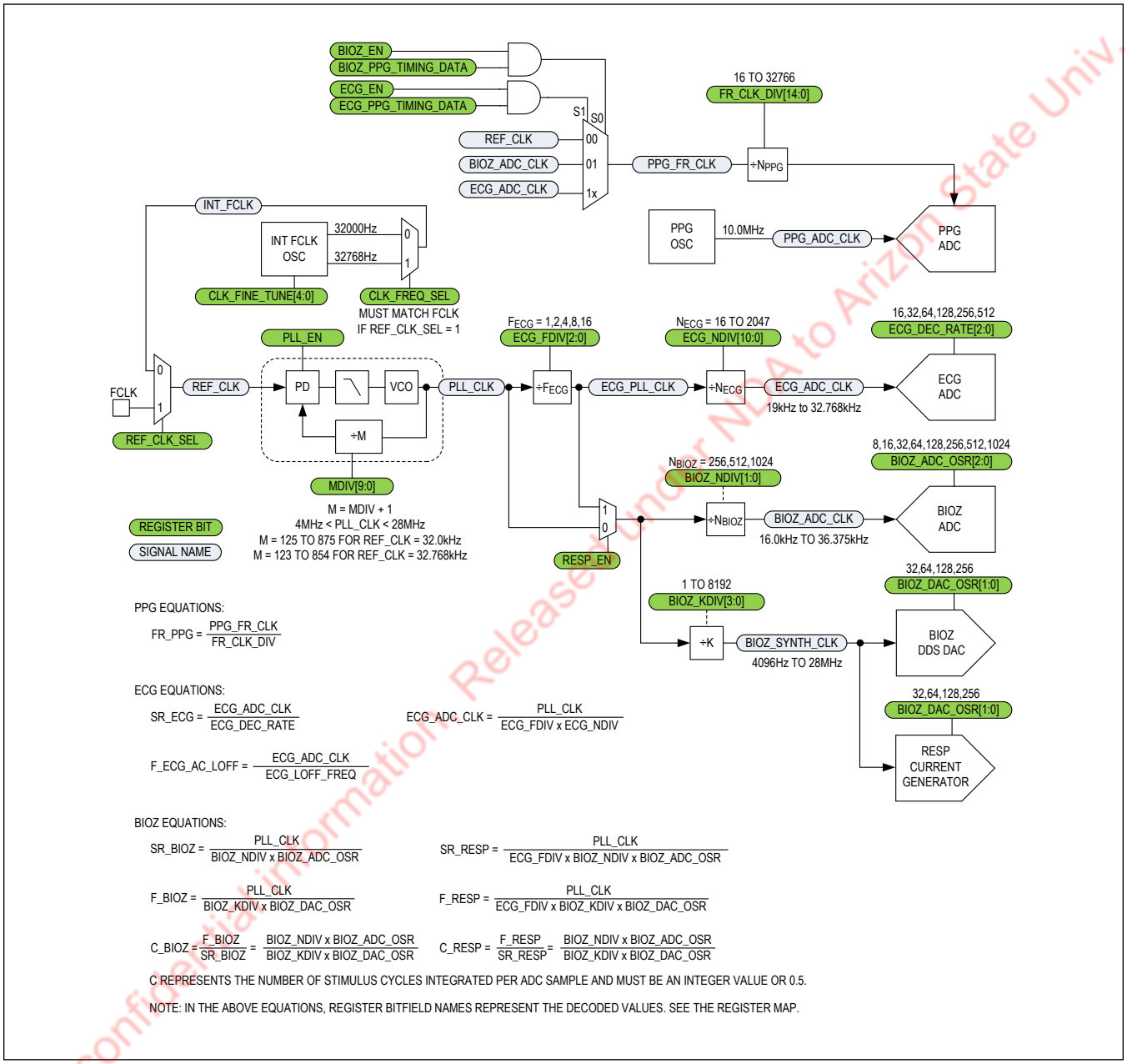



*Figure 1. Timing Subsystem*

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## Clock Sources

The MAX86178 timing system incorporates two internal slow oscillators, 32.0kHz and 32.768kHz, and the user can select the oscillator with the desired frequency by setting CLK_FREQ_SEL[5](0x1D). If REF_CLK_SEL[6](0x1D) = 0, the internal slow oscillator selected by CLK_FREQ_SEL provides the reference clock to the PLL circuit used as the time base for the ECG and BioZ channels and generates the clock for the PPG frame rate (PPG_FR_CLK). If REF_CLK_SEL = 1, the oscillator on the FCLK (either 32.0kHz or 32.768kHz) becomes the source of REF_CLK. Even when using the external oscillator on FCLK, the CLK_FREQ_SEL must be set according to the frequency of the external clock source.

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It is recommended to use a low-jitter external oscillator with < 347ppm of frequency accuracy to meet IEC60601-2-47 timing accuracy compliance. The FCLK source must be an active-drive clock, not only a crystal.

The two internal slow oscillators, 32.0kHz and 32.768kHz in MAX86178 are factory trimmed and exhibit a drift with temperature (primary cause of drift) of less than ±1% over the temperature range of 0°C to +50°C. If this level of stability is inadequate, then the MAX86178 offers a fine adjust register CLK_FINE_TUNE[4:0](0x1D), which can be used in combination with a highly stable crystal based real-time clock (RTC) oscillator in the host microcontroller to trim out the drift of the on-chip slow oscillator. By counting the time between the MAX86178 generated interrupts using the microcontroller based RTC, it is possible to compute the error in the slow oscillator frequency and trim it to within ±0.0625% (typ) of the microcontroller based RTC. Using this approach, it is possible to achieve accuracy near that of a crystal oscillator since the phase noise of the MAX86178 slow oscillator is low and the drift is primarily due to temperature.

The MAX86178 also has a 10MHz on-chip fast oscillator, which is the clock source for PPG ADC and is factory trimmed to better than 1% accuracy at room temperature and 2% accuracy over the entire operating temperature range.

### PLL and PLL Synchronization

The MAX86178 timing subsystem shown in Figure 1 allows the use of an internal PLL synchronized to either an internal or external clock source that is used by the ECG and BioZ channels.

The PLL generates an output clock (PLL_CLK) that operates over a 4MHz to 28MHz frequency range. The frequency of PLL_CLK is selected by the frequency of REF_CLK and the M divider value, which is set in MDIV[9:0](0x18, 0x19), where M = MDIV + 1.

The 10-bit MDIV register field must be set such that the PLL output frequency (PLL_CLK) is between 4MHz and 28MHz. For a reference clock of 32.768kHz, this would mean a valid MDIV range is 122 to 853 (M = 123 to 854). For a reference clock of 32.0kHz, this would mean a valid MDIV range of 124 to 874 (M = 125 to 875).

Soft-reset using RESET[0](0x11) is not allowed when PLL is enabled (PLL_EN[0](0x18) = 1). If the ECG and BioZ reference is not enabled before PLL_EN is set to 1, it automatically turns on, but can take up to 6ms to settle.

### Sequence of Operation when PLL is used

When enabling or disabling PLL, the proper sequence of operations must be followed. This section describes the recommended sequence of operations for various scenarios when PLL is used.

### Enabling and Disabling the PLL

The following sequence is recommended when enabling and disabling the PLL.

- Disable ECG, PPG and BioZ, if enabled.
- Enable PLL by setting PLL_EN to 1.
- Wait for PLL to lock using either the FREQ_LOCK[3](0x02) or PHASE_LOCK[2](0x02) status bits.
- Enable ECG, PPG, BioZ as needed.
- Disable ECG, PPG, and BioZ when data collection is done.
- Disable PLL by setting PLL_EN to 0.

### Entering and Exiting Shutdown

The following sequence is recommended when putting the device into a shutdown state and to exit the shutdown state.

- Disable ECG, PPG, and BioZ if enabled.
- Disable PLL by setting PLL_EN to 0 if enabled.
- Set SHDN to 1, to enter shutdown mode.
- ...
- Set SHDN to 0 to enter normal mode.
- Enable PLL by setting PLL_EN to 1.
- Enable ECG, PPG, and BioZ as needed.
- ...

### Soft-Reset Sequence

The following sequence is recommended when resetting the device using the RESET bit.

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- Disable PLL by setting PLL_EN to 0 if enabled.
- Set RESET = 1 to reset all registers.
- Enable PLL by setting PLL_EN to 1.
- ...

### PLL Synchronization

The MAX86178 provides a PLL synchronization feature for use with multiple MAX86176, MAX30005, MAX86178, or Maxim bioimpedance AFEs in a system. This allows the PLLs of the multiple AFEs to remain synchronized and output synchronized samples. PLL synchronization uses either the TRIG pin or the broadcast feature. Both options are discussed as following.

### PLL Synchronization Using the TRIG Pin

When the TRIG pin is used for PLL synchronization, one AFE is set up to act as a controller and initiates the PLL and timing subsystem synchronization process, while the other(s) act as a target(s). Alternatively, all the AFEs together act as targets and the microcontroller acts as the controller of this process. In either case, all AFEs should use the same external reference clock for the PLLs and all the PLLs must be enabled and locked before generating a timing system reset pulse. The timing when using the MAX86178 in a controller configuration is shown in Figure 2. When acting as a controller, TRIG_FCFG[7:5](0x13) is set to 0x2 and TIMING_SYS_RESET[7](0x10) is set to 1. TIMING_SYS_RESET is a self clearing bit and gets reset to 0 on the second rising edge of FCLK after it is asserted. Once it is cleared, the MAX86178 sends out a timing system reset pulse on the TRIG output after an internal time delay. On the first FCLK rising edge after the TRIG pin is pulled high by the timing system reset pulse, the ECG N-divider, ECG F-divider, and the BioZ N-divider of the controller (if the AFE is the controller) restart their counts at 0.

The timing when using the MAX86178 in a target configuration is shown in Figure 3. When acting as a target, TRIG_FCFG[7:5] (0x13) is set to 0x1. In this mode, MAX86178 receives a timing system reset pulse on the TRIG pin. After the TRIG input is pulled high, the timing system reset signal is latched by the target on the first falling edge of FCLK as shown in Figure 3. On the first FCLK rising edge after the timing system reset pulse is latched on the TRIG input, the ECG N-divider, ECG F-divider, and the BioZ N-divider of the target AFE device(s) restart their count at 0. Since the FCLK input to multiple devices use the same clock, PLLs of all the devices in the system are synchronized automatically.

This is the recommended sequence for synchronizing PLLs on multiple devices using the TRIG pin:

- Disable BioZ by setting ECG_BIOZ_BG_EN[2](0xA0) to 0 and BIOZ_EN[1:0](0xA0) to 0x0 if enabled.
- Disable ECG by settling ECG_EN[0](0x80) to 0 if enabled.
- Disable PPG by setting MEASx_EN (x = 1 to 6) (0x20) to 0 if enabled.
- Reset the part and flush the FIFO by setting SHDN[1](0x11) to 1.
- Then set SHDN = 0 on all the parts .
- Wait for 6ms.
- Program the configuration registers for BioZ, ECG, PPG as needed.
  - For synchronizing multiple devices, the PLLs should be configured such that the PPG frame rate, ECG sample rate, and BioZ sample rate match across all devices.
  - Set ECG_PPG_TIMING_DATA = 1, BIOZ_PPG_TIMING_DATA = 1, ECG_BIOZ_TIMING_DATA = 1, on all the parts (optional).
  - The broadcast feature can be used to program the common registers.
- Set TRIG_FCFG to 0x2 on the controller device, and set TRIG_FCFG to 0x1 on all the target devices.
- Program FIFO_A_FULL[7:0](0x0D) as desired, and A_FULL_EN1[7](0xC0) = 1 on the controller device.
- Enable PLL by setting PLL_EN[0](0x18) to 1 on all the parts (using the broadcast feature or separately).
- Wait for PLL to lock on all the devices.
- Enable BioZ by setting ECG_BIOZ_BG_EN to 1, and BIOZ_EN to 0x1 or 0x2 as needed.
- Enable ECG by setting ECG_EN to 1 if needed.
- Enable PPG by setting MEASx_EN (x = 1 to 6) to 1 if needed.
- Set TIMING_SYS_RESET to 1 on the controller device. All the devices reset the ECG N-divider, ECG F-divider and the BioZ N-divider, and restart the state machines within the current clock cycle.
  - Note: If both ECG_PPG_TIMING_DATA = 0 and BIOZ_PPG_TIMING_DATA = 0, PPG does not use the PLL, and its state machine does not restart.
- Disable BioZ, ECG, and PPG when data collection is done.

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- Disable PLL by setting PLL_EN to 0.

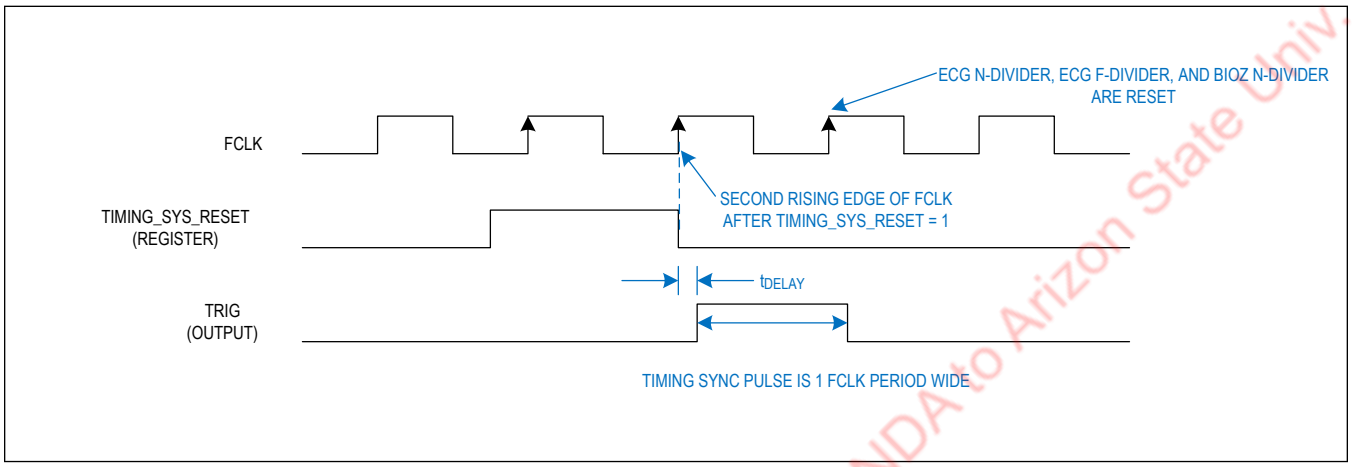



*Figure 2. Timing System Synchronization with MAX86178 as a Controller*

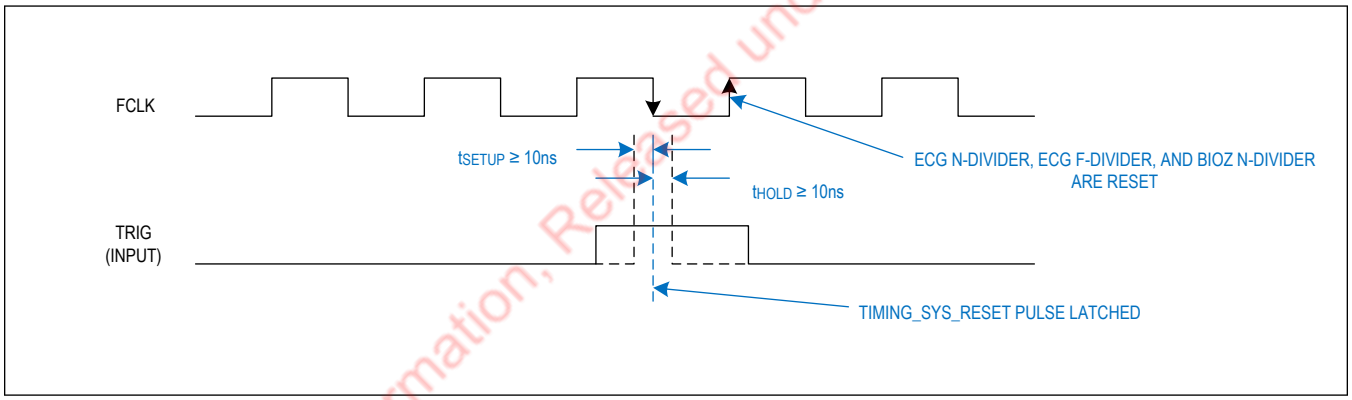



*Figure 3. Timing System Synchronization with MAX86178 as a Target*

### PLL Synchronization Using the Broadcast Command

PLLs on multiple AFEs can be synchronized using the broadcast feature if the TIMING_SYS_RESET[7](0x10) bit is at the same address in all the devices, and if all the PLLs use the same external reference clock for their PLLs.

**Using the I²C Serial Interface:**

Set I2C_BCAST_EN to 1, and I2C_BCAST_ADDR to the upper 7 bits of the broadcast address chosen for the system on all the AFEs whose PLLs need to be synchronized. Using this address as the Target Address, write 1 to the TIMING_SYS_RESET bit.

**Using the SPI Serial Interface:**

Write 1 to the TIMING_SYS_RESET bits of all the AFEs in a single transaction by asserting their CSB inputs at the same time.

The internal timing reset pulse resets the ECG N-divider, ECG F-divider and the BioZ N-divider in all the AFEs at the same time on the third rising edge of the FCLK after TIMING_SYS_RESET bit is set to 1; thus, synchronizing all the PLLs.

This is the recommended sequence for synchronizing PLLs on multiple MAX86178 devices using the broadcast feature:

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- Disable BioZ by setting ECG_BIOZ_BG_EN[2](0xA0) to 0 and BIOZ_EN[1:0](0xA0) to 0x0 if enabled.
- Disable ECG by setting ECG_EN[0[(0x80) to 0 if enabled.
- Disable PPG by setting MEASx_EN (x = 1 to 6) to 0.
- Reset the part and flush the FIFO by setting SHDN[1](0x11) to 1.
- Then set SHDN = 0 on all the parts.
- Wait for 6ms.
- Program the configuration registers for BioZ, ECG, and PPG as needed.
  - For synchronizing multiple devices, the PPG frame rate, ECG sample rate, and BioZ sample rate should be the same on all the parts.
  - Set ECG_PPG_TIMING_DATA = 1, BIOZ_PPG_TIMING_DATA = 1, and ECG_BIOZ_TIMING_DATA = 1 on all the parts (optional).
  - The broadcast feature can be used to program the common registers.
- Program FIFO_A_FULL[7:0](0x0D) as desired, and A_FULL_EN1[7](0xC0) = 1 on one of the parts, which becomes the primary part (others parts are secondary parts).
- Enable PLL by setting PLL_EN[0](0x18) to 1 on all the devices (using the broadcast feature or separately).
- Wait for PLL to lock on all devices.
- Enable BioZ by setting ECG_BIOZ_BG_EN to 1 and BIOZ_EN[1:0] to 0x1 or 0x2 as needed using broadcast feature.
- Enable ECG by setting ECG_EN to 1 as needed, using the broadcast feature.
- Enable PPG by setting MEASx_EN (x = 1 to 6) to 1 as needed using the broadcast feature
- Using the broadcast feature, set TIMING_SYS_RESET to 1 on all the devices.
  - All the devices reset the ECG N-divider, ECG F-divider and the BIOZ N-divider, and restart the state machines within the current clock cycle.

    Note: If ECG_PPG_TIMING_DATA = 0 and BIOZ_PPG_TIMING_DATA = 0, PPG does not use the PLL, and its state machine does not restart.
- Disable BioZ, ECG, and PPG when data collection is done.

  Note: TIMING_SYS_RESET can be set to 1 before or after enabling BioZ, ECG, and PPG.
- Disable PLL by setting PLL_EN to 0.

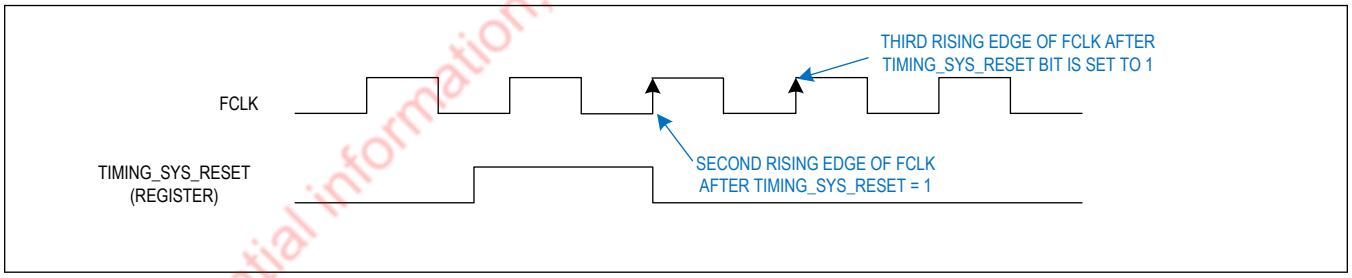



*Figure 4. Timing System Synchronization for MAX86178 Using the Broadcast Feature*

### PPG Frame Rate

The PPG frame rate (FR_PPG) is set by this equation, and some example frame rates are shown in Table 1.

$$FR_PPG = \frac{PPG_FR_CLK}{FR_CLK_DIV[14:0]}$$

When using Sample Averaging (SMP_AVE[6:4](0x22) > 0), the data output rate is reduced by the Frame rate/Sample Averaging factor.

$$Data_Output_Rate = \frac{FR_PPG}{2^{SMP_AVE}}$$

The PPG frame rate clock (PPG_FR_CLK) has several sources to allow for synchronized sampling with ECG or BioZ channels.

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1. By default, the reference clock (as selected by REF_CLK_SEL[6](0x1D)) provides the PPG_FR_CLK.
2. When both ECG and ECG to PPG timing data are enabled (ECG_EN[0](0x80) = 1 and ECG_PPG_TIMING_DATA[5](0x11) = 1), ECG_ADC_CLK is selected as the PPG_FR_CLK to allow ECG samples and PPG frames to be synchronized.
3. When both BioZ and BioZ to PPG timing data are enabled (BIOZ_EN[1:0](0xA0) = 1 and BIOZ_PPG_TIMING_DATA[4](0x11) = 1), BIOZ_ADC_CLK is selected as the PPG_FR_CLK to allow BioZ samples and PPG frames to be synchronized.
4. In case all of the above (ECG_EN, ECG_PPG_TIMING_DATA, BIOZ_EN, and BIOZ_PPG_TIMING_DATA) are enabled, PPG_FR_CLK is sourced from ECG_ADC_CLK.

## Table 1. PPG Frame Rate Examples

| PPG_FR_CLK (kHz) | FR_CLK_DIV[14:0](0x28, 0x29) | FR_PPG (fps) |
|---|---|---|
| 32.768 | 17 | 1927.5 |
| 32.768 | 256 | 128 |
| 32.868 | 32766 | 1.000 |
| 32.0 | 17 | 1882 |
| 32.0 | 256 | 125 |
| 32.0 | 32766 | 0.977 |

The FR_CLK_DIV should be programmed so that it is greater than the product of $t_{MEASUREMENT}$ and PPG_FR_CLK where $t_{MEASUREMENT}$ is the time to complete all the enabled measurement conversions selected in the MEASx_EN (x = 1 to 6) in PPG Configuration 1 (0x20) registers and is calculated as follows.

- $t_{MEASUREMENT} = t_{INIT1} + t_{MEAS1} + t_{MEAS2} + t_{MEAS3} + .....+ t_{MEAS6}$
- $t_{INIT1} = 10 \times t_{FRAME}$
- $t_{MEASx} = [t_{INIT} + MEASx_TINT \times (2 \times 2^{MEASx_AVER} + 1) + 2 \times 2^{MEASx_AVER} \times MEASx_PD_SETLNG] \times MEASx_EN$ when using CDM or SINC3 filter.
- $t_{MEASx} = [t_{INIT} + 2 \times MEASx_TINT + MEASx_PD_SETLNG] \times MEASx_EN$ when using FDM and SINC3 is disabled.

where

$t_{FRAME}$ = Frame clock period, 1/PPG_FR_CLK

$t_{INIT} = 4 \times t_{FRAME}$

MEASx_TINT = Integration time defined in each measurement

MEASx_AVER = Measurement average for each measurement

MEASx_PD_SETLNG = Photodiode settling time defined in each measurement

MEASx_EN = 1 for each enabled measurement.

For details on all these parameters, refer to the Register Map.

If $t_{MEASUREMENT}$ calculated as above results in a frame period longer than $t_{FRAME}$, then a timing error occurs and INVALID_PPG_CFG[7](0x01) is set to 1.

If FR_CLK_DIV[14:0] < 0x0010, it is automatically overwritten to 0x0010, which is the period for the smallest frame period. Note that FR_CLK_DIV[14:0] = 0x7FFF is reserved.

The PPG frame rate when using external oscillator depends on the frequency on FCLK and FR_CLK_DIV[14:0]. PPG measurement and frame timing is driven by the on-chip fast oscillator (PPG OSC provides timing for the PPG channel) and remains the same whether using internal or external reference clock.

If PPG configuration registers (0x28 to 0x5F) are modified during measurement, any active frame is aborted and a new frame is started immediately following each register write. This feature prevents corrupted frames from being pushed to the FIFO, but results in a discontinuity in the frame rate because the frame clock divider is reset. If this behavior is not desired, an external frame trigger on the TRIG pin (with PPG_SYNC_MODE[5](0x21) = 1) should be used to maintain consistent frame timing. To avoid dropped frames, register writes should be performed after the completion of a frame and before the start of the next frame.

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### PPG Synchronization Modes

The MAX86178 supports two synchronization modes for PPG frame rate. The two synchronization modes are internal frame sync and external frame sync, and are controlled by PPG_SYNC_MODE[5](0x21) as shown in Table 2.

## Table 2. Frame Triggers in Synchronization Modes

| PPG_SYNC_MODE | PPG FRAME RATE |
|---|---|
| 0 | PPG_FR_CLK / FR_CLK_DIV |
| 1 | External pulse on TRIG |

Internal frame sync mode is the free running mode of operation for PPG. It is also the default mode. In this mode, the MAX86178 uses the PPG_FR_CLK as a frame rate clock, and FR_CLK_DIV[14:0](0x28, 0x29) as the internal user programmable divider to set the time between subsequent frames or the PPG frame rate. External frame sync mode is a one-shot mode of operation for PPG. Setting TRIG_FCFG[7:5](0x13) to 0x0 and PPG_SYNC_MODE[5](0x21) to 1 enables the TRIG input to trigger a PPG frame. On either the falling edge (TRIG_ICFG[4](0x13) = 0) or the rising edge (TRIG_ICFG[4](0x13) = 1) of the TRIG input, a frame cycle begins. A frame cycle includes a power-up cycle and execution of each enabled measurement from MEAS1 to MEAS6.

### ECG Sample Rate and AC Lead-Off Frequency

The ECG sample rate depends on the state of the following:

- CLK_FREQ_SEL[5](0x1D) defines the frequency of the reference clock
- REF_CLK_SEL[6](0x1D) selects the reference clock for input to PLL
- MDIV[9:0](0x18, 0x19) is the M divider
- ECG_FDIV[3:0](0x1B) is the ECG F divider
- ECG_NDIV[10:0] (0x1B, 0x1C) is the ECG N divider
- ECG_DEC_RATE[3:1](0x80) sets the decimation factor for ECG ADC

The ECG sample rate is calculated as follows.

- $SR_ECG = \frac{ECG_ADC_CLK}{ECG_DEC_RATE}$
- $ECG_ADC_CLK = \frac{PLL_CLK}{ECG_FDIV \times ECG_NDIV}$ (must be between 19.0kHz and 32.768kHz)
- PLL_CLK = REF_CLK × *M* (must be between 4.0MHz and 28MHz)

REF_CLK is either 32.0kHz or 32.768kHz depending on the state of the CLK_FREQ_SEL and REF_CLK_SEL bits, and M = MDIV +1.

The AC lead-off stimulus frequency is calculated as follows:

- $F_ECG_AC_LOFF = \frac{ECG_ADC_CLK}{ECG_LOFF_FREQ}$

Table 3 shows some common ECG sample rates optimized for minimum supply-current consumption. To minimize supply-current consumption for continuous ECG monitoring, PLL_CLK should be set to the lowest frequency possible between 4MHz and 28MHz. ECG_FDIV should be set to the highest value possible while maintaining ECG_ADC_CLK between 19kHz and 32.768kHz.

## Table 3. Common ECG Sample Rates

| SR_ECG (sps) | REF_CLK (kHz) | M = MDIV + 1 | PLL_CLK (MHz) | ECG_FDIV | ECG_NDIV | ECG_ADC_CLK (kHz) | ECG_DEC_RATE |
|---|---|---|---|---|---|---|---|
| 125 | 32.768 | 125 | 4.096 | 8 | 16 | 32.0 | 256 |
| 128 | 32.768 | 128 | 4.194 | 8 | 16 | 32.768 | 256 |
| 200 | 32.768 | 200 | 6.554 | 16 | 16 | 25.4 | 128 |
| 250 | 32.768 | 125 | 4.096 | 8 | 16 | 32.0 | 128 |
| 256 | 32.768 | 128 | 4.194 | 8 | 16 | 32.768 | 128 |
| 300 | 32.768 | 150 | 4.915 | 16 | 16 | 19.2 | 64 |
| 360 | 32.768 | 180 | 5.898 | 16 | 16 | 23.04 | 64 |

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## Table 3. Common ECG Sample Rates (continued)

| SR_ECG (sps) | REF_CLK (kHz) | M = MDIV + 1 | PLL_CLK (MHz) | ECG_FDIV | ECG_NDIV | ECG_ADC_CLK (kHz) | ECG_DEC_RATE |
|---|---|---|---|---|---|---|---|
| 500 | 32.768 | 125 | 4.096 | 8 | 16 | 32.0 | 64 |
| 512 | 32.768 | 128 | 4.194 | 8 | 16 | 32.768 | 64 |
| 1000 | 32.768 | 125 | 4.096 | 8 | 16 | 32.0 | 32 |
| 1024 | 32.768 | 128 | 4.194 | 8 | 16 | 32.768 | 32 |
| 2000 | 32.768 | 125 | 4.096 | 8 | 16 | 32.0 | 16 |
| 2048 | 32.768 | 128 | 4.194 | 8 | 16 | 32.768 | 16 |

### BioZ Sample Rate and Stimulus Frequency

To make a BioZ measurement, the parameters listed below must be set, and then either I or Q measurements enabled by BIOZ_EN[1:0](0xA0).

The BioZ sample rate and stimulus frequencies depend on the state of the following fields:

- MDIV[9:0](0x18, 0x19)
- BIOZ_NDIV[7:6](0x1A)
- BIOZ_KDIV[3:0](0x1A)
- BIOZ_ADC_OSR[5:3](0xA0)
- BIOZ_DAC_OSR[7:6](0xA0)
- CLK_FREQ_SEL[5](0x1D)
- REF_CLK_SEL[6](0x1D)

The BioZ sample rate is calculated as follows.

- $SR_BIOZ = \frac{PLL_CLK}{BIOZ_NDIV \times BIOZ_ADC_OSR}$
- $BIOZ_ADC_CLK = \frac{PLL_CLK}{BIOZ_NDIV}$ (must be between 16.0kHz and 36.375kHz)
- PLL_CLK = *M*xREF_CLK (must be between 4MHz and 28MHz)

REF_CLK is either 32.0kHz or 32.768kHz depending on the state of the CLK_FREQ_SEL and REF_CLK_SEL bits, and M = MDIV +1.

The BioZ stimulus frequency is set by the following equation.

- $F_BIOZ = \frac{PLL_CLK}{BIOZ_KDIV \times BIOZ_DAC_OSR}$
- $BIOZ_SYNTH_CLK = \frac{PLL_CLK}{BIOZ_KDIV}$ (must be between 4096Hz and 28MHz)

The ratio of F_BIOZ to SR_BIOZ must be an integer value or 0.5, so that each BioZ sample is integrated over a given number of stimulus half-cycles. This ratio, C_BIOZ, is calculated by the following equation.

- $C_BIOZ = \frac{F_BIOZ}{SR_BIOZ} = \frac{BIOZ_NDIV \times BIOZ_ADC_OSR}{BIOZ_KDIV \times BIOZ_DAC_OSR}$

The procedure for setting the BioZ timing parameters is as follows:

First decide the target stimulus frequency (F_BIOZ) for the BioZ measurement.

If F_BIOZ < 54,668Hz

1. Set BIOZ_DAC_OSR = 256.
2. Set BIOZ_KDIV to get PLL_CLK in range (see Note below Table 6).
3. Calculate MDIV + 1 = ROUND(PLL_CLK / REF_CLK).
4. Set BIOZ_NDIV to get BIOZ_ADC_CLK in range.
5. Set BIOZ_ADC_OSR so that C is an integer or 0.5.

If F_BIOZ > 54,668Hz

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1. Set BIOZ_KDIV = 1.
2. Set BIOZ_DAC_OSR to get PLL_CLK in range.
3. Calculate MDIV + 1 = ROUND(PLL_CLK / REF_CLK).
4. Set BIOZ_NDIV to get BIOZ_ADC_CLK in range.
5. Set BIOZ_ADC_OSR so that C is an integer or 0.5.

Examples are shown in Table 4 and Table 5

## Table 4. Example Calculations of BioZ Configuration Parameters for F_BIOZ < 54688Hz

| STEP | APPLICATION | EX1 | EX2 | EX3 | EX4 | EX5 |
|---|---|---|---|---|---|---|
| | EDA/GSR | X | X | X | | |
| | BIA/BIS | | | X | X | X |
| | RESP | | | | | X |
| | ICG | | | | | |
| | CLK_REF | 32,768 | 32,768 | 32,768 | 32,768 | 32,768 |
| Target | F_BIOZ | 8 | 100 | 1,000 | 10,000 | 40,000 |
| Step 1 | BIOZ_DAC_OSR[2:0] | 256 | 256 | 256 | 256 | 256 |
| Target | BIOZ_SYNTH_CLK | 2,048 | 25,600 | 256,000 | 2,560,000 | 10,240,000 |
| Step 2 | BIOZ_KDIV[3:0] | 8192 | 1024 | 64 | 8 | 2 |
| Target | PLL_CLK | 16,777,216 | 26,214,400 | 16,384,000 | 20,480,000 | 20,480,000 |
| Step 3 | MDIV[9:0] + 1 | 512 | 800 | 500 | 625 | 625 |
| Step 4 | BIOZ_NDIV[3:0] | 1024 | 1024 | 512 | 1024 | 1024 |
| Target | BIOZ_ADC_CLK | 16,384 | 25,600 | 32,000 | 20,000 | 20,000 |
| Step 5 | BIOZ_ADC_OSR[2:0] | 1024 | 512 | 128 | 128 | 128 |
| | C | 0.5 | 2 | 4 | 64 | 256 |
| Actual | PLL_CLK | 16,777,216 | 26,208,000 | 16,384,000 | 20,480,000 | 20,480,000 |
| Actual | BIOZ_ADC_CLK | 16,384 | 25,600 | 32,000 | 20,000 | 20,000 |
| Actual | BIOZ_SYNTH_CLK | 2,048 | 25,600 | 256,000 | 2,560,000 | 10,240,000 |
| Actual | SR_BIOZ | 16 | 50 | 250 | 156.25 | 156.25 |
| Actual | F_BIOZ | 8 | 100 | 1,000 | 10,000 | 40,000 |
| Actual | Error | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |

## Table 5. Example Calculations of BioZ Configuration Parameters for F_BIOZ > 54688Hz

| STEP | APPLICATION | EX1 | EX2 | EX3 | EX4 | EX5 |
|---|---|---|---|---|---|---|
| | EDA/GSR | | | | | |
| | BIA/BIS | X | X | X | X | X |
| | RESP | X | X | X | | |
| | ICG | X | X | X | | |
| | CLK_REF | 32,768 | 32,768 | 32,768 | 32,768 | 32,768 |
| Target | F_BIOZ | 54688 | 100000 | 150,000 | 250,000 | 500,000 |
| Step 1 | BIOZ_KDIV[3:0] | 1 | 1 | 1 | 1 | 1 |

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## Table 5. Example Calculations of BioZ Configuration Parameters for F_BIOZ > 54688Hz (continued)

| STEP | APPLICATION | EX1 | EX2 | EX3 | EX4 | EX5 |
|---|---|---|---|---|---|---|
| Target | BIOZ_DAC_OSR[2:0] | 256 | 256 | 128 | 64 | 32 |
| Step 2 | BIOZ_SYNTH_CLK | 14000128 | 25600000 | 19200000 | 16000000 | 16000000 |
| Target | PLL_CLK | 14,000,128 | 25,600,000 | 19,200,000 | 16,000,000 | 16,000,000 |
| Step 3 | MDIV[9:0] + 1 | 427 | 781 | 586 | 488 | 488 |
| Step 4 | BIOZ_NDIV[3:0] | 512 | 1024 | 1024 | 512 | 512 |
| Target | BIOZ_ADC_CLK | 27,344 | 25,000 | 18,750 | 31,250 | 31,250 |
| Step 5 | BIOZ_ADC_OSR[2:0] | 128 | 128 | 128 | 256 | 256 |
|  | C | 427 | 512 | 1024 | 2048 | 4096 |
| Actual | PLL_CLK | 13,991,936 | 25,591,808 | 19,202,048 | 15,990,784 | 15.990,784 |
| Actual | BIOZ_ADC_CLK | 27,328 | 24,992 | 18,752 | 31,232 | 31,232 |
| Actual | BIOZ_SYNTH_CLK | 13,991,936 | 25,591,808 | 19,202,048 | 15,990,784 | 15,990,784 |
| Actual | SR_BIOZ | 213.5 | 195.25 | 146.5 | 122.0 | 122 |
| Actual | F_BIOZ | 54,656 | 99,968 | 150,016 | 249,856 | 499,712 |
| Actual | Error | -0.06% | -0.03% | 0.01% | -0.06% | -0.06% |

Some common stimulus frequencies are shown in Table 6 for REF_CLK frequencies of 32.768kHz. In the BioZ receive channel, the demodulation is done at the same frequency as the stimulus.

## Table 6. Common BioZ Stimulus Frequencies and Sample Rates with REF_CLK = 32.768kHz

| REF_CLK (Hz) | M (MDIV + 1) | PLL_CLK (Hz) | BIOZ_ KDIV | BIOZ_ DAC_OSR | F_BIOZ (Hz) | BIOZ_NDIV | BIOZ_ ADC_OSR | INTEGRATION CYCLES | SR_BIOZ (sps) |
|---|---|---|---|---|---|---|---|---|---|
| 32768 | 790 | 25886720 | 1 | 32 | 808960 | 1024 | 128 | 4096 | 197.50 |
| 32768 | 706 | 23134208 | 1 | 32 | 722944 | 1024 | 128 | 4096 | 176.50 |
| 32768 | 634 | 20774912 | 1 | 32 | 649216 | 1024 | 128 | 4096 | 158.50 |
| 32768 | 568 | 18612224 | 1 | 32 | 581632 | 512 | 128 | 2048 | 284.00 |
| 32768 | 488 | 15990784 | 1 | 32 | 499712 | 512 | 128 | 2048 | 244.00 |
| 32768 | 458 | 15007744 | 1 | 32 | 468992 | 512 | 128 | 2048 | 229.00 |
| 32768 | 822 | 26935296 | 1 | 64 | 420864 | 1024 | 128 | 2048 | 205.50 |
| 32768 | 738 | 24182784 | 1 | 64 | 377856 | 1024 | 128 | 2048 | 184.50 |
| 32768 | 662 | 21692416 | 1 | 64 | 338944 | 1024 | 128 | 2048 | 165.50 |
| 32768 | 594 | 19464192 | 1 | 64 | 304128 | 1024 | 128 | 2048 | 148.50 |
| 32768 | 533 | 17465344 | 1 | 64 | 272896 | 512 | 128 | 1024 | 266.50 |
| 32768 | 488 | 15990784 | 1 | 64 | 249856 | 512 | 128 | 1024 | 244.00 |
| 32768 | 479 | 15695872 | 1 | 64 | 245248 | 512 | 128 | 1024 | 239.50 |
| 32768 | 430 | 14090240 | 1 | 64 | 220160 | 512 | 128 | 1024 | 215.00 |
| 32768 | 781 | 25591808 | 1 | 128 | 199936 | 1024 | 128 | 1024 | 195.25 |
| 32768 | 691 | 22642688 | 1 | 128 | 176896 | 1024 | 128 | 1024 | 172.75 |
| 32768 | 621 | 20348928 | 1 | 128 | 158976 | 1024 | 128 | 1024 | 155.25 |

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## Table 6. Common BioZ Stimulus Frequencies and Sample Rates with REF_CLK = 32.768kHz (continued)

| REF_CLK (Hz) | M (MDIV + 1) | PLL_CLK (Hz) | BIOZ_ KDIV | BIOZ_ DAC_OSR | F_BIOZ (Hz) | BIOZ_NDIV | BIOZ_ ADC_OSR | INTEGRATION CYCLES | SR_BIOZ (sps) |
|---|---|---|---|---|---|---|---|---|---|
| 32768 | 559 | 18317312 | 1 | 128 | 143104 | 512 | 128 | 512 | 279.50 |
| 32768 | 512 | 16777216 | 1 | 128 | 131072 | 512 | 128 | 512 | 256.00 |
| 32768 | 449 | 14712832 | 1 | 128 | 114944 | 512 | 128 | 512 | 224.50 |
| 32768 | 781 | 25591808 | 1 | 256 | 99968 | 1024 | 128 | 512 | 195.25 |
| 32768 | 727 | 23822336 | 1 | 256 | 93056 | 1024 | 128 | 512 | 181.75 |
| 32768 | 648 | 21233664 | 1 | 256 | 82944 | 1024 | 128 | 512 | 162.00 |
| 32768 | 641 | 21004288 | 1 | 256 | 82048 | 1024 | 128 | 512 | 160.25 |
| 32768 | 586 | 19202048 | 1 | 256 | 75008 | 1024 | 128 | 512 | 146.50 |
| 32768 | 523 | 17137664 | 1 | 256 | 66944 | 512 | 128 | 256 | 261.50 |
| 32768 | 469 | 15368192 | 1 | 256 | 60032 | 512 | 128 | 256 | 234.50 |
| 32768 | 844 | 27656192 | 2 | 256 | 54016 | 1024 | 128 | 256 | 211.00 |
| 32768 | 781 | 25591808 | 2 | 256 | 49984 | 1024 | 128 | 256 | 195.25 |
| 32768 | 672 | 22020096 | 2 | 256 | 43008 | 1024 | 128 | 256 | 168.00 |
| 32768 | 641 | 21004288 | 2 | 256 | 41024 | 1024 | 128 | 256 | 160.25 |
| 32768 | 609 | 19955712 | 2 | 256 | 38976 | 1024 | 256 | 512 | 76.13 |
| 32768 | 547 | 17924096 | 2 | 256 | 35008 | 512 | 256 | 256 | 136.75 |
| 32768 | 484 | 15859712 | 2 | 256 | 30976 | 512 | 256 | 256 | 121.00 |
| 32768 | 438 | 14352384 | 2 | 256 | 28032 | 512 | 256 | 256 | 109.50 |
| 32768 | 781 | 25591808 | 4 (See Note) | 256 | 24992 | 1024 | 256 | 256 | 97.63 |
| 32768 | 719 | 23560192 | 4 (See Note) | 256 | 23008 | 1024 | 256 | 256 | 89.88 |
| 32768 | 625 | 20480000 | 4 (See Note) | 256 | 20000 | 1024 | 256 | 256 | 78.13 |
| 32768 | 563 | 18448384 | 4 (See Note) | 256 | 18016 | 512 | 256 | 128 | 140.75 |
| 32768 | 500 | 16384000 | 4 (See Note) | 256 | 16000 | 512 | 256 | 128 | 125.00 |
| 32768 | 469 | 15368192 | 4 (See Note) | 256 | 15008 | 512 | 256 | 128 | 117.25 |
| 32768 | 438 | 14352384 | 4 (See Note) | 256 | 14016 | 512 | 256 | 128 | 109.50 |
| 32768 | 813 | 26640384 | 8 | 256 | 13008 | 1024 | 256 | 128 | 101.63 |
| 32768 | 750 | 24576000 | 8 | 256 | 12000 | 1024 | 256 | 128 | 93.75 |
| 32768 | 688 | 22544384 | 8 | 256 | 11008 | 1024 | 256 | 128 | 86.00 |
| 32768 | 625 | 20480000 | 8 | 256 | 10000 | 1024 | 256 | 128 | 78.13 |
| 32768 | 563 | 18448384 | 8 | 256 | 9008 | 1024 | 256 | 128 | 70.38 |
| 32768 | 500 | 16384000 | 8 | 256 | 8000 | 512 | 256 | 64 | 125.00 |
| 32768 | 438 | 14352384 | 8 | 256 | 7008 | 512 | 256 | 64 | 109.50 |

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## Table 6. Common BioZ Stimulus Frequencies and Sample Rates with REF_CLK = 32.768kHz (continued)

| REF_CLK (Hz) | M (MDIV + 1) | PLL_CLK (Hz) | BIOZ_ KDIV | BIOZ_ DAC_OSR | F_BIOZ (Hz) | BIOZ_NDIV | BIOZ_ ADC_OSR | INTEGRATION CYCLES | SR_BIOZ (sps) |
|---|---|---|---|---|---|---|---|---|---|
| 32768 | 750 | 24576000 | 16 | 256 | 6000 | 1024 | 256 | 64 | 93.75 |
| 32768 | 625 | 20480000 | 16 | 256 | 5000 | 1024 | 256 | 64 | 78.13 |
| 32768 | 500 | 16384000 | 16 | 256 | 4000 | 512 | 256 | 32 | 125.00 |
| 32768 | 500 | 16384000 | 32 | 256 | 2000 | 512 | 512 | 32 | 62.50 |
| 32768 | 500 | 16384000 | 64 | 256 | 1000 | 512 | 1024 | 32 | 31.25 |
| 32768 | 500 | 16384000 | 128 | 256 | 500 | 512 | 1024 | 16 | 31.25 |
| 32768 | 500 | 16384000 | 256 | 256 | 250 | 512 | 1024 | 8 | 31.25 |
| 32768 | 500 | 16384000 | 512 | 256 | 125 | 512 | 1024 | 4 | 31.25 |
| 32768 | 500 | 16384000 | 1024 | 256 | 63 | 512 | 1024 | 2 | 31.25 |
| 32768 | 500 | 16384000 | 2048 | 256 | 31 | 512 | 1024 | 1 | 31.25 |
| 32768 | 500 | 16384000 | 4096 | 256 | 16 | 1024 | 1024 | 1 | 15.63 |
| 32768 | 500 | 16384000 | 8192 | 256 | 8 | 1024 | 1024 | 0.5 | 15.63 |

Note: Setting BIOZ_KDIV = 4 results in a larger offset in the receive channel. If a calibration is applied to the measurement, this effect is negated by the offset measurement and subtraction. If no calibration is applied and accurate impedance is desired, it is recommended to avoid using this setting, and instead lowering MDIV by a factor of two to achieve the desired frequency with BIOZ_KDIV = 2.

### Respiration Sample Rate and Stimulus Frequency

To enable respiration mode, set RESP_EN[0](0xB6). In this mode, the stimulus frequency and sample rate have slightly different control registers than other BioZ modes.

The respiration sample rate and stimulus frequencies depend on the state of the following fields:

- MDIV[9:0](0x18, 0x19)
- ECG_FDIV[3:0](0x1B)
- BIOZ_NDIV[7:6](0x1A)
- BIOZ_KDIV[3:0](0x1A)
- BIOZ_ADC_OSR[5:3](0xA0)
- BIOZ_DAC_OSR[7:6](0xA0)
- CLK_FREQ_SEL[5](0x1D)
- REF_CLK_SEL[6](0x1D)

The respiration sample rate is calculated as follows.

- $SR_RESP = \frac{PLL_CLK}{ECG_FDIV \times BIOZ_NDIV \times BIOZ_ADC_OSR}$
- $BIOZ_ADC_CLK = \frac{PLL_CLK}{BIOZ_NDIV \times ECG_FDIV}$ (must be between 16.0kHz and 36.375kHz)
- PLL_CLK = *M*xREF_CLK (must be between 4MHz and 28MHz)

REF_CLK is either 32.0kHz or 32.768kHz depending on the state of the CLK_FREQ_SEL and REF_CLK_SEL bits, and M = MDIV +1.

The respiration stimulus frequency is calculated as follows.

- $F_RESP = \frac{PLL_CLK}{ECG_FDIV \times BIOZ_KDIV \times BIOZ_DAC_OSR}$
- $BIOZ_SYNTH_CLK = \frac{PLL_CLK}{BIOZ_KDIV \times ECG_FDIV}$ (must be between 4096Hz and 28MHz)

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The ratio of F_RESP to SR_RESP must be an integer value or 0.5, so that each respiration sample is integrated over a given number of stimulus half-cycles. This ratio (C_RESP) is calculated by the following equation.

- $C_RESP = \frac{F_RESP}{SR_RESP} = \frac{BIOZ_NDIV \times BIOZ_ADC_OSR}{BIOZ_KDIV \times BIOZ_DAC_OSR}$

Table 7 shows some common combinations of ECG sample rates and respiration sample rates since the two features are commonly used together. To minimize supply current consumption for continuous ECG and respiration monitoring, PLL_CLK should be set to the lowest frequency between 4MHz and 28MHz allowed by the BIOZ_ADC_CLK range (16.0kHz to 36.375kHz).

## Table 7. Common ECG and Respiration Sample Rates

| SR_ECG (sps) | REF_CLK (kHz) | M (MDIV + 1) | PLL_CLK (MHz) | ECG_FDIV | ECG_NDIV | ECG_ADC_CLK (kHz) | ECG_DEC_RATE | BIOZ_NDIV | BIOZ_ADC_CLK (kHz) | BIOZ_ADC_OSR | SR_RESP (sps) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 128 | 32.768 | 128 | 4.194 | 1 | 128 | 32.768 | 256 | 256 | 16.384 | 512 | 32 |
| 125 | 32.768 | 125 | 4.096 | 1 | 128 | 32 | 256 | 256 | 16 | 512 | 31.25 |
| 200 | 32.768 | 200 | 6.554 | 1 | 256 | 25.6 | 128 | 256 | 25.6 | 1024 | 25 |
| 250 | 32.768 | 125 | 4.096 | 1 | 128 | 32 | 128 | 256 | 16 | 512 | 31.25 |
| 256 | 32.768 | 128 | 4.194 | 1 | 128 | 32.768 | 128 | 256 | 16.384 | 512 | 32 |
| 300 | 32.768 | 150 | 4.915 | 1 | 256 | 19.2 | 64 | 256 | 19.2 | 512 | 37.5 |
| 360 | 32.768 | 180 | 5.898 | 1 | 256 | 23.04 | 64 | 256 | 23.04 | 512 | 45 |
| 500 | 32.768 | 250 | 4.096 | 1 | 128 | 32 | 64 | 256 | 16 | 256 | 50 |
| 512 | 32.768 | 256 | 4.194 | 1 | 128 | 32.768 | 64 | 256 | 16.384 | 256 | 64 |

Note: Setting BIOZ_KDIV = 4 results in a larger offset in the receive channel. It is recommended to avoid this setting for respiration applications, and instead increase MDIV or FDIV by a factor of two to achieve the desired frequency with BIOZ_KDIV = 2 or 8.

### Timing Data in the FIFO

The MAX86178 supports the timing reference of samples in the FIFO for the separate data paths (PPG, ECG, and BioZ) with respect to each other. Below are the details on ECG-to-PPG timing reference, BioZ-to-PPG timing reference, and ECG-to-BioZ timing reference.

### ECG-to-PPG Timing Data

The timing reference of PPG samples with respect to the ECG samples in the FIFO is supported by setting ECG_PPG_TIMING_DATA[5](0x11) to 1. When this bit is set to 1, PPG timing data is saved in the FIFO at the start of the first measurement in a PPG frame. This timing data is the number of PPG frame clock (PPG_FR_CLK) cycles, starting from the time when the last ECG sample was saved in the FIFO until the first PPG measurement in a frame starts. The timing data is 10 bits wide and saturates at 0x3FF.

See the *FIFO Description* section for information on the tag used for ECG-to-PPG timing data. Figure 5, Figure 6, and Figure 7 show some examples.

The timing of each measurement after the PPG Timing Data is saved in the FIFO is shown in Figure 8.

Timing calculations for measurements with MEASx_SINC3 = 1 (SINC3 mode) or MEASx_FILT_SEL = 0 (CDM mode) are:

$t_1$ = 64 fast clocks (10MHz)

$t_{2n}$ = MEASx_TINT x (2 x MEASx_AVER + 1) + MEASx_PD_SETLNG x 2 x MEASx_AVER

$t_{3n}$ = Roundup (( $t_1$ + $t_{2n}$) / $t_{FRAME}$ ) + 3 x PPG_FR_CLK cycles

Timing calculations for measurements with MEASx_SINC3 = 0 (POR) and MEASx_FILT_SEL = 1 (FDM mode) are:

$t_1$ = 64 fast clocks (10 MHz)

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$t_{2n}$ = 2 x MEASx_TINT + MEASx_PD_SETLNG

$t_{3n}$ = Roundup (( $t_1$ + $t_{2n}$) / $t_{FRAME}$ ) + 3 x PPG_FR_CLK cycles

where,

$t_{FRAME}$ = Frame Clock period (32.0kHz/32.768kHz)

MEASx_TINT = Integration time defined for each measurement (x = 1 to 6)

MEASx_AVER = Number of averages defined for each measurement (x = 1 to 6)

MEASx_PD_SETLNG = Photodiode settling time defined for each measurement (x = 1 to 6)

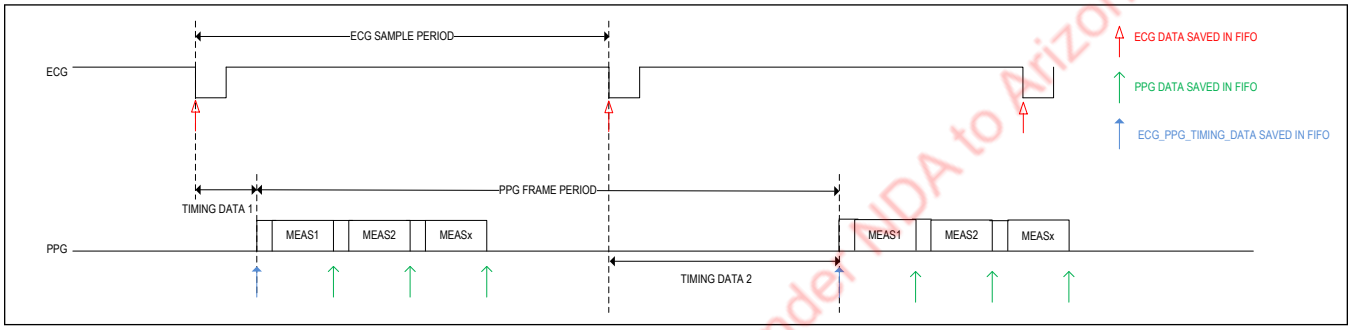



*Figure 5. Example 1: ECG Sample Rate Faster than PPG Frame Rate*

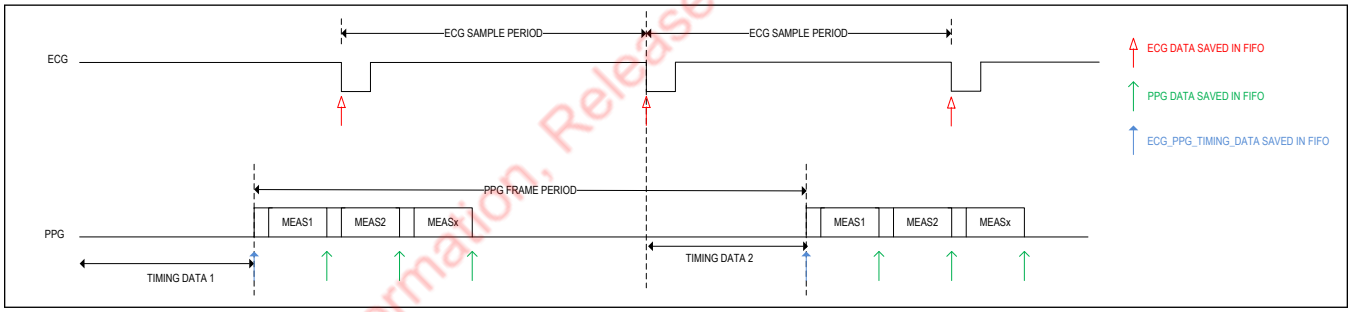



*Figure 6. Example 2: ECG Sample Rate Faster than PPG Frame Rate*

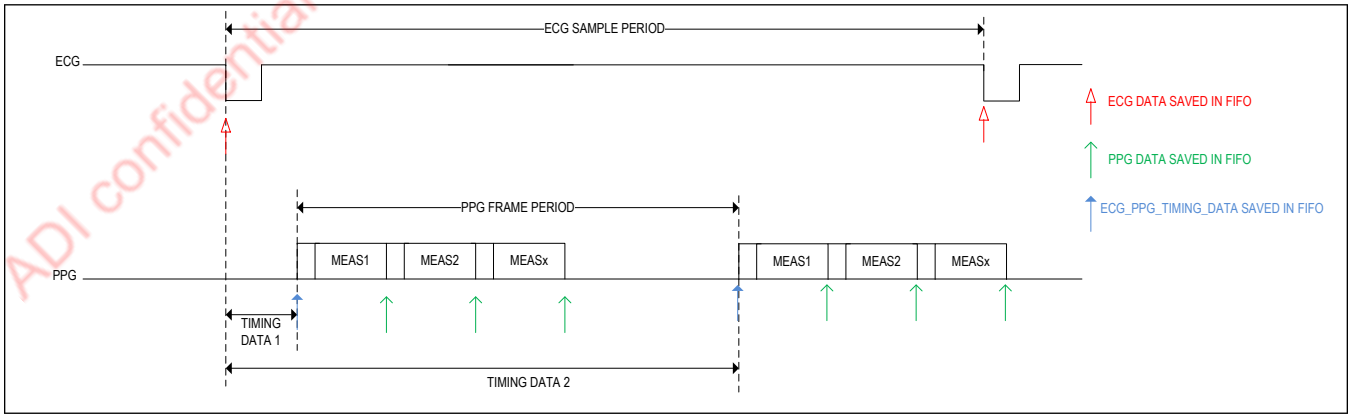



*Figure 7. Example 3: ECG Sample Rate Slower than PPG Frame Rate*

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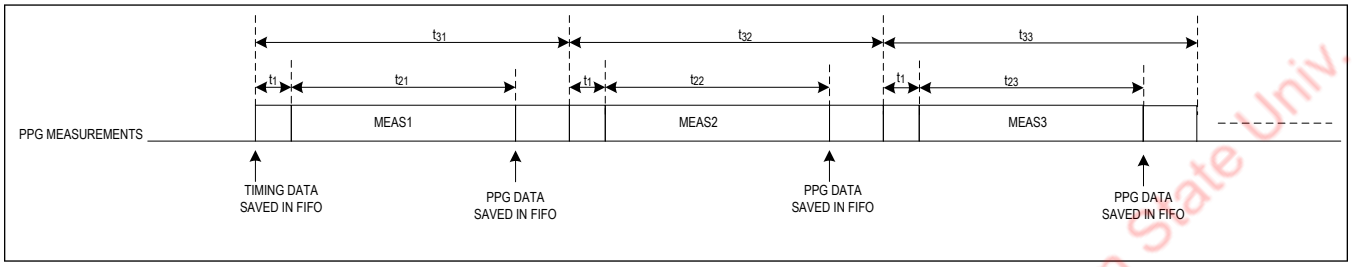



*Figure 8. PPG Measurement Timing in a Frame*

### BioZ-to-PPG Timing Data

The timing reference of the PPG samples with respect to the BioZ samples in the FIFO is supported by setting BIOZ_PPG_TIMING_DATA[4](0x11) to 1. When this bit is set to 1, PPG timing data is saved in the FIFO at the start of the first measurement in a PPG frame. This timing data is the number of PPG frame clock (PPG_FR_CLK) cycles, starting from the time when the last BioZ sample was saved in the FIFO until the first PPG measurement in a frame starts. The timing data is 14 bits wide and saturates at 0x3FFF.

See the *FIFO Description* section for information on the tag used for BioZ-to-PPG timing data. Figure 9, Figure 10, and Figure 11 show some examples.

The timing calculation for measurements is explained in the *ECG to PPG Timing Data* section.

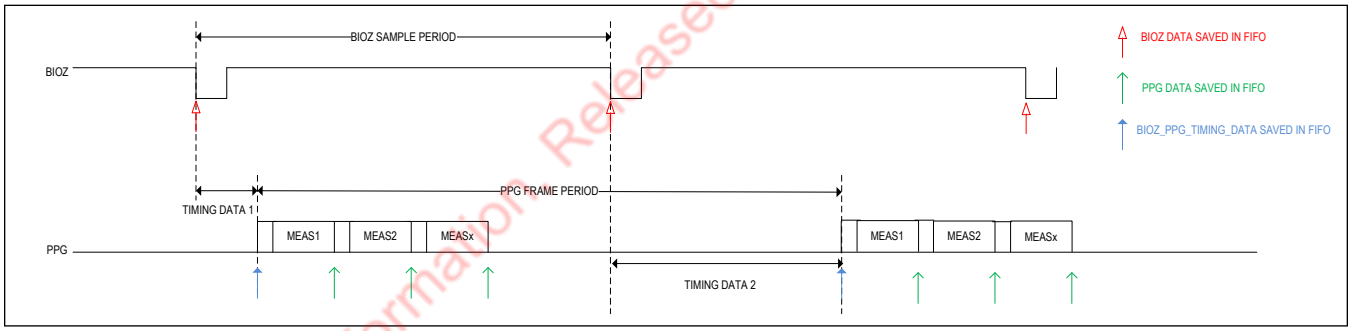



*Figure 9. Example 1: BioZ Sample Rate Faster than PPG Frame Rate*

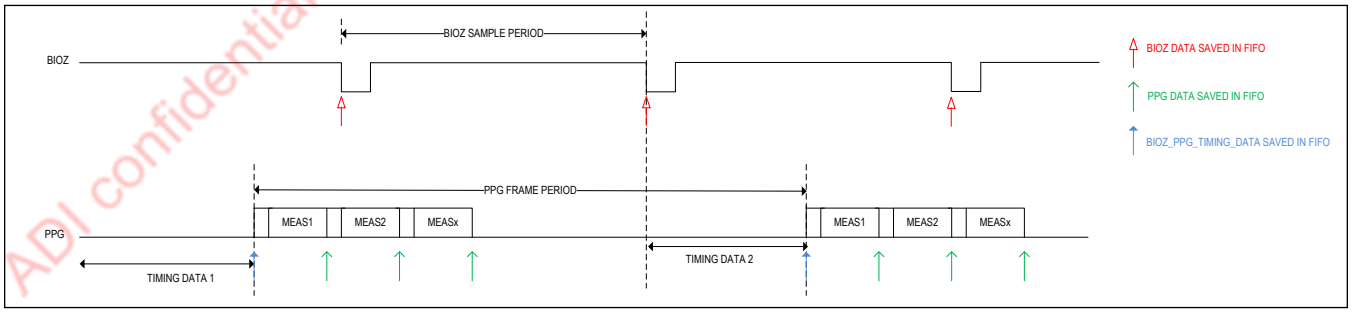



*Figure 10. Example 2: BioZ Sample Rate Faster than PPG Frame Rate*

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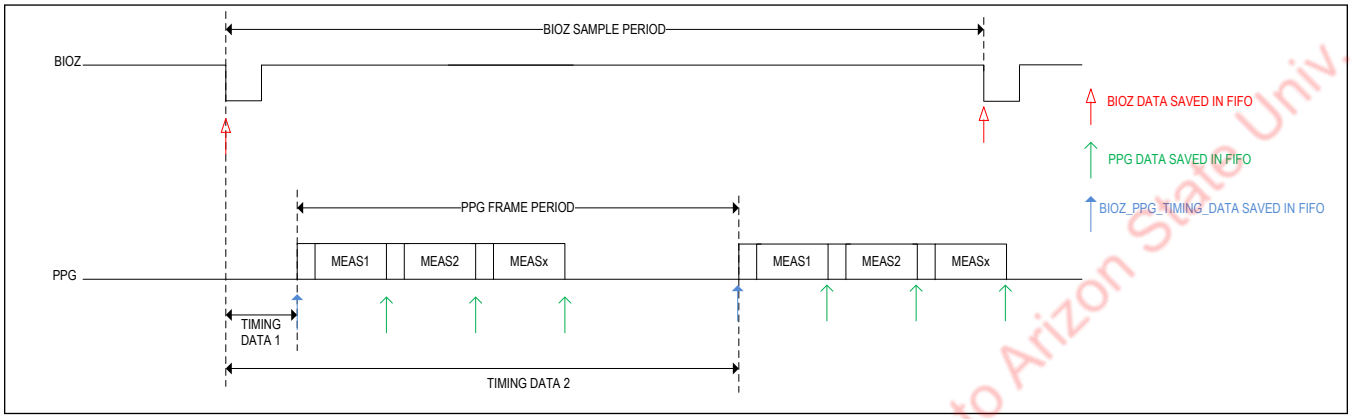



*Figure 11. Example 3: BioZ Sample Rate Slower than PPG Frame Rate*

### ECG-to-BioZ Timing Data

The timing reference of BioZ samples with respect to ECG samples in the FIFO is supported by setting ECG_BIOZ_TIMING_DATA[3](0x11) to 1. When this bit is set to 1, BioZ timing data is saved in the FIFO when the BioZ data is ready to be saved in the FIFO. This timing data is the number of PLL reference clock (REF_CLK) cycles, starting from the time when the last ECG sample was saved to FIFO until the BioZ data is ready to be saved in the FIFO. The timing data is 10 bits wide and saturates at 0x3FF.

See the *FIFO Description* section for information on the tag used for ECG-to-BioZ timing data. Figure 12, Figure 13, and Figure 14 show some examples.

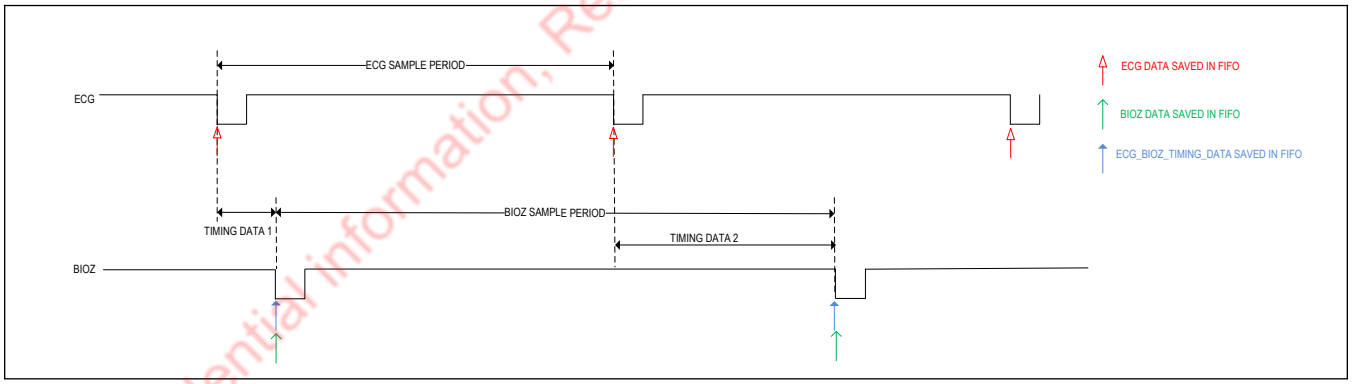



*Figure 12. Example 1: ECG Sample Rate Faster than BioZ Sample Rate*

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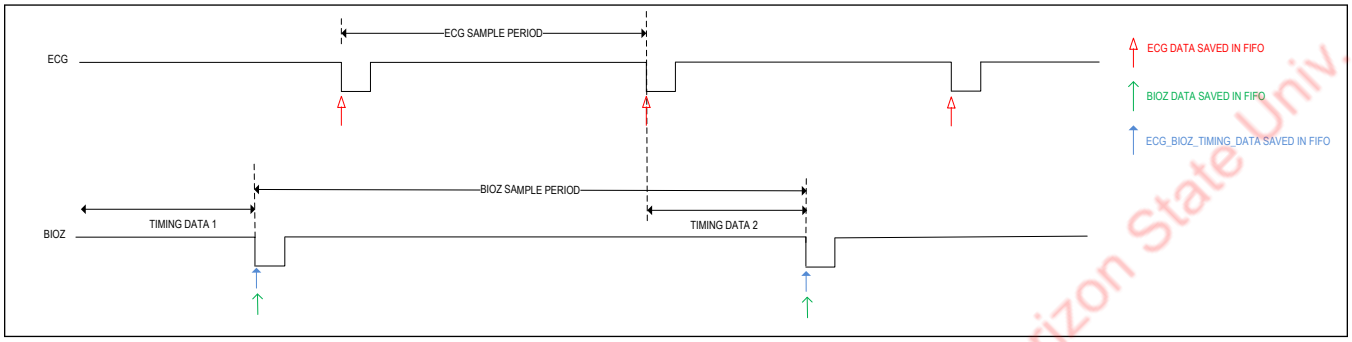



*Figure 13. Example 2: ECG Sample Rate Faster than BioZ Sample Rate*

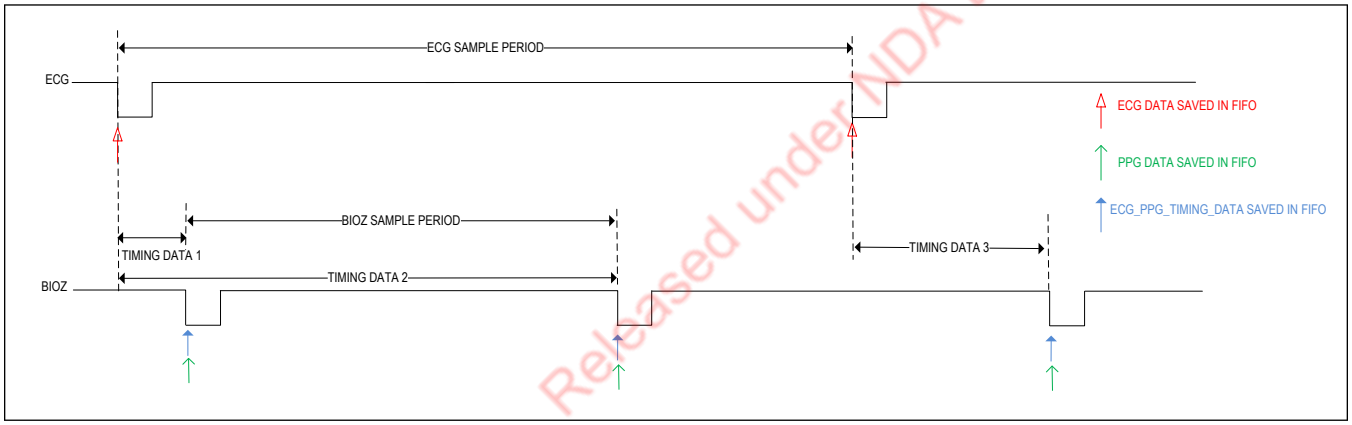



*Figure 14. Example 3: ECG Sample Rate Slower than BioZ Sample Rate*

### PPG, ECG, and BioZ Signal Alignment

In order to temporally align the PPG, ECG, and BioZ analog signals with high precision, the latency of the ECG and BioZ signal chains need to be considered in addition to the digital sample alignment provided by the FIFO timing data. Because PPG samples are integrated over individual time windows, their timing is well defined by the timing subsystem with very little analog latency. However, the ECG and BioZ channels have multiple sources of latency between the analog signal entering the MAX86178 and digitized samples entering the FIFO. These include the analog HPF, INA, PGA, anti-aliasing filter, and the delta-sigma ADC with its low-pass decimation filter. This latency results in the digitized ECG and BioZ signals lagging behind the digitized PPG signal, as shown in Figure 15. If precise signal alignment is needed, the relative latencies between each pair of channels should be characterized at the sample rates intended for the application.

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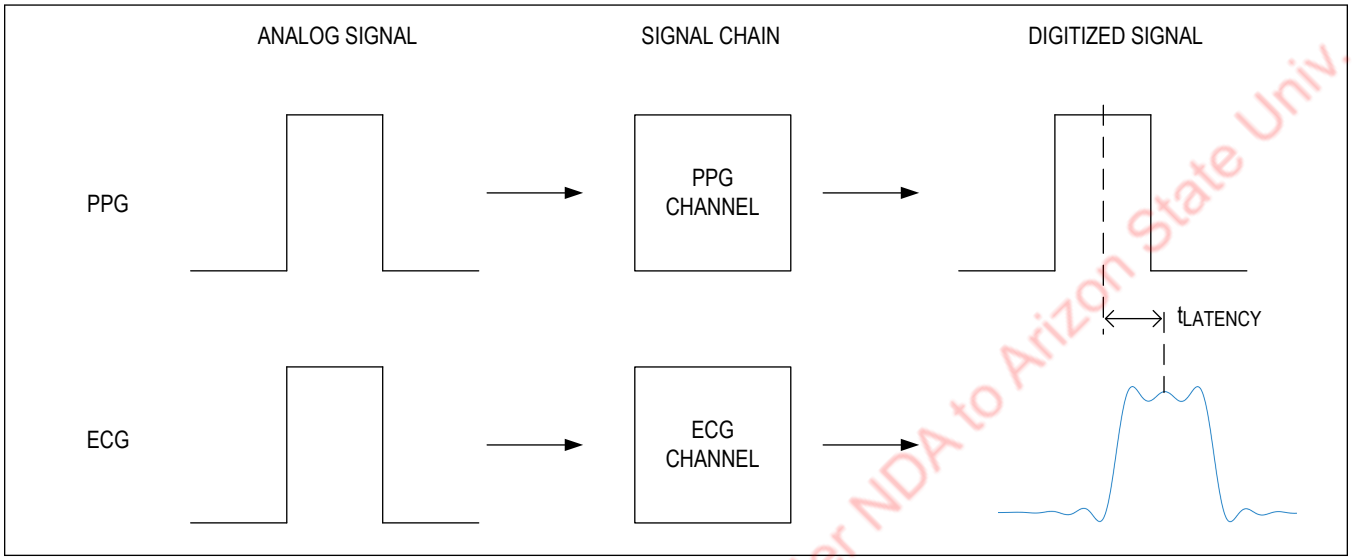



*Figure 15. ECG and PPG Simultaneous Pulse Response*

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## PPG

The PPG system in MAX86178 has two high-current LED drivers that support a total of 6 LED drive-output pins through two on-chip 6 x 1 multiplexers and control logic. Each driver has an 8-bit programmable DAC that can be programmed for a wide range of LED drive-current settings. The optical receiver has two high resolution readout channels that operate simultaneously and are connected to 4 PD input pins through on-chip control logic. Each individual channel has robust ambient-light cancellation in analog and digital domains. The PPG data from both ADCs in the two channels is stored in the 256-word FIFO, which can be analyzed and used in a number of different applications. The PPG system in MAX86178 can operate on a single supply voltage ranging between 3.1V and 5.5V that supplies $V_{LED}$ to the LED driver as well as input to the on-chip LDO at LDO_IN. In this case, LDO_OUT is connected to AVDD and DVDD; thereby, providing +1.8V for the AFE. Alternatively, the AFE main supply at AVDD and DVDD can be connected to an external +1.8V without using on-chip LDO.

The PPG system in MAX86178 operates in a dynamic power-down mode, always powering down between frames to minimize power consumption. For more details on the power consumption at various sample rates, refer to the *Electrical Characteristics* table.

Keeping the voltage ripple on the $V_{LED}$ as low as possible ensures the highest SNR is achieved. If a regulated supply is not available, the switching frequency on $V_{LED}$ should be kept between 100kHz and 3MHz. By ensuring the switching frequency stays within the recommended range, along with a good load-transient response, high SNR can be maintained at the heaviest loads (high LED drive-current applications).

The MAX20345 or the smaller MAX20343 are recommended solutions for a buck-boost supplying $V_{LED}$. Both offer highly efficient buck-boost regulators with a very small load ripple, fast load-transient responses, and have load-pulse consistencies that provide more than 90dB SNR in a white card DC SNR test. Higher SNR is achieved by enabling the DAC offset in addition to utilizing the on-chip averaging and off-chip low-pass or band-pass filtering, allowing for 113dB SNR to be achieved.

The various blocks and features in the PPG system of MAX86178 are discussed in detail in the following sections.

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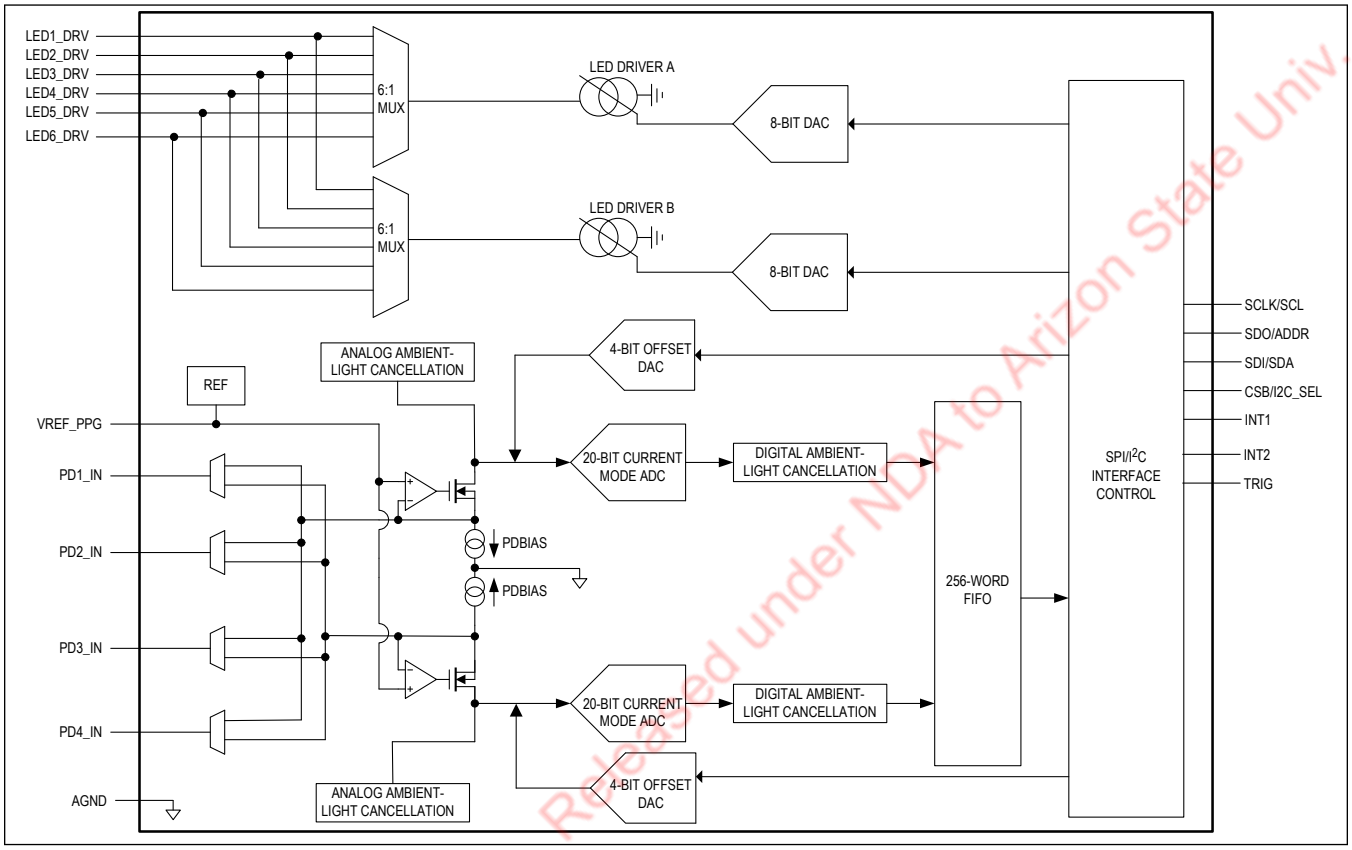



*Figure 16. PPG Detailed Block Diagram*

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### Optical Transmitter

The PPG block in MAX86178 has two independent precision LED current drivers that are connected to six LED-driver pins through two 6 x 1 muxes. Two LED-current DACs modulate LED pulses for a variety of optical measurements. The two LED-current DACs have 8-bit dynamic range with four programmable full-scale range settings of 32mA, 64mA, 96mA, and 128mA (typ). The configuration of the LED drivers can be uniquely set for each measurement. The MEASx_Selects registers (see the Register Map section) define how each LED driver is connected for that particular measurement. Thus, the configuration of the LED drivers can be uniquely set for each measurement and each measurement can drive one or both LED drivers.

## Table 8. LED Driver and LED Mux Configuration

| MEASx_DRVA/MEASx_DRVB | LEDn_DRV PIN CONNECTED TO LED DRIVER |
|---|---|
| 0 | LED1_DRV |
| 1 | LED2_DRV |
| 2 | LED3_DRV |
| 3 | LED4_DRV |
| 4 | LED5_DRV |
| 5 | LED6_DRV |

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This configuration of LED driver and LED mux is highly flexible, allowing for not only any combination of the six LED-driver pins to be used at any one time, but also allows for any pin to sink up to 256mA by combining both drivers to generate a higher output current. Figure 17 shows how the two LED drivers are connected to the six LED driver pins.

Both LED drivers are low-dropout current sources allowing for low-noise and power-supply independent LED currents to be sourced with minimal voltage overhead; thereby, minimizing LED power consumption. Four full-scale range settings are provided to allow for the optimization of LED driver noise, and dropout voltage on the LEDn_DRV pins. Table 9 illustrates this trade-off.

## Table 9. LED Driver Full-Scale Range Trade-Off

| FULL-SCALE RANGE (mA) | RECOMMENDED MINIMUM $V_{LEDn_DRV}$ (mV) | MAX SINGLE SHOT SNR (dB) |
|---|---|---|
| 32 | 300 | 84 |
| 64 | 500 | 88 |
| 96 | 700 | 90 |
| 128 | 900 | 91 |

The LED on-time is the sum of the receiver integration time selected in MEASx_TINT, and the LED settling time selected in MEASx_LED_SETLNG. The duty cycle is set by the ratio of the LED on-time and the frame period. The average LED supply current is calculated as the product of the programmed LED current and the duty cycle.

The voltage on $V_{LED}$ depends on the forward voltage ($V_F$) of the LEDs driven by the MAX86178. The AFE requires a minimum of 3.1V applied to the $V_{LED}$ pin (see the *Electrical Characteristics* table). Additionally, the minimum required $V_{LED}$ voltage in a system is determined by the sum of forward voltage ($V_F$) of each LED at maximum LED current and LED driver headroom voltage ($V_{LEDn_DRV}$ as shown in Table 9)

The $V_{LED}$ voltage must be above this minimum to avoid compression, and allow for enough headroom to supply the LED drive current as needed; otherwise, the $I_{LED}$ can be reduced and become more sensitive to $V_{LED}$ supply changes. The voltage on LEDn_DRV pins can be measured during an exposure to ensure the minimum headroom voltage is maintained during each LED exposure (LED on-time). The minimum $V_{LED}$ can be simply calculated by using the formulas below:

$V_{LED} \geq 3.1V$ and $V_{LED} \geq V_F + V_{LEDn_DRV}$

where, $V_F$ is a function of maximum LED current for the system.

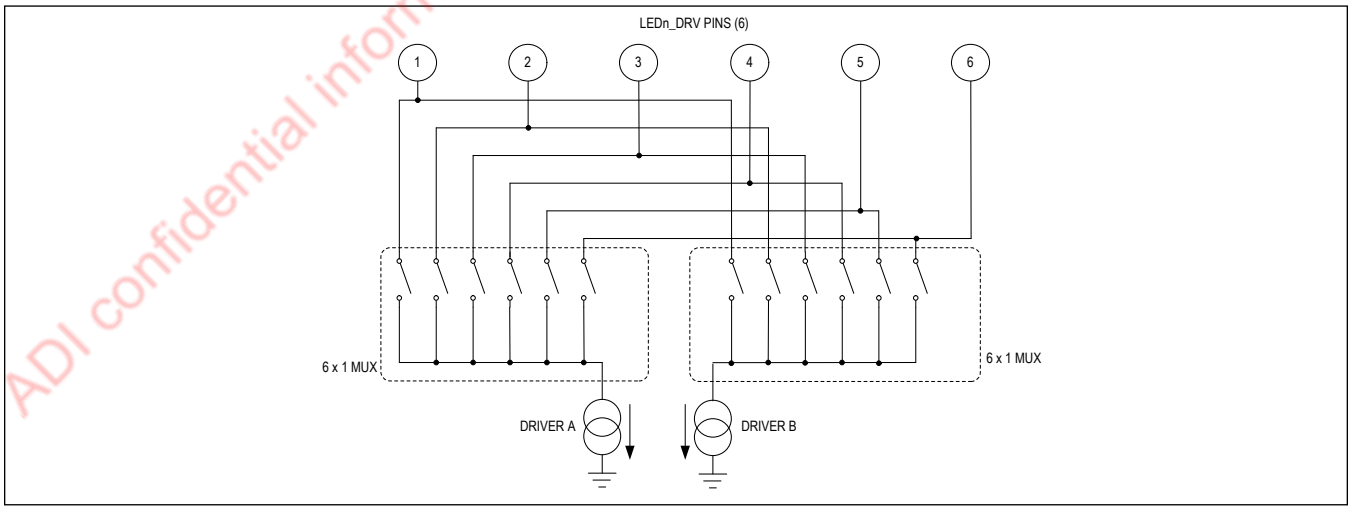



*Figure 17. LED Drivers*

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### Optical Receiver

The optical path in the PPG block of MAX86178 is composed of a front-end photodiode biasing circuit with an analog ambient-light cancellation (ALC) sample and hold circuit that nulls the ambient-light photodiode current at the input of the ADC. This front-end biasing circuit is followed by a current integrating, continuous-time sigma-delta ADC with a proprietary discrete time filter. This discrete time filter uses multiple dark and exposure samples to generate an accurate 19-bit effective exposure output signal with excellent low- and high-frequency ambient-light rejection.

The MAX86178 PPG block incorporates dual signal paths and has four photodiode input pins. Each photodiode input is either connected to one of the two PPG signal paths or is left open by configuring MEASx_PDy_SEL (x = 1 to 6, y = 1 to 4) as needed, so that input to the two PPG ADCs is received simultaneously. For applications requiring only one optical signal path to be active (for example, lower power consumption), either one of the two channels is powered down by setting PPG1_PWRDN or PPG2_PWRDN (register 0x10) to 1. In this way, the MAX86178 is used in a single-channel configuration. By default, the MAX86178 PPG has dual-channel configuration.

Each signal path supports four full-scale range settings of 4µA, 8µA, 16µA, and 32µA set in the MEASx_PPGy_ADC_RGE (x = 1 to 6, y = 1, 2) field in each of the measurement configurations block. Also supported are four options for integration time, which effectively modulate the channel bandwidth, allowing for a trade-off between LED power consumption and PPG signal quality.

Each PPG signal path also incorporates a 4-bit offset DAC for extending the optical dynamic range by sourcing some of the exposure current to the offset DAC. The current offset to DAC is selected in the MEASx_PPGy_DACOFF (x = 1 to 6, y = 1, 2) register bit field. This feature is especially useful under certain conditions that occur when attempting to limit the exposure ADC counts, for example, when avoiding saturation while increasing the exposure-signal perfusion index. The optical paths also support multiple photodiode and LED settling time settings in order to support flexible multiparameter measurements for different types of photodiode/LED wavelength combinations.

Most significantly, each signal path supports up to six unique combinations of the above configurations as needed. This allows a single optical AFE to support multiple optical measurements in a compact, energy-efficient design.

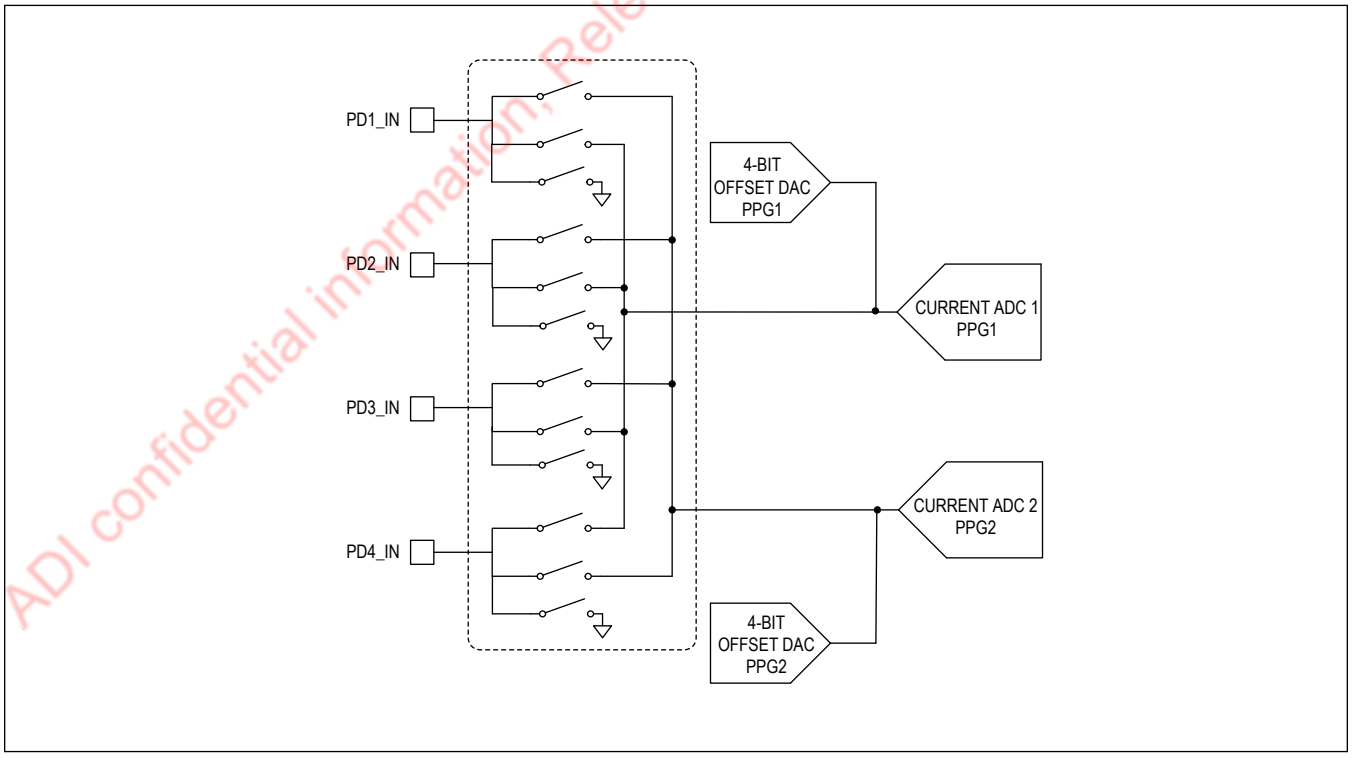



*Figure 18. Optical Receive Channel*

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### Photodiode Biasing

There are three photodiode biasing options in MAX86178 PPG to support a large range of photodiode capacitance. Each photodiode input can have a separate bias setting; thereby, allowing for different photodiodes to be used. The PDm_BIAS (register 0x24) settings adjust the PDm_IN bias point impedance to ensure that each photodiode settles rapidly enough to support the sample timing.

## Table 10. Recommended PD Bias Settings Based on the Photodiode Capacitance

| PDm_BIAS (m = 1 to 4) | PHOTODIODE CAPACITANCE (pF) |
|---|---|
| 0 | Do not use |
| 1 | 0 to 125 |
| 2 | 125 to 250 |
| 3 | 250 to 500 |

The PDm_BIAS impacts the dark current noise of the MAX86178. The relationship between PDm_BIAS and noise with increasing photodiode capacitance is shown in Table 11. Because of the increased noise with a higher PDm_BIAS setting, the lowest recommended PDm_BIAS value should be used for any given photodiode capacitance.

## Table 11. Dark Noise vs PD Bias Setting and Integration Time

| PDm_BIAS | DARK NOISE ($pA_{RMS}$) MEASx_TINT = 14.6µs | DARK NOISE ($pA_{RMS}$) MEASx_TINT = 29.2µs | DARK NOISE ($pA_{RMS}$) MEASx_TINT = 58.6µs | DARK NOISE ($pA_{RMS}$) MEASx_TINT = 117.1µs |
|---|---|---|---|---|
| 0x1 | 380 | 206 | 152 | 104 |
| 0x2 | 512 | 276 | 186 | 131 |
| 0x3 | 590 | 405 | 305 | 220 |

### Measurement Configuration and Timing

A measurement is essentially one combination of LED (or LEDs) and PD (or PDs) that results in an optical measurement. The MAX86178 supports six individual measurements, each of which can be configured independently. Each measurement can be configured by the group of registers named PPG MEASx Setup (x = 1 to 6). These registers set up a number of parameters for each measurement independently as listed below.

- Connection of each of the two LED drivers to one of the six LED driver pins
- Connection of each of the four photodiode inputs to one of the two PPG channels
- Ambient measurement
- LED driver range
- LED drive current for each driver
- LED settling time
- PD settling time
- Number of burst averages
- Ambient rejection scheme (CDM or FDM)
- On-chip decimation filter selection (COI, COI2, or SINC3)
- ADC integration time
- ADC range for each channel
- DAC offset for each channel

In addition, the following parameters are set for all enabled measurements:

- PPG synchronization mode
- Dual-channel or single-channel mode
- Proximity mode
- Number of sample averages
- Bias setting for all PD inputs

A measurement can be configured to pulse one or two LED drivers sequentially at multiple wavelengths as is done in pulse-oximetry measurements or simultaneously to drive multiple LEDs such as is done with heart-rate measurements on the wrist. A measurement is also configurable to measure the direct ambient level. If the direct ambient is enabled in a measurement through MEASx_AMB (see the Register Map section), it must be the last measurement in the sequence

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of enabled measurements.

Figure 19 represents one measurement with only one of the LED drivers active. No averaging is used. As seen in Figure 19 only MEASx_DRVA is pulsing during the exposure time. In this mode, each driver pulse results in a single optical-sampled value for each PD input to be stored in the FIFO. For example, in a single-channel configuration, only one ADC data sample is stored in the FIFO for each driver pulse, but for a dual-channel configuration, two ADC data samples are stored in the FIFO for each driver pulse. This configuration can be used when heart rate is being measured with a single LED.

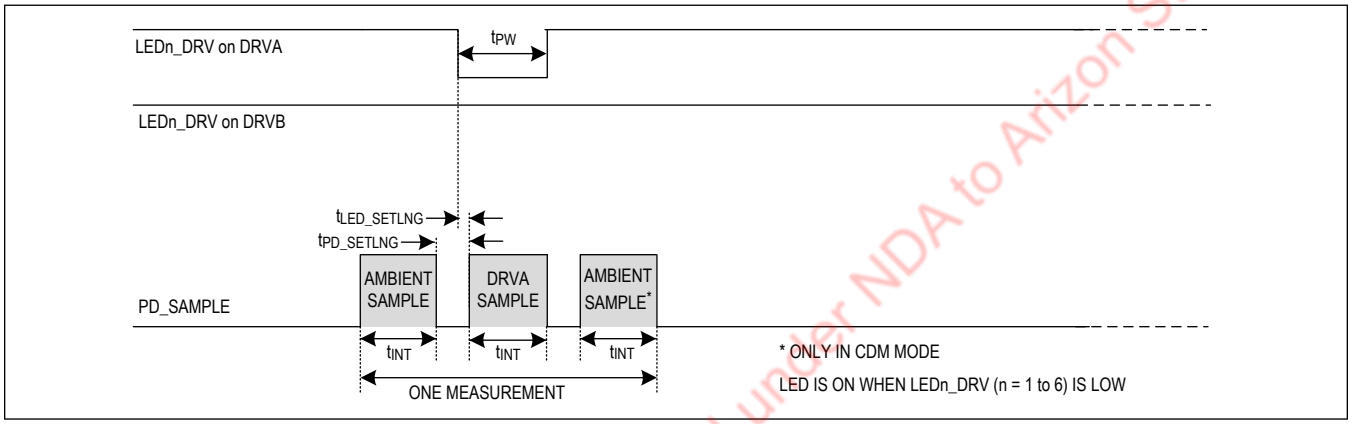



*Figure 19. Measurement with One LED Driver*

### Frame

A frame is a combination of one (min) to six (max) measurements configured in MEASx_EN (x = 1 to 6). The frame rate defines how frequently a frame is repeated and is expressed in units of frames per second (fps). For details on the PPG frame-rate configuration, see the *PPG Frame Rate* section.

Any combination of measurements can be enabled, but measurements are done in a numerical order inside a frame. For example, it is valid to enable MEAS1, MEAS2, and MEAS5 while MEAS3 and MEAS4 are skipped. But if a measurement of direct ambient is configured, then this measurement must be configured as the last measurement in the frame.

Figure 20 represents the timing diagram for one measurement in each frame. This measurement has only one LED driver pulsing. See Figure 19 for details about what comprises M1.

Figure 21 illustrates the timing for six measurements in each frame. Each measurement can be configured independently using the MEASx Selects Registers (see the Register Map section). Alternatively, all measurements share the configuration of M1 by setting MEAS1_CONFIG_SEL [0] (0x22) to 1. Each measurement is comprised of ambient (LED off) sample(s) as well as exposure (LED on) sample as shown in Figure 19.

Figure 22 illustrates the timing with burst averaging in each frame with MEAS3 configured to have an on-chip burst average of 2 (MEAS3_AVER[2:0](0x41) = 0x1). The result of MEAS3 in each frame is only one FIFO data pushed into FIFO. This FIFO data is the average of the two PD samples labeled F1M3a and F1M3b. The burst averaging factor is configured by MEASx_AVER (x = 1 to 6) and it is available only with CDM (MEASx_FILT_SEL = 0). The ambient (LED off) samples as well as the exposure samples in case of burst averaging are shown in Figure 22. In this case, the maximum valid frame rate is limited by the burst averaging factor for enabled measurements.

Figure 23 shows timing with sample averaging of 2 in each frame (SMP_AVE[6:4](0x22) = 0x1). Each individual measurement in frame 1 (F1) and frame 2 (F2) is averaged and the result is saved in FIFO as a single sample for each measurement. For example, the sample in FIFO for measurement 1 is (F1M1 + F2M1) / 2. Each enable measurement within every frame has its ambient sample(s) and exposure sample as detailed in Figure 19. When using sample averaging, the data output rate is reduced by the frame rate / sample averaging factor. Therefore, sample averaging also limits the bandwidth of the PPG data and helps improve SNR of the AFE.

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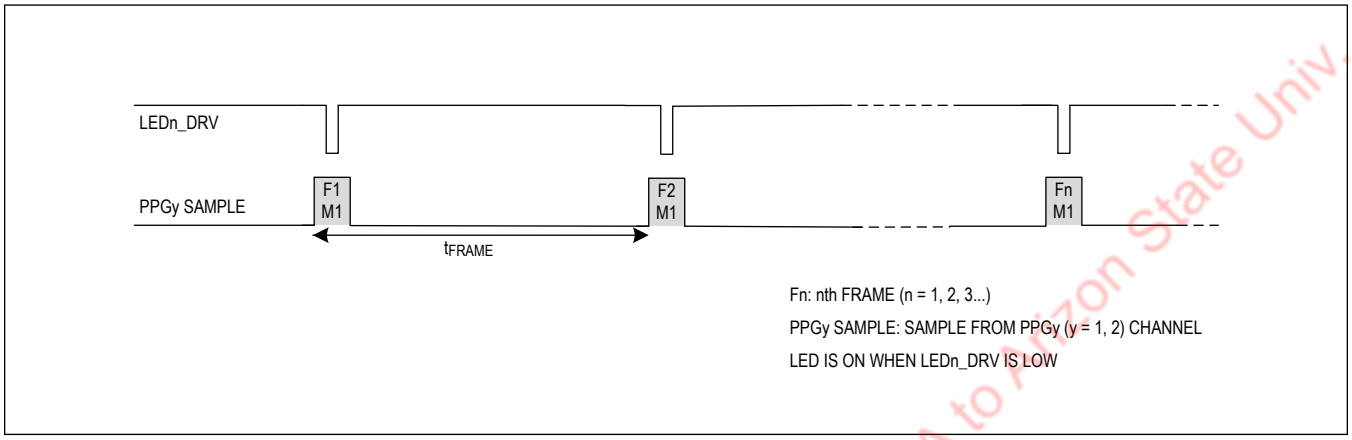



*Figure 20. Frame with One Measurement*

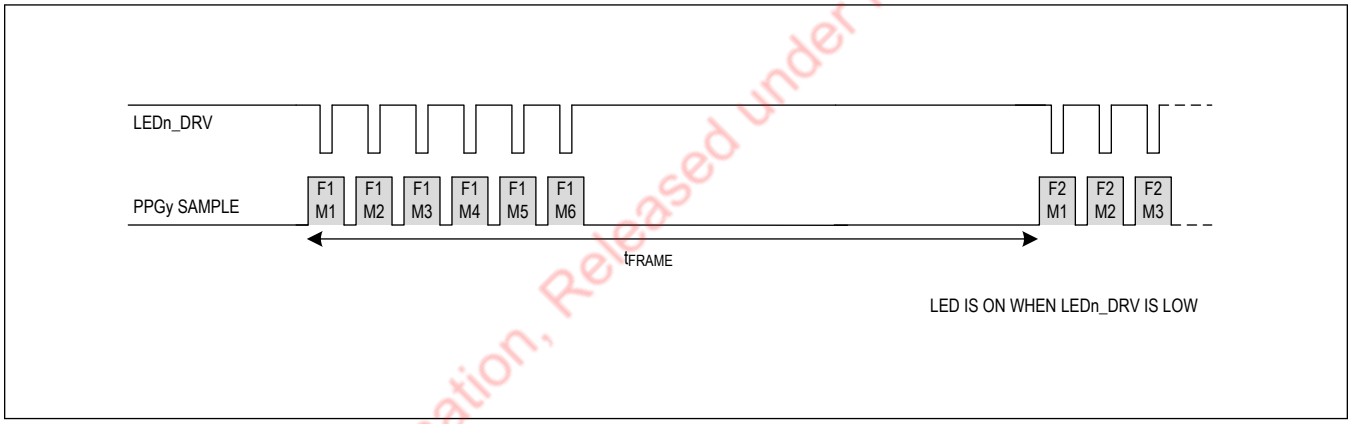



*Figure 21. Frame with Six Measurements*

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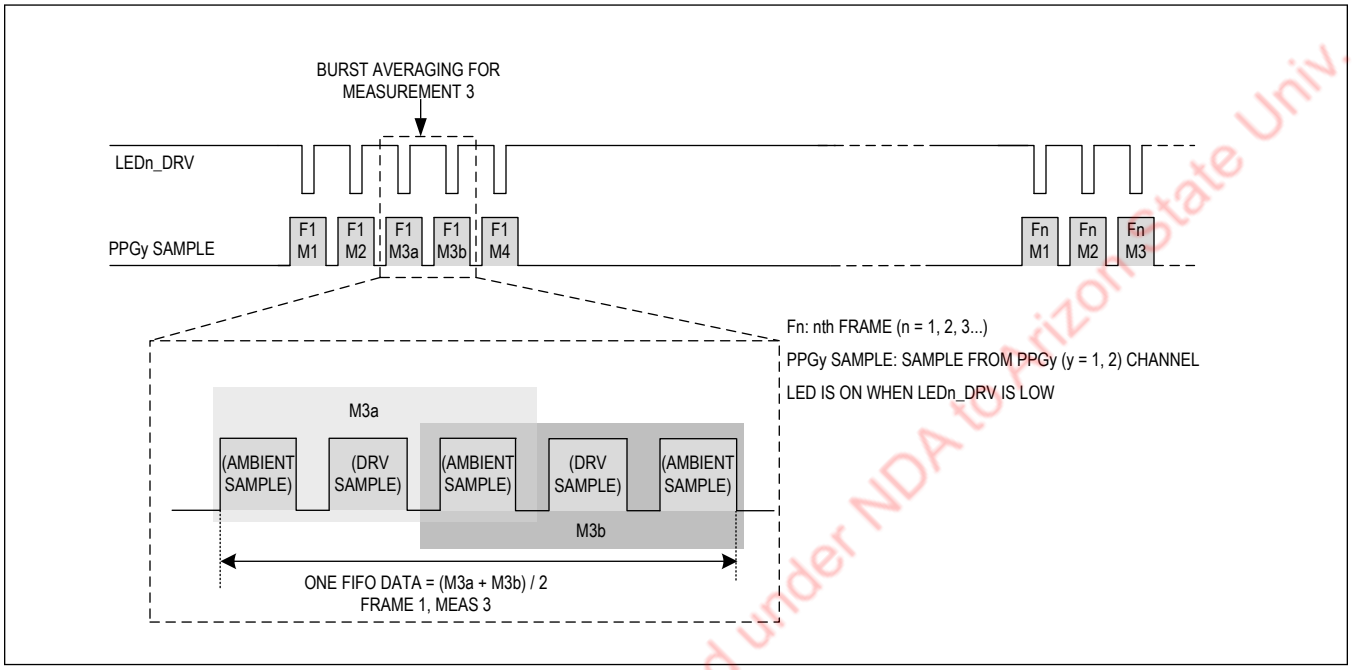



*Figure 22. Frame with Burst Averaging*

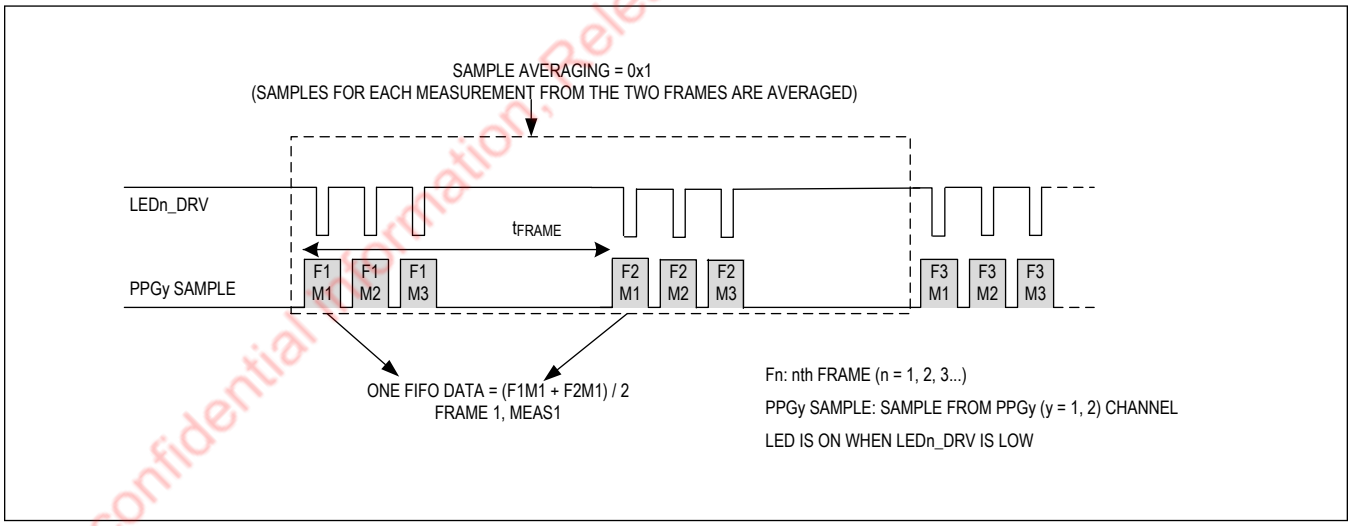



*Figure 23. Frame with Sample Averaging*

Note: If the internal frame trigger (PPG_SYNC_MODE=b0, default) is used, then changing the PPG frame rate (registers FR Clock Divider MSB[7:0] and FR Clock Divider LSB[7:0]) restarts the finite state machine (FSM) which forces the start of a new frame of measurements. Therefore, a glitch in the frame rate can be expected between the frame measured just before and just after registers, FR Clock Divider are written to.

The restart of the FSM also happens when other PPG configuration settings (like MEASx_PPGn_ADC_RGE, MEASx_AVER, MEASx_DRVm_PA, and others) are modified. This restart of the FSM does not disrupt the PPG frame rate if using the external frame trigger (SYNC_MODE= 0x1) so if a consistent frame rate is required, then using an

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external frame trigger is the ideal setup. If an external trigger was not available, writing to the PPG configuration registers just before the new frame starts would minimize the frame rate glitch, but the timing may be complicated as to when to start the write, how long the write takes, etc. To ensure that full frames are pushed to the FIFO, write to the PPG configuration registers only after a complete frame is ready (i.e., after FRAME_READY interrupt or A_FULL interrupt, where FIFO_A_FULL is in multiples of full frames) and the FIFO is read, otherwise the FSM restarts before all measurements in the frame have been completed, thus leaving the FIFO with a partial frame. This is the case even when using an external frame trigger. Additionally, when using the internal trigger and multiple PPG configuration registers are written, flush the FIFO after the last register write because if the writes are slow, then unwanted measurements may be pushed to the FIFO and these are cleared by the flush. Optionally setting self-clearing bit FIFO_MARK pushes a special tag (code 0xFFFFFE) to the FIFO. This feature can be used for differentiating the data in the FIFO before and after the tag and potentially highlighting when the Frame rate or PPG configuration was changed. The ECG sample rate, BIOZ sample rate, PPG-ECG timing data, and PPG-BIOZ timing data are not affected by any PPG configuration register or current code register writes.

### Ambient Rejection

The MAX86178 implements ambient-light cancellation in two steps, a coarse cancellation and a fine cancellation. Each PPG channel has its own dedicated ambient-light cancellation circuits. Also, each MEASx has its own ambient-light cancellation configuration. The coarse cancellation is in the analog domain. It is enabled by default and can be disabled by setting the ALC_DISABLE[4](0x22) to 1. The fine cancellation is a digital cancellation scheme and is configured as either central difference method (CDM) or forward difference method (FDM) using the MEASx_FILT_SEL bit.

ALC is an analog sample and hold scheme, which cancels up to 200µA of DC photodiode current. Any drift or residual from ALC is cancelled by CDM/FDM cancellation.

One CDM sample is comprised of 3 ADC conversions, with 2 ambient and 1 exposure conversion. One FDM sample is comprised of 2 ADC conversions, with 1 ambient only and 1 exposure conversion.

The ambient and exposure samples are used for digital cancellation of any residual error (from ALC) or drift (in ambient signal). The final computed value is the effective exposure signal, which is stored in the FIFO.

See Figure 24 for the timing diagram for CDM and Figure 25 for the timing diagram for FDM.

The MAX86178 features various decimation filtering options for the delta-sigma PPG ADC. By default, the MAX86178 uses a second-order decimation filter (COI2). Alternatively, the user can select a third-order decimation filter (COI3). The second-order COI2 filter has a narrower noise-equivalent bandwidth as compared to the third-order COI3, and therefore, results in lower-noise performance. The selection for either COI2 or COI3 is done by programming MEASx_FILT2_SEL for each individual measurement. However, the second order decimation filter COI2 is available only with the longest integration time setting (MEASx_TINT = 0x3). Using COI2 instead of COI3 for the longest integration-time setting provides about 1 dB higher SNR for PPG results. Both the COI2 and COI3 filters have a non-symmetrical impulse response with no ripple in the pass band or the stop band. A third option is to use the SINC3 decimation filter for the delta-sigma PPG ADC, which is enabled for each measurement individually by setting MEASx_SINC3_SEL. This filter provides improved high-frequency roll-off, 60dB/dec roll-off for out of band frequencies, that significantly improves the high-frequency ambient-light rejection. In comparison, the COI2 and COI3 filters provide only about 20dB/dec roll-off at high frequencies but provide excellent quantization performance.

Ambient-light cancellation in low-frequency ambient light improves when using a burst average by setting MEASx_AVER ≥ 0x2.

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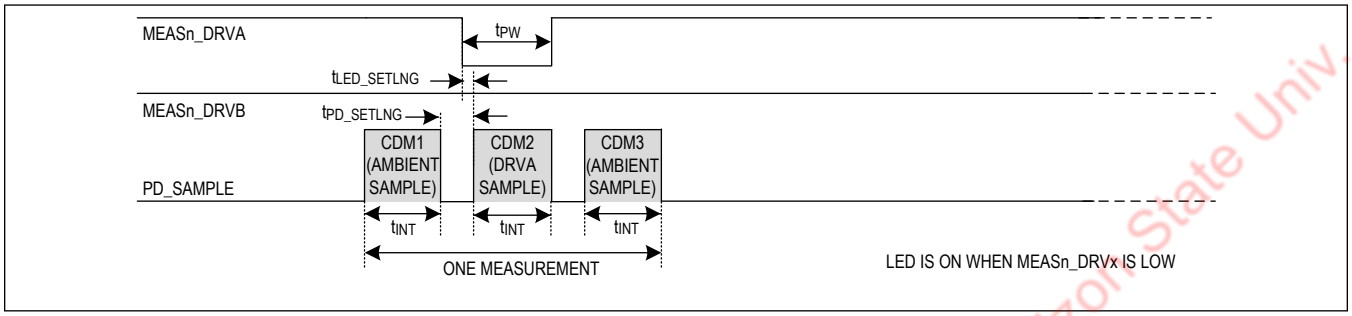



*Figure 24. Central Difference Method (CDM)*

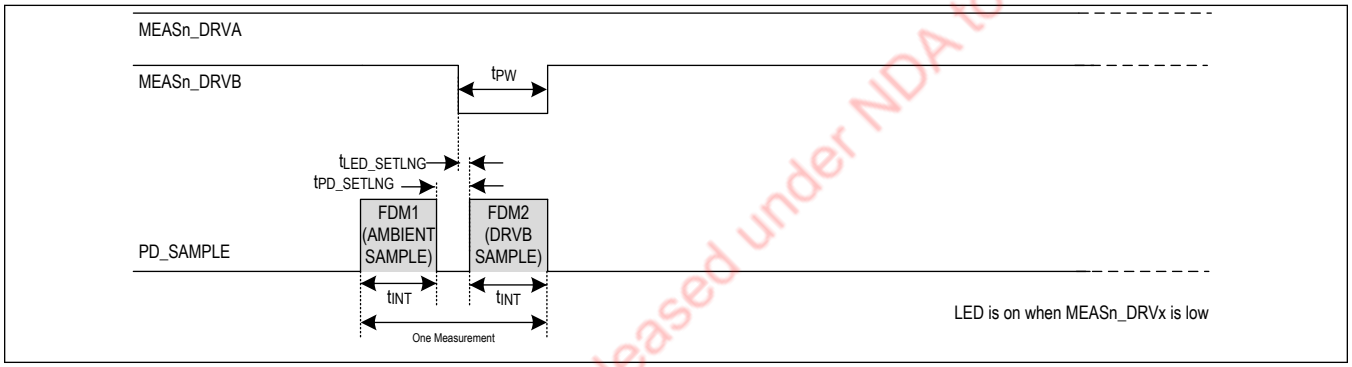



*Figure 25. Forward Difference Method (FDM)*

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### Threshold-Detect Function

The MAX86178 includes a threshold-detect function that enables users to detect ADC counts higher than a specified range or lower than a specified range.

The threshold-detect function is used in proximity mode (see the *Automatic Proximity Detect Mode* section) to reduce energy consumption and extend battery life when the sensor is not in contact with skin. There are two separate instances of threshold-detect functions available in MAX86178, Threshold1 and Threshold2. Both are disabled by default.

The threshold-detect function is enabled by selecting a measurement in THRESHx_MEAS_SEL(x = 1, 2) in register 0x70. The threshold-detect function is set up for either PPG1 or PPG2 by configuring THRESHx_PPG_SEL (x = 1, 2) in register 0x71. In order to configure the threshold-detect function, both an upper limit and a lower limit must be set. These can be configured in the PPG Threshold Interrupts registers (0x72 to 0x75).

In addition, two features are available to make the threshold-detect function more adaptable for various system and application requirements. These are time hysteresis and level hysteresis and are configurable through the TIME_HYST[4:3] and LEVEL_HYST[2:0] in register 0x71. Time hysteresis sets the number of consecutive samples that must be outside the limits defined by THRESHOLDx_UPPER and THRESHOLDx_LOWER in order to assert the threshold interrupt. Level hysteresis defines the sample variation around THRESHOLDx_UPPER and THRESHOLDx_LOWER. This value is in ADC counts and is applied at ±0.5 x LEVEL_HYST around THREHSOLDx_UPPER as well as ±0.5 x LEVEL_HYST around THRESHOLDx_LOWER. Specifically, in order for a threshold interrupt to be asserted, a sample must either transition above the THRESHOLDx_UPPER + 0.5 x LEVEL_HYST and stay above THRESHOLDx_UPPER - 0.5 x LEVEL_HYST for the number of samples defined in TIME_HYST or transition below THRESHOLDx_LOWER - 0.5 x LEVEL_HYST and stay below THRESHOLDx_LOWER + 0.5 x LEVEL_HYST for the number of samples defined in TIME_HYST as shown in Figure 26.

If a threshold-detect function instance is enabled, the corresponding THRESHx_HILO interrupt bit (register 0x00) is asserted and threshold mode is activated when the ADC counts of the assigned measurement on the specified

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PPG channel drop below the lower limit, or exceed the upper limit (in consideration with the LEVEL_HYST and TIME_HYST settings). The upper threshold check is disabled by setting THRESHx_UPPER (0x72, 0x74) to 0xFF. The PPG ADC reading, if negative, is clipped to 0x00000 before comparing with the threshold limits. Therefore, programming THRESHOLDx_LOWER to 0x00 effectively disables the lower threshold check.

The threshold-detect function enables low power consumption while in automatic proximity detect mode. Alternatively, the LED configuration during the threshold-detect active mode is determined by the firmware settings as needed for each application. The lower settings of the LED current, ADC integration time, and frame rate result in reducing power consumption during situations when there is no reflective returned signal.

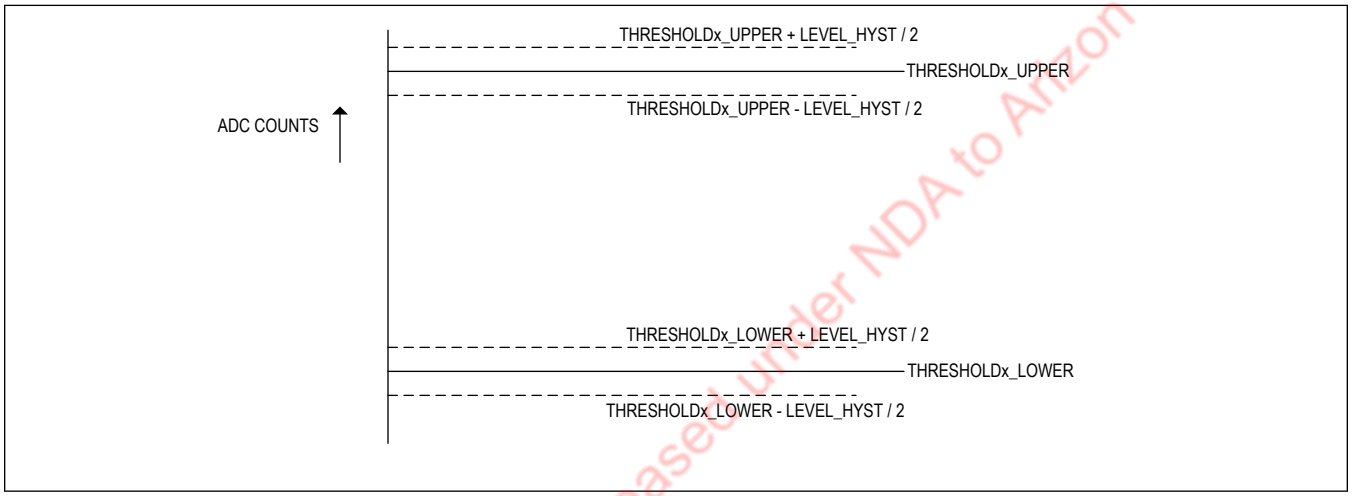



*Figure 26. Threshold Limits with LEVEL_HYST*

### Automatic Proximity-Detect Mode

The MAX86178 includes a proximity detect mode that switches the device automatically between proximity-detect mode (PROX mode) and normal mode. Using the threshold-detect function (see the *Threshold-Detect Function* section), the proximity mode significantly reduces energy consumption; thereby, extending battery life when the sensor is not in contact with skin.

When the PROX_AUTO[3](0x23) is set to 1, PROX mode is enabled. In this mode, PPG measurement 6 is reserved for proximity function and is automatically enabled even if MEAS6_EN[5](0x20) is programmed to 0. Threshold1 is used for proximity detect, but THRESH1_MEAS_SEL[3:0](0x70) is ignored. Instead, measurement 6 is considered for this threshold instance. The PROX mode is implemented on the PPG channel selected in THRESH1_PPG_SEL[6](0x71).

The device enters PROX mode when the measurement 6 ADC reading is below the threshold defined by THRESHOLD1_LOWER[7:0](0x73). Otherwise, it is in normal mode. THRESHOLD1_UPPER[7:0](0x72) is not used. The device switches between PROX mode and normal mode automatically. In PROX mode, the frame rate automatically changes to 8fps, and only measurement 6 is selected for ADC conversions. While in normal mode, measurement 6 and all the enabled measurements 1 to 5 are selected for ADC conversions, and the frame rate is configured by FR_CLK_DIV[14:0](0x28, 0x29). In order to reduce power consumption while in PROX mode, the MEAS6 LED drive current should be as low as possible. The LED current for MEAS6 is configured through MEAS6_DRVA_PA[7:0](0x5E) and MEAS6_DRVB_PA[7:0](0x5F).

If THRESH1_HILO interrupt is enabled using THRESH1_HILO_EN1[1](0xC0) or THRESH1_HILO_EN2[1](0xC5), an interrupt is asserted on the INT1 or INT2 pin, respectively, when the part switches from normal mode to PROX mode and also for each ADC conversion while in PROX mode. There is no interrupt when the part switches from PROX mode to normal mode.

In order to get an interrupt when switching from PROX mode to normal mode, a second threshold instance is enabled using THRESH2_HILO_EN1[2](0xC0) or THRESH2_HILO_EN2[2](0xC5). THRESHOLD2_UPPER[7:0](0x74) should be

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programmed to be equal to THRESHOLD1_LOWER[7:0](0x73), and THRESHOLD2_LOWER[7:0](0x75) should be zero. THRESH2_MEAS_SEL[7:4](0x70) should be programmed to select MEAS6. THRESH2_PPG_SEL[7](0x71) should be programmed the same as THRESH1_PPG_SEL[6](0x71).When configured this way, Threshold 2 generates an interrupt when switching from PROX mode to normal mode and does not generate an interrupt when the part switches from normal mode to PROX mode.

To reduce the number of interrupts, THRESH1_HILO interrupt should be enabled when the part is in normal mode, and THRESH2_HILO should be enabled when the part is in PROX mode. If interrupts are not needed for detecting switching between PROX and normal modes, Threshold 2 Interrupt registers need not be programmed.

False detection of PROX mode and normal mode can be avoided by configuring TIME_HYST[4:3] and LEVEL_HYST[2:0] in register 0x71 as desired.

See the *Threshold-Detect Function* section for more details on threshold interrupts.

## ECG

The ECG channel is a complete signal chain with all of the critical features necessary to collect high-quality ECG data such as flexible gain, critical filtering, low noise, high input impedance, and multiple lead-biasing options. Additional features, such as fast recovery, AC and DC lead-off detection, ultra-low-power lead-on detection, and right leg drive, enable robust operation in demanding applications such as wrist-worn devices with dry electrodes. The analog-signal chain drives an 18-bit sigma-delta ADC with a wide range of user-selected output sample rates. The following sections describe each of these features in detail and provide the information needed to implement them.

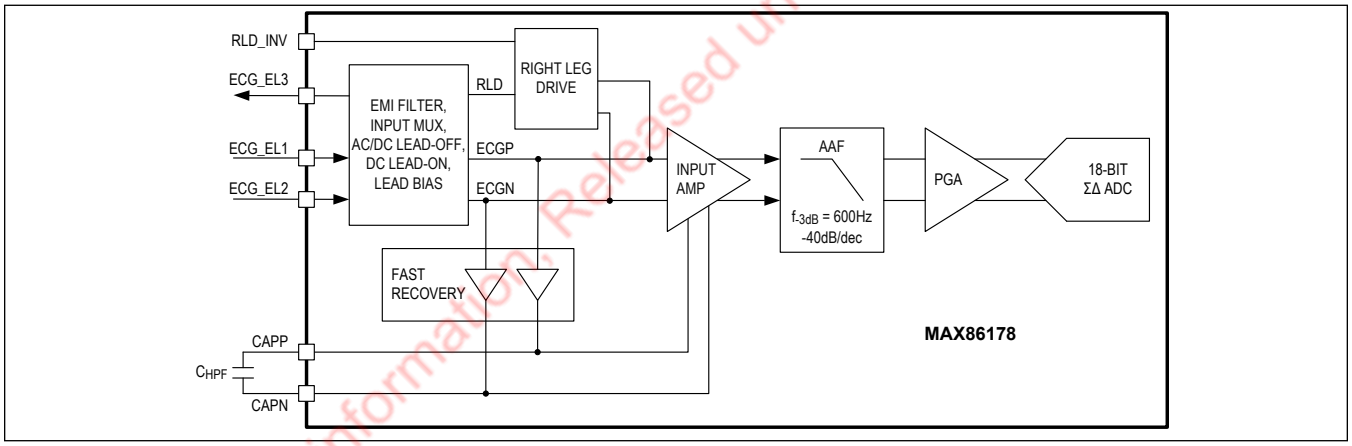



*Figure 27. Simplified ECG Channel Signal Chain*

### Analog-Signal Chain and ADC

The ECG-signal chain shown in Figure 27 consists of an input amplifier with high-pass filtering and fast recovery, an anti-aliasing filter (AAF), a programmable gain amplifier (PGA), and a delta-sigma ADC.

The input amplifier (INA) has a wide input range of 150mV to 1.35V to allow for large electrode offsets and changes in body bias, as well as offering programmable gain. The DC bias on ECGP and ECGN as well as low frequency signals below the corner frequency of the analog HPF are removed by the INA. The AC portion of the signal occupying frequencies above the corner frequency is amplified and passed to the PGA and ADC. This differential AC signal, as shown in Figure 28 can be as large as $200mV_{P-P}$. The AC signal is amplified further by the PGA and then passed to the ADC, which has a full-scale range of $2V_{P-P}$. To avoid saturating the ADC, the combined gain of the INA and PGA should be chosen such that the AC input signal amplitude is less than $2V_{P-P}$ / (INA gain x PGA gain). The INA gain is set by ECG_INA_GAIN[1:0](0x81) and ECG_INA_RGE[3:2](0x81) and ranges from 2.5V/V to 60V/V. The PGA gain is set by ECG_PGA_GAIN[6:4](0x81) and ranges from 1V/V to 16V/V. Both gain settings are located in the ECG Configuration 2 (0x81) register.

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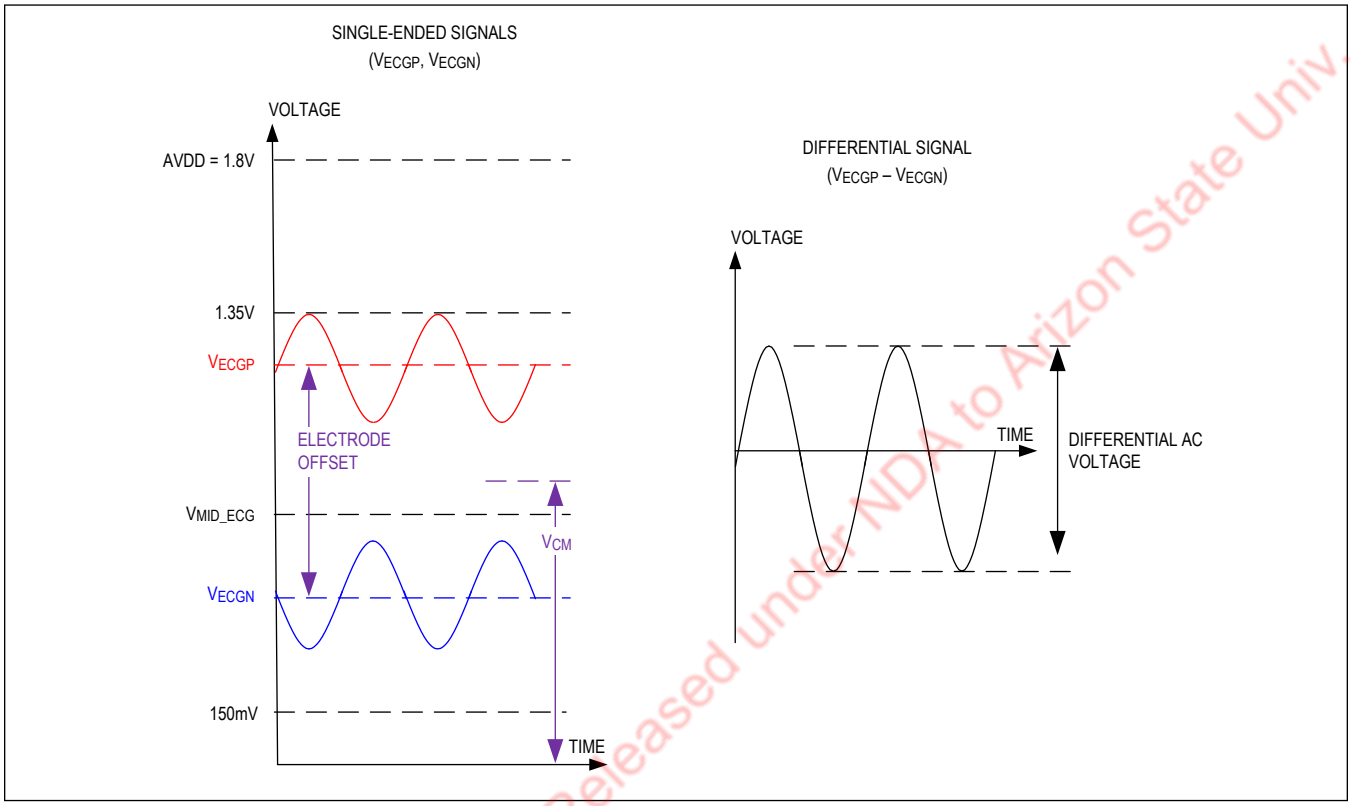



*Figure 28. ECG Channel Input Ranges*

The INA utilizes an external capacitor ($C_{HPF}$) connected between CAPP and CAPN to accomplish built-in high-pass filtering, as shown in Figure 29. The INA buffers the voltage on ECGP and ECGN onto CAPP and CAPN through buffers with 200kΩ each by default, for a combined resistance of 400kΩ ($R_{HPF}$) to form a low-pass filtered voltage across $C_{HPF}$. The INA then subtracts the voltage on $C_{HPF}$ from the input voltage and outputs the amplified difference. The end result is a high-pass filter with the corner frequency set by the value of $C_{HPF}$. To guarantee that the corner frequency is lower than the desired value across the operating temperature range, the minimum value of $R_{HPF}$ (240kΩ min) and the minimum capacitor tolerance of $C_{HPF}$ must be considered in the corner frequency calculation:

$$f_{-3dB} = 1 / (2 \times \pi \times R_{HPF} \times C_{HPF})$$

Table 12 shows the minimum effective capacitance needed to guarantee common corner frequencies. The resistance of $R_{HPF}$ can either be fixed at 400kΩ (default) or vary with the input amplifier range (see ECG_IMP_HI[3](0x82)).

## Table 12. ECG Analog HPF Corner Frequency Selection with $R_{HPF}$ = 400kΩ

| HPF CORNDER FREQUENCY (Hz) | MINIMUM EFFECTIVE CAPACITANCE (µF) | STANDARD 10% CAPACITOR VALUE (µF) |
|---|---|---|
| ≤ 5 | 0.13 | 0.15 |
| ≤ 0.67 | 0.99 | 1.1 |
| ≤ 0.5 | 1.3 | 1.5 |
| ≤ 0.05 | 13 | 15 |

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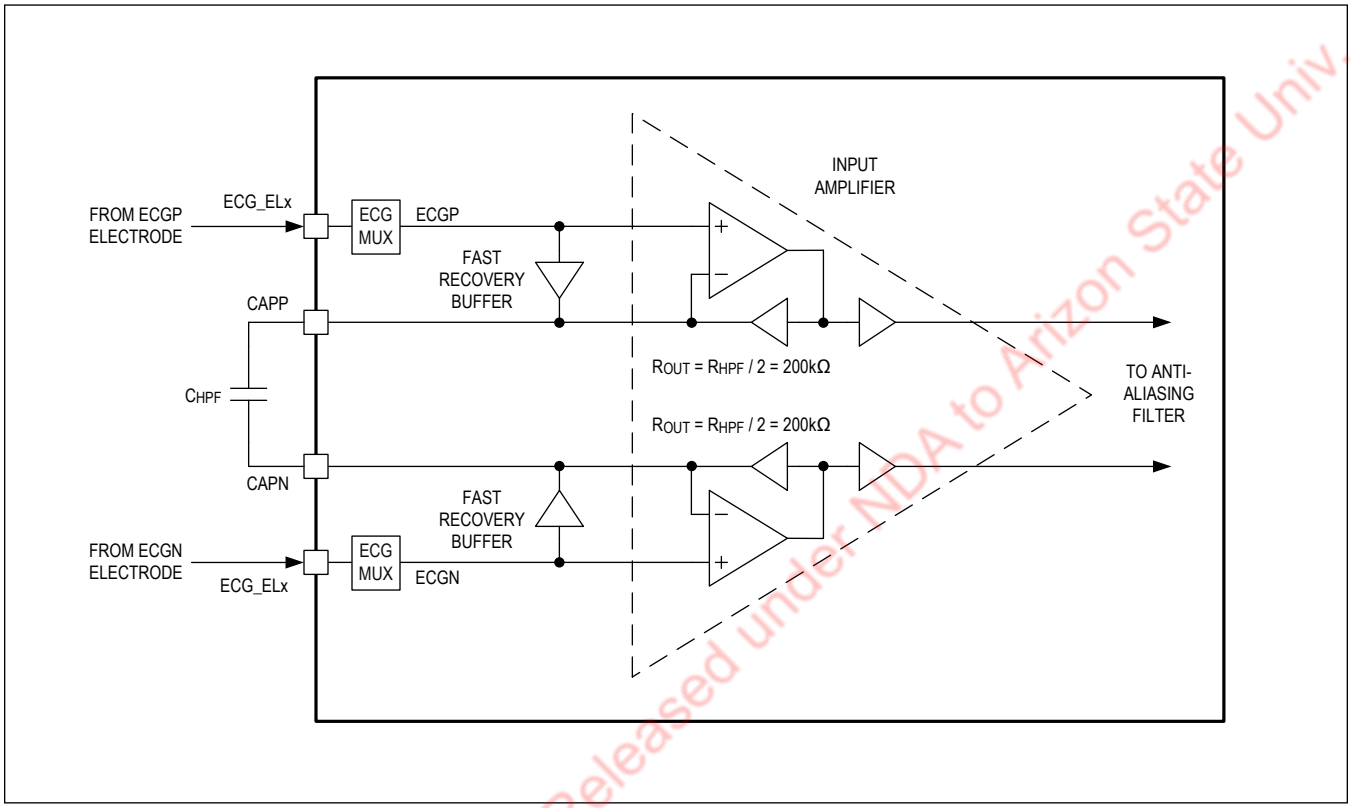



*Figure 29. Input Amplifier Detailed Block Diagram*

Following the input amplifier is a 2-pole active anti-aliasing filter with a 600Hz corner frequency that provides 57dB of attenuation at half the modulator sampling rate (approximately 16kHz). Following the PGA is a delta-sigma ADC with an input sampling rate set by the ECG ADC sample clock (32.0kHz or 32.768kHz (see the *Timing Subsystem* section). This input sampling rate is divided by the value set by ECG_DEC_RATE[2:0](0x80) to set the ECG sample rate between 62.5sps and 2048sps. The decimation filter has a SINC3 CIC architecture, which ensures adequate attenuation above the Nyquist frequency of the digitized signal, but also reduces the bandwidth of the pass-band to approximately 0.26 x SR_ECG. At higher sample rates, the INA, PGA, and AAF also limit the ECH channel bandwidth. Table 13 shows the -1dB and -3dB bandwidths at common sample rates (see Table 3).

## Table 13. ECG Channel Bandwidth

| SAMPLE RATE (sps) | -1dB BANDWIDTH (Hz) | -3dB BANDWIDTH (Hz) |
|---|---|---|
| 128 | 19 | 33 |
| 200 | 30 | 52 |
| 204.8 | 31 | 53 |
| 250 | 38 | 65 |
| 256 | 38 | 66 |
| 500 | 74 | 128 |
| 512 | 76 | 131 |
| 1024 | 144 | 250 |
| 2048 | 246 | 432 |

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### Fast Recovery Modes

The input amplifier (INA) has the ability to rapidly recover from an excessive overdrive event such as a defibrillation pulse, high-voltage external pacing, and electro-surgery interference, or a rapid change in electrode offset due to motion. These events create a large change in the differential offset between ECGP and ECGN, causing the input amplifier or ADC to saturate. The analog HPF would eventually converge and bring the offset voltage back into range, but this can take a long time. Fast recovery greatly speeds up this process by enabling strong buffers to rapidly charge $C_{HPF}$, effectively raising the HPF corner frequency momentarily. These buffers are shown in Figure 29.

Fast recovery can be activated by three methods:

1. **Analog automatic mode:** An analog circuit detects if the INA is saturated, and briefly enables the fast recovery buffers.
2. **Digital automatic mode:** A digital function detects if the ADC counts are outside of a programmable threshold, and momentarily enables the fast recover buffers.
3. **Manual mode:** The fast recovery buffers are controlled manually by the host microcontroller.

Analog automatic mode is enabled by ECG_AUTO_REC[2](0x82) and monitors the INA for saturation, which occurs when the offset voltage changes rapidly. If the INA is saturated for more than approximately 2ms, the fast recovery buffers are enabled and remain active for approximately 32ms. After the buffers are disabled, there is a lockout period of approximately 96ms, after which saturation monitoring resumes. ECG samples taken while this mode is triggered are not marked with the fast recovery flag.

Digital automatic mode is enabled by setting EN_ECG_FAST_REC[7:6](0x83) to 0x2, and compares the ECG ADC counts to the symmetrical threshold set by ECG_FAST_REC_THRESHOLD[5:0](0x83). If the ADC counts are outside of the threshold for more than approximately 125ms, the fast recovery buffers are enabled and remain active for approximately 500ms. This behavior is shown in Figure 30. ECG samples taken while this mode is triggered are marked with the fast recovery flag (see the *FIFO Description* section).

In manual mode, the fast recovery buffers are enabled by setting EC_ECG_FAST_REC to 0x1, and disabled by setting it to 0x0. This mode allows the host microcontroller to implement custom fast-recovery logic, such as modified time durations or rising and falling thresholds. For example, the host could monitor the ADC counts and activate fast recovery if the counts saturate (e.g., > ±95% FSR), and keep fast recovery enabled until the counts fall to within a certain threshold (e.g., < ±10% FSR). This logic scheme would allow for much shorter fast recovery times in response to minor disturbances. ECG samples taken while this mode is active are marked with the fast recovery flag (see the *FIFO Description* section).

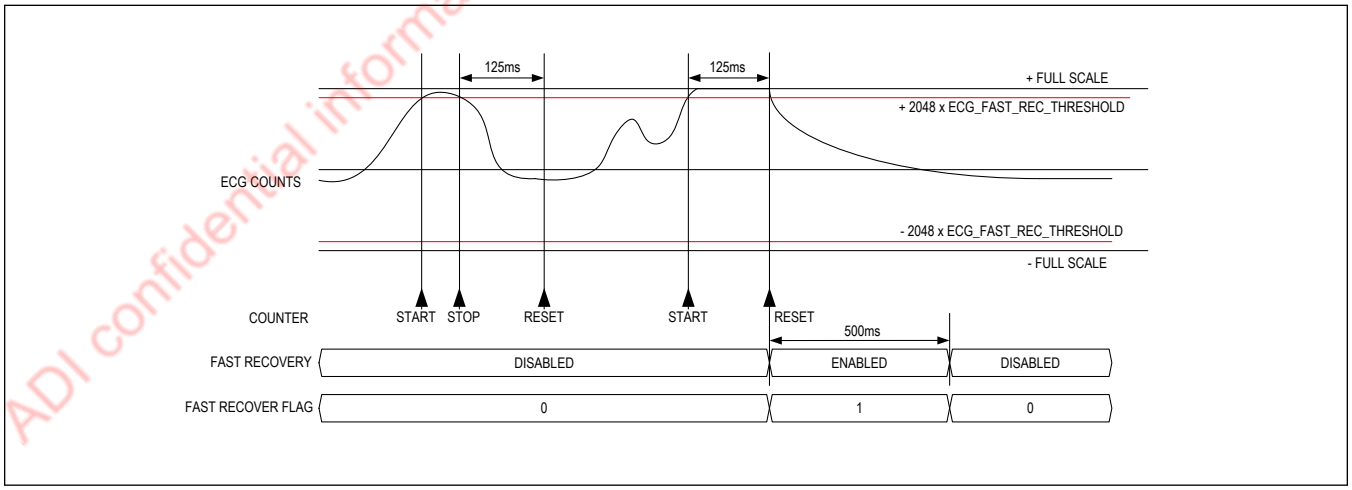



*Figure 30. Digital Automatic Fast-Recovery Behavior*

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### ECG Input MUX

The biopotential channel input MUX shown in Figure 31 contains ECG electrode assignment switches, EMI filtering, AC and DC lead-off detect current sources, ultra-low-power DC lead-on detect, lead biasing, and a programmable calibration-voltage source to enable channel built-in self-test. The series switches in the MAX86178 isolate ECGP and ECGN pins from the internal signal path. The series switches are disconnected by default, and must be connected by clearing OPEN_P[7](0x86) and OPEN_N[6](0x86) to record an ECG signal. ECG_IPOL[7](0x81) controls the polarity of the pin connections, and is noninverted by default. The RLD Block is described in the *Right Leg Drive and Body Bias* section.

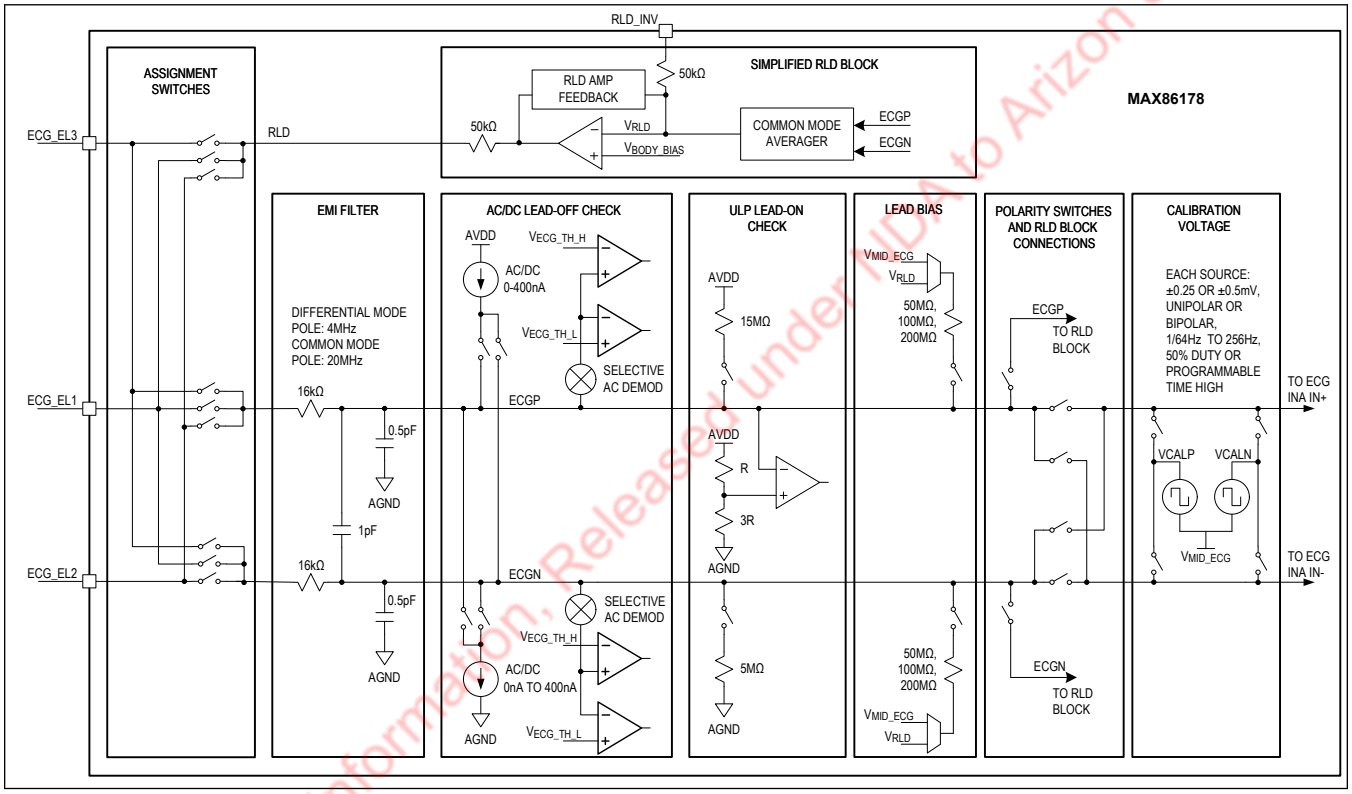



*Figure 31. ECG Channel Input MUX*

### ECG EMI Filtering

The EMI filter on the ECGP and ECGN inputs consists of 16kΩ resistors connected to ECGP and ECGN followed by a 1pF differential mode capacitor and 0.5pF common-mode capacitors. These form a single-pole, low-pass, differential- and common-mode filter with the differential-mode pole located at approximately 4MHz and the common-mode pole located at approximately 20MHz. Additional external EMI filters are not recommended for applications with dry electrodes in order to maintain high input impedance and common-mode rejection ratio, which helps mitigate the impact of electrode-impedance mismatch. With lower input impedance and CMRR, the electrode impedance mismatch translates into additional common-mode-to-differential-mode voltage conversions. Applications with wet electrodes can be used with external EMI filters with high precision components to minimize electrode impedance mismatch. In this case, the differential-mode pole can be set as low as the desired signal bandwidth, and the common-mode pole should be set at least a decade below the AM radio band (535kHz).

The ECG_ELx pins have ESD protection compliance with 2kV HBM. For IEC ESD compliance, external ESD diodes must be added (see *Typical Applications Circuits*).

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### ECG DC Lead-On Detection

The MAX86178 features an ultra-low powered (ULP) lead-on detection circuit for the ECG electrodes. The ECG channel must be disabled (ECG_EN[0](0x80) = 0) when the ULP lead-on is enabled.

The ULP lead-on detect circuit operates by pulling ECGN low with a pulldown resistance of 5MΩ (typ) and pulling ECGP high with a pullup resistance of 15MΩ (typ). A low-power comparator determines if ECGP is pulled below 0.75 x AVDD (typ), and asserts the ECG_LON[7](0x04) status bit if ECGP remains below the threshold for at least 128ms. This circuit is shown in Figure 31.

There are several conditions that can pull ECGP below the threshold and trigger a lead-on status:

- The total impedance between ECGP and ECGN is below 40MΩ (typ) due to both electrodes contacting the body.
- The total impedance between ECGP and AGND is below 45MΩ (typ) due to the the ECGP electrode contacting a body that is coupled to AGND. For example, if the MAX86178 system is coupled to earth ground through a power or data cable, and the body is also coupled to earth ground, then a low-impedance path could pull the ECGP electrode low.
- The ECGP electrode is contacting the body and has a large half-cell potential. The half-cell potential can push the ECGP voltage below the threshold.

If the ECG_LON interrupt is enabled by ECG_LON_EN1[7](0xC4) or ECG_LON_EN2[7](0xC8), an interrupt is generated to alert the host microcontroller of the lead-on condition. This interrupt allows the microcontroller to sleep when the system is not in use, and only to wake up when the user touches the device electrodes. Upon receiving an interrupt and waking up, the microcontroller should read the ECG_LON status register to determine if a lead-on condition has occurred. Because of the bit clear-on-read behavior, the status register should be read a second time to determine if the lead-on condition persists.

### ECG DC Lead-Off Detection

The MAX86178 features DC lead-off detection, which is enabled through the Lead Detect Configuration 1 (0x88) register. DC lead-off detection is used while ECG is active and cannot be used at the same time as AC lead-off detection.

A matching source and sink current is injected into the ECGP and ECGN electrodes. This current, when the electrodes are connected properly, flows through a first electrode-tissue interface, through the body and then through a second electrode-tissue interface. The electrode-tissue interface impedance is dependent on the type of electrode used (wet or dry), as well as the materials of the electrodes. The current flow develops a differential voltage across the two input pins. If one or both electrode-tissue interfaces have poor contact with the body, the current path has much higher impedance and the resulting voltage is large. If the electrodes are properly connected, then this voltage is small. DC lead-off detection can be used to evaluate the viability of both wet and dry electrode tissue interfaces.

The programmable stimulus current magnitudes are programmed using the Lead Detect Configuration 2 (0x89) register. The LOFF_IMAG[4:0] bits in the Lead Detect Configuration 2 register are used to set the nominal current between 0nA and 2.8µA. More details on setting the DC lead-off current are available in the register description.

There is also a voltage threshold setting that sets the threshold of the comparators and is used to determine whether a DC lead-off condition is encountered. The threshold setting is programmable using the ECG_LOFF_THRESH bit within the Lead Detect Configuration 2 register (0x89). The voltage threshold is centered around $V_{MID_ECG}$.

When using DC lead-off detection, there are two approaches available for alerting the microcontroller and user if the viability of the electrode tissue interfaces has been compromised: (1) dual comparators indicating if ECGP or ECGN voltages exceeds the programmable high threshold or programmable low threshold; and, (2) utility mode measurements can be used to digitize the voltage drop (the DC lead-off voltage) across the two electrodes and the body impedance (see the *Utility Mode Measurements* section).

The dual comparators can be used to generate a hardware interrupt using the Interrupt Enable 4 registers (0xC8, 0xC8) if the DC lead-off voltage exceeds the threshold (i.e., a minimum continuous violation) for 128ms before asserting the ECG_LOFF interrupt flags. See Figure 32 diagram below. There are four bits in the Status 4 (0x03) register, which when asserted indicate a lead-off condition has been detected. ECG_LOFF_PH, ECG_LOFF_PL, ECG_LOFF_NH, and ECG_LOFF_NL all give detailed information of the lead-off condition that has been detected. More details are available in the register map.

When using DC lead-off detection, the BioZ channel ADC can be repurposed to measure a buffered copy of either the differential ($V_{ECGP}$ - $V_{ECGN}$) voltage or a buffered copy of the ($V_{CAPP}$ - $V_{CAPN}$) voltage by asserting the

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EN_UTIL_MODE bit in the BioZ Configuration 4 register at address 0xA3. Refer to the *Utility Mode Measurements* section for details.

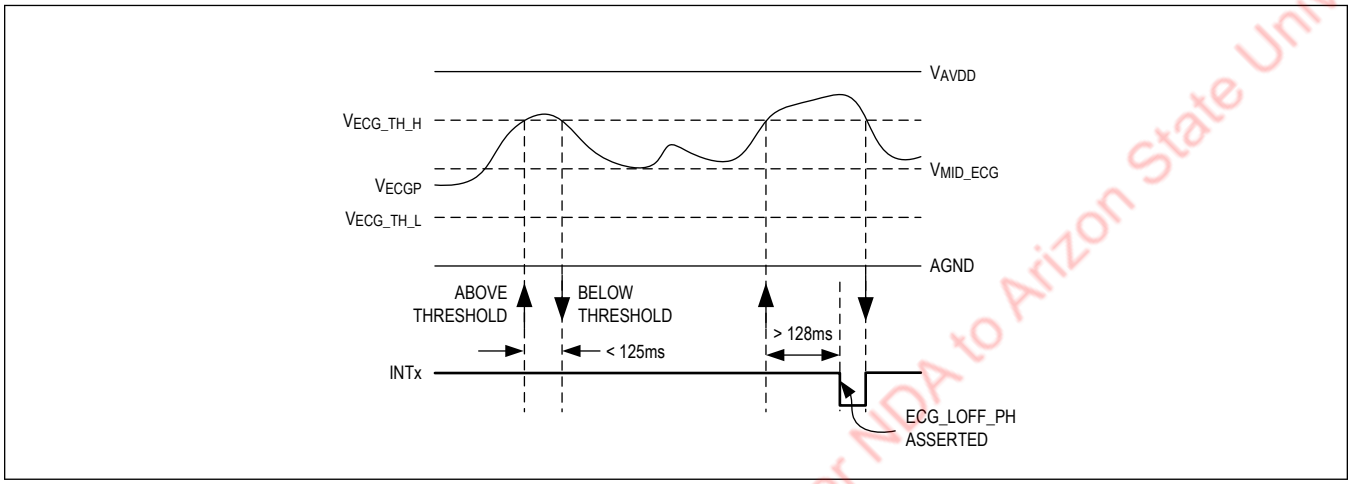



*Figure 32. DC Lead-Off Detection Behavior*

Table 14 shows calculated current and threshold settings to be used for a given contact impedance. For different contact impedances, the table can be calculated using using Ohm's Law (V = I x R), where I is the current generator setting and R is the contact impedance. Contact impedance is the body impedance in combination with the two electrode tissue interfaces. The table is meant for reference based on nominal current generator settings. For current generator accuracy refer to the electrical characteristics table. When using dry electrodes, it is recommended to use higher threshold values to accommodate larger electrode offset voltages.

## Table 14. Recommended Lead-Off Current and Threshold Values

| | CONTACT IMPEDANCE | | | | | | |
|---|---|---|---|---|---|---|---|
| **LOFF_IMAG CURRENT [nA]** | **10kΩ** | **100kΩ** | **1MΩ** | **5MΩ** | **10MΩ** | **20MΩ** | **40MΩ** |
| 1.25 | All Settings of $V_{TH}$ | | | | | $V_{TH}$ > 300mV | |
| 20 | All Settings of $V_{TH}$ | | | | | $V_{TH}$ > 300mV | $V_{TH}$ > 400mV |
| 40 | All Settings of $V_{TH}$ | | | | | $V_{TH}$ > 320mV | $V_{TH}$ > 480mV |
| 120 | All Settings of $V_{TH}$ | | | $V_{TH}$ > 260mV | $V_{TH}$ > 380mV | $V_{TH}$ > 460mV | Do not use |
| 200 | All Settings of $V_{TH}$ | | | $V_{TH}$ > 400mV | $V_{TH}$ > 440mV | Do not use | Do not use |
| 400 | All Settings of $V_{TH}$ | | | $V_{TH}$ > 440mV | Do not use | Do not use | Do not use |
| 1200 | All Settings of $V_{TH}$ | | $V_{TH}$ > 360mV | Do not use | Do not use | Do not use | Do not use |
| 2800 | All Settings of $V_{TH}$ | | Do not use | Do not use | Do not use | Do not use | Do not use |

### ECG AC Lead-Off Detection

The MAX86178 offers ECG AC lead-off detection as an alternative to DC lead-off detection. Available frequencies and current stimulus magnitudes are described in the ECG Lead Detect Configuration 1 and 2 registers (0x88 and 0x89). If ECG_LOFF_MODE is set to 1, then AC lead-off is selected. This circuit operates much the same as the DC lead-off circuit, except a clock is applied to chop the DC output current into a square wave. The ECG_LOFF_IMAG[2:0] field is used for setting both the DC and AC lead-off current magnitudes. On the receive side, there is a small AC-to-DC converter that is switched on for AC mode. The rectified DC output is then compared to the same thresholds that are used for DC lead-off detection. The AC stimulus frequency is set by ECG_LOFF_FREQ[3:0]—details are available in the *ECG Sample Rate and AC Lead-Off Frequency* section.

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### Utility Mode Measurements

The MAX86178 includes a utility mode that repurposes the BioZ receive channel to measure ($V_{ECGP}$ - $V_{ECGN}$), ($V_{ECGP}$ - $V_{MID_ECG}$), ($V_{ECGN}$ - $V_{MID_ECG}$), or ($V_{CAPP}$ - $V_{CAPN}$). Utility mode measurements are used to determine the DC biases of the ECG inputs since the ECG channel only measures the high-pass-filtered differential-input voltage. The DC bias information can be used to determine if the inputs are within range, or to directly measure the DC lead-off voltage. Utility mode measurements and BioZ measurements cannot be performed simultaneously.

Utility mode is enabled by EN_UTIL_MODE[7](0xA3), and the measurement is selected by BIOZ_EN[1:0](0xA0). BIOZ_DM_DIS[2](0xA5) must be set to 1 to disable the demodulator, and BIP_ASSIGN[7:6](0xAC) and BIN_ASSIGN[5:4](0xAC) must each be set to 0x3 to connect the selected signal. The selected signal is applied to the inputs of the BioZ PGA as shown in Figure 33. The PGA gain is set by BIOZ_GAIN[1:0](0xA5) and can be set to 1V/V, 2.5V/V, or 5V/V. The utility mode sample rate is equal to the BioZ sample rate, as discussed in the *BioZ Sample Rate and Stimulus Frequency* section. Utility mode samples are 20 bits, and marked with unique data tags in the FIFO (see the *FIFO Description* section). The input-referred voltage of the selected measurement can be calculated with the following equation.

$V_{UTIL}(V) = ADC_COUNT \times V_{REF_ECG} / (2^{19} \times BIOZ_PGA_GAIN)$

where,

ADC_COUNT = ADC counts in signed magnitude format

$V_{REF_ECG}$ = 1V (typ)

BIOZ_PGA_GAIN = PGA gain set by BIOZ_GAIN

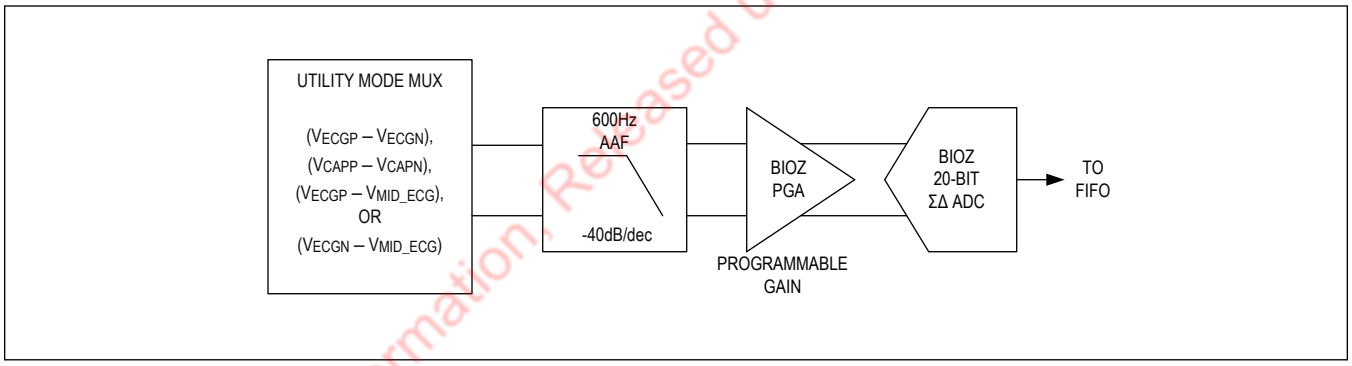



*Figure 33. Utility Mode Signal Chain*

### ECG Calibration Voltage Sources

Calibration voltage sources are available to provide rectangular pulse-train signals for internal signal-chain validation. Each of the two sources, VCALP and VCALN, can be unipolar or bipolar with respect to $V_{MID_ECG}$, with amplitudes of ±0.25mV or ±0.5mV. Each input can be connected to either of the two sources or $V_{MID_ECG}$ for differential mode amplitudes between 0.25m$V_{P-P}$ and 2.0m$V_{P-P}$ or common-mode amplitudes between 0.25m$V_{P-P}$ and 1m$V_{P-P}$.

Figure 34 illustrates the possible calibration waveforms. Frequency selections are available in 4x increments from 15.625mHz to 256Hz with pulse widths between 30.5µs and 31.723ms or 50% duty cycle. When applying calibration voltage sources with the device connected to a subject, the series input switches must be disconnected so as not to drive signals into the subject. See registers ECG CAL Configuration 1/2/3 (0x84, 0x85, 0x86) to select a configuration.

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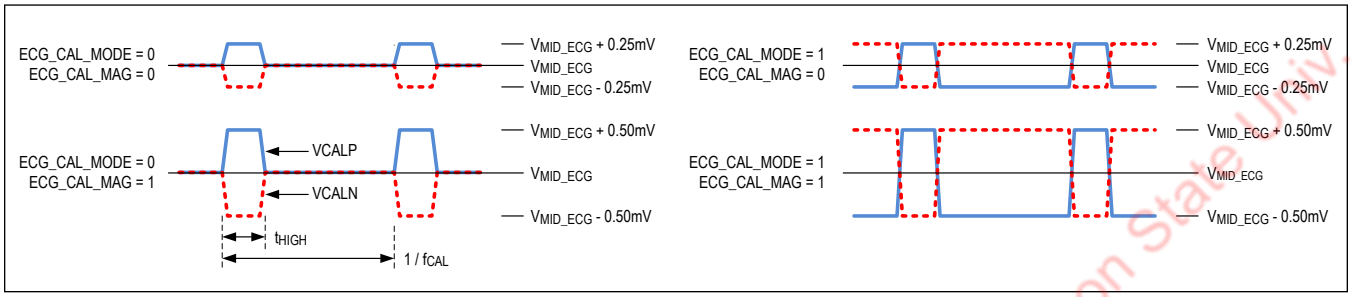



*Figure 34. Calibration Voltage Source Options*

### Converting Digitized ECG Samples to Voltage

Biopotential channel samples are recorded in 18-bit, left justified, 2's complement format. After converting to signed magnitude format, the biopotential channel differential input voltage, $V_{ECG}$ (mV) = ($V_{ECGP}$ - $V_{ECGN}$), is calculated by the following equation:

$$V_{ECG}\ (mV) = ADC_COUNT \times V_{REF_ECG} / (2^{17} \times ECG_GAIN)$$

ADC_COUNT is the ADC counts in signed magnitude format, $V_{REF_ECG}$ is 1000mV (typ, see the *Electrical Characteristics* section) and ECG_GAIN = ECG_INA_GAIN x ECG_PGA_GAIN. ECG_INA_GAIN is programmable between 2.5V/V and 60V/V; and ECG_PGA_GAIN is programmable at 1V/V, 2V/V, 4V/V, 8V/V, or 16V/V. These two gains are set in the ECG Configuration 2 register (0x81).

### ECG Noise Measurements

Table 15, Table 16, Table 17, and Table 18 show the typical noise performance of the biopotential channel of MAX86178 referred to the ECG inputs for the four INA range settings. Most applications use gains of 10V/V or higher, but lower gain ranges are included for an expanded AC differential-input range to accommodate large baseline wander. These lower gain settings exhibit higher input-referred noise compared to gains larger than 10V/V. The input-referred noise is given over three different bandwidths: 40Hz, 100Hz, and 150Hz. Note that this low-pass filtering is assumed to be done as a post-processed step by the microcontroller and its firmware, as there is no on-chip digital low-pass filtering. The filters used for the data in this table are 12-tap digital FIR filters with 20dB of stop-band attenuation. This filter design is optimized for low signal latency when displaying the ECG signal in real time. Higher-order digital filters are suitable for ECG recording systems where latency is not an issue and a better filter response is preferred. This data was taken at 512sps, with the RMS and P-P values measured over 1 a second window.

## Table 15. Biopotential Channel Noise Performance with ECG_INA_RGE = 0x0

| ECG_INA_RGE | PGA GAIN (V/V) | INA GAIN (V/V) | TOTAL GAIN (V/V) | BAND-WIDTH (Hz) | NOISE ($\mu V_{RMS}$) | NOISE ($\mu V_{P-P}$) | SNR (dB) | ENOB (bits) |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 10 | 10 | 40 | 1.04 | 5.52 | 96.6 | 15.8 |
| | | | | 100 | 1.24 | 7.41 | 95.1 | 15.5 |
| | | | | 150 | 1.32 | 8.13 | 94.6 | 15.4 |
| | | 20 | 20 | 40 | 0.64 | 3.34 | 94.8 | 15.5 |
| | | | | 100 | 0.94 | 5.95 | 91.5 | 14.9 |
| | | | | 150 | 1.04 | 6.66 | 90.6 | 14.8 |
| | | 40 | 40 | 40 | 0.74 | 4.09 | 87.5 | 14.2 |
| | | | | 100 | 0.95 | 4.75 | 85.4 | 13.9 |
| | | | | 150 | 1.03 | 5.65 | 84.7 | 13.8 |
| | | 60 | 60 | 40 | 0.67 | 3.56 | 84.9 | 13.8 |
| | | | | 100 | 0.88 | 4.65 | 82.6 | 13.4 |

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## Table 15. Biopotential Channel Noise Performance with ECG_INA_RGE = 0x0 (continued)

| ECG_INA_RGE | PGA GAIN (V/V) | INA GAIN (V/V) | TOTAL GAIN (V/V) | BAND-WIDTH (Hz) | NOISE ($\mu V_{RMS}$) | NOISE ($\mu V_{P-P}$) | SNR (dB) | ENOB (bits) |
|---|---|---|---|---|---|---|---|---|
| | | | | 150 | 0.98 | 5.60 | 81.6 | 13.3 |
| | 2 | 10 | 20 | 40 | 0.88 | 4.17 | 92.1 | 15.0 |
| | | | | 100 | 1.07 | 6.70 | 90.4 | 14.7 |
| | | | | 150 | 1.14 | 6.87 | 89.8 | 14.6 |
| | | 20 | 40 | 40 | 0.68 | 3.53 | 88.2 | 14.4 |
| | | | | 100 | 0.89 | 5.36 | 85.9 | 14.0 |
| | | | | 150 | 0.98 | 6.09 | 85.1 | 13.8 |
| | | 40 | 80 | 40 | 0.71 | 3.68 | 81.8 | 13.3 |
| | | | | 100 | 0.91 | 5.34 | 79.7 | 12.9 |
| | | | | 150 | 1.01 | 6.53 | 78.9 | 12.8 |
| | | 60 | 120 | 40 | 0.64 | 3.64 | 79.3 | 12.9 |
| | | | | 100 | 0.85 | 4.50 | 76.8 | 12.5 |
| | | | | 150 | 0.94 | 4.95 | 76.0 | 12.3 |
| | 4 | 10 | 40 | 40 | 0.75 | 4.03 | 87.4 | 14.2 |
| | | | | 100 | 0.98 | 5.42 | 85.1 | 13.8 |
| | | | | 150 | 1.09 | 6.34 | 84.2 | 13.7 |
| | | 20 | 80 | 40 | 0.70 | 4.40 | 82.0 | 13.3 |
| | | | | 100 | 0.87 | 4.79 | 80.1 | 13.0 |
| | | | | 150 | 0.97 | 5.60 | 79.2 | 12.9 |
| | | 40 | 160 | 40 | 0.68 | 4.29 | 76.3 | 12.4 |
| | | | | 100 | 0.87 | 5.10 | 74.2 | 12.0 |
| | | | | 150 | 0.96 | 5.53 | 73.3 | 11.9 |
| | | 60 | 240 | 40 | 0.63 | 3.19 | 73.3 | 11.9 |
| | | | | 100 | 0.82 | 4.40 | 71.1 | 11.5 |
| | | | | 150 | 0.91 | 5.69 | 70.2 | 11.4 |
| | 8 | 10 | 80 | 40 | 0.78 | 3.69 | 81.1 | 13.2 |
| | | | | 100 | 1.01 | 6.33 | 78.8 | 12.8 |
| | | | | 150 | 1.11 | 6.17 | 78.0 | 12.7 |
| | | 20 | 160 | 40 | 0.61 | 2.98 | 77.1 | 12.5 |
| | | | | 100 | 0.86 | 4.84 | 74.2 | 12.0 |
| | | | | 150 | 0.97 | 5.79 | 73.1 | 11.9 |
| | | 40 | 320 | 40 | 0.68 | 3.90 | 70.2 | 11.4 |
| | | | | 100 | 0.88 | 5.32 | 68.0 | 11.0 |
| | | | | 150 | 0.95 | 5.99 | 67.3 | 10.9 |
| | | 60 | 480 | 40 | 0.65 | 3.58 | 67.1 | 10.9 |
| | | | | 100 | 0.86 | 4.78 | 64.6 | 10.4 |
| | | | | 150 | 0.94 | 5.70 | 63.9 | 10.3 |
| | 16 | 10 | 160 | 40 | 0.78 | 4.66 | 75.1 | 12.2 |
| | | | | 100 | 0.95 | 6.51 | 73.4 | 11.9 |

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## Table 15. Biopotential Channel Noise Performance with ECG_INA_RGE = 0x0 (continued)

| ECG_INA_RGE | PGA GAIN (V/V) | INA GAIN (V/V) | TOTAL GAIN (V/V) | BAND-WIDTH (Hz) | NOISE ($\mu V_{RMS}$) | NOISE ($\mu V_{P-P}$) | SNR (dB) | ENOB (bits) |
|---|---|---|---|---|---|---|---|---|
| | | | | 150 | 1.02 | 7.00 | 72.7 | 11.8 |
| | | 20 | 320 | 40 | 0.58 | 3.00 | 71.6 | 11.6 |
| | | | | 100 | 0.81 | 4.43 | 68.8 | 11.1 |
| | | | | 150 | 0.88 | 5.12 | 68.0 | 11.0 |
| | | 40 | 640 | 40 | 0.70 | 3.84 | 63.9 | 10.3 |
| | | | | 100 | 0.83 | 5.04 | 62.4 | 10.1 |
| | | | | 150 | 0.92 | 5.96 | 61.6 | 9.9 |
| | | 60 | 960 | 40 | 0.74 | 3.85 | 60.0 | 9.7 |
| | | | | 100 | 0.93 | 5.01 | 57.9 | 9.3 |
| | | | | 150 | 1.02 | 6.08 | 57.2 | 9.2 |

Note: $SNR = 20\log\left(\frac{V_{IN(RMS)}}{V_{N(RMS)}}\right)$, $ENOB = (SNR - 1.76) / 6.02$

Note: $V_{IN(P-P)}$ = 2V / TOTAL GAIN, which represents the largest signal that can be input into the ECG channel. $V_{IN(RMS)} = V_{IN(P-P)} / (2\sqrt{2})$.

## Table 16. Biopotential Channel Noise Performance with ECG_INA_RGE = 0x1

| ECG_INA_RGE | PGA GAIN (V/V) | INA GAIN (V/V) | TOTAL GAIN (V/V) | BAND-WIDTH (Hz) | NOISE ($\mu V_{RMS}$) | NOISE ($\mu V_{P-P}$) | SNR (dB) | ENOB (bits) |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 7.5 | 7.5 | 40 | 1.13 | 5.55 | 98.5 | 16.1 |
| | | | | 100 | 1.39 | 7.49 | 96.6 | 15.8 |
| | | | | 150 | 1.48 | 8.24 | 96.1 | 15.7 |
| | | 15 | 15 | 40 | 0.94 | 5.21 | 94.0 | 15.3 |
| | | | | 100 | 1.15 | 7.20 | 92.3 | 15.0 |
| | | | | 150 | 1.21 | 8.19 | 91.8 | 15.0 |
| | | 30 | 30 | 40 | 0.79 | 3.79 | 89.5 | 14.6 |
| | | | | 100 | 1.01 | 5.48 | 87.4 | 14.2 |
| | | | | 150 | 1.08 | 5.77 | 86.8 | 14.1 |
| | | 45 | 45 | 40 | 0.78 | 4.35 | 86.1 | 14.0 |
| | | | | 100 | 0.97 | 5.71 | 84.2 | 13.7 |
| | | | | 150 | 1.09 | 6.38 | 83.1 | 13.5 |
| | 2 | 7.5 | 15 | 40 | 0.96 | 5.38 | 93.8 | 15.3 |
| | | | | 100 | 1.23 | 6.72 | 91.7 | 14.9 |
| | | | | 150 | 1.34 | 7.65 | 90.9 | 14.8 |
| | | 15 | 30 | 40 | 0.86 | 4.91 | 88.7 | 14.4 |
| | | | | 100 | 1.09 | 6.88 | 86.7 | 14.1 |
| | | | | 150 | 1.18 | 7.44 | 86.0 | 14.0 |
| | | 30 | 60 | 40 | 0.80 | 4.68 | 83.4 | 13.6 |
| | | | | 100 | 1.02 | 6.35 | 81.3 | 13.2 |
| | | | | 150 | 1.09 | 6.50 | 80.7 | 13.1 |
| | | 45 | 90 | 40 | 0.72 | 4.63 | 80.8 | 13.1 |

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## Table 16. Biopotential Channel Noise Performance with ECG_INA_RGE = 0x1 (continued)

| ECG_INA_RGE | PGA GAIN (V/V) | INA GAIN (V/V) | TOTAL GAIN (V/V) | BAND-WIDTH (Hz) | NOISE ($\mu V_{RMS}$) | NOISE ($\mu V_{P-P}$) | SNR (dB) | ENOB (bits) |
|---|---|---|---|---|---|---|---|---|
| | | | | 100 | 0.94 | 5.76 | 78.5 | 12.7 |
| | | | | 150 | 1.04 | 6.38 | 77.6 | 12.6 |
| | 4 | 7.5 | 30 | 40 | 0.91 | 5.14 | 88.3 | 14.4 |
| | | | | 100 | 1.12 | 6.20 | 86.4 | 14.1 |
| | | | | 150 | 1.21 | 7.55 | 85.8 | 14.0 |
| | | 15 | 60 | 40 | 0.70 | 3.55 | 84.5 | 13.7 |
| | | | | 100 | 0.93 | 4.90 | 82.0 | 13.3 |
| | | | | 150 | 1.05 | 5.48 | 81.0 | 13.2 |
| | | 30 | 120 | 40 | 0.75 | 4.09 | 77.9 | 12.6 |
| | | | | 100 | 0.96 | 5.82 | 75.8 | 12.3 |
| | | | | 150 | 1.06 | 5.85 | 74.9 | 12.1 |
| | | 45 | 180 | 40 | 0.77 | 3.79 | 74.2 | 12.0 |
| | | | | 100 | 0.95 | 5.98 | 72.4 | 11.7 |
| | | | | 150 | 1.02 | 6.69 | 71.7 | 11.6 |
| | 8 | 7.5 | 60 | 40 | 0.82 | 4.40 | 83.1 | 13.5 |
| | | | | 100 | 1.09 | 5.60 | 80.7 | 13.1 |
| | | | | 150 | 1.18 | 7.05 | 80.0 | 13.0 |
| | | 15 | 120 | 40 | 0.80 | 4.37 | 77.3 | 12.6 |
| | | | | 100 | 1.05 | 5.98 | 75.0 | 12.2 |
| | | | | 150 | 1.12 | 6.84 | 74.5 | 12.1 |
| | | 30 | 240 | 40 | 0.79 | 3.94 | 71.4 | 11.6 |
| | | | | 100 | 0.95 | 5.66 | 69.8 | 11.3 |
| | | | | 150 | 1.06 | 6.27 | 68.9 | 11.1 |
| | | 45 | 360 | 40 | 0.81 | 4.95 | 67.7 | 11.0 |
| | | | | 100 | 1.02 | 6.92 | 65.7 | 10.6 |
| | | | | 150 | 1.14 | 7.55 | 64.7 | 10.5 |
| | 16 | 7.5 | 120 | 40 | 0.81 | 3.96 | 77.2 | 12.5 |
| | | | | 100 | 1.10 | 5.96 | 74.6 | 12.1 |
| | | | | 150 | 1.19 | 6.49 | 73.9 | 12.0 |
| | | 15 | 240 | 40 | 0.85 | 4.91 | 70.8 | 11.5 |
| | | | | 100 | 1.04 | 5.98 | 69.1 | 11.2 |
| | | | | 150 | 1.13 | 6.44 | 68.3 | 11.1 |
| | | 30 | 480 | 40 | 0.81 | 4.67 | 65.1 | 10.5 |
| | | | | 100 | 1.02 | 6.35 | 63.2 | 10.2 |
| | | | | 150 | 1.13 | 6.51 | 62.3 | 10.1 |
| | | 45 | 720 | 40 | 0.79 | 4.14 | 61.9 | 10.0 |
| | | | | 100 | 1.00 | 6.13 | 59.9 | 9.6 |
| | | | | 150 | 1.06 | 6.53 | 59.3 | 9.6 |

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Note: SNR = $20\log\left(\frac{V_{IN(RMS)}}{V_{N(RMS)}}\right)$, ENOB = (SNR − 1.76) / 6.02

Note: $V_{IN(P-P)}$ = 2V / TOTAL GAIN, which represents the largest signal that can be input into the ECG channel. $V_{IN(RMS)} = V_{IN(P-P)} / (2\sqrt{2})$.

## Table 17. Biopotential Channel Noise Performance with ECG_INA_RGE = 0x2

| ECG_INA_RGE | PGA GAIN (V/V) | INA GAIN (V/V) | TOTAL GAIN (V/V) | BAND-WIDTH (Hz) | NOISE ($\mu V_{RMS}$) | NOISE ($\mu V_{P-P}$) | SNR (dB) | ENOB (bits) |
|---|---|---|---|---|---|---|---|---|
| 2 | 1 | 5 | 5 | 40 | 1.63 | 9.24 | 98.7 | 16.1 |
| | | | | 100 | 2.00 | 10.90 | 97.0 | 15.8 |
| | | | | 150 | 2.14 | 11.59 | 96.4 | 15.7 |
| | | 10 | 10 | 40 | 1.14 | 5.61 | 95.8 | 15.6 |
| | | | | 100 | 1.45 | 8.07 | 93.7 | 15.3 |
| | | | | 150 | 1.56 | 9.48 | 93.1 | 15.2 |
| | | 20 | 20 | 40 | 0.90 | 5.58 | 91.9 | 15.0 |
| | | | | 100 | 1.21 | 6.91 | 89.3 | 14.5 |
| | | | | 150 | 1.35 | 7.36 | 88.3 | 14.4 |
| | | 30 | 30 | 40 | 1.22 | 6.72 | 85.7 | 14.0 |
| | | | | 100 | 1.41 | 8.61 | 84.5 | 13.7 |
| | | | | 150 | 1.51 | 9.26 | 83.9 | 13.6 |
| | 2 | 5 | 10 | 40 | 1.38 | 6.89 | 94.2 | 15.4 |
| | | | | 100 | 1.65 | 10.51 | 92.6 | 15.1 |
| | | | | 150 | 1.78 | 10.86 | 92.0 | 15.0 |
| | | 10 | 20 | 40 | 1.11 | 6.28 | 90.1 | 14.7 |
| | | | | 100 | 1.39 | 8.12 | 88.1 | 14.3 |
| | | | | 150 | 1.54 | 9.64 | 87.2 | 14.2 |
| | | 20 | 40 | 40 | 1.05 | 6.19 | 84.5 | 13.7 |
| | | | | 100 | 1.31 | 8.39 | 82.6 | 13.4 |
| | | | | 150 | 1.42 | 8.93 | 81.9 | 13.3 |
| | | 30 | 60 | 40 | 0.98 | 4.91 | 81.6 | 13.3 |
| | | | | 100 | 1.22 | 7.13 | 79.7 | 12.9 |
| | | | | 150 | 1.37 | 8.77 | 78.7 | 12.8 |
| | 4 | 5 | 20 | 40 | 1.09 | 6.65 | 90.2 | 14.7 |
| | | | | 100 | 1.50 | 9.05 | 87.4 | 14.2 |
| | | | | 150 | 1.63 | 9.31 | 86.7 | 14.1 |
| | | 10 | 40 | 40 | 1.00 | 5.45 | 85.0 | 13.8 |
| | | | | 100 | 1.32 | 8.03 | 82.6 | 13.4 |
| | | | | 150 | 1.42 | 8.56 | 81.9 | 13.3 |
| | | 20 | 80 | 40 | 1.10 | 6.16 | 78.1 | 12.7 |
| | | | | 100 | 1.34 | 7.89 | 76.4 | 12.4 |
| | | | | 150 | 1.44 | 8.92 | 75.7 | 12.3 |
| | | 30 | 120 | 40 | 0.90 | 5.63 | 76.3 | 12.4 |
| | | | | 100 | 1.19 | 7.53 | 73.9 | 12.0 |
| | | | | 150 | 1.30 | 7.55 | 73.2 | 11.9 |

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## Table 17. Biopotential Channel Noise Performance with ECG_INA_RGE = 0x2 (continued)

| ECG_INA_RGE | PGA GAIN (V/V) | INA GAIN (V/V) | TOTAL GAIN (V/V) | BAND-WIDTH (Hz) | NOISE ($\mu V_{RMS}$) | NOISE ($\mu V_{P-P}$) | SNR (dB) | ENOB (bits) |
|---|---|---|---|---|---|---|---|---|
| | 8 | 5 | 40 | 40 | 1.01 | 6.13 | 84.8 | 13.8 |
| | | | | 100 | 1.34 | 8.53 | 82.4 | 13.4 |
| | | | | 150 | 1.52 | 9.40 | 81.3 | 13.2 |
| | | 10 | 80 | 40 | 1.07 | 6.14 | 78.3 | 12.7 |
| | | | | 100 | 1.41 | 8.28 | 76.0 | 12.3 |
| | | | | 150 | 1.53 | 8.96 | 75.3 | 12.2 |
| | | 20 | 160 | 40 | 0.89 | 6.16 | 73.9 | 12.0 |
| | | | | 100 | 1.18 | 7.13 | 71.5 | 11.6 |
| | | | | 150 | 1.31 | 8.48 | 70.6 | 11.4 |
| | | 30 | 240 | 40 | 1.06 | 6.81 | 68.9 | 11.2 |
| | | | | 100 | 1.31 | 8.63 | 67.0 | 10.8 |
| | | | | 150 | 1.39 | 8.57 | 66.5 | 10.8 |
| | 16 | 5 | 80 | 40 | 1.17 | 5.92 | 77.6 | 12.6 |
| | | | | 100 | 1.42 | 8.55 | 75.9 | 12.3 |
| | | | | 150 | 1.53 | 9.08 | 75.2 | 12.2 |
| | | 10 | 160 | 40 | 1.08 | 5.61 | 72.2 | 11.7 |
| | | | | 100 | 1.39 | 7.75 | 70.1 | 11.3 |
| | | | | 150 | 1.50 | 8.44 | 69.4 | 11.2 |
| | | 20 | 320 | 40 | 0.86 | 4.88 | 68.2 | 11.0 |
| | | | | 100 | 1.16 | 6.72 | 65.6 | 10.6 |
| | | | | 150 | 1.31 | 8.76 | 64.6 | 10.4 |
| | | 30 | 480 | 40 | 1.02 | 5.70 | 63.2 | 10.2 |
| | | | | 100 | 1.31 | 7.09 | 61.0 | 9.8 |
| | | | | 150 | 1.43 | 8.29 | 60.3 | 9.7 |

Note: $SNR = 20\log\left(\frac{V_{IN(RMS)}}{V_{N(RMS)}}\right)$, $ENOB = (SNR - 1.76) / 6.02$

Note: $V_{IN(P-P)}$ = 2V / TOTAL GAIN, which represents the largest signal that can be input into the ECG channel. $V_{IN(RMS)} = V_{IN(P-P)} / (2\sqrt{2})$.

## Table 18. Biopotential Channel Noise Performance with ECG_INA_RGE = 0x3

| ECG_INA_RGE | PGA GAIN (V/V) | INA GAIN (V/V) | TOTAL GAIN (V/V) | BAND-WIDTH (Hz) | NOISE ($\mu V_{RMS}$) | NOISE ($\mu V_{P-P}$) | SNR (dB) | ENOB (bits) |
|---|---|---|---|---|---|---|---|---|
| 3 | 1 | 2.5 | 2.5 | 40 | 3.19 | 18.67 | 99.0 | 16.2 |
| | | | | 100 | 3.81 | 24.08 | 97.4 | 15.9 |
| | | | | 150 | 3.99 | 24.94 | 97.0 | 15.8 |
| | | 5 | 5 | 40 | 2.13 | 11.22 | 96.4 | 15.7 |
| | | | | 100 | 2.64 | 15.90 | 94.6 | 15.4 |
| | | | | 150 | 2.83 | 17.91 | 94.0 | 15.3 |
| | | 10 | 10 | 40 | 2.01 | 10.29 | 90.9 | 14.8 |
| | | | | 100 | 2.50 | 15.47 | 89.0 | 14.5 |

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## Table 18. Biopotential Channel Noise Performance with ECG_INA_RGE = 0x3 (continued)

| ECG_INA_RGE | PGA GAIN (V/V) | INA GAIN (V/V) | TOTAL GAIN (V/V) | BAND-WIDTH (Hz) | NOISE ($\mu V_{RMS}$) | NOISE ($\mu V_{P-P}$) | SNR (dB) | ENOB (bits) |
|---|---|---|---|---|---|---|---|---|
| | | | | 150 | 2.65 | 18.48 | 88.5 | 14.4 |
| | | 15 | 15 | 40 | 2.07 | 11.73 | 87.2 | 14.2 |
| | | | | 100 | 2.45 | 16.32 | 85.7 | 13.9 |
| | | | | 150 | 2.55 | 18.00 | 85.3 | 13.9 |
| | 2 | 2.5 | 5 | 40 | 2.58 | 14.89 | 94.8 | 15.5 |
| | | | | 100 | 3.15 | 18.70 | 93.0 | 15.2 |
| | | | | 150 | 3.26 | 19.79 | 92.7 | 15.1 |
| | | 5 | 10 | 40 | 2.22 | 11.78 | 90.1 | 14.7 |
| | | | | 100 | 2.69 | 14.87 | 88.4 | 14.4 |
| | | | | 150 | 2.82 | 15.70 | 88.0 | 14.3 |
| | | 10 | 20 | 40 | 1.81 | 9.20 | 85.8 | 14.0 |
| | | | | 100 | 2.27 | 14.19 | 83.9 | 13.6 |
| | | | | 150 | 2.39 | 14.97 | 83.4 | 13.6 |
| | | 15 | 30 | 40 | 1.66 | 9.01 | 83.1 | 13.5 |
| | | | | 100 | 2.12 | 15.00 | 80.9 | 13.1 |
| | | | | 150 | 2.35 | 15.75 | 80.0 | 13.0 |
| | 4 | 2.5 | 10 | 40 | 2.22 | 11.75 | 90.1 | 14.7 |
| | | | | 100 | 2.74 | 15.08 | 88.2 | 14.4 |
| | | | | 150 | 3.02 | 16.57 | 87.4 | 14.2 |
| | | 5 | 20 | 40 | 1.82 | 11.28 | 85.8 | 14.0 |
| | | | | 100 | 2.26 | 14.08 | 83.9 | 13.6 |
| | | | | 150 | 2.45 | 15.47 | 83.2 | 13.5 |
| | | 10 | 40 | 40 | 1.80 | 10.23 | 79.9 | 13.0 |
| | | | | 100 | 2.27 | 13.15 | 77.8 | 12.6 |
| | | | | 150 | 2.43 | 14.95 | 77.2 | 12.5 |
| | | 15 | 60 | 40 | 1.53 | 7.91 | 77.7 | 12.6 |
| | | | | 100 | 1.99 | 9.81 | 75.4 | 12.2 |
| | | | | 150 | 2.18 | 11.18 | 74.6 | 12.1 |
| | 8 | 2.5 | 20 | 40 | 2.12 | 12.93 | 84.4 | 13.7 |
| | | | | 100 | 2.51 | 17.03 | 83.0 | 13.5 |
| | | | | 150 | 2.73 | 17.35 | 82.2 | 13.4 |
| | | 5 | 40 | 40 | 1.79 | 10.80 | 79.9 | 13.0 |
| | | | | 100 | 2.24 | 13.57 | 77.9 | 12.7 |
| | | | | 150 | 2.40 | 14.57 | 77.3 | 12.6 |
| | | 10 | 80 | 40 | 1.65 | 9.66 | 74.6 | 12.1 |
| | | | | 100 | 2.09 | 11.69 | 72.5 | 11.8 |
| | | | | 150 | 2.26 | 13.31 | 71.8 | 11.6 |
| | | 15 | 120 | 40 | 1.76 | 11.21 | 70.5 | 11.4 |
| | | | | 100 | 2.21 | 11.97 | 68.5 | 11.1 |

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## Table 18. Biopotential Channel Noise Performance with ECG_INA_RGE = 0x3 (continued)

| ECG_INA_RGE | PGA GAIN (V/V) | INA GAIN (V/V) | TOTAL GAIN (V/V) | BAND-WIDTH (Hz) | NOISE ($\mu V_{RMS}$) | NOISE ($\mu V_{P-P}$) | SNR (dB) | ENOB (bits) |
|---|---|---|---|---|---|---|---|---|
| | | | | 150 | 2.36 | 13.98 | 67.9 | 11.0 |
| | 16 | 2.5 | 40 | 40 | 2.05 | 11.85 | 78.7 | 12.8 |
| | | | | 100 | 2.56 | 16.60 | 76.8 | 12.5 |
| | | | | 150 | 2.66 | 17.07 | 76.5 | 12.4 |
| | | 5 | 80 | 40 | 2.19 | 12.51 | 72.1 | 11.7 |
| | | | | 100 | 2.56 | 18.25 | 70.8 | 11.5 |
| | | | | 150 | 2.58 | 19.19 | 70.7 | 11.5 |
| | | 10 | 160 | 40 | 1.89 | 10.11 | 67.4 | 10.9 |
| | | | | 100 | 2.24 | 13.51 | 65.9 | 10.7 |
| | | | | 150 | 2.37 | 13.97 | 65.4 | 10.6 |
| | | 15 | 240 | 40 | 1.64 | 8.82 | 65.1 | 10.5 |
| | | | | 100 | 2.23 | 13.42 | 62.4 | 10.1 |
| | | | | 150 | 2.39 | 15.23 | 61.8 | 10.0 |

Note: $SNR = 20\log\left(\frac{V_{IN(RMS)}}{V_{N(RMS)}}\right)$, $ENOB = (SNR - 1.76) \Big/ 6.02$

Note: $V_{IN(P-P)}$ = 2V / TOTAL GAIN, which represents the largest signal that can be input into the ECG channel.
$V_{IN(RMS)} = V_{IN(P-P)} / (2\sqrt{2})$.

### Right Leg Drive and Body Bias

The right leg drive (RLD) circuit enables improved system-level common-mode rejection of signals coupled to the user from the environment, primarily 50Hz or 60Hz power-line interference. When RLD is enabled through RLD_EN[7](0x92), the circuit senses the AC common-mode input signal from the input electrodes, inverts and amplifies the signal, and drives it onto the body through a third electrode. This has the effect of attenuating the common-mode signal at the inputs and driving them toward a selectable reference voltage, typically $V_{MID_ECG}$. This also ensures proper common-mode biasing of the electrodes, allowing the internal lead bias resistors to be disabled. Alternatively, the RLD circuit acts as a DC body-bias buffer when RLD_MODE[6](0x92) is set to 0. When RLD_EN = 0, the output of the RLD amplifier and the RLD pin are high-impedance.

The RLD circuit, shown in Figure 35, has three main parts: a common-mode averager, an amplifier, and an out-of-range detector. The common-mode detector selects a combination of the input voltage, with ACTV_CM_P[3](0x92) and ACTV_CM_N[2](0x92) enabling the positive and negative inputs to the detector, and RLD_SEL_ECG[6](0x93) and RLD_SEL_BIOZ[7](0xA7) selecting ECGP and ECGN, CAPP and CAPN, or BIP and BIN. The voltages at CAPP and CAPN are low-pass filtered, so these are less effective at attenuating higher frequency common-mode signals such as power-line interference. The use of BIP and BIN voltages might be considered in case one is simultaneously using the biopotential channel for ECG and the bioimpedance channel for BIA/BIS work.

The amplifier applies inverting gain to the common-mode signal and applies negative feedback with a selectable reference and bandwidth. The default reference setting is $V_{MID_ECG}$, but this can be adjusted up or down to compensate for electrode offset voltages by setting BODY_BIAS_DAC[3:0](0x93). The amplifier can use internal or external feedback with RLD_EXT_RES[7](0x93) connecting or disconnecting the internal feedback network. The internal feedback network sets the gain to 12V/V, 24V/V, 48V/V, or 97V/V according to RLD_GAIN[1:0](0x92). The external feedback network cannot be disconnected internally, so RLD_INV should be left disconnected if using internal feedback only. If using external feedback, the gain is equal to (100kΩ + $R_{RLDFB}$) / 150kΩ when averaging both inputs. The higher gain settings provide the best results for electrodes with high contact impedance, including dry electrodes.

The amplifier bandwidth can be adjusted with RLD_BW[5:4](0x93), but the lowest bandwidth setting is adequate for

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power-line frequencies. Higher bandwidth settings consume more power. If enabled by EN_RLD_OOR[4](0x92), the out-of-range detector detects if the RLD output voltage goes outside of 0.127 x $V_{AVDD}$ and 0.870 x $V_{AVDD}$ for 128ms, and asserts the RLD_OOR[3](0x04) status bit. If BYP_DLY[7](0x12) is enabled, the 128ms delay is bypassed and RLD_OOR is asserted immediately. This function indicates that the total impedance between the RLD electrode and the input electrodes is too high to maintain the feedback loop, usually because electrodes are off or have poor contact on the person.

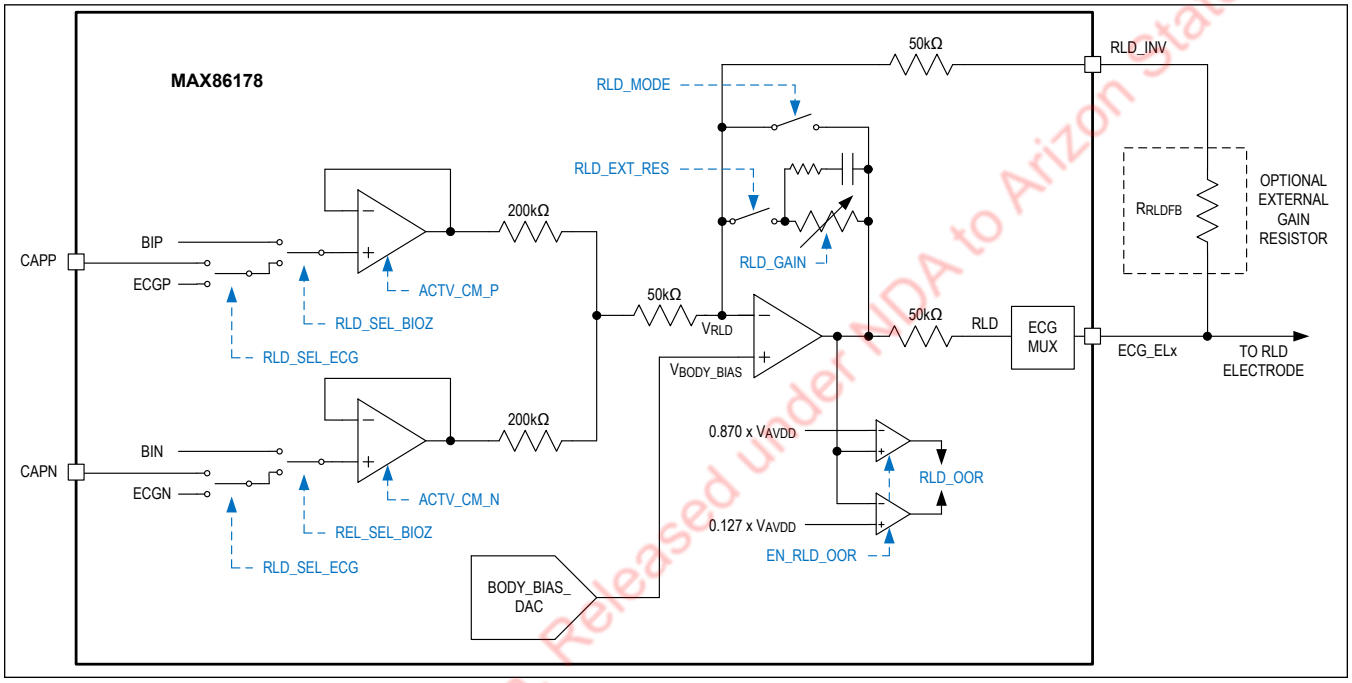



*Figure 35. Detailed Right Leg Drive (RLD) Block Diagram*

### ECG Lead Bias

The MAX86178 limits the ECGP and ECGN DC input common-mode range between 0.25V and ($V_{AVDD}$ - 0.5V). This range can be maintained either through external or internal lead biasing.

Internal DC lead biasing consists of 50MΩ, 100MΩ, or 200MΩ selectable resistors connected from ECGP and ECGN to either $V_{MID_ECG}$ or $V_{RLD}$ that bias the MAX86178 to the proper potential relative to the body in battery-powered systems. By matching the voltage of $V_{MID_ECG}$ to the body, the lead bias ensures that the common-mode input voltage of ECGP and ECGN are within the DC input range of the ECG channel. Lead bias is only effective when the MAX86178 system has high galvanic isolation from earth ground. See ECG_RBIAS_VALUE[3:2](0x90) to select a resistance value, and EN_ECG_RBIASP[1](0x90) and EN_ECG_RBIASN[1](0x90) to enable lead bias. The lead bias voltage is selected by RLD_RBIAS[5](0x92). Selecting $V_{RLD}$ (the output of the right leg drive common-mode averager) as the lead bias voltage can improve the ECG channel CMRR without needing a third electrode. If the ECG and BioZ receive channels share electrodes, only one channel should have the lead bias enabled.

## BioZ

The BioZ system shown in Figure 36 primarily consists of a transmit (TX) channel, a receive (RX) channel, and an input/output MUX. The BioZ system supports calibration using internal or external calibration resistors, enabling 0.1% accuracy in the I/Q channel. The flexible I/O MUX, lead-on and lead-off detection, adjustable amplifier bias, and lead bias allow for flexible hardware designs capable of multiple measurement types with low power consumption. The stimulus frequency and sample rate are determined by the timing subsystem (see the *Timing Subsystem* section).

The BioZ channel can measure impedances across multiple application areas with a wide range of frequencies and magnitudes as shown in Figure 37. Each type of stimulus can operate over frequencies between 16Hz and 500kHz, as

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well as DC current-mode measurements.

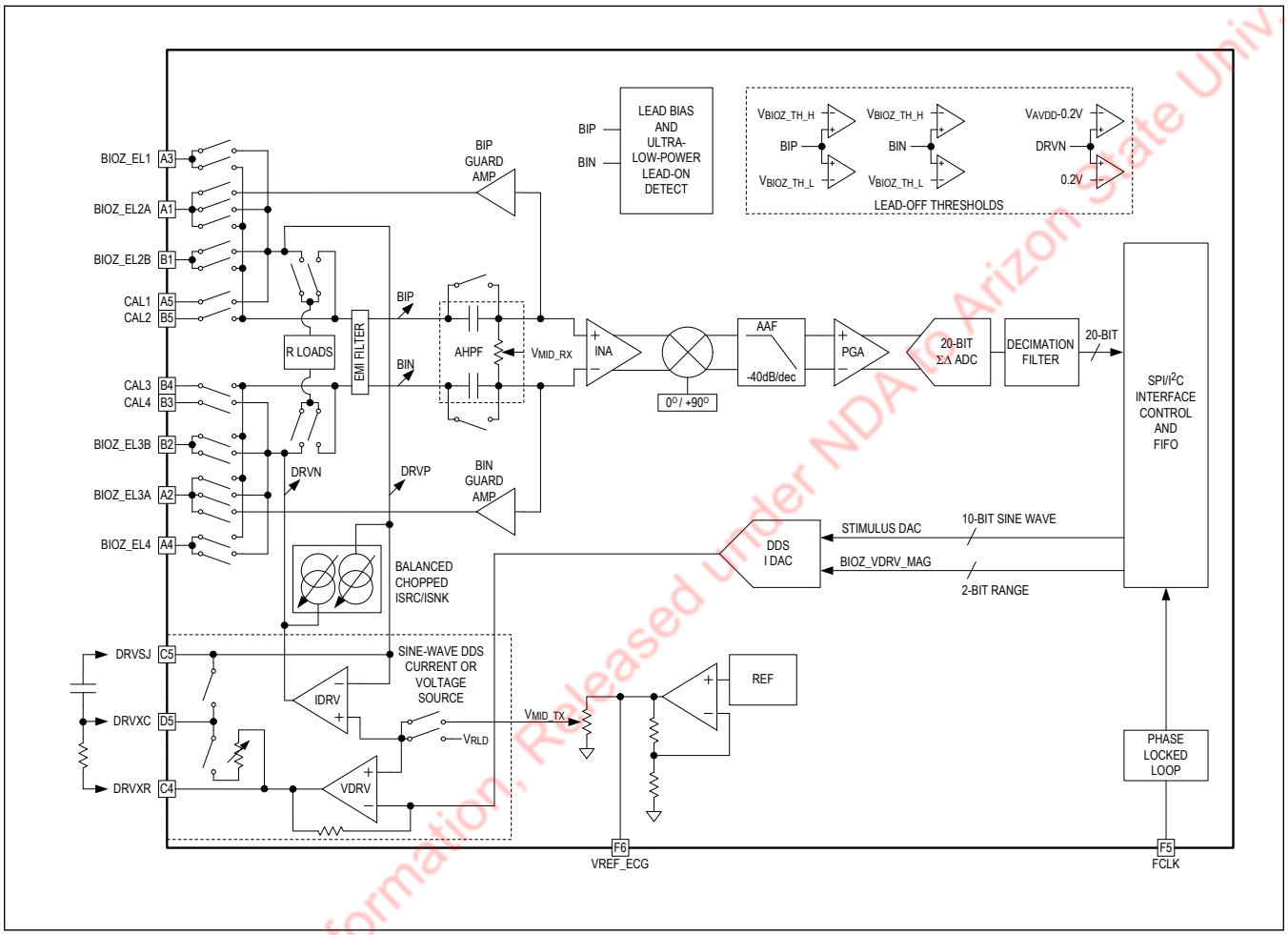



*Figure 36. BioZ System Block Diagram*

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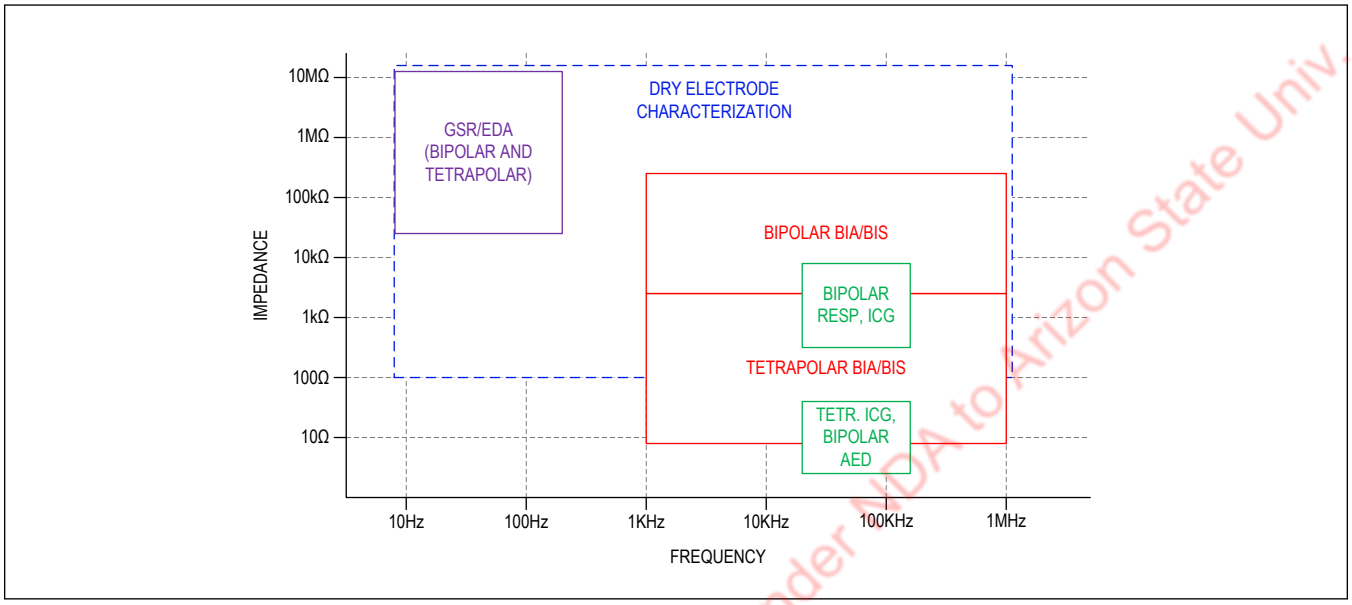



*Figure 37. BioZ Typical Application Areas*

### BioZ Startup and Shutdown

To power on the BioZ subsystem, set ECG_BIOZ_BG_EN[2](0xA0) to 1, which enables the bandgap reference shared by the ECG and BioZ channels, $V_{REF_ECG}$. After a startup time of approximately 512ms, BioZ measurements can be initiated by selecting in-phase (I) or quadrature (Q) in BIOZ_EN[1:0](0xA0). BioZ data begins within 2ms of enabling the measurement.

When the measurement is finished, the BioZ subsystem can be shut down by writing ECG_BIOZ_BG_EN and BIOZ_EN to 0x0. Alternatively, if another measurement is set to follow shortly after, setting BIOZ_DRV_MODE[1:0](0xA2) to 0x3 can place the BioZ TX channel into standby mode, while ECG_BIOZ_BG_EN and BIOZ_EN remain set to their active values. Standby mode disables the stimulus but maintains the DC bias of the drive electrodes, preventing any settling time associated with charging the electrode and body capacitance during the next measurement. In standby mode, the receive channel stops sampling and enters low-power mode unless BIOZ_STBYON[4](0xA7) is set. Setting BIOZ_STBYON = 1 can further reduce settling time during the next measurement by maintaining the receive channel in an active state.

Note that if the BioZ channel is disabled while a sample is being pushed to the FIFO, the FIFO will contain an extra sample containing the value 0x4000, which should be ignored. This can be avoided by disabling the BioZ channel immediately after a FIFO_DATA_RDY or A_FULL interrupt.

### BioZ Transmit Channel

The MAX86178 can generate four types of stimuli: a balanced square-wave source/sink current, a sine-wave current, a sine-wave voltage, and an H-bridge voltage square wave. Each of these modes are described in the following sections. The stimulus generator can be put into a low-power standby mode by setting BIOZ_DRV_MODE[1:0](0xA2) to 0x3. In this mode, the DDS circuit remains active, but the BIOZ_DRV_RESET switch is closed, driving DRVN to $V_{MID_TX}$. The amplifiers are put into a low-power state to reduce power consumption, and the H-bridge driver is disabled.

### Balanced Square-Wave Current Source/Sink Stimulus

The primary stimulus mode recommended for respiration measurements and ICG measurements is the balanced square-wave source/sink current generator, shown in Figure 38. To select this mode, set BIOZ_EN[1:0](0xA0) to 0x1 or 0x2 and set RESP_EN[0](0xB6) to 1.

Current amplitudes between $8\mu A_{PK}$ to $384\mu A_{PK}$ are selectable using this mode of operation. The magnitude of the

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CG stimulus is controlled by RESP_CG_MAG[6:4](0xB1) and RESP_CG_MAG_4X[7](0xB1). The range of frequencies intended for use with this mode is 10kHz to 131kHz. For patient safety according to IEC 60601-1, do not assert the RESP_CG_MAG_4X bit unless the stimulus frequency is greater than 10kHz. The current amplitude should also be chosen to not exceed 1000m$V_{P-P}$ at the BIP and BIN inputs based on the network impedance at the current injection frequency. 47nF DC blocking capacitors are required between the patient drive electrodes and the pins assigned to DRVP and DRVN, typically BIOZ_EL1 and BIOZ_EL4.

The current generators have several options for common-mode feedback, which corrects for current source/sink mismatch and maintains the common mode bias of the current generators. The type of common-mode feedback is selected by CG_MODE[2:1](0xB6), and the trade-offs of each mode are shown in Table 19. Dynamic matching (or chopping) is used to reduce current source/sink mismatch, and greatly reduces 1/f flicker noise. Low-pass common-mode filtering reduces the dynamic matching bandwidth to maintain higher common-mode source impedance, which is important when using a two electrode configuration or sharing electrodes with the ECG channel. The bandwidth of the LPF is controlled by CG_LPF_DUTY[7:5](0xB6).

## Table 19. Current Generator Common Mode Feedback Options

| CG_MODE | Common Mode Feedback (CMFB) Operation |
|---|---|
| 0x0 | Dynamic matching disabled with analog low-pass filter. Higher 1/f noise with higher common-mode impedance at 50Hz/60Hz. |
| 0x1 | Dynamic matching enabled without analog low-pass filter. Lower 1/f noise with lower common-mode impedance at 50Hz/60Hz. |
| 0x2 | Dynamic matching enabled with analog low-pass filter. Lower 1/f noise with higher common-mode impedance. This is the recommended mode, but stability is not guaranteed across all circumstances due to the limited common mode feedback bandwidth. |
| 0x3 | Dynamic matching enabled with internal resistive common-mode load. Lower 1/f noise with 5MΩ resistors connecting each current source/sink output to $V_{MID_TX}$. This mode has high common-mode impedance at 50Hz and 60Hz, but has lower input impedance than can be achieved with the CMFB amplifier. |

The user should consider the CG_MODE selection as a toolbox for optimizing the BioZ measurement results. Further optimization of the CG_MODE = 0x0 and CG_MODE = 0x2 options, which use the low-pass filter within the CMFB circuit, can be achieved by adjusting CG_LPF_DUTY that sets the CMFB circuit bandwidth. The CMFB bandwidth can be set to various values of CG_LPF_DUTY, which varies the number of FCLK cycles in the setting of the duty cycle feedback signal, acting as a low-pass filter. A CG_LPF_DUTY of 1 FCLK cycle yields a 0.98Hz bandwidth. Each doubling of the the CG_LPF_DUTY doubles the bandwidth. See the Register Map for details. The bandwidth should be chosen according to the application; for instance, a respiration application might use the 7.79Hz or 15.54Hz setting, and an ICG application might use the 30.89Hz or 61.08Hz setting.

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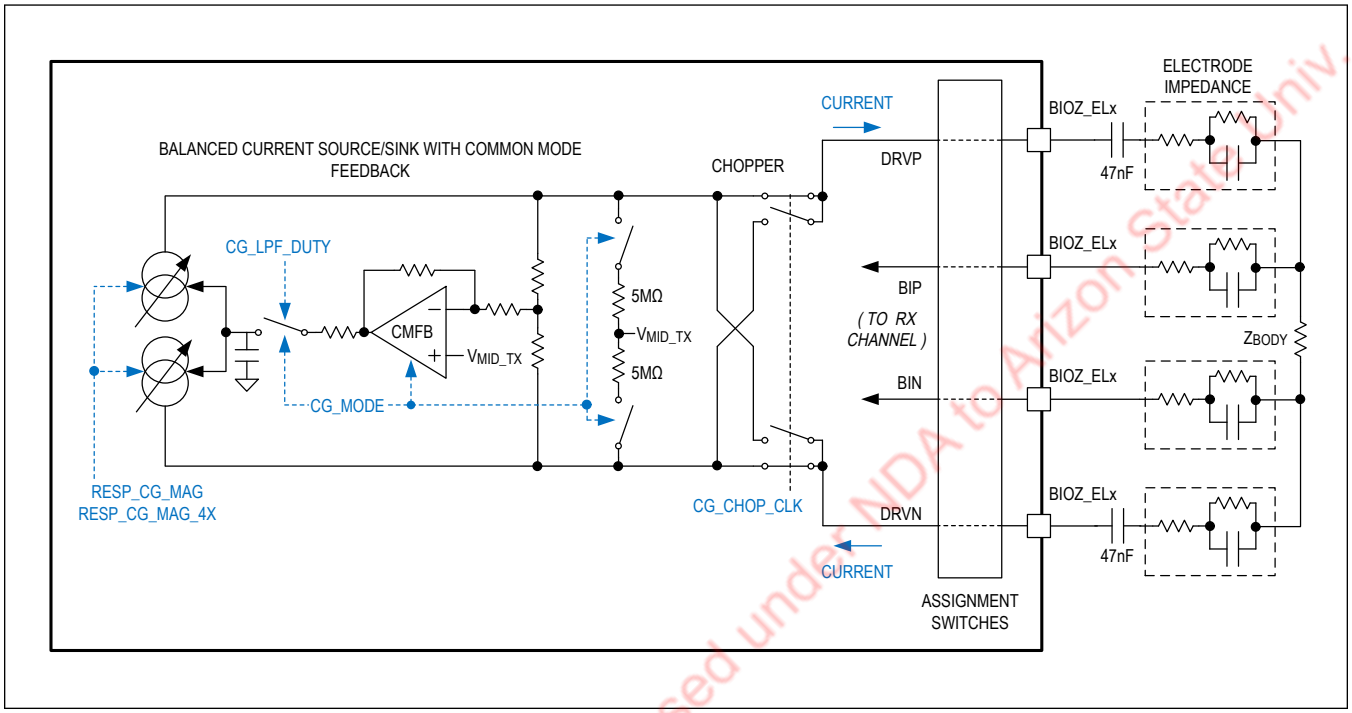



*Figure 38. Bioimpedance Stimulus Generator—Balanced Square Wave Current Mode*

### Sine-Wave Current Stimulus

To select the sine-wave current mode, set BIOZ_DRV_MODE[1:0](0xA2) to 0x0 and RESP_EN[0](0xB6) to 0. When generating a sine-wave current stimulus, the AC current is injected into the body using electrodes assigned to the DRVP and DRVN (drive) functions with the bioimpedance sensed differentially through the electrodes assigned to the BIP and BIN (bioimpedance receive) functions. Two and four electrode configurations are supported for typical wet or dry electrode impedances. Figure 39 shows the stimulus signal path with a four electrode configuration. A sine-wave current stimulus is generated by a direct digital synthesis (DDS) circuit with the help of a 10-bit current DAC (I DAC). The VDRV amplifier converts this sine-wave current into a sine-wave voltage. One of four range resistors should be selected using BIOZ_IDRV_RGE[3:2](0xA2). The current range can alternatively be set with an external resistor by enabling BIOZ_EXT_RES[7](0xA2). The sine-wave voltage appears on one side of this resistor and the other side is held at $V_{MID_TX}$ by the operation of the IDRV amplifier; thus, creating the sine-wave current stimulus in the IDRV amplifier feedback loop. This current flows through the range resistor, the electrodes, and the body impedance, then back into the IDRV amplifier output terminal.

A blocking capacitor ($C_{EXT}$) connected between the DRVXC and DRVSJ pins is required to avoid the DC current from being driven through the body. A 47nF capacitor is recommended for all applications.

Both amplifiers in the signal chain have adjustable range and bandwidth to optimize power consumption for the required performance. BIOZ_AMP_RGE[3:2](0xA6) sets the amplifier range, and BIOZ_AMP_BW[1:0](0xA6) sets the gain-bandwidth product. When using the MAX86178 for impedance cardiography (ICG), it is recommended to set the BIOZ_AMP_RGE and BIOZ_AMP_BW to higher values. It is generally acceptable to leave these settings at the lowest value for other applications.

When selecting a stimulus current magnitude, there are several restrictions that must be followed. The stimulus current is set by a combination of BIOZ_IDRV_RGE[3:2](0xA2) and BIOZ_VDRV_MAG[5:4](0xA2), and Table 20 shows the stimulus current options available for MAX86178.

1. To ensure patient safety, some current amplitude and frequency combinations are not allowed (see Table 20). If an off-limits setting is selected, the BIOZ_VDRV_MAG and BIOZ_IDRV_RGE fields are automatically overwritten

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to the highest allowed value based on the frequency settings. It is the responsibility of the end application device manufacturer to ensure that the MAX86178 is programmed properly and in conformance with *IEC60601-1 Medical electrical equipment—Part 1: General requirements for basic safety and essential performance* with regards to patient auxiliary current limitations.

2. When using stimulus currents greater than 640μ$A_{RMS}$, BIOZ_EL1 and BIOZ_EL4 must be used for DRVP and DRVN, respectively. Electrode pins BIOZ_EL2A, BIOZ_EL2B, BIOZ_EL3A, and BIOZ_EL3B are not designed to support currents above 640μ$A_{RMS}$. Assigning the wrong pins does not damage the MAX86178, but switch resistance is higher and degrades measurement accuracy.
3. The current amplitude should be chosen to not exceed 1000m$V_{P-P}$ at the BIP and BIN pins based on the network impedance at the current injection frequency.

## Table 20. Stimulus Current Options

| STEP | BIOZ_IDRV_RGE | RANGE RESISTOR | BIOZ_VDRV_MAG | RMS CURRENT | FREQUENCY RANGE (Hz) | RECOMMENDED BIOZ_AMP_RGE |
|---|---|---|---|---|---|---|
| 1 | 1 (0x0) | 552.5kΩ | low (0x0) | 16nA | All frequencies | Low |
| 2 | 1 (0x0) | 552.5kΩ | low mid (0x1) | 32nA | All frequencies | Low |
| 3 | 1 (0x0) | 552.5kΩ | high mid (0x2) | 80nA | All frequencies | Low |
| 4 | 1 (0x0) | 552.5kΩ | high (0x3) | 160nA | All frequencies | Medium-Low |
| 5 | 2 (0x1) | 110.5kΩ | low (0x0) | 320nA | All frequencies | Medium-Low |
| 6 | 2 (0x1) | 110.5kΩ | low mid (0x1) | 640nA | All frequencies | Medium-Low |
| 7 | 2 (0x1) | 110.5kΩ | high mid (0x2) | 1.6μA | All frequencies | Medium-Low |
| 8 | 2 (0x1) | 110.5kΩ | high (0x3) | 3.2μA | All frequencies | Medium-Low |
| 9 | 3 (0x2) | 5.525kΩ | low (0x0) | 6.4μA | All frequencies | Medium-Low |
| 10 | 3 (0x2) | 5.525kΩ | low mid (0x1) | 12.8μA | All frequencies | Medium-High |
| 11 | 3 (0x2) | 5.525kΩ | high mid (0x2) | 32μA | All frequencies | Medium-High |
| 12 | 3 (0x2) | 5.525kΩ | high (0x3) | 64μA | All frequencies | Medium-High |
| 13 | 4 (0x3) | 276.25Ω | low (0x0) | 128μA | ≥ 512 | High |
| 14 | 4 (0x3) | 276.25Ω | low mid (0x1) | 256μA | ≥ 2048 | High |
| 15 | 4 (0x3) | 276.25Ω | high mid (0x2) | 640μA | ≥ 8192 | High |
| 16 | 4 (0x3) | 276.25Ω | high (0x3) | 1.28mA | ≥ 16384 | High |

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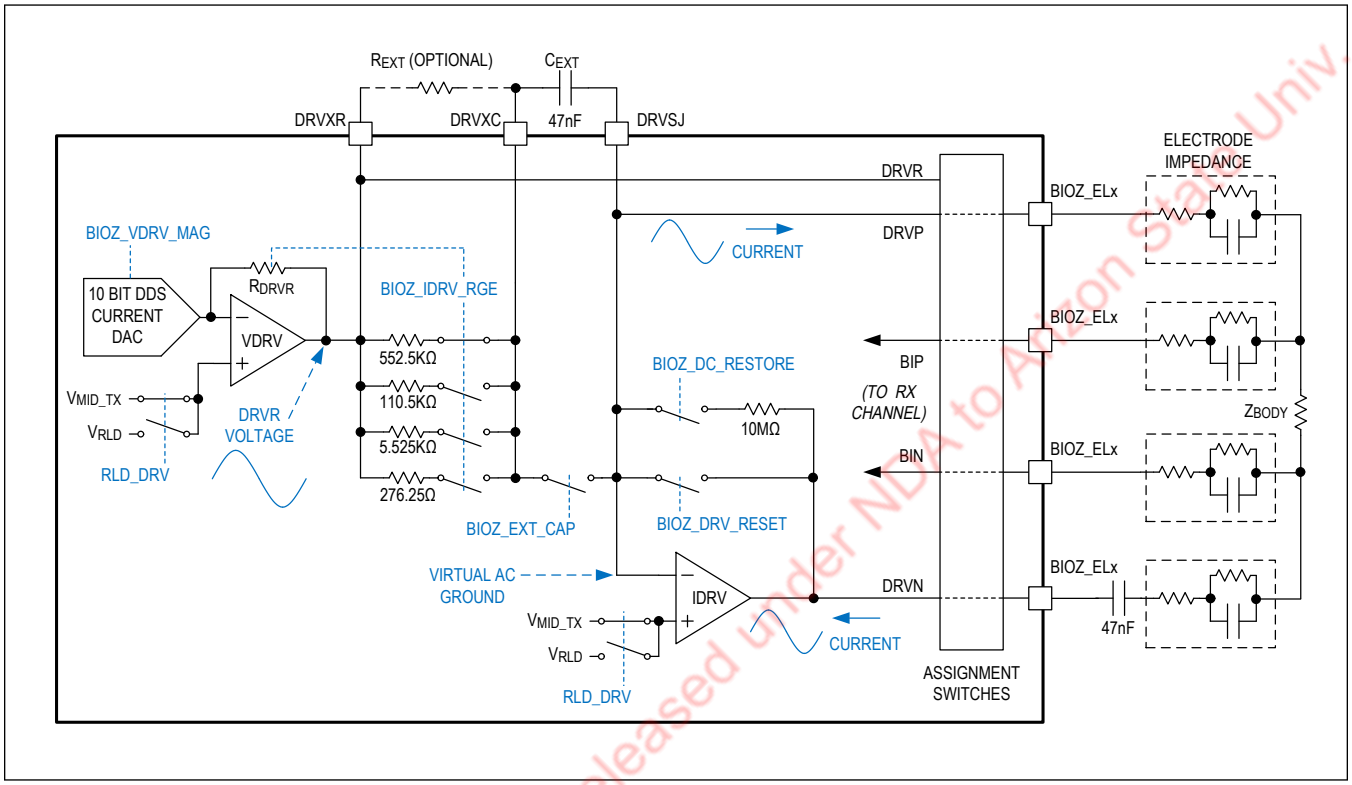



*Figure 39. Bioimpedance Stimulus Generator Diagram—Sine-Wave Current Mode*

### Sine-Wave Current Stimulus Compliance Voltage

The sine-wave current generates a voltage between the ELx pins connected to DRVP and DRVN.

This voltage equals:

$V_{PEAK} = \sqrt{2} * I_{RMS} * (Z_{BODY} + ZE_{DRVP} + ZE_{DRVN})$

where: $ZE_{DRVP}$ and $ZE_{DRVN}$ = Electrode impedance at DRVP and DRVN

Too high a voltage can cause the current driver to exceed its Compliance Voltage limits or the Compliance Monitor Threshold limits, which are shown in the EC table. If ($V_{MID_TX}$ - $V_{PEAK}$) is less than the minimum Compliance Voltage or if ($V_{MID_TX}$ + $V_{PEAK}$) is greater than the maximum Compliance Voltage, then the BioZ drive current waveform is distorted and gives incorrect results. If ($V_{MID_TX}$ - $V_{PEAK}$) or ($V_{MID_TX}$ + $V_{PEAK}$) exceed the Compliance Monitor Thresholds and EN_DRV_OOR = 1, then the DRV_OOR status bit is asserted. To resolve this issue, either use a lower drive current or reduce the electrode impedance.

### Sine-Wave Voltage Stimulus

To select sine-wave voltage mode, set BIOZ_DRV_MODE[1:0](0xA2) to 0x1 and RESP_EN[0](0xB6) to 0. Figure 40 shows the stimulus signal path for a four electrode configuration. The voltage output from the VDRV amplifier, available on the DRVXR node, is applied directly to the BIOZ_EL1 electrode by means of a switch within the MAX86178 I/O MUX. In this mode of operation, a switch is closed around the IDRV amplifier so that the amplifier becomes a voltage follower and drives the BIOZ_EL4 node to $V_{MID_TX}$. In this mode, external resistors on the BIOZ_EL1 and BIOZ_EL4 nodes are used to limit the patient current to $V_{RMS}$ / (2 x $R_{SERIES}$) where $V_{RMS}$ is the voltage output amplitude out of the VDRV amplifier set by BIOZ_VDRV_MAG[5:4](0xA2). When using this voltage stimulus mode, BIOZ_EL1 must be selected for DRVP, and BIOZ_EL4 must be selected for DRVN. Electrode pins BIOZ_EL2A, BIOZ_EL2B, BIOZ_EL3A, and BIOZ_EL3B are not available for voltage mode stimulation.

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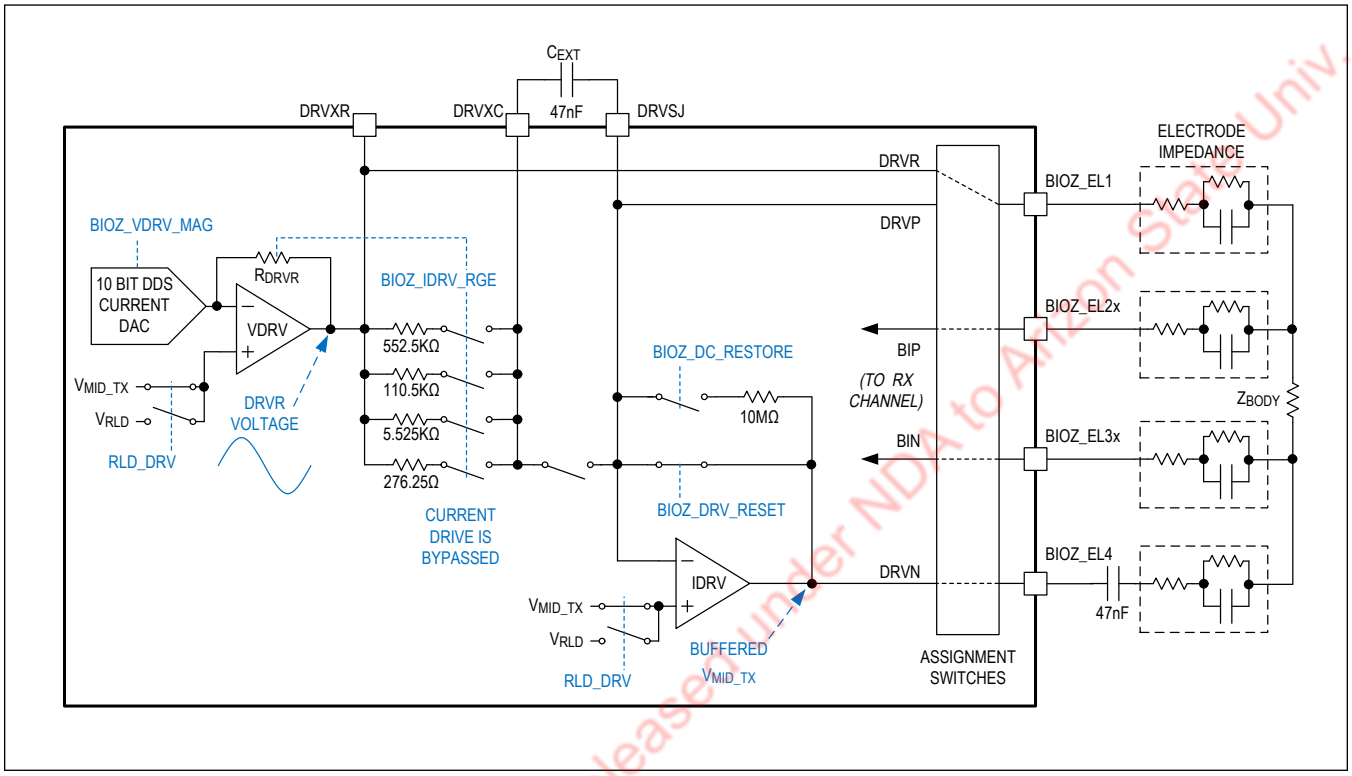



*Figure 40. Bioimpedance Stimulus Generator Diagram—Sine-Wave Voltage Mode*

### Square-Wave Voltage (H-Bridge) Stimulus

To select H-bridge square-wave voltage stimulus mode, set BIOZ_DRV_MODE[1:0](0xA2) to 0x2 and RESP_EN to 0. Figure 41 shows the signal path for H-bridge mode. An H-bridge is used to alternately switch AVDD and then AGND onto BIOZ_EL1, and AGND and then AVDD onto BIOZ_EL4. In this case, the DDS circuit and IDRV amplifier are disabled and the range resistor switches are all opened. When using this mode of operation, there must be series precision resistors in both the BIOZ_EL1 and BIOZ_EL4 paths to limit the current to $V_{AVDD}$ / (2 x $R_{SERIES}$). The current from the H-bridge flows through $R_{SERIES_EL1}$, a 1st electrode impedance, body impedance with its variable component, a 2nd electrode impedance, and $R_{SERIES_EL4}$. The applied patient current is set by selecting the appropriate $R_{SERIES}$ resistance value. The two $R_{SERIES}$ resistors and the body form an impedance divider, and the portion of the AC voltage signal across the body impedance is sensed by the bioimpedance AFE receive channel with the inputs selected from BIOZ_EL2A, BIOZ_EL2B, BIOZ_EL3A, and BIOZ_EL4B. DRVP must be assigned to BIOZ_EL1 and DRVN must be assigned to BIOZ_EL4.

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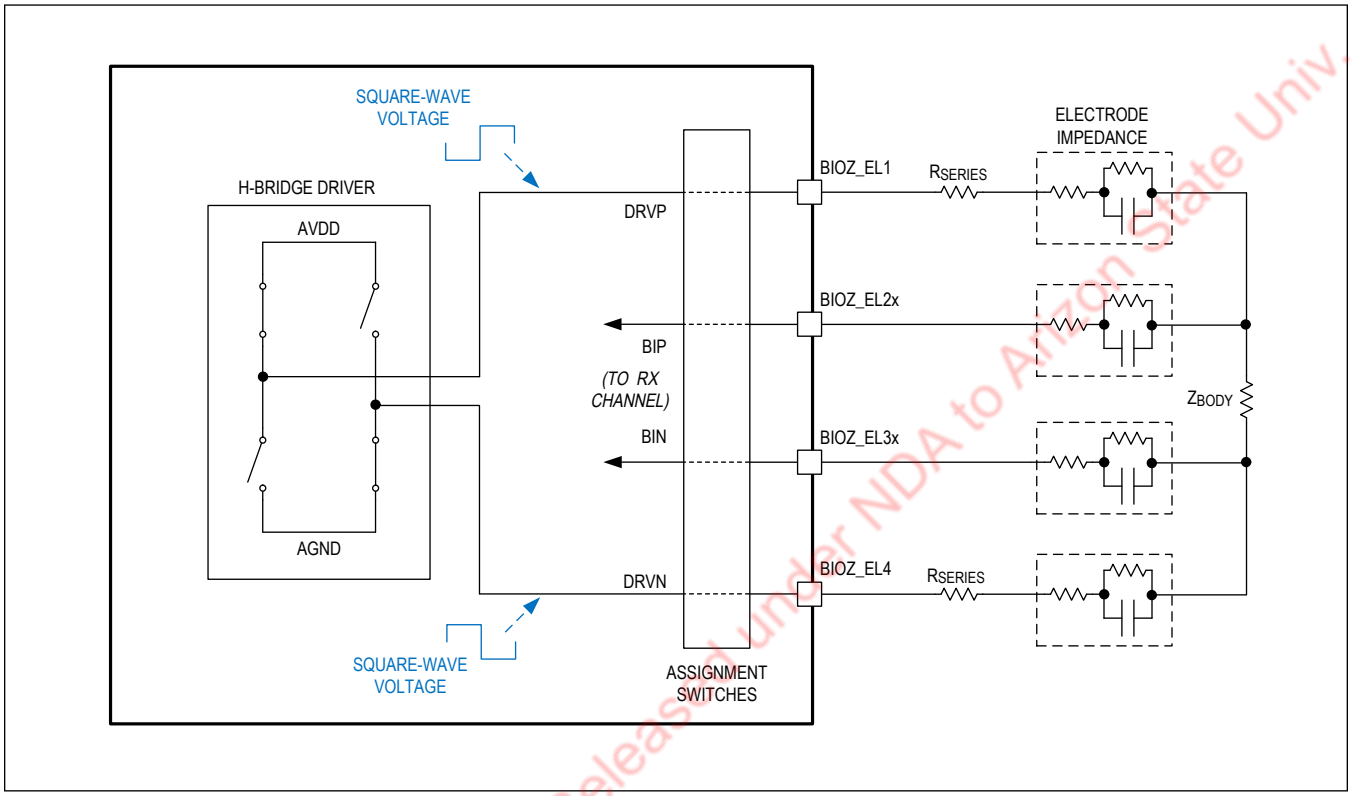



*Figure 41. Bioimpedance Stimulus Generator Diagram—H-Bridge Square-Wave Voltage Mode*

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### BioZ Receive Channel

Figure 42 illustrates the BioZ receive channel block diagram. The channel is comprised of an input MUX, a bypassable and programmable analog high-pass filter, an instrumentation amplifier with programmable gain, demodulator, anti-alias filter, another programmable gain amplifier, and an analog-to-digital converter (ADC). The input MUX includes several features such as EMI filtering, programmable electrode assignment switches, lead biasing, DC lead-off detection, and ultra-low power lead-on detection.

The MAX86178 BioZ receive channel instrumentation amplifier (INA) provides low-noise amplification of the differential signal, rejects differential DC voltage due to the analog high-pass filter, rejects common-mode interference such as AC mains interference, and provides high input impedance to guarantee high CMRR even in the presence of severe electrode impedance mismatch. The total channel gain can be set to 1V/V, 2V/V, 5V/V, or 10V/V, and is set by BIOZ_GAIN[1:0](0xA5), which affects both the INA gain and the PGA gain. The demodulator multiplies the received signal by a square wave with the same frequency $f_{STIM}$ to down-convert the measurement frequency to DC. The phase of the demodulator $f_{DEMOD}$ signal is chosen by BIOZ_EN[1:0](0xA0). Following the PGA amplifier is a 2-pole, active low-pass anti-aliasing filter (AAF) with a 600Hz, -3dB frequency that provides approximately 57dB of attenuation at half the sigma-delta ADC input sampling rate (BIOZ_ADC_CLK). After the AAF is a 20-bit sigma-delta ADC. The effective bits of the ADC depend on the value of BIOZ_ADC_OSR[5:3](0xA0) with higher oversampling ratios resulting in more effective bits (see the *Electrical Characteristics* sections).

When AC coupling the BioZ receive channel or using the internal analog high-pass filter (HPF), the AC differential range is > $1000mV_{P-P}$ with an INA gain of 1V/V. When DC-coupling the bioimpedance receive channel, the usable common-mode range of the bioimpedance receive channel is 0.5V to $V_{AVDD}$ - 0.75V. Internal lead biasing is used to achieve these requirements (see the *BioZ Lead Bias* section).

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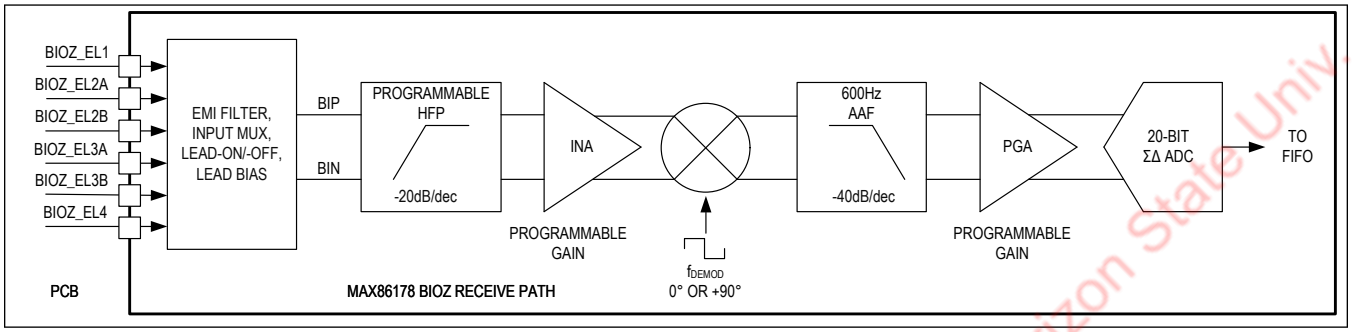



*Figure 42. BioZ Receive Channel*

### BioZ Decimation and Digital Filters

The decimation filter is used along with the sigma-delta modulator within the ADC to reduce the sample rate of the BIOZ_ADC_CLK to a smaller programmable output rate (SR_BIOZ). The decimation filter has a SINC3 response with a corner frequency at approximately 0.26 x SR_BIOZ. The decimation filter is followed by a programmable digital filter to implement HPF and LPF selections.

The programmable digital high-pass filter scales with the ADC clock rate and can be set to either 0.00025 x SR_BIOZ, 0.002 x SR_BIOZ, or bypassed by setting BIOZ_DHPF[7:6](0xA1). Similarly, the programmable digital low-pass filter scales with the ADC clock rate and can be set to 0.005 x SR_BIOZ, 0.02 x SR_BIOZ, 0.08 x SR_BIOZ, 0.25 x SR_BIOZ, or bypassed using BIOZ_LPF[5:3](0xA1).

### Converting Digitized BioZ Samples to Voltage and Impedance

BioZ channel samples are recorded in 20-bit left-justified 2's complement format. These samples represent the voltage at the ADC, which has passed through the INA, demodulator, PGA, and AAF. The INA and PGA apply a combined gain of 1V/V, 2V/V, 5V/V, or 10V/V as set by BIOZ_GAIN[1:0](0xA5). The demodulator multiplies the incoming sine wave or square wave by a square wave with the same frequency as $f_{STIM}$. The AAF is a two-pole low-pass filter with a 600Hz corner frequency. The decimation filter in the ADC has a bandwidth of approximately 0.26 x SR_BIOZ.

When performing absolute impedance measurements for applications such as BIA/BIS and GSR/EDA, the DC component of the demodulated voltage represents the measured impedance. When $f_{STIM}$ >> 600Hz or when SR_BIOZ << $f_{STIM}$, the harmonics resulting from the square-wave demodulation can be ignored, and the digitized samples represent the DC component of the demodulated voltage. For sine-wave stimulation, the square-wave demodulation applies a scaling factor of 2 / π to the DC component, as shown in Figure 43.

The DC component of the demodulated voltage is converted by the ADC, and represents load impedance in current-stimulus mode according to the following equations.

Sine-Wave Stimulus: $Z_{BIOZ}(\Omega) = ADC_COUNT \times V_{REF_ECG} / (2^{19} \times BIOZ_GAIN \times 2 / \pi \times I_{MAG})$

Square-Wave Stimulus: $Z_{BIOZ}(\Omega) = ADC_COUNT \times V_{REF_ECG} / (2^{19} \times BIOZ_GAIN \times I_{MAG})$

where,

ADC_COUNT = ADC counts in signed magnitude format

$V_{REF_ECG}$ = 1V (typ, see the *Electrical Characteristics* section)

BIOZ_GAIN = Options1V/V, 2V/V, 5V/V, and 10V/V.

$I_{MAG}$ = Stimulus current in $A_{PK}$ set by BIOZ_VDRV_MAG[5:4](0xA2) and BIOZ_IDRV_RGE[3:2](0xA2), or RESP_CG_MAG[6:4](0xB1) and RESP_CG_MAG_4X[7](0xB1)

The input-referred voltage amplitude can likewise be calculated with the following equations.

Sine-Wave Stimulus: $V_{BIOZ}(V_{PK}) = ADC_COUNT \times V_{REF_ECG} / (2^{19} \times BIOZ_GAIN \times 2 / \pi)$

Square-Wave Stimulus: $V_{BIOZ}(V_{PK}) = ADC_COUNT \times V_{REF_ECG} / (2^{19} \times BIOZ_GAIN)$

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For voltage stimulus modes, the impedance can be calculated from an impedance divider with the series resistors.

For respiration and ICG applications, the signal of interest is contained in the time-varying impedance signal, so the DC component is not as important. The respiration signal band is typically 0.05Hz to 4Hz, and the ICG signal band is typically DC to 64Hz. For these applications, the BioZ sample rate and digital filters can be adjusted to select the signal band of interest, considering the decimation filter 0.26 x SR_BIOZ bandwidth. The above impedance or voltage calculations can be performed for these applications, but these conversions are not strictly necessary.

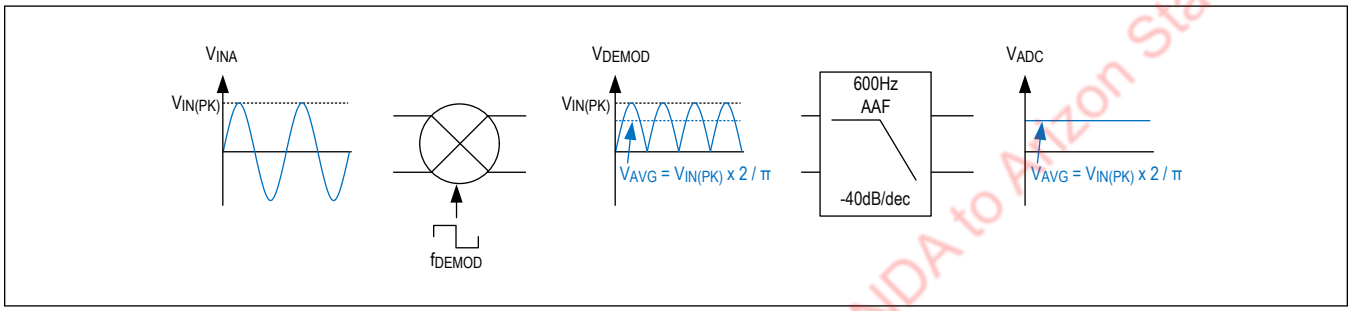



*Figure 43. Square-Wave Demodulation for a Sine-Wave Stimulus (INA and PGA Gain Not Shown)*

### BioZ Noise Measurements

Table 21 shows the input referred voltage noise of the BioZ receive channel measured with the inputs shorted and the frequency settings shown in Table 22 and BIOZ_AHPF[7:4](0xA5) = 5kHz.

## Table 21. BioZ Receive Channel Input-Referred Noise, 256sps

| BIOZ_GAIN (V/V) | BIOZ_DLPF (Hz) | NOISE ($\mu V_{RMS}$) | NOISE ($\mu V_{P-P}$) | SNR (dB) | ENOB |
|---|---|---|---|---|---|
| 1 | bypass | 10.05 | 51.50 | 90.9 | 14.8 |
| | 1.28 | 2.44 | 7.63 | 103.2 | 16.9 |
| | 5.12 | 4.17 | 17.17 | 98.6 | 16.1 |
| | 20.48 | 5.75 | 28.61 | 95.8 | 15.6 |
| | 64 | 9.36 | 47.68 | 91.5 | 14.9 |
| 2 | bypass | 4.41 | 25.75 | 98.1 | 16.0 |
| | 1.28 | 1.88 | 4.77 | 105.5 | 17.2 |
| | 5.12 | 1.98 | 6.68 | 105.0 | 17.2 |
| | 20.48 | 3.08 | 14.31 | 101.2 | 16.5 |
| | 64 | 4.86 | 26.70 | 97.2 | 15.9 |
| 5 | bypass | 2.73 | 14.11 | 94.3 | 15.4 |
| | 1.28 | 0.97 | 2.67 | 103.3 | 16.9 |
| | 5.12 | 0.84 | 4.58 | 104.5 | 17.1 |
| | 20.48 | 1.70 | 8.39 | 98.4 | 16.0 |
| | 64 | 2.27 | 12.21 | 95.9 | 15.6 |
| 10 | bypass | 2.41 | 12.21 | 89.4 | 14.6 |
| | 1.28 | 0.76 | 2.67 | 99.4 | 16.2 |
| | 5.12 | 0.94 | 3.62 | 97.5 | 15.9 |
| | 20.48 | 1.28 | 6.29 | 94.9 | 15.5 |
| | 64 | 1.21 | 6.10 | 95.3 | 15.5 |

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Note: SNR = $20\log\left(\frac{V_{IN(RMS)}}{V_{N(RMS)}}\right)$, ENOB =( SNR -1.76)/6.02

Note: $V_{IN(P-P)}$ = ($2V_{P-P}$ / BIOZ_GAIN) or $1V_{P-P}$, whichever is smaller. This represents the maximum signal of the receive channel: $V_{IN(RMS)} = \frac{V_{IN(P-P)}}{2\sqrt{2}}$

## Table 22. Input-Referred Noise Frequency Settings

| REF_CLK (Hz) | M | PLL_CLK (Hz) | BIOZ_ KDIV | BIOZ_DAC_OSR | F_BIOZ (Hz) | BIOZ_ NDIV | ADC_OSR | INTEGRATION CYCLES | ADC_SR (sps) | BIOZ_AHPF (kHz) |
|---|---|---|---|---|---|---|---|---|---|---|
| 32768 | 256 | 8388608 | 1 | 128 | 65536 | 256 | 128 | 256 | 256 | 5 |

### BioZ Input/Output MUX

The BioZ input MUX shown in Figure 44 has many helpful circuits to support BioZ applications. For the electrodes assigned to the receive channel, this circuitry contains integrated EMI protection, DC lead-off detect current sources, lead biasing, programmable resistor loads as well as a programmable high-pass filter (HPF). The electrodes assigned to provide the stimulus have compliance monitors. The input/output MUX assigns physical electrodes to the available BioZ channel functions. For example, BIOZ_EL1 can be assigned to DRVP (positive drive), BIOZ_EL2B to BIP (positive input), BIOZ_EL3B to BIN (negative input), and BIOZ_EL4 to DRVN (negative drive). The BIOZ_EL2 and BIOZ_EL3 electrode inputs have A and B pins, allowing one board to support both GSR/EDA and BIA/BIS in the same application. In this case, the A inputs should be used for GSR/EDA and need external AC coupling capacitors, and the B inputs should be used for BIA/BIS and do not need external capacitors. The MUX also has 4-wire calibration port (CAL1 to CAL4) for in-situ calibration to one or more precision external resistors with more details shown in Figure 50.

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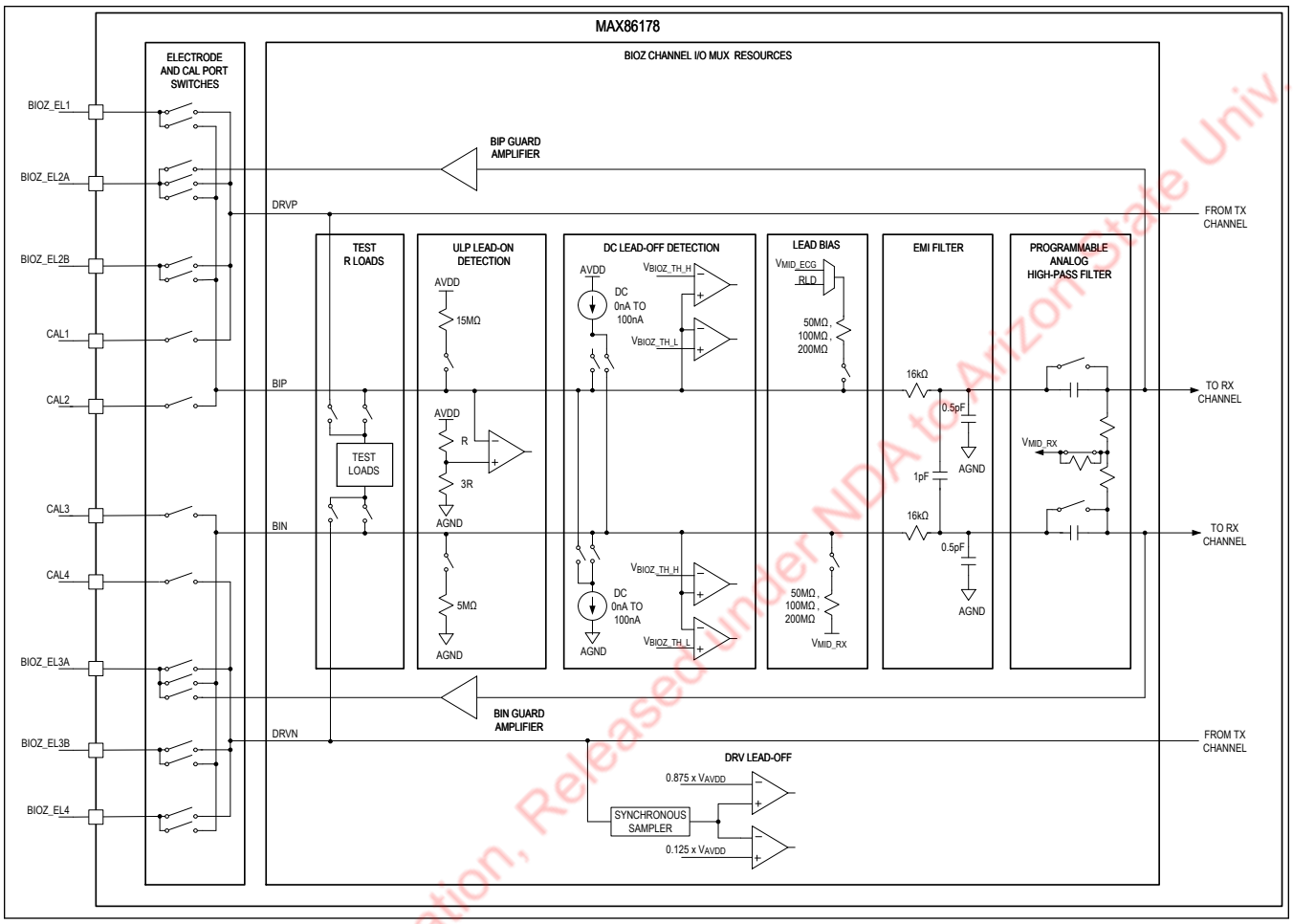



*Figure 44. BioZ Input/Output MUX*

### BioZ EMI Filtering and ESD Protection

The EMI filter on the BIP and BIN internal inputs consists of 16kΩ resistors connected to BIP and BIN followed by a 1pF differential-mode capacitor and 0.5pF common-mode capacitors. These form a single pole, low-pass, differential- and common-mode filter with the differential-mode pole located at approximately 4MHz and the common-mode pole located at approximately 20MHz. Additional external EMI filters are not recommended for applications with dry electrodes in order to maintain high input impedance, which helps mitigate the impact of electrode-impedance mismatch. With lower input impedance, the electrode impedance mismatch translates into increased conversion of common-mode voltage to differential-mode voltage. Applications with wet electrodes can use external EMI filters with high-precision components to minimize electrode impedance mismatch. In this case, the differential-mode pole can be set as low as the desired signal bandwidth, and the common-mode pole is set at least a decade below the AM radio band (535kHz).

The BIOZ_ELx and CALx pins have ESD protection compliance with 2kV HBM. For IEC ESD compliance, external ESD diodes must be added (see *Typical Applications Circuits*).

### BioZ Lead Bias

The MAX86178 limits the BIP and BIN DC input range to 0.5V to $V_{AVDD}$ - 0.75V. This range can be maintained either through external or internal lead biasing.

Internal DC lead biasing consists of 50MΩ, 100MΩ, or 200MΩ selectable resistors from BIP and BIN to either $V_{MID_RX}$ or $V_{RLD}$ that bias the MAX86178 to the proper potential relative to the body in battery-powered systems. By matching the voltage of $V_{MID_RX}$ to the body, lead bias ensures that the common-mode input voltage of the BIP and BIN

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are within the DC input range of the BioZ receive channel. Lead bias is only effective when the MAX86178 system has high galvanic isolation from earth ground. See BIOZ_RBIAS_VALUE[3:2](0xB4) to select a resistance value, and EN_BIOZ_RBIASP[1](0xB4) and EN_BIOZ_RBIASN[0](0xB4) to enable lead bias. The lead bias voltage is selected by RLD_RBIAS[5](0x92). Selecting $V_{RLD}$ (the output of the right leg drive common mode averager) as the lead bias voltage can improve the BioZ receive channel CMRR without needing a dedicated RLD electrode. If the ECG and BioZ receive channels share electrodes, only one channel should have lead bias enabled.

### Programmable BioZ Resistor Load

The programmable resistive load allows a built-in self-test of the current generator (CG) and the BioZ receive channel. Refer to Figure 45 for implementation details. For BIA/BIS applications, there is a selection of low-impedance loads from which to choose: 200Ω, 400Ω, 800Ω, and 5kΩ. For GSR/EDA applications, there is a selection of high-impedance loads from which to choose: 25kΩ, 100kΩ, 500kΩ, and 1MΩ.

See register fields BMUX_RSEL[7:6](0xAA), BMUX_BIST_EN[5](0xAA), BMUX_GSR_RSEL[7:6](0xAB) and GSR_LOAD_EN[5](0xAB) to set the resistor value.

The BIA/BIA resistive loads can also be used as internal calibration resistors. See BIST_R_ERROR[7:0](0xAD) for details.

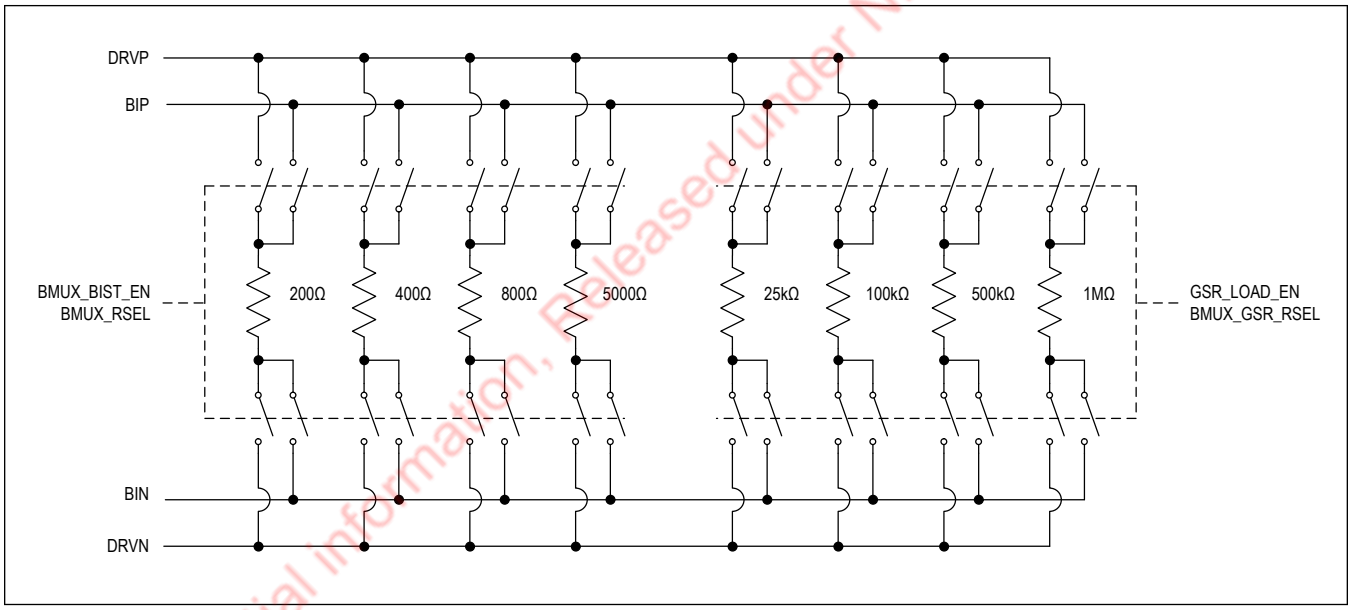



*Figure 45. Programmable Resistor Load*

### BioZ Lead-Off Detection

The MAX86178 has three techniques that can be used for determining if there is one or more electrode lead-off condition(s). Lead-off is a term used to indicate that one or more electrodes used during the bioimpedance measurements has/have become open. In other words, the electrode-tissue interface impedance is too high and the bioimpedance measurement might become unreliable. These three lead-off techniques are DC lead-off, DRV lead-off, and programmable thresholds (AC lead-off) circuit. The relevant techniques depend on the electrode configuration (bipolar or tetrapolar) and which electrode is off. The first two techniques use circuits that are contained within the I/O MUX circuitry and the final technique is accomplished using digital circuitry after the receive channel ADC.

There is also an ultra-low power (ULP) lead-on detect circuit that is only intended to be used to wake up the microcontroller that is controlling the MAX86178. This circuit should never be used while making bioimpedance measurements and should not be confused with the lead-off circuit functionality.

The DC lead-off detection circuit provides matched source and sink currents that are injected into the BIP and BIN

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electrodes. This current, when the electrodes are connected properly, flows through a first electrode-tissue interface, through the body and then through a second electrode-tissue interface. This current flow develops a differential voltage across the two input pins. If one or both electrode-tissue interfaces has a poor connection with the body, then this current path has much higher impedance and this voltage is large. If the electrodes are properly connected, then this voltage is small. The DC lead-off circuitry provides two sets of dual comparators to test if the differential voltage is too high.

This feature is enabled by EN_BIOZ_LOFF[6](0xB0), and the stimulus current magnitude (5nA to 100nA) is set by BIOZ_LOFF_IMAG[2:0](0xB0). The current magnitude should be chosen to match the expected acceptable maximum impedance of the specific electrodes used in the application.

When using DC lead-off detection, the microcontroller and user can be alerted if the viability of the electrode tissue interface electrodes has been compromised by dual comparators indicating if BIP or BIN voltages exceeds either a programmable high limit or a programmable low limit. The dual comparators can be used to generate a hardware interrupt if the DC lead-off voltage exceeds the threshold (i.e., a minimum continuous violation) for an interval exceeding either 115ms or 140ms depending on the setting of FCLK before asserting one of the BIOZ_LOFF interrupt flags. The comparator threshold is controlled by BIOZ_LOFF_THRESH[3:0](0xB1). See Figure 46 for an example of the threshold and timing behavior.

For applications without external AC-coupling capacitors, the DC lead-off detection is applied directly to the BIP and BIN electrodes as shown in Figure 47. However, when external AC-coupling capacitors are used, such as in GSR/EDA applications, DC lead-off detection must be applied externally. In this case, BIP and BIN must be assigned to BIOZ_EL2A and BIOZ_EL3A, and DC lead-off detection is applied externally through BIOZ_EL2B and BIOZ_EL3B. To enable this feature, BIOZ_EL2B and BIOZ_EL3B must be connected outside of the AC-coupling capacitors as shown in Figure 48, and EN_EXT_BIOZ_LOFF[5](0xB0) must be enabled in addition to EN_BIOZ_LOFF.

The DRV lead-off circuit checks the BioZ current stimulus path to determine whether or not the DRVP and DRVN connections are in place when using a tetrapolar electrode configuration. In this case, one or both of the DRVP and DRVN electrodes are compromised and are not reliably connected to the body. When the impedance between the DRVP and DRVN electrodes becomes too high, the magnitude of this amplifier output signal starts to approach one or both rails (AVDD or AGND), getting very close to amplifier saturation. When the DRV lead-off circuit is enabled, a sample-and-hold circuit followed by a dual-comparator samples the IDRV amplifier output signal and determines if it is outside of 0.125 x $V_{AVDD}$ and 0.875 x $V_{AVDD}$. If it is, the comparator trips and a DRV lead-off condition is flagged. To turn on the DRV lead-off circuit, set EN_BIOZ_DRV_OOR[4](0xB0) to 1. An out-of-range condition must be exceeded for either 125ms or 128ms, depending on CLK_FREQ_SEL[5](0x1D), before the BIOZ_DRV_OOR status bit is asserted.

The BioZ threshold (AC lead-off) detection circuit monitors the output of the BioZ ADC with programmable high or low thresholds, and is enabled by EN_BIOZ_THRESH[0](0xA1). The thresholds are set by BIOZ_LO_THRESH[7:0](0xA8) and BIOZ_HI_THRESH[7:](0xA9). If the digitized output remains over BIOZ_HI_THRESH or under BIOZ_LO_THRESH for longer than 128ms, then the BIOZ_OVER[6](0x04) or BIOZ_UNDR[5](0x04) are asserted. This behavior is described graphically in Figure 49.

Table 23 shows the lead-off techniques suitable for each electrode configuration and lead-off condition. The internal HPF is suitable for stimulus frequencies above 1kHz, and external capacitors are need for stimulus frequencies under 1kHz.

## Table 23. BioZ Lead-Off Cases

| CONFIGURATION | CONDITION | DRVP | DRVN | BIP | BIN | MEASURED SIGNAL | LEAD-OFF TECHNIQUES |
|---|---|---|---|---|---|---|---|
| Bipolar, internal HPF | DRVP/BIP off | DDS sine | large signal | DDS sine | large signal | large | DC Lead-Off, DRVN Lead-Off, AC Lead-Off (Over) |
| Bipolar, internal HPF | DRVP/BIN off | DDS sine | large signal | DDS sine | large signal | large | DC Lead-Off, DRVN Lead-Off, AC Lead-Off (Over) |
| Bipolar, external capacitors | DRVP/BIP off | DDS sine | large signal | DDS sine | large signal | large | DRVN Lead-Off, AC Lead-Off (Over) |
| Bipolar, external capacitors | DRVN/BIN off | DDS sine | large signal | DDS sine | large signal | large | DRVN Lead-Off, AC Lead-Off (Over) |
| Tetrapolar, internal HPF | DRVP off | DDS sine | large signal | large signal | large signal | indeterminant | DRVN Lead-Off |

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## Table 23. BioZ Lead-Off Cases (continued)

| CONFIGURATION | CONDITION | DRVP | DRVN | BIP | BIN | MEASURED SIGNAL | LEAD-OFF TECHNIQUES |
|---|---|---|---|---|---|---|---|
| Tetrapolar, internal HPF | DRVN off | DDS sine | railed | railed | railed | railed | DRVN Lead-Off, AC Lead-Off (Over) |
| Tetrapolar, internal HPF | BIP off | $V_{MID_TX}$ | normal | $V_{MID_RX}$ | normal | 1/2 of normal size | DC Lead-Off, AC Lead-Off (Under) |
| Tetrapolar, internal HPF | BIN off | $V_{MID_TX}$ | normal | $V_{MID_RX}$ | $V_{MID_RX}$ | near DC | DC Lead-Off, AC Lead-Off (Under) |
| Tetrapolar, internal HPF | BIP and BIN off | $V_{MID_TX}$ | normal | $V_{MID_RX}$ | $V_{MID_RX}$ | near DC | DC Lead-Off, AC Lead-Off (Under) |
| Tetrapolar, internal HPF | DRVP and BIP off | DDS sine | large signal | $V_{MID_RX}$ | large signal | large | DC Lead-Off, DRVN Lead-Off, AC Lead-Off (Over) |
| Tetrapolar, internal HPF | DRVP and BIN off | DDS sine | large signal | DDS sine | $V_{MID_RX}$ | DDS sine magnitude | DC Lead-Off, DRVN Lead-Off |
| Tetrapolar, internal HPF | DRVN and BIP off | DDS sine | large signal | $V_{MID_RX}$ | large signal | large | DC Lead-Off, DRVN Lead-Off, AC Lead-Off (Over) |
| Tetrapolar, internal HPF | DRVN and BIN off | DDS sine | large signal | DDS sine | $V_{MID_RX}$ | DDS sine magnitude | DC Lead-Off, DRVN Lead-Off |
| Tetrapolar, external capacitors | DRVP off | DDS sine | large signal | large signal | DDS sine | large | DRVN Lead-Off, AC Lead-Off (Over) |
| Tetrapolar, external capacitors | DRVN off | DDS sine | large signal | large signal | large signal | indeterminant | AC Lead-Off (Over) |
| Tetrapolar, external capacitors | BIP off | $V_{MID_TX}$ | normal | $V_{MID_RX}$ | normal | DDS sine magnitude | DC Lead-Off |
| Tetrapolar, external capacitors | BIN off | $V_{MID_TX}$ | normal | normal | $V_{MID_RX}$ | DDS sine magnitude | DC Lead-Off |
| Tetrapolar, external capacitors | BIP and BIN off | $V_{MID_TX}$ | normal | $V_{MID_RX}$ | $V_{MID_RX}$ | near DC | DC Lead-Off, AC Lead-Off (Under) |
| Tetrapolar, external capacitors | DRVP and BIP off | DDS sine | large signal | $V_{MID_RX}$ | large signal | large | DC Lead-Off, DRVN Lead-Off, AC Lead-Off (Over) |
| Tetrapolar, external capacitors | DRVP and BIN off | DDS sine | large signal | large signal | $V_{MID_RX}$ | large | DC Lead-Off, DRVN Lead-Off, AC Lead-Off (Over) |
| Tetrapolar, external capacitors | DRVN and BIP off | DDS sine | large signal | $V_{MID_RX}$ | DDS sine | DDS sine magnitude | DC Lead-Off |
| Tetrapolar, external capacitors | DRVN and BIN off | DDS sine | large signal | DDS sine | $V_{MID_RX}$ | DDS sine magnitude | DC Lead-Off |

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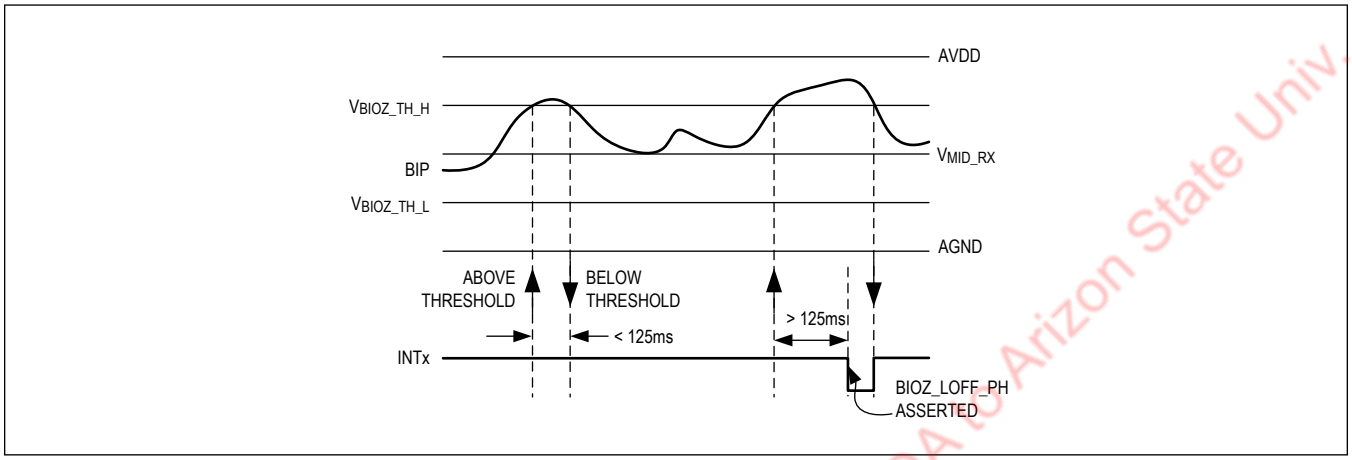



*Figure 46. BioZ DC Lead-Off Behavior*

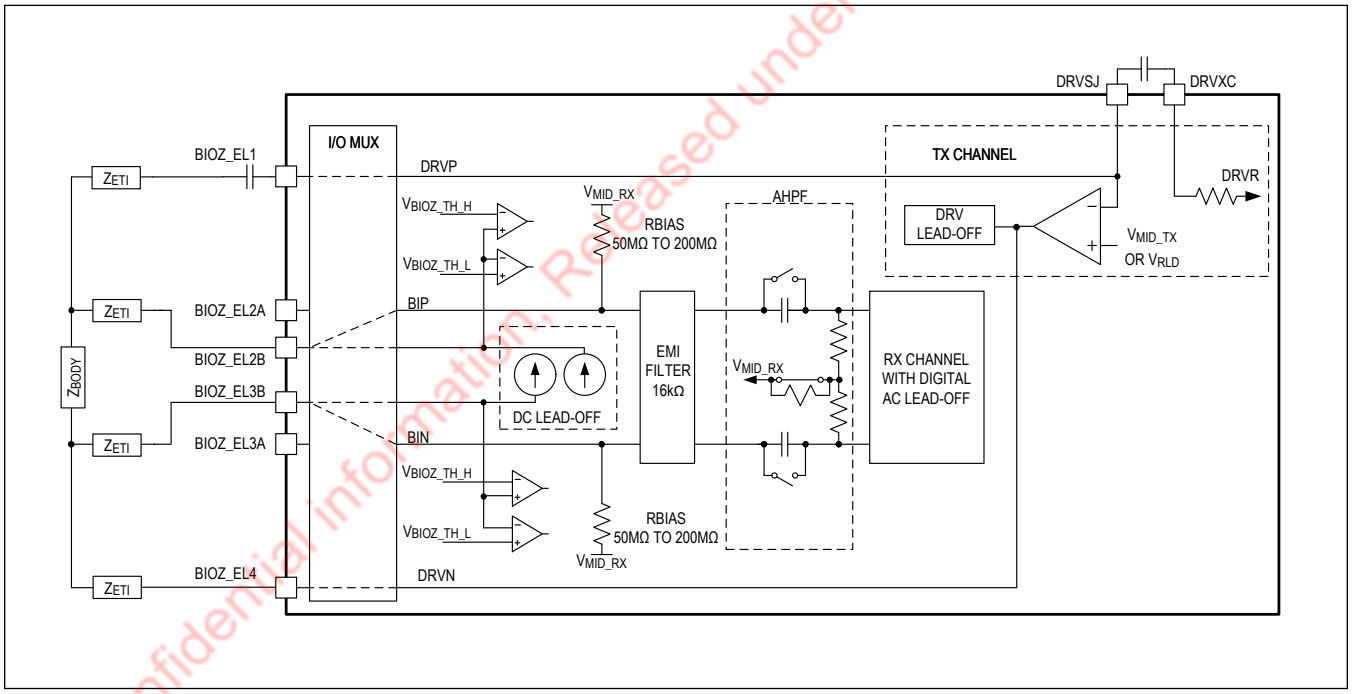



*Figure 47. BioZ DC Lead-Off Detection with Internal Sense*

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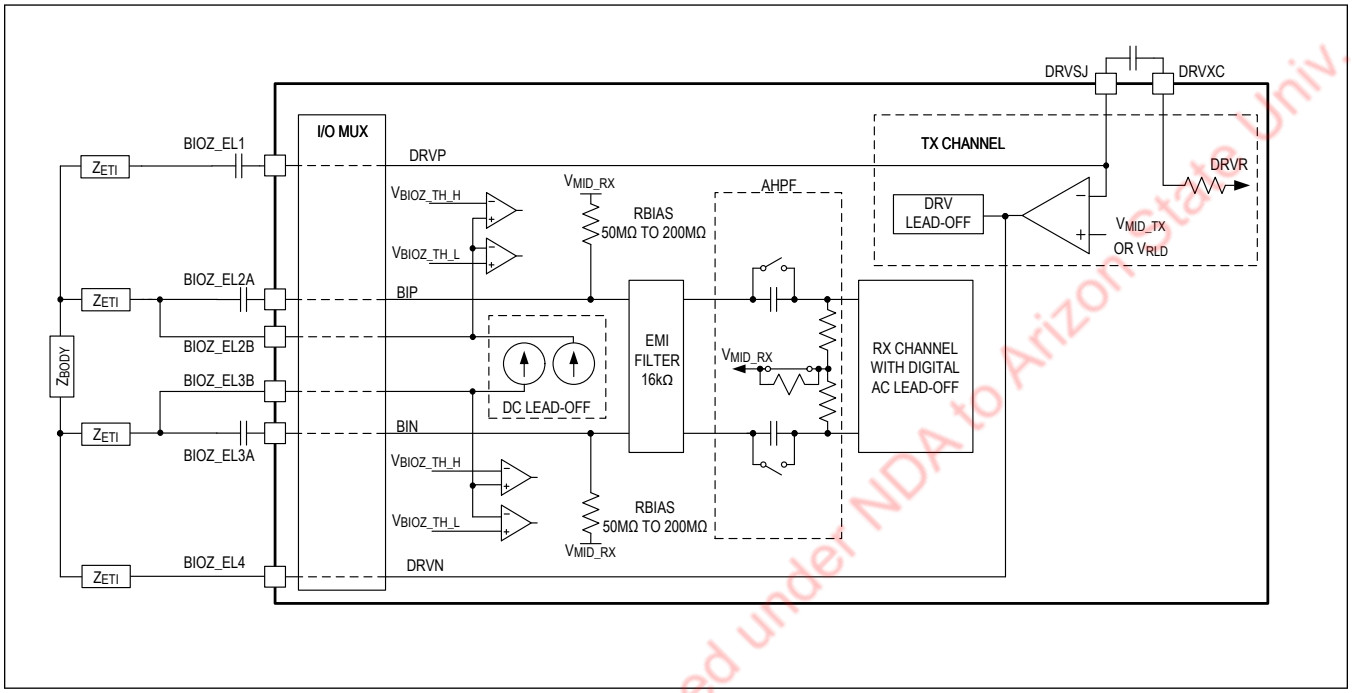



*Figure 48. BioZ DC Lead-Off Detection with External Sense*

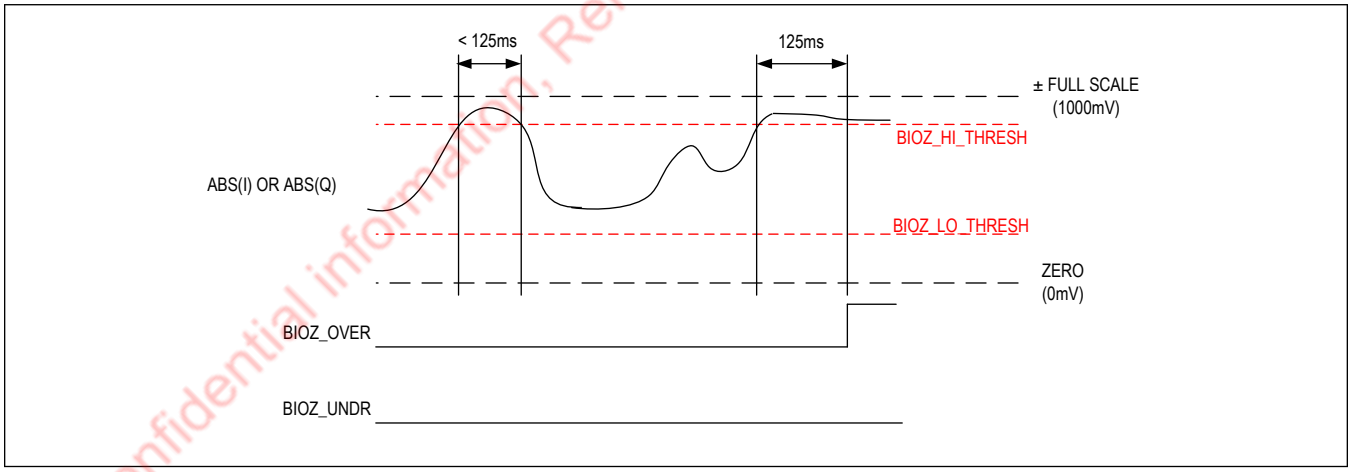



*Figure 49. BioZ Threshold Behavior*

### BioZ Ultra-Low-Power (ULP) Lead-On Detection

The MAX86178 features an ultra-low-powered (ULP) lead-on detection circuit for the BioZ input electrodes. The BioZ channel must be disabled (BIOZ_EN[1:0](0xA0) = 0x0) when ULP lead-on detection is enabled.

The ULP lead-on detect circuit operates by pulling BIN low with a pulldown resistance of 5MΩ (typ) and pulling BIP high with a pullup resistance of 15MΩ (typ). A low-power comparator determines if BIP is pulled below 0.75 x AVDD (typ), and asserts the BIOZ_LON[7](0x05) status bit if BIP remains below the threshold for at least 128ms. This circuit is shown in Figure 44. Because this circuit relies on DC current flowing through the electrodes, it does not work when external

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AC-coupling capacitors are used on the pins assigned to BIP and BIN.

There are several conditions that can pull BIP below the threshold and trigger a lead-on status:

- The total impedance between BIP and BIN is below 40MΩ (typ) due to both electrodes contacting the body.
- The total impedance between BIP and AGND is below 45MΩ (typ) due to the the BIP electrode contacting a body that is coupled to AGND. For example, if the MAX86178 system is coupled to earth ground through a power or data cable, and the body is also coupled to earth ground, then a low-impedance path could pull the BIP electrode low.
- The ECGP electrode is contacting the body and has a large half-cell potential. The half-cell potential can push the ECGP voltage below the threshold.

If the BIOZ_LON interrupt is enabled by BIOZ_LON_EN1[7](0xC5) or BIOZ_LON_EN2[7](0xC9), an interrupt is generated to alert the host microcontroller of the lead-on condition. This interrupt allows the microcontroller to sleep when the system is not in use, and only wake up when the user touches the device electrodes. Upon receiving an interrupt and waking up, the microcontroller should read the BIOZ_LON status register to determine if a lead-on condition has occurred. Because of the bit's clear-on-read behavior, the status register should be read a second time to determine if the lead-on condition persists.

### BioZ Calibration

The MAX86178 can be calibrated and can achieve impedance magnitude errors of 0.1% and impedance phase errors of 0.1°. The calibration can be performed at the factory by applying a precision resistor to the device electrodes, or on-board by connecting a precision resistor to the CALx pins. To achieve 0.1% accuracy, the calibration resistor(s) must have 0.05% tolerance, or be measured by an external reference with 0.05% accuracy. If using an on-board calibration, the reference resistor ($R_{CAL}$) should have a low temperature coefficient and should be connected to the calibration port as shown in Figure 50. To connect the on-board $R_{CAL}$, assert BIOZ_MUX_EN[1](0xAA) and BIOZ_CAL_EN[0](0xAA).

The calibration consists of measuring the I and Q offsets and magnitude and phase coefficients at each measurement frequency, according to the following steps. Registers not mentioned in the steps below should be set to the values intended for use during measurement.

1. Set the synthesis frequency to the desired frequency.
2. Measure the I and Q offsets:
    1. Set the stimulus current magnitude to the minimum $16nA_{RMS}$ by setting BIOZ_VDRV_MAG[5:4](0xA2) and BIOZ_IDRV_RGE[3:2](0xA2) to 0x0.
    2. Enable BIOZ_DRV_RESET[5](0xA6) to apply a short-circuit across the load.
    3. Set BIOZ_EN[1:0](0xA0) to 0x1 to enable I phase measurement.
    4. Record data until the impedance signal has settled, and then record the average impedance in Ohms (I_offset [Ω]). Settling time varies with sample rate, filter selections, and other settings.
    5. Set BIOZ_EN[1:0](0xA0) to 0x2 to enable Q phase measurement.
    6. Record data until the impedance signal has settled, and then record the average impedance in Ohms (Q_offset [Ω]).
    7. Note: I_offset and Q_offset should be calculated using the intended measurement current magnitude, not the minimum $16nA_{RMS}$.
3. Measure the calibration resistor I and Q impedances:
    1. Set the stimulus current to the desired value by adjusting BIOZ_VDRV_MAG[5:4](0xA2) and BIOZ_IDRV_RGE[3:2](0xA2).
    2. Disable BIOZ_DRV_RESET[5](0xA6).
    3. Set BIOZ_EN[1:0](0xA0) to 0x1 to enable I phase measurement.
    4. Record data until the impedance signal has settled, and then record the average impedance in Ohms (I_resistor [Ω]).
    5. Set BIOZ_EN[1:0](0xA0) to 0x2 to enable Q phase measurement.
    6. Record data until the impedance signal has settled, and then record the average impedance in Ohms (Q_resistor [Ω]).
4. Subtract the offsets from the resistor measurements:
    1. I_calibration [Ω] = I_resistor - I_offset
    2. Q_calibration [Ω]= Q_resistor - Q_offset
5. Calculate the calibration magnitude and phase coefficients:

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1. Mag_calibration [Ω] = √(I_calibration$^2$ + Q_calibration$^2$)
2. Mag_coef = Mag_calibration / $R_{CAL}$
3. Phase_coef [°] = arctan(Q_calibration / I_calibration) x 180° / π

To apply the calibration coefficients to a measured impedance, follow the steps below.

1. Measure I and Q load impedances (I_load [Ω] and Q_load [Ω]).
2. Subtract the offsets from the load impedances:
   1. I_load-offset [Ω] = I_load - I_offset
   2. Q_load-offset [Ω] = Q_load - Q_offset
3. Calculate the load impedance magnitude and phase:
   1. Mag_load [Ω]= √(I_load-offset$^2$ + Q_load-offset$^2$)
   2. Phase_load [°]= arctan(Q_load-offset / I_load-offset) x 180° / π
4. Apply the calibration coefficients:
   1. Mag_calibrated [Ω] = Mag_load / Mag_coef
   2. Phase_calibrated [°] = Phase_load - Phase_coef
   3. I_calibrated [Ω] = Mag_calibrated x cos(Phase_calibrated x π / 180°)
   4. Q_calibrated [Ω] = Mag_calibrated x sin(Phase_calibrated x π / 180°)

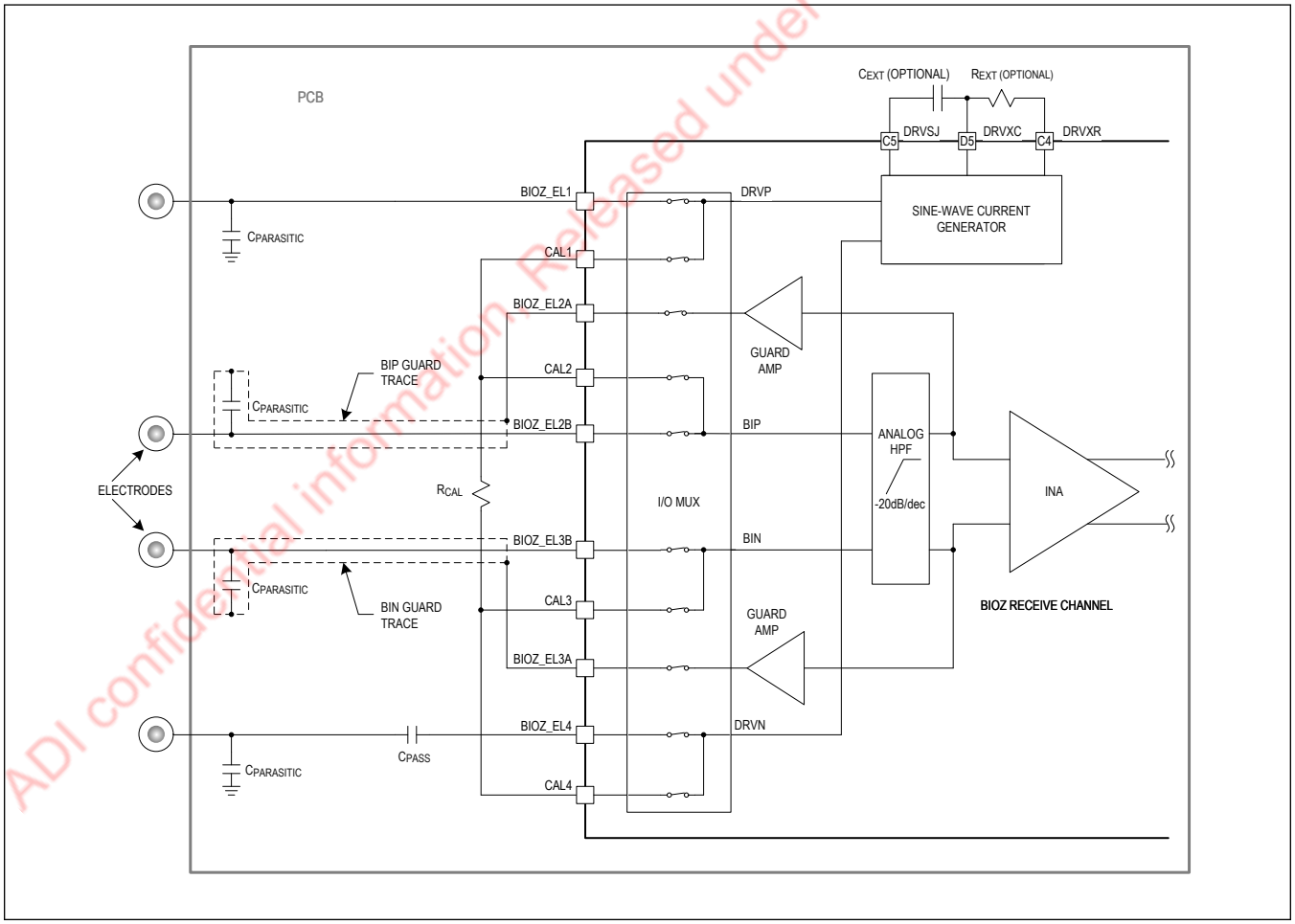



*Figure 50. Calibration Port Connections*

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### Improving Accuracy of BIA/BIS Measurements

The parasitic capacitance on the pins assigned to DRVP and DRVN can be calibrated out using the procedure described above. However, the parasitic capacitances on the pins assigned to BIP and BIN are more difficult to manage. The MAX86178 has two features that can be used to minimize the effect of these receive-channel parasitic capacitances.

First, it is important to realize that differences in BIP and BIN electrode tissue interface impedances working against these parasitic BIP and BIN capacitances can create phase accuracy issues, for instance, if the two PCB traces used to route BIP and BIN to the MAX86178 are carefully managed so that their parasitic capacitances are nearly equal, but the BIP electrode has a higher electrode tissue interface impedance than the BIN electrode. In such a case, the BIP side has more phase lag than the BIN side, creating a potential source of inaccuracy.

This problem can be mitigated by use of driven guard amplifiers and BIP and BIN guard traces. The BIP and BIN driven guard amplifiers can be enabled using EN_EXT_INLOAD[1](0xAB). The outputs of the BIP and BIN guard amplifiers are offered at pins BIOZ_EL2A and BIOZ_EL3A, respectively. Guard traces remove the effect of the parasitic PCB capacitances by driving a guard signal, that moves at the same frequency, amplitude, and phase as the input signal of interest; one can then use these guard signal routes to surround the BIP and BIN nets, usually parallel to the nets and on both sides; since there is no AC voltage drop across these parasitic capacitances, there is no associated AC currents needed to charge or discharge them.

To help mitigate the effects of parasitic capacitances within the MAX86178 itself, assert the EN_INT_INLOAD[0](0xAB), which enables an inverse capacitive load on each input.

## FIFO Description

The FIFO holds a maximum of 256 samples and supports various data types from the ECG, PPG, and BioZ channels. Each sample in the FIFO is three bytes wide and contains the tag and data. The tag embedded in the FIFO_DATA[23:0] identifies the source and type of each sample. The data in FIFO_DATA[23:0] is right justified for all data types. Table 24 shows each data type in the FIFO along with the associated tags.

### Table 24. FIFO Tags

| DATA TYPE | TAG AND DATA | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BIT | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| PPG1/2 MEAS 1 | 0 | 0 | 0 | 0 | PPG_MEAS1_DATA[19:0] | | | | | | | | | | | | | | | | | | | |
| PPG1/2 MEAS 2 | 0 | 0 | 0 | 1 | PPG_MEAS2_DATA[19:0] | | | | | | | | | | | | | | | | | | | |
| PPG1/2 MEAS 3 | 0 | 0 | 1 | 0 | PPG_MEAS3_DATA[19:0] | | | | | | | | | | | | | | | | | | | |
| PPG1/2 MEAS 4 | 0 | 0 | 1 | 1 | PPG_MEAS4_DATA[19:0] | | | | | | | | | | | | | | | | | | | |
| PPG1/2 MEAS 5 | 0 | 1 | 0 | 0 | PPG_MEAS5_DATA[19:0] | | | | | | | | | | | | | | | | | | | |
| PPG1/2 MEAS 6 | 0 | 1 | 0 | 1 | PPG_MEAS6_DATA[19:0] | | | | | | | | | | | | | | | | | | | |
| PPG1/2 DARK | 0 | 1 | 1 | 0 | PPG_DARK_DATA[19:0] | | | | | | | | | | | | | | | | | | | |
| PPG1/2 ALC OVF | 0 | 1 | 1 | 1 | PPG_ALC_OVF_DATA[19:0] | | | | | | | | | | | | | | | | | | | |
| PPG1/2 EXP OVF | 1 | 0 | 0 | 0 | PPG_EXP_OVF_DATA[19:0] | | | | | | | | | | | | | | | | | | | |
| BIOZ I | 1 | 0 | 0 | 1 | BIOZ_I_DATA[19:0] | | | | | | | | | | | | | | | | | | | |
| BIOZ Q | 1 | 0 | 1 | 0 | BIOZ_Q_DATA[19:0] | | | | | | | | | | | | | | | | | | | |
| ECG and Fast Recover Flag | 1 | 0 | 1 | 1 | 0 | F | ECG_DATA[17:0] | | | | | | | | | | | | | | | | | |
| ECGP ECGN Differential | 1 | 1 | 0 | 0 | ECGP_DATA[19:0]/ECGN_DATA[19:0]/ECGPN_DIFF_DATA[19:0] | | | | | | | | | | | | | | | | | | | |
| CAPP CAPN Differential | 1 | 1 | 0 | 1 | CAPPN_DIFF_DATA[19:0] | | | | | | | | | | | | | | | | | | | |
| ECG to PPG Timing | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | ECG_PPG_TIMING_DATA[9:0] | | | | | | | | | |
| BIOZ to PPG Timing | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | BIOZ_PPG_TIMING_DATA[13:0] | | | | | | | | | | | | | |

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## Table 24. FIFO Tags (continued)

| DATA TYPE | TAG AND DATA | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ECG to BIOZ Timing | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | ECG_BIOZ_TIMING_DATA[9:0] | | | | | | | | | | |
| Marker | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |
| Invalid data | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

**PPG Tags**

PPG tags are four bits. The PPG data is in two's complement format and is 20 bits. The MSB of the PPG data FIFO_DATA[19:0] is the sign bit. The sequencing of exposures is controlled by MEAS1_EN through MEAS6_EN bits in PPG Configuration 1 register 0x20. The ADC conversion sequence cycles through the enabled measurements starting from MEAS1.

The exposures selected for each of the six measurements, and their configuration are setup in the MEAS1 Setup to MEAS6 Setup registers (see the Register Map for details). If a measurement has both PPG channels enabled, the two resulting samples have the same tag and the FIFO position determines the corresponding channel. The first sample is from channel 1 and the second sample is from channel 2.

When COLLECT_RAW_DATA[1](0x22) is disabled, the computed data is stored as a single MEASx sample in the FIFO (see the *Ambient Rejection* section). When COLLECT_RAW_DATA is enabled, the raw exposure sample and the dark (ambient) sample(s) are stored as separate samples in the FIFO. The raw exposure sample is tagged with its corresponding MEASx tag, but all dark samples are tagged with the same DARK tag, regardless of which measurement it came from.

When ALC_OVF[4](0x00) is detected for any measurement, the computed data for that measurement is tagged with the PPG_ALC_OVF_DATA tag, which is 0x7. When EXP_OVF[3](0x00) is detected for any measurement, the computed data for that measurement is tagged with the PPG_EXP_OVF_DATA tag, which is 0x8. The positions of the data with the PPG_ALC_OVF_DATA or PPG_EXP_OVF_DATA tags indicate which measurement and which optical channel the data belongs to. When both ALC_OVF and EXP_OVF are detected on the same measurement, the sample is tagged with the ALC OVF tag.

**BioZ Tags:**

BioZ tags are four bits. Tags for I channel data and Q channel data are 0x9 and 0xA, respectively.

Note that if the BioZ channel is disabled while a sample is being pushed to the FIFO, the FIFO will contain an extra sample containing the value 0x4000, which should be ignored. This can be avoided by disabling the BioZ channel immediately after a FIFO_DATA_RDY or A_FULL interrupt.

**ECG Tag:**

The ECG tag is six bits, of which the LSB indicates if the data corresponds to normal mode (F = 0) or fast recovery mode (F = 1). The tag for normal mode is 0x2C and for fast recovery mode is 0x2D.

**Timing Data Tags:**

The timing data tags are six bits and the timing data is right justified. The tag for ECG to PPG timing data is 0x38, and the data is ten bits. The tag for BioZ to PPG timing data is 0x39, and the data is fourteen bits. The tag for ECG to BioZ timing data is 0x3A and the data is ten bits. See the *Timing Data in the FIFO* section for more information.

**Utility Data Tags:**

Utility data tags are four bits. These are used for ECGP to ECGN differential input and CAPP-to-CAPN differential input measurements. The tags used for these are 0xC and 0xD, respectively.

Other Tags:

Invalid data and marker tags are 24 bits each.

An attempt to read an empty FIFO returns the INVALID_DATA tag, which is 0xFFFFFF.

MAX86178 provides a feature of saving a FIFO marker when needed in an application. For example, in order to distinguish data saved in the FIFO before and after a configuration change, a marker can be saved in the FIFO just before the configuration change. The marker tag is 0xFFFFFE.

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For details on FIFO configuration, see the Register Map (registers 0x08 to 0x0E).

**FIFO Configuration Examples**

FIFO configuration is best explained by a few examples.

***Example 1: PPG only***

Assume it is desired to perform an $SpO_2$ measurement along with heart-rate monitoring using dual optical channels and also monitor the ambient level on the photodiode to adjust the LED intensity. To perform this measurement, configure the following registers.

```
//PLL Configuration

Set REF_CLK_SEL as desired          //Use internal oscillator or external clock signal
Set CLK_FREQ_SEL as desired          //Select 32kHz or 32.768kHz REF_CLK

//PPG Frame Rate Configuration
Set FR_CLK_DIV as desired           //Set the frame rate divider

//PPG configuration
Set PPG1_PWRDN = 0                  //Do not power down PPG1 ADC
Set PPG2_PWRDN = 0                  //Do not power down PPG2 ADC
Set MEAS1_EN = 1                   //Enable measurement 1 to 4
Set MEAS2_EN = 1
Set MEAS3_EN = 1
Set MEAS4_EN = 1

//MEASUREMENT 1 Configuration
Set MEAS1_AMB = 0                  //Ambient measurement off
Set MEAS1_DRVA = 0x0                //LED Driver A driving green LED on LED1_DRV
Set MEAS1_AVER as desired           //Number of LED pulses in each frame
Set MEAS1_PPG1_ADC_RGE as desired     //Gain range control for PPG1 ADC
Set MEAS1_PPG2_ADC_RGEas desired      //Gain range control for PPG2 ADC
Set MEAS1_PPG_TINT as desired        //ADC integration time control
Set MEAS1_LED_RGE as desired         //LED drive current range
Set MEAS1_PD_SETLNG as desired        //Settling time for photodiodes
Set MEAS1_LED_SETLNG as desired      //LED settling time
Set MEAS1_DRVA_PA as desired         //LED driver A current driving the green LED on LED1_DRV
Set MEAS1_DRVB_PA = 0x00            //LED driver B current should be set to 0
Set MEAS1_PD1_SEL = 0x2             //Photodiode 1 selected on optical channel 1
Set MEAS1_PD3_SEL = 0x3             //Photodiode 3 selected on optical channel 2

//MEASUREMENT 2 Configuration
Set MEAS2_AMB = 0                  //Ambient measurement off
Set MEAS2_DRVA = 0x1                //LED Driver A driving IR LED on LED2_DRV
Set MEAS2_AVER as desired           //Number of LED pulses in each frame
Set MEAS2_PPG1_ADC_RGE as desired     //Gain range control for PPG1 ADC
Set MEAS2_PPG2_ADC_RGEas desired      //Gain range control for PPG2 ADC
Set MEAS2_PPG_TINT as desired        //ADC integration time control
Set MEAS2_LED_RGE as desired         //LED drive current range
Set MEAS2_PD_SETLNG as desired        //Settling time for photodiodes
Set MEAS2_LED_SETLNG as desired      //LED settling time
Set MEAS2_DRVA_PA as desired         //LED driver A current driving the IR LED on LED2_DRV
Set MEAS2_DRVB_PA = 0x00            //LED driver B current should be set to 0
```

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```
Set MEAS2_PD2_SEL = 0x2          //Photodiode 2 selected on optical channel 1

//MEASUREMENT 3 Configuration
Set MEAS3_AMB = 0                //Ambient measurement off
Set MEAS3_DRVB = 0x2             //LED Driver B driving red LED on LED3_DRV
Set MEAS3_AVER as desired        //Number of LED pulses in each frame
Set MEAS3_PPG1_ADC_RGE as desired    //Gain range control for PPG1 ADC
Set MEAS3_PPG2_ADC_RGEas desired     //Gain range control for PPG2 ADC
Set MEAS3_PPG_TINT as desired        //ADC integration time control
Set MEAS3_LED_RGE as desired         //LED drive current range
Set MEAS3_PD_SETLNG as desired       //Settling time for photodiodes
Set MEAS3_LED_SETLNG as desired      //LED settling time
Set MEAS3_DRVA_PA = 0x00         //LED driver A current should be set to 0
Set MEAS3_DRVB_PA as desired         //LED driver B current driving the red LED on LED3_DRV
Set MEAS3_PD2_SEL = 0x2          //Photodiode 2 selected on optical channel 1

//MEASUREMENT 4 Configuration
Set MEAS4_AMB = 1                //Ambient measurement selected
Set MEAS4_PPG1_ADC_RGE as desired    //Gain range control for PPG1 ADC
Set MEAS4_PPG2_ADC_RGE as desired    //Gain range control for PPG2 ADC
Set MEAS4_PPG_TINT as desired        //ADC integration time control
Set MEAS4_PD1_SEL = 0x2          //Photodiode 1 selected on optical channel 1
Set MEAS4_PD3_SEL = 0x3          //Photodiode 3 selected on optical channel 2
```

With this configuration the sample sequence and the data format in the FIFO follows the following time/location sequence.

| TAG[23:20] | DATA[19:0] |
|---|---|
| 0b0000 | PPG1 Measurement 1 data |
| 0b0000 | PPG2 Measurement 1 data |
| 0b0001 | PPG1 Measurement 2 data |
| 0b0001 | PPG2 Measurement 2 data |
| 0b0010 | PPG1 Measurement 3 data |
| 0b0010 | PPG2 Measurement 3 data |
| 0b0011 | PPG1 Measurement 4 data |
| 0b0011 | PPG2 Measurement 4 data |
| . | |
| . | |
| . | |
| . | |
| . | |
| 0b0000 | PPG1 Measurement 1 data |
| 0b0000 | PPG2 Measurement 1 data |
| 0b0001 | PPG1 Measurement 2 data |
| 0b0001 | PPG2 Measurement 2 data |
| 0b0010 | PPG1 Measurement 3 data |
| 0b0010 | PPG2 Measurement 3 data |
| 0b0011 | PPG1 Measurement 4 data |
| 0b0011 | PPG2 Measurement 4 data |

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where:

PPGm Measurement 1 data = Ambient corrected exposure data from green LED in optical channel m

PPGm Measurement 2 data = Ambient corrected exposure data from IR LED in optical channel m

PPGm Measurement 3 data = Ambient corrected exposure data from red LED in optical channel m

PPGm Measurement 4 data = Direct ambient sample in optical channel m

m is 1 for optical channel 1, and 2 for optical channel 2

The number of bytes of data for the two optical channels in one frame is given by: 2 x 3 x N where:

N = the number of measurements enabled

***Example 2: ECG with three dry electrodes***

This setup performs ECG measurements at 512sps and assumes three dry electrodes are in contact with the body, so it applies right leg drive and AC lead-off detection. To perform this measurement, configure the following registers.

```
//PLL Configuration (SR_ECG = 512sps)

Set REF_CLK_SEL as desired          //Use internal oscillator or external clock signal
Set CLK_FREQ_SEL = 1                //Select 32.768kHz clock frequency

Set MDIV = 0x07F                    //PLL M divider set to 128 (PLL CLK = 4.19MHz)
Set ECG_NDIV = 0x010                //ECG N divider set to 16
Set ECG_FDIV = 0x4                  //ECG F divider set to 8
Set PLL_EN = 1                      //Enable the PLL



//ECG Configuration (SR_ECG = 512sps)
Set ECG_DEC_RATE = 0x2              //ECG decimation ratio is 64
Set ECG_PGA_GAIN = 0x0              //PGA gain is 1V/V
Set ECG_INA_RGE = 0x0               //INA gain is 10V/V
Set ECG_INA_GAIN = 0x0              //INA gain is 10V/V
Set ECG_AUTO_REC = 1                //Automatic INA recovery enabled
Set EN_ECG_FAST_REC = 0x2           //Automatic fast recovery enabled
Set ECG_FAST_REC_THRESHOLD = 0x3D   //Fast recovery threshold set to 95% of full scale
Set EN_ECG_RBIASP = 0               //Bias resistor on ECGP is disabled
Set EN_ECG_RBIASN = 0               //Bias resistor on ECGN is disabled
Set ECG_MUX_SEL as desired          //Enable the ECG MUX and assign the proper electrodes
Set ECG_OPEN_P = 0                  //Connect ECGP to the selected electrode
Set ECG_OPEN_N = 0                  //Connect ECGN to the selected electrode

Set ECG_EN = 1                      //ECG channel is enabled

//AC Lead-Off Detection Configuration (8192Hz)
Set ECG_LOFF_FREQ = 0x1             //ACLO N divider set to 4
Set ECG_LOFF_IMAG = 0x6             //ACLO current = 200nA_PK, or as desired to match Z_ETI
Set ECG_LOFF_THRESH = 0x7           //ACLO threshold = V_MID_ECG ±200mV, or as desired
Set ECG_LOFF_MODE = 1               //AC lead-off detection mode

Set EN_ECG_LOFF = 1                 //Enable lead-off detection

//RLD Configuration
Set RLD_MODE = 1                    //Closed loop RLD (AC Common Mode feedback)
```

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```
Set EN_RLD_OOR = 1              //Out of Range detection is enabled
Set ACTV_CM_P = 1               //Positive input buffer is enabled
Set ACTV_CM_N = 1               //Negative input buffer is enabled
Set RLD_GAIN = 0x3              //RLD Gain is 97V/V
Set RLD_EXT_RES = 0             //Internal gain resistor is used
Set RLD_SEL_ECG = 1             //Input voltages from ECGP and ECGN
Set RLD_BW = 0x0                //Low bandwidth
Set BODY_BIAS_DAC = 0x0         //Reference is VMID_ECG

Set RLD_EN = 1                  //Right Leg Drive is enabled
```

In this case, the FIFO only contains ECG samples in the following format:

| TAG[23:18] | DATA[17:0] |
|---|---|
| 0b10110F | ECG data |
| 0b10110F | ECG data |
| 0b10110F | ECG data |
| ... | |

***Example 3: ECG with two wet electrodes, respiration, and PPG with one LED in single-channel mode***

ECG to PPG timing data and ECG to BioZ timing data are also enabled. This setup performs ECG measurements at 512sps and assumes two wet electrodes are attached to the body, so it applies internal lead bias and uses DC lead-off detection. Respiration shares the ECG electrodes, and measures at 128sps with a stimulus frequency of 65.536kHz. PPG enables one green LED on pin 1 with a single-channel configuration. ECG to PPG timing data indicates the time difference between measurement 1 in each frame and the last ECG sample saved in the FIFO. ECG to BioZ timing data indicates the time difference between the BioZ samples entering the FIFO and the last ECG samples saved in the FIFO.

```
//System Configuration
Set ECG_PPG_TIMING_DATA = 1     //ECG to PPG timing data is enabled
Set ECG_BIOZ_TIMING_DATA = 1    //ECG to BioZ timing data is enabled

//PLL Configuration (SR_ECG = 512sps, SR_RESP = 128sps, F_RESP = 65.563kHz)

Set REF_CLK_SEL as desired      //Use internal oscillator or external clock signal
Set CLK_FREQ_SEL = 1            //Select 32.768kHz clock frequency
Set MDIV = 0x1FF                //PLL M divider set to 512
Set ECG_NDIV = 0x200            //ECG N divider set to 512
Set ECG_DEC_RATE = 0x2          //ECG ADC OSR = 64
Set ECG_FDIV = 0x1              //ECG/Respiration F divider set to 1
Set BIOZ_KDIV = 0x0             //BioZ K divider set to 1
Set BIOZ_NDIV = 0x1             //BioZ N divider set to 512
Set BIOZ_DAC_OSR = 0x3          //BioZ DAC OSR set to 256
Set BIOZ_ADC_OSR = 0x5          //BioZ ADC OSR set to 256
Set PLL_EN = 1                  //Enable the PLL


//PPG Frame Rate Clock Configuration (PPG_FR_CLK = ECG_ADC_CLK = 32.768kHz)
Set FR_CLK_DIV = 0x0100         //FR divider set to 256, FR_PPG = 128fps

//PPG Configuration

Set PPG1_PWRDN = 0              //Do not power down PPG1 ADC
Set PPG2_PWRDN = 1              //PPG2 ADC is powered down
Set MEAS1_EN = 1                //Only measurement 1 is enabled
```

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```
//MEASUREMENT 1 Configuration
Set MEAS1_AMB = 0                  //Ambient measurement off
Set MEAS1_DRVA = 0x0               //LED driver A driving green LED on LED1_DRV
Set MEAS1_AVER as desired          //Number of LED pulses in each frame
Set MEAS1_PPG1_ADC_RGE as desired  //Gain range control for PPG1 ADC
Set MEAS1_PPG2_ADC_RGEas desired   //Don't care because PPG2 is powered down
Set MEAS1_PPG_TINT as desired      //ADC integration time control
Set MEAS1_LED_RGE as desired       //LED drive current range
Set MEAS1_PD_SETLNG as desired     //Settling time for photodiodes
Set MEAS1_LED_SETLNG as desired    //LED settling time
Set MEAS1_DRVA_PA as desired       //LED driver A current driving the green LED on LED1_DRV
Set MEAS1_DRVB_PA = 0x00           //LED driver B current should be set to 0
Set MEAS1_PPG1_PDSEL = 0x0         //Photodiode 1 selected on optical channel 1
Set MEAS1_PPG2_PDSEL               //Don't care because PPG2 is powered down

//ECG Configuration (SR_ECG = 512sps)
Set ECG_DEC_RATE = 0x3             //ECG ADC OSR = 128
Set ECG_PGA_GAIN = 0x0             //PGA gain is 1V/V
Set ECG_INA_GAIN = 0x1             //INA gain is 20V/V
Set ECG_INA_RGE = 0x0              //INA gain is 20V/V
Set ECG_AUTO_REC = 1               //Automatic fast recovery is enabled
Set ECG_FAST_REC_THRESHOLD = 0x3D  //Fast recovery threshold set to 95% of full scale
Set ECG_RBIAS_VALUE = 0x1          //Bias resistor value is 100MΩ
Set EN_ECG_RBIASP = 1              //Bias resistor on ECGP is enabled
Set EN_ECG_RBIASN = 1              //Bias resistor on ECGN is enabled
Set ECG_MUX_EN as desired          //Enable the ECG MUX and assign the proper electrodes

Set ECG_EN = 1                     //ECG channel is enabled

//ECG DC Lead-Off Detection Configuration
Set EN_ECG_LOFF = 1                //DC lead-off detection enabled
Set ECG_LOFF_IMAG = 0x4            //Current magnitude is 50nA
Set ECG_LOFF_THRESH = 0x5          //Threshold is V_MID_ECG ±300mV

//BioZ and Respiration Configuration (SR_RESP = 128sps, F_RESP = 65.536kHz)
Set ECG_BIOZ_BG_EN = 1             //Enable the ECG/BioZ reference
Set BIOZ_ADC_OSR = 0x5             //BioZ ADC OSR set to 256
Set CG_CHOP_CLK = 0x0              //Respiration CG clock divider set to 256
Set BIOZ_DHPF = 0x1                //Set the digital HPF to SR_BIOZ x 0.00025 = 0.032Hz
Set BIOZ_AHPF = 0x5                //Set the analog HPF to 5kHz
Set BIOZ_GAIN = 0x0                //Set the receive channel gain to 1V/V, or as desired
Set BIOZ_CH_FSEL = 0               //F_BIOZ is not equal to BIOZ_ADC_CLK/16
Set RESP_CG_MAG_4X as desired      //Match the current magnitude to the expected load impedance

Set RESP_CG_MAG as desired         //Match the current magnitude to the expected load impedance
Set CG_MODE = 0x2                  //Enable dynamic matching with LPF
Set CG_LPF_DUTY = 0x3              //Set the LPF bandwidth to approximately 7.79Hz
Set DRVP_ASSIGN = 0x0              //DRVP is set to BIOZ_EL1
Set DRVN_ASSIGN = 0x0              //DRVN is set to BIOZ_EL4
Set BIP_ASSIGN = 0x2               //BIP is set to BIOZ_EL2B
Set BIN_ASSIGN = 0x2               //BIN is set to BIOZ_EL3B
Set BIOZ_MUX_EN = 1                //Enable the MUX
Set RESP_EN = 1                    //Enable respiration mode
```

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```
Set BIOZ_EN = 0x1                //Enable in-phase BioZ measurements
```

In this case, the FIFO contains ECG, BioZ, and PPG samples, along with ECG to PPG timing data and ECG to BioZ timing data in the following format:

| TAG | DATA |
|---|---|
| ... | |
| 0b10110F | ECG data |
| 0b10110F | ECG data |
| 0b111010 | ECG to BioZ Timing data |
| 0b1001 | BioZ I data |
| 0b10110F | ECG data |
| 0b111000 | ECG to PPG Timing data |
| 0b0000 | PPG1 Measurement 1 data |
| 0b10110F | ECG data |
| 0b10110F | ECG data |
| ... | |

***Example 4: BIA measurement at 10kHz***

This setup measures BioZ at 78.13sps and a stimulus frequency of 10.0kHz, and assumes a tetrapolar electrode configuration. For a typical bioimpedance analysis (BIA), this measurement would be repeated for the I and Q phases at multiple frequencies with the TX channel placed in standby mode between measurements.

```
//PLL Configuration (SR_BIOZ = 78.13sps, F_BIOZ = 10.0kHz)

Set REF_CLK_SEL as desired       //Use internal oscillator or external clock signal
Set CLK_FREQ_SEL = 1             //Select 32.768kHz clock frequency

Set MDIV = 0x270                 //PLL M divider set to 625
Set BIOZ_KDIV = 0x3              //BioZ K divider set to 8
Set BIOZ_NDIV = 0x2              //BioZ N divider set to 1024
Set PLL_EN = 1                   //Enable the PLL



//BioZ Configuration (SR_BIOZ = 78.13sps, F_BIOZ = 10.0kHz)
Set ECG_BIOZ_BG_EN = 1           //Enable the ECG/BioZ reference
Set BIOZ_ADC_OSR = 0x5           //BioZ ADC OSR set to 256
Set BIOZ_DAC_OSR = 0x3           //BioZ DAC OSR set to 256
Set BIOZ_AHPF = 0x2              //Set the analog HPF to 500Hz
Set BIOZ_GAIN = 0x0              //Set the receive channel gain to 1V/V, or as desired
Set BIOZ_CH_FSEL = 0             //F_BIOZ is not equal to BIOZ_ADC_CLK/16
Set BIOZ_DRV_MODE = 0x0          //Sine wave current mode
Set BIOZ_VDRV_MAG = 0x2          //Set current magnitude to 32μA_RMS

Set BIOZ_IDRV_RGE = 0x2          //Set current magnitude to 32μA_RMS
Set BIOZ_AMP_RGE = 0x2           //Match the amplifier range with the current range
Set BIOZ_AMP_BW = 0x2            //Set amplifier BW to mid-high
Set BIOZ_INA_MODE = 0            //INA is in low-noise mode
Set BIOZ_INT_INLOAD = 1          //Enable internal input load cancellation
Set BIOZ_EXT_INLOAD = 1          //Enable guard amplifiers (if guard traces are routed)
Set DRVP_ASSIGN = 0x0            //DRVP is set to BIOZ_EL1
Set DRVN_ASSIGN = 0x0            //DRVN is set to BIOZ_EL4
```

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```
Set BIP_ASSIGN = 0x2          //BIP is set to BIOZ_EL2B
Set BIN_ASSIGN = 0x2          //BIN is set to BIOZ_EL3B
Set BIOZ_MUX_EN = 1           //Enable the MUX
Set BIOZ_EN = 0x1             //Enable in-phase BioZ measurements
```

In this case, the FIFO contains only BioZ I data in the following format:

| TAG | DATA |
|---|---|
| ... | |
| 0b1001 | BioZ I data |
| 0b1001 | BioZ I data |
| 0b1001 | BioZ I data |
| 0b1001 | BioZ I data |
| ... | |

If the application always responds much faster than the selected sample rate, it could just read 256 minus FIFO_A_FULL[7:0] items when it gets an A_FULL interrupt, and be assured that all data from the FIFO are read. However, if there is a need to calculate the number of items available in the FIFO one can perform the following pseudo-code:

```
read the OVF_COUNTER register
read the FIFO_DATA_COUNT registers
if OVF_COUNTER == 0 //no overflow occurred
   NUM_AVAILABLE_SAMPLES = FIFO_DATA_COUNT
else
   NUM_AVAILABLE_SAMPLES = 256   // overflow occurred and data has been lost
endif
```

## LDO

The MAX86178 includes a low dropout (LDO) linear regulator, enabling operation from a single supply in size-constrained applications, or applications powered by a primary coin-cell battery. The single supply voltage should be selected according to the highest LED forward voltage plus the minimum voltage overhead (see the *Optical Transmitter* section). This supply is then connected to $V_{LED}$, the LED anodes, and LDO_IN. LDO_OUT provides 1.825V (typ) to power AVDD and DVDD (and IOVDD if needed), but must not be used to power other components or ICs due to the thermal constraints of the MAX86178 package. This configuration is shown in *Typical Application Circuit 1* and *Typical Application Circuit 2*. If the LDO is not used, connect LDO_IN to AGND and leave LDO_OUT unconnected as shown in *Typical Application Circuit 3.*

## Digital Interface

The MAX86178 supports $I^2C$ and SPI serial interfaces. The CSB/I2C_SEL pin selects the interface being used at any one time. When the I2C_SEL pin is high using an external pullup resistor, the interface is in $I^2C$ mode and idles looking for a start condition on the SCL and SDA pins, while the SPI interface is held in a reset state. When the CSB/I2C_SEL pin is low, the $I^2C$ interface is disabled and the SPI interface is activated. In the following sections, timings and protocols for both interfaces are described.

### SPI Interface

The SPI interface on the MAX86178 is SPI-/QSPI-/microwire-/DSP-compatible consisting of a serial data input (SDI), a serial data output (SDO), a serial clock line (SCLK), and a chip select (CSB). In SPI mode, the SDI/SDA pin operates as SDI and the SCLK/SCL pin operates as SCLK. The timing of the SPI interface is shown in Figure 51. Data is strobed on the SCLK rising edge while clocked out on the SCLK falling edge. All single-word SPI read and write operations are done in a 3-byte, 24-clock-cycle SPI instruction framed by a CSB low interval. The content of the SPI operation consists of a

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one-byte register address (A[7:0]) followed by a one-byte command word, which defines the transaction as write or read, followed by a single-byte data word either written to or read from the register location provided in the first byte.

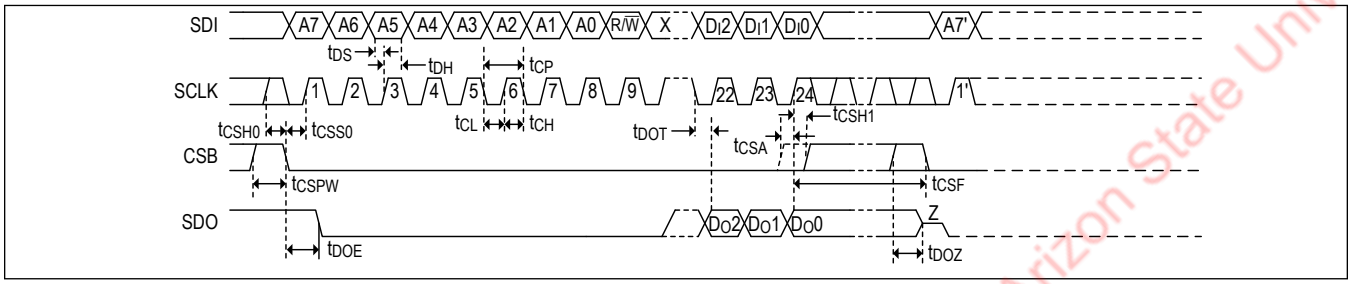



*Figure 51. Detailed SPI Timing Diagram*

### Single-Word SPI Register Read and Write Transactions

SPI write mode operations for MAX86178 are executed on the 24th SCLK rising edge using the first three bytes of data available. In write mode, any data supplied after the 24th SCLK rising edge is ignored as shown in Figure 52. Subsequent writes require CSB to deassert high and then assert low for the next write command. A rising CSB edge preceding the 24th rising edge of SCLK by $t_{CSA}$ as shown in Figure 51, results in the transaction being aborted.

Read mode operations access the requested data on the 16th SCLK rising edge, and present the MSB of the requested data on the following SCLK falling edge, allowing the microcontroller to latch the data MSB on the 17th SCLK rising edge as shown in Figure 53. Configuration and status registers are available using normal-mode read-back sequences. FIFO reads must be done with a burst mode FIFO read (see the *SPI Burst Mode Read Transaction* section). In a normal read sequence, any SCLK rising edges after the 24th SCLK rising edge are ignored and if more than 24 SCLK rising edges are provided, the device reads back zeros.

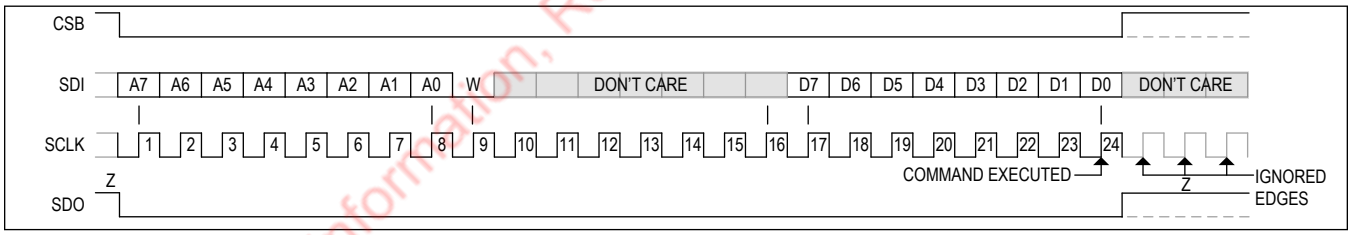



*Figure 52. SPI Write Transaction*

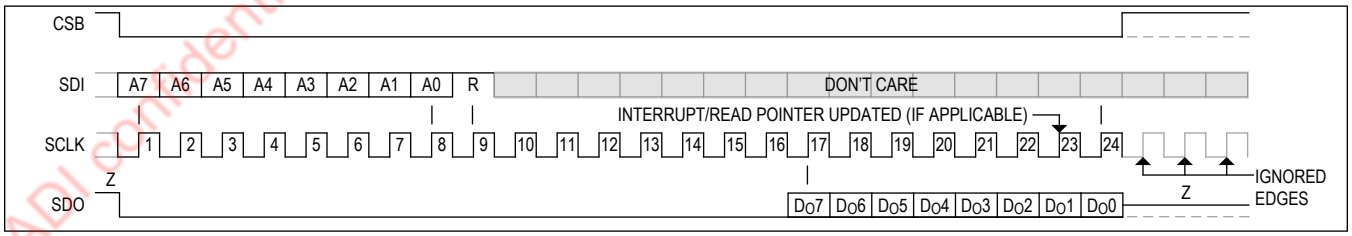



*Figure 53. SPI Read Transaction*

### SPI Burst Mode Read Transaction

The MAX86178 has a FIFO burst-read mode to increase data transfer efficiency. The first 16 SCLK cycles operate exactly as described for the normal read mode, where the first byte is the register address and the second is the read command. The subsequent SCLKs consist of FIFO data, 24 SCLKs per word. All words in the FIFO should be read with

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a single FIFO burst-read command.

Each FIFO sample consists of 3 bytes per sample, and thus, requires 24 SCLKs per sample to read out. The first byte (SCLK 17 to 24) consists of a tag indicating the data type of the subsequent bits as well as the MSBs of the data. The next two bytes (SCLK 24 to 40) consist of data. For example, Figure 54 shows a FIFO burst read consisting of three PPG samples in FIFO, labeled A through C, each with a 4-bit tag and 20-bit data. The number of words in the FIFO depends on the FIFO configuration. See the *FIFO Description* section for more details about the FIFO configuration and readout.

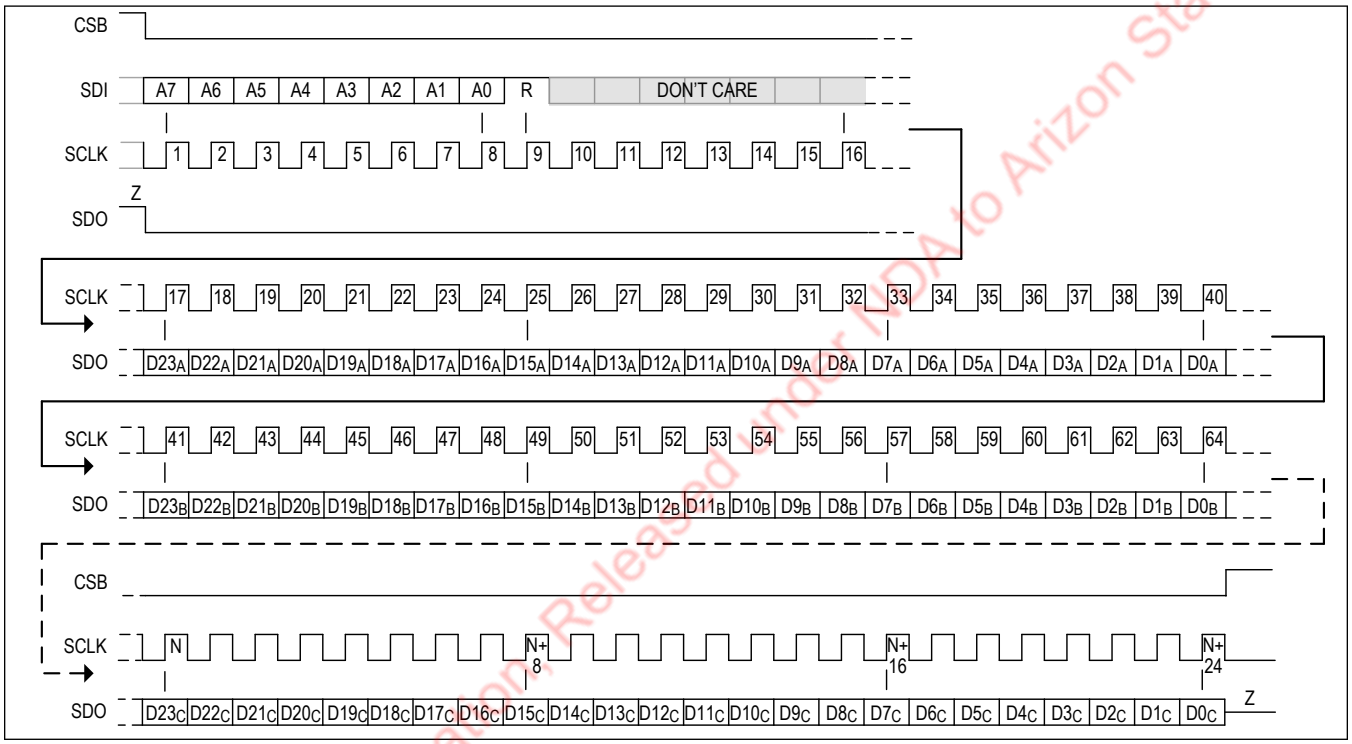



*Figure 54. SPI FIFO Burst Mode Read Transaction*

### I²C-/SMBus-Compatible Serial Interface

The I²C interface on the MAX86178 is an I²C-/SMBus-compatible, 2-wire serial interface consisting of a serial data line (SDA) and a serial clock line (SCL). In I²C mode, the SDI/SDA pin operates as SDA and the SCLK/SCL pin operates as SCL. These two pins are used for the communication between the MAX86178 and the controller at clock rates up to 400kHz. Figure 55 shows the 2-wire interface timing diagram. The controller generates SCL and initiates data transfer on the bus. The controller device writes data to the MAX86178 by transmitting the proper target address followed by the register address and then the data word. Each transmit sequence is framed by a START (S) or REPEATED START (Sr) condition and a STOP (P) condition. Each word transmitted to the MAX86178 is 8-bits long and is followed by an acknowledge clock pulse. A controller reading data from the MAX86178 transmits the proper target address followed by a series of nine SCL pulses. The MAX86178 transmits data on SDA in sync with the controller-generated SCL pulses. The controller acknowledges receipt of each byte of data. Each read sequence is framed by a START (S) or REPEATED START (Sr) condition, a not acknowledge (NACK), and a STOP (P) condition. SDA operates as both an input and an open-drain output. A pullup resistor is required on SDA. SCL operates only as an input. A pullup resistor is required on SCL if there are multiple controllers on the bus, or if the single controller has an open-drain SCL output. Series resistors in line with SDA and SCL are optional. Series resistors protect the digital inputs from high-voltage spikes on the bus lines, and minimize crosstalk and undershoot of the bus signals.

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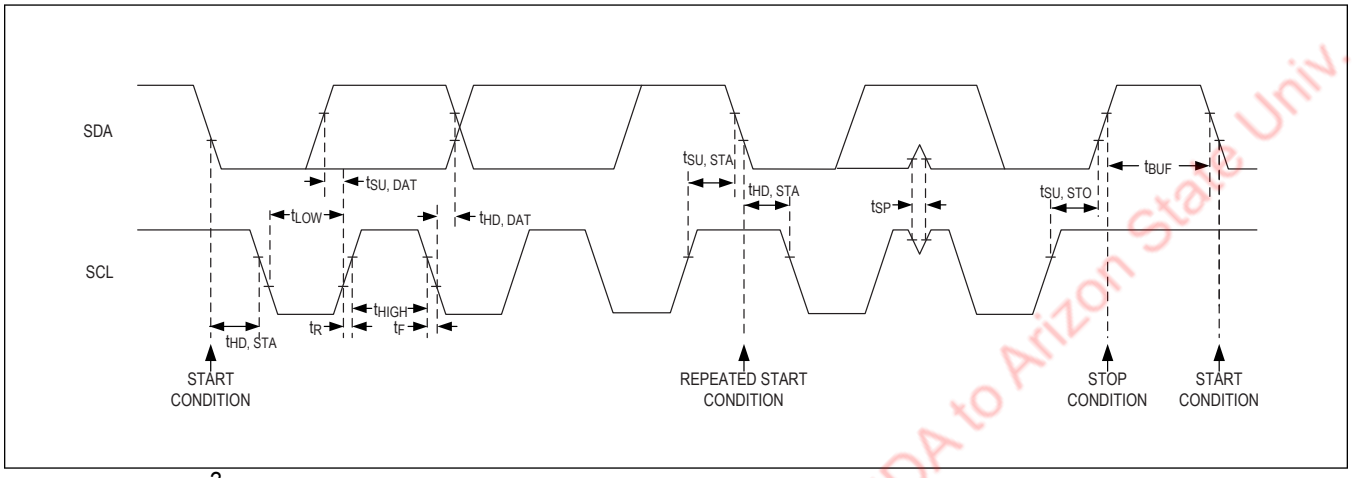



*Figure 55. Detailed I²C Timing Diagram*

### Bit Transfer

One data bit is transferred during each SCL cycle. The data on SDA must remain stable during the high period of the SCL pulse. Changes in SDA while SCL is high are control signals (see the *START and STOP Conditions* section).

### START and STOP Conditions

SDA and SCL idle high when the bus is not in use. A controller initiates communication by issuing a START condition, which indicates the beginning of a transmission to the MAX86178. A START condition is a high-to-low transition on SDA while SCL is high as shown in Figure 56. The controller terminates transmission, and frees the bus, by issuing a STOP condition. A STOP condition is a low-to-high transition on SDA while SCL is high as shown in Figure 56. The bus remains active if a REPEATED START condition is generated instead of a STOP condition. A REPEATED START condition is the same as a START condition (high-to-low transition with SCL high), but it is sent after a START condition.

The MAX86178 recognizes a STOP condition at any point during data transmission except if the STOP condition occurs in the same high pulse as a START condition. For proper operation, do not send a STOP condition during the same SCL high pulse as the START condition.

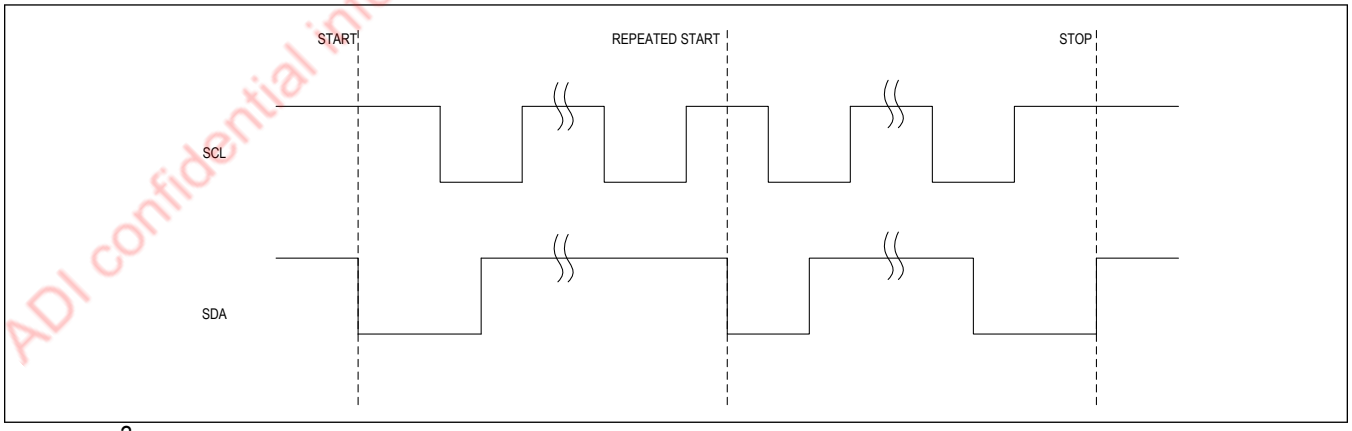



*Figure 56. I²C START, STOP, and REPEATED START Conditions*

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### I²C Target Address

In I²C mode, the SDO/ADDR pin acts as the device address selector pin. The I²C target address has two values, selected by SDO/ADDR when the I2C_SEL pin is pulled high. When ADDR is pulled low, either by a pulldown resistor or by the host controller, the target address is 0xD8 (write) and 0xD9 (read), or 0b1101100 + R/$\overline{W}$. When ADDR is pulled high, the target address is 0xDA (write) and 0xDB (read), or 0b1101101 + R/$\overline{W}$.

## Table 25. I²C Addresses for MAX86178

| ADDR PIN | WRITE ADDRESS | READ ADDRESS |
|---|---|---|
| Low | 0xD8 | 0xD9 |
| High | 0xDA | 0xDB |

### Acknowledge Bit

The acknowledge bit (ACK) is a clocked 9th bit that the MAX86178 uses to handshake-receipt each byte of data when in write mode as shown in Figure 57. The MAX86178 pulls down SDA during the entire controller-generated 9th clock pulse if the previous byte is successfully received. Monitoring ACK allows for detection of unsuccessful data transfers. An unsuccessful data transfer occurs if a receiving device is busy or if a system fault has occurred. In the event of an unsuccessful data transfer, the bus controller retries communication. The controller pulls down SDA during the 9th clock cycle to acknowledge receipt of data when the MAX86178 is in read mode. An acknowledge is sent by the controller after each read byte to allow data transfer to continue. A not-acknowledge is sent when the controller reads the final byte of data from the MAX86178, followed by a STOP condition.

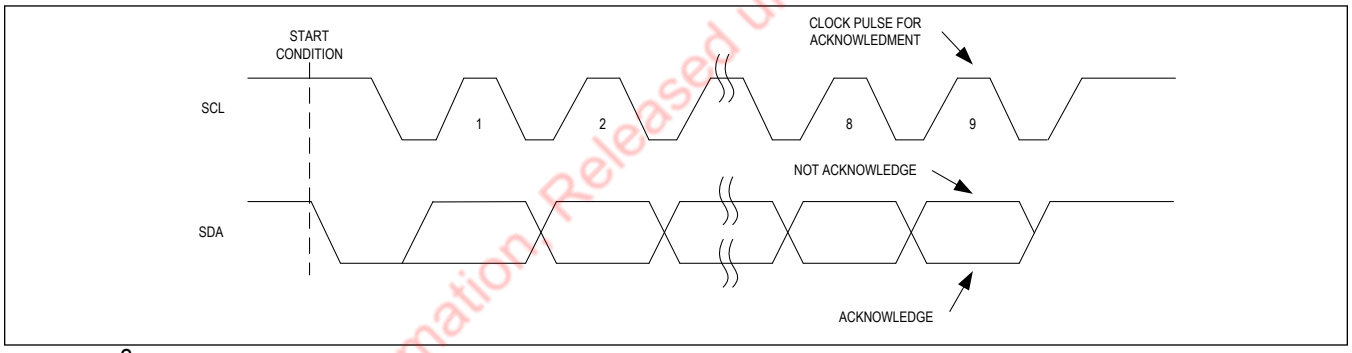



*Figure 57. I²C Acknowledge Bit*

### I²C Write Data Format

A write to the MAX86178 includes transmission of a START condition, the target address with the R/$\overline{W}$ bit set to 0, one byte of data to configure the internal register-address pointer, one or more bytes of data, and a STOP condition. Figure 58 illustrates the proper frame format for writing one byte of data to the MAX86178. Figure 59 illustrates the frame format for writing multiple bytes of data to the MAX86178.

The target address with the R/$\overline{W}$ bit set to 0 indicates that the controller intends to write data to the MAX86178. The device acknowledges receipt of the address byte during the controller-generated 9th-SCL pulse.

The second byte transmitted from the controller configures the internal register-address pointer of the MAX86178. The pointer tells the device where to write the next byte of data. An acknowledge pulse is sent by the MAX86178 upon receipt of the address-pointer data.

The third byte sent to the MAX86178 contains the data to be written to the pointed register. An acknowledge pulse from the MAX86178 signals receipt of the data byte. The address pointer auto increments to the next register address after each received data byte. This auto-increment feature allows a controller to write to sequential registers within one continuous frame. The controller signals the end of transmission by issuing a STOP condition. The auto-increment feature is disabled when there is an attempt to write to the FIFO Data register(0x0C).

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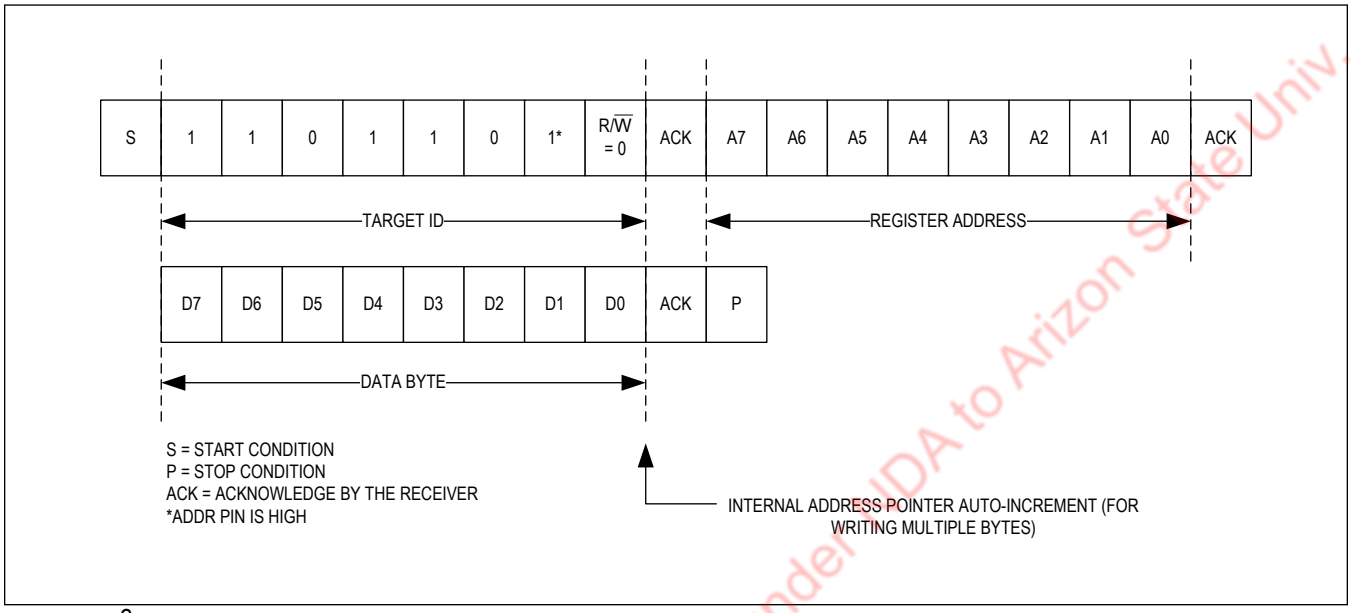



*Figure 58. I²C Single-Byte Write Transaction*

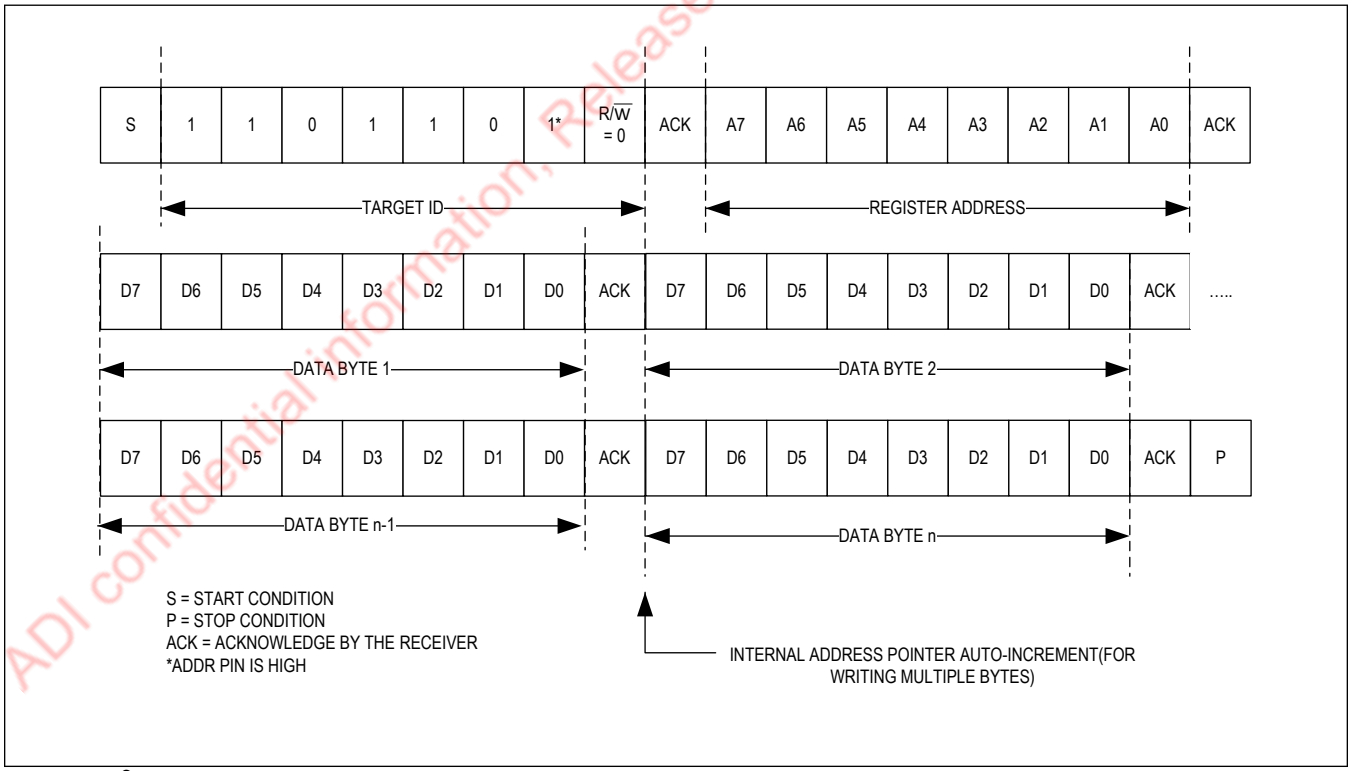



*Figure 59. I²C Multi-Byte Write Transaction*

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### I²C Read Data Format

A read from the MAX86178 includes sending the target address with the R/$\overline{W}$ bit set to 1 to initiate a read operation. The MAX86178 acknowledges receipt of the target address by pulling SDA low during the 9th SCL clock pulse. A START command followed by a read command resets the address pointer to register 0x00.

The first byte transmitted from the MAX86178 is the content of register 0x00. Transmitted data is valid on the rising edge of SCL. The address pointer auto-increments after each read data byte. This auto-increment feature allows all registers to be read sequentially within one continuous frame. The auto-increment feature is disabled when there is an attempt to read from the FIFO Data register (0x0C). A STOP condition can be issued after any number of read data bytes. If a STOP condition is issued followed by another read operation, the first data byte to be read is from register 0x00.

The address pointer can be preset to a specific register before a read command is issued. The controller presets the address pointer by first sending the MAX86178 target address with the R/$\overline{W}$ bit set to 0 followed by the register address. A REPEATED START condition is then sent followed by the target address with the R/$\overline{W}$ bit set to 1. The MAX86178 then transmits the content of the specified register. The address pointer auto-increments after transmitting the first byte.

The controller acknowledges receipt of each read byte during the acknowledge clock pulse. The controller must acknowledge all correctly received bytes except the last byte. The final byte must be followed by a not acknowledge from the controller and then a STOP condition. Figure 60 illustrates the frame format for reading one byte from the MAX86178. Figure 61 illustrates the frame format for reading multiple bytes from the MAX86178.

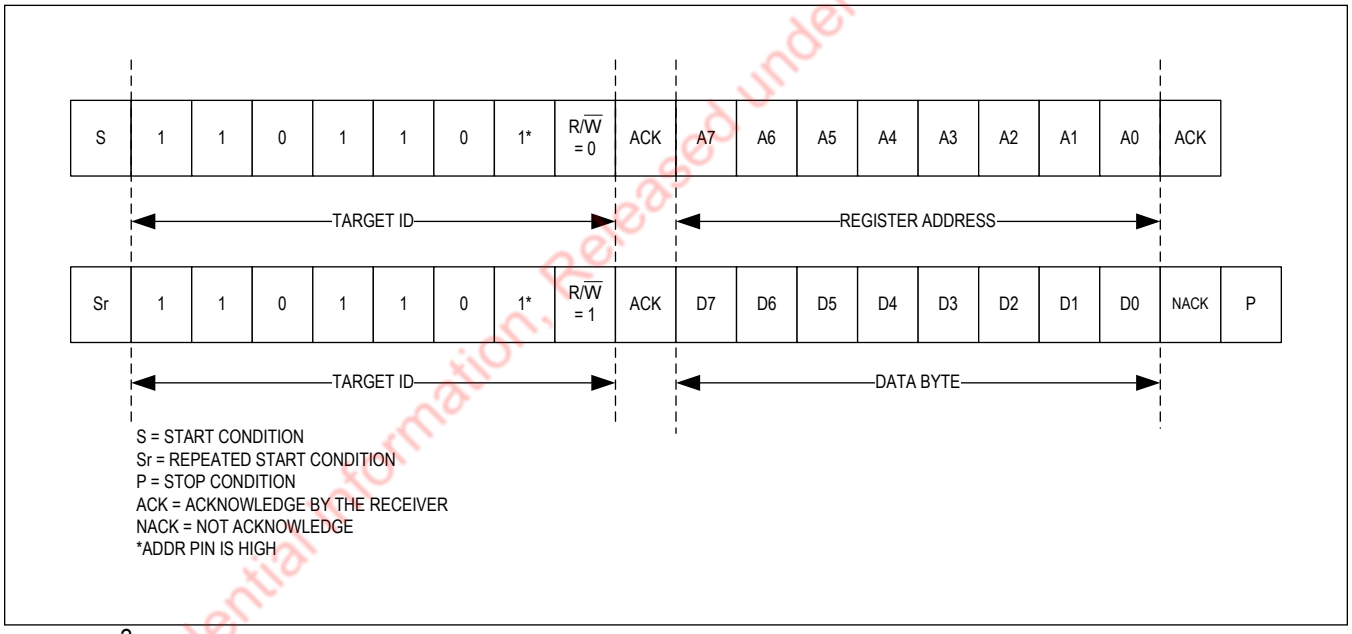



*Figure 60. I²C Single-Byte Read Transaction*

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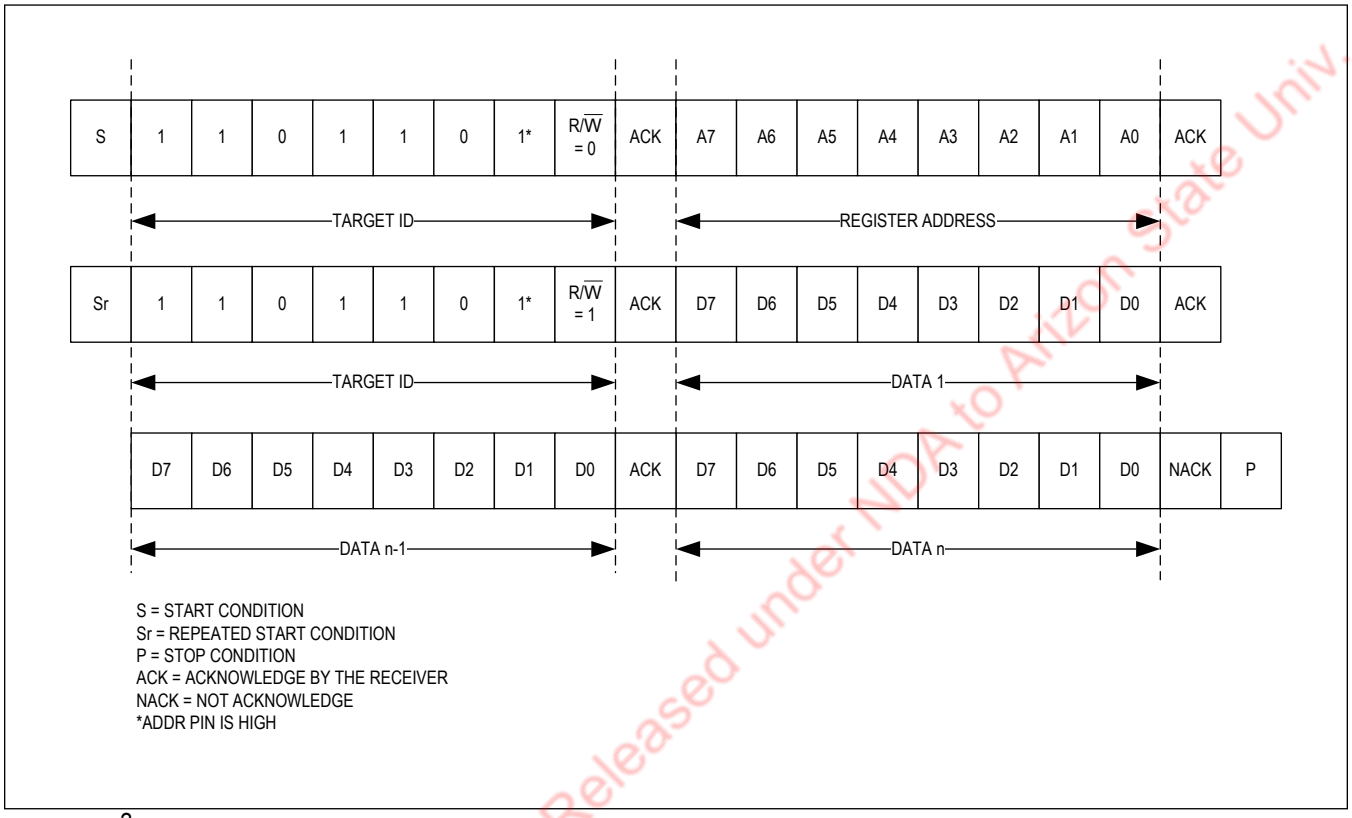



*Figure 61. I²C Multi-Byte Read Transaction*

### I²C Broadcast

The MAX86178 provides a feature of I²C broadcast write transactions to multiple devices simultaneously using the I²C serial interface. The host microcontroller uses the address programmed in I2C_BCAST_ADDR[7:1](0x15) to send a write command to multiple devices and the target devices respond with an ACK. To use the broadcast feature, I2C_BCAST_EN[0[(0x15) must be set to 1.

This feature is especially useful for:

1. Synchronizing PLLs on multiple devices using the TIMING_SYS_RESET[7](0x10) bit; thereby, avoiding any external connections.

2. Programming the same configuration to multiple devices at the same time.

Read transactions in broadcast mode are not supported. If a host sends out a read command using the I²C broadcast address, the device responds with a NACK.

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## Register Map

### User Register Map

| ADDRESS | NAME | MSB | | | | | | | LSB |
|---|---|---|---|---|---|---|---|---|---|
| **Status** | | | | | | | | | |
| 0x00 | Status 1[7:0] | A_FULL | PPG_FRAME_RDY | FIFO_DATA_RDY | ALC_OVF | EXP_OVF | PPG_THRESH2_HILO | PPG_THRESH1_HILO | PWR_RDY |
| 0x01 | Status 2[7:0] | INVALID_PPG_CFG | – | LED6_COMPB | LED5_COMPB | LED4_COMPB | LED3_COMPB | LED2_COMPB | LED1_COMPB |
| 0x02 | Status 3[7:0] | – | – | – | FREQ_UNLOCK | FREQ_LOCK | PHASE_UNLOCK | PHASE_LOCK | – |
| 0x03 | Status 4[7:0] | ECG_LON | – | ECG_FAST_REC | RLD_OOR | ECG_LOFF_PH | ECG_LOFF_PL | ECG_LOFF_NH | ECG_LOFF_NL |
| 0x04 | Status 5[7:0] | BIOZ_LON | BIOZ_OVER | BIOZ_UNDR | BIOZ_DRV_OOR | BIOZ_LOFF_PH | BIOZ_LOFF_PL | BIOZ_LOFF_NH | BIOZ_LOFF_NL |
| **FIFO** | | | | | | | | | |
| 0x08 | FIFO Write Pointer[7:0] | FIFO_WR_PTR[7:0] | | | | | | | |
| 0x09 | FIFO Read Pointer[7:0] | FIFO_RD_PTR[7:0] | | | | | | | |
| 0x0A | FIFO Counter 1[7:0] | FIFO_DATA_COUNT[8] | OVF_COUNTER[6:0] | | | | | | |
| 0x0B | FIFO Counter 2[7:0] | FIFO_DATA_COUNT[7:0] | | | | | | | |
| 0x0C | FIFO Data Register[7:0] | FIFO_DATA[7:0] | | | | | | | |
| 0x0D | FIFO Configuration 1[7:0] | FIFO_A_FULL[7:0] | | | | | | | |
| 0x0E | FIFO Configuration 2[7:0] | – | – | FIFO_MARK | FLUSH_FIFO | FIFO_STAT_CLR | A_FULL_TYPE | FIFO_RO | – |
| **System Control** | | | | | | | | | |
| 0x10 | System Sync[7:0] | TIMING_SYS_RESET | – | – | – | – | – | – | – |
| 0x11 | System Configuration 1[7:0] | – | DISABLE_I2C | ECG_PPG_TIMING_DATA | BIOZ_PPG_TIMING_DATA | ECG_BIOZ_TIMING_DATA | – | SHDN | RESET |
| 0x12 | System Configuration 2[7:0] | BYP_DLY | – | – | ECG_SAMP_SYNC_FREQ[4:0] | | | | |
| 0x13 | Pin Functional Configuration[7:0] | TRIG_FCFG[2:0] | | | TRIG_ICFG | INT2_FCFG[1:0] | | INT1_FCFG[1:0] | |
| 0x14 | Output Pin Configuration[7:0] | TRIG_OCFG[1:0] | | – | – | INT2_OCFG[1:0] | | INT1_OCFG[1:0] | |
| 0x15 | I2C Broadcast Address[7:0] | I2C_BCAST_ADDR[6:0] | | | | | | | I2C_BCAST_EN |

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| **PLL** | | | | | | | | | |
| 0x18 | PLL Configuration 1[7:0] | MDIV[9:8] | | – | – | – | – | PLL_LOCK_WNDW | PLL_EN |
| 0x19 | PLL Configuration 2[7:0] | MDIV[7:0] | | | | | | | |
| 0x1A | PLL Configuration 3[7:0] | BIOZ_NDIV[1:0] | | – | – | BIOZ_KDIV[3:0] | | | |
| 0x1B | PLL Configuration 4[7:0] | ECG_NDIV[10:8] | | | – | – | ECG_FDIV[2:0] | | |
| 0x1C | PLL Configuration 5[7:0] | ECG_NDIV[7:0] | | | | | | | |
| 0x1D | PLL Configuration 6[7:0] | – | REF_CLK_SEL | CLK_FREQ_SEL | CLK_FINE_TUNE[4:0] | | | | |
| **PPG Setup** | | | | | | | | | |
| 0x20 | PPG Configuration 1[7:0] | – | – | MEAS6_EN | MEAS5_EN | MEAS4_EN | MEAS3_EN | MEAS2_EN | MEAS1_EN |
| 0x21 | PPG Configuration 2[7:0] | – | – | PPG_SYNC_MODE | – | PPG2_PWRDN | PPG1_PWRDN | – | – |
| 0x22 | PPG Configuration 3[7:0] | – | SMP_AVE[2:0] | | | ALC_DISABLE | – | COLLECT_RAW_DATA | MEAS1_CONFIG_SEL |
| 0x23 | PPG Configuration 4[7:0] | – | – | – | PROX_DATA_EN | PROX_AUTO | – | – | – |
| 0x24 | Photodiode Bias[7:0] | PD4_BIAS[1:0] | | PD3_BIAS[1:0] | | PD2_BIAS[1:0] | | PD1_BIAS[1:0] | |
| **PPG Frame Rate Clock** | | | | | | | | | |
| 0x28 | FR Clock Divider MSB[7:0] | – | FR_CLK_DIV[14:8] | | | | | | |
| 0x29 | FR Clock Divider LSB[7:0] | FR_CLK_DIV[7:0] | | | | | | | |
| **PPG MEAS1 Setup** | | | | | | | | | |
| 0x30 | MEAS1 Selects[7:0] | – | MEAS1_AMB | MEAS1_DRVB[2:0] | | | MEAS1_DRVA[2:0] | | |
| 0x31 | MEAS1 Configuration 1[7:0] | MEAS1_SINC3_SEL | MEAS1_FILT2_SEL | MEAS1_FILT_SEL | MEAS1_TINT[1:0] | | MEAS1_AVER[2:0] | | |
| 0x32 | MEAS1 Configuration 2[7:0] | – | – | MEAS1_PPG2_ADC_RGE[1:0] | | – | – | MEAS1_PPG1_ADC_RGE[1:0] | |
| 0x33 | MEAS1 Configuration 3[7:0] | MEAS1_PPG2_DACOFF[3:0] | | | | MEAS1_PPG1_DACOFF[3:0] | | | |
| 0x34 | MEAS1 Configuration 4[7:0] | MEAS1_PD_SETLNG[1:0] | | – | – | MEAS1_LED_SETLNG[1:0] | | MEAS1_LED_RGE[1:0] | |
| 0x35 | MEAS1 Configuration 5[7:0] | MEAS1_PD4_SEL[1:0] | | MEAS1_PD3_SEL[1:0] | | MEAS1_PD2_SEL[1:0] | | MEAS1_PD1_SEL[1:0] | |
| 0x36 | MEAS1 LEDA Current[7:0] | MEAS1_DRVA_PA[7:0] | | | | | | | |
| 0x37 | MEAS1 LEDB Current[7:0] | MEAS1_DRVB_PA[7:0] | | | | | | | |
| **PPG MEAS2 Setup** | | | | | | | | | |
| 0x38 | MEAS2 Selects[7:0] | – | MEAS2_AMB | MEAS2_DRVB[2:0] | | | MEAS2_DRVA[2:0] | | |

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| 0x39 | MEAS2 Configuration 1[7:0] | MEAS2_SINC3_SEL | MEAS2_FILT2_SEL | MEAS2_FILT_SEL | MEAS2_TINT[1:0] | | MEAS2_AVER[2:0] | | |
| 0x3A | MEAS2 Configuration 2[7:0] | – | – | MEAS2_PPG2_ADC_RGE[1:0] | | – | – | MEAS2_PPG1_ADC_RGE[1:0] | |
| 0x3B | MEAS2 Configuration 3[7:0] | MEAS2_PPG2_DACOFF[3:0] | | | | MEAS2_PPG1_DACOFF[3:0] | | | |
| 0x3C | MEAS2 Configuration 4[7:0] | MEAS2_PD_SETLNG[1:0] | | – | – | MEAS2_LED_SETLNG[1:0] | | MEAS2_LED_RGE[1:0] | |
| 0x3D | MEAS2 Configuration 5[7:0] | MEAS2_PD4_SEL[1:0] | | MEAS2_PD3_SEL[1:0] | | MEAS2_PD2_SEL[1:0] | | MEAS2_PD1_SEL[1:0] | |
| 0x3E | MEAS2 LEDA Current[7:0] | MEAS2_DRVA_PA[7:0] | | | | | | | |
| 0x3F | MEAS2 LEDB Current[7:0] | MEAS2_DRVB_PA[7:0] | | | | | | | |
| **PPG MEAS3 Setup** | | | | | | | | | |
| 0x40 | MEAS3 Selects[7:0] | – | MEAS3_AMB | MEAS3_DRVB[2:0] | | | MEAS3_DRVA[2:0] | | |
| 0x41 | MEAS3 Configuration 1[7:0] | MEAS3_SINC3_SEL | MEAS3_FILT2_SEL | MEAS3_FILT_SEL | MEAS3_TINT[1:0] | | MEAS3_AVER[2:0] | | |
| 0x42 | MEAS3 Configuration 2[7:0] | – | – | MEAS3_PPG2_ADC_RGE[1:0] | | – | – | MEAS3_PPG1_ADC_RGE[1:0] | |
| 0x43 | MEAS3 Configuration 3[7:0] | MEAS3_PPG2_DACOFF[3:0] | | | | MEAS3_PPG1_DACOFF[3:0] | | | |
| 0x44 | MEAS3 Configuration 4[7:0] | MEAS3_PD_SETLNG[1:0] | | – | – | MEAS3_LED_SETLNG[1:0] | | MEAS3_LED_RGE[1:0] | |
| 0x45 | MEAS3 Configuration 5[7:0] | MEAS3_PD4_SEL[1:0] | | MEAS3_PD3_SEL[1:0] | | MEAS3_PD2_SEL[1:0] | | MEAS3_PD1_SEL[1:0] | |
| 0x46 | MEAS3 LEDA Current[7:0] | MEAS3_DRVA_PA[7:0] | | | | | | | |
| 0x47 | MEAS3 LEDB Current[7:0] | MEAS3_DRVB_PA[7:0] | | | | | | | |
| **PPG MEAS4 Setup** | | | | | | | | | |
| 0x48 | MEAS4 Selects[7:0] | – | MEAS4_AMB | MEAS4_DRVB[2:0] | | | MEAS4_DRVA[2:0] | | |
| 0x49 | MEAS4 Configuration 1[7:0] | MEAS4_SINC3_SEL | MEAS4_FILT2_SEL | MEAS4_FILT_SEL | MEAS4_TINT[1:0] | | MEAS4_AVER[2:0] | | |
| 0x4A | MEAS4 Configuration 2[7:0] | – | – | MEAS4_PPG2_ADC_RGE[1:0] | | – | – | MEAS4_PPG1_ADC_RGE[1:0] | |
| 0x4B | MEAS4 Configuration 3[7:0] | MEAS4_PPG2_DACOFF[3:0] | | | | MEAS4_PPG1_DACOFF[3:0] | | | |
| 0x4C | MEAS4 Configuration 4[7:0] | MEAS4_PD_SETLNG[1:0] | | – | – | MEAS4_LED_SETLNG[1:0] | | MEAS4_LED_RGE[1:0] | |
| 0x4D | MEAS4 Configuration 5[7:0] | MEAS4_PD4_SEL[1:0] | | MEAS4_PD3_SEL[1:0] | | MEAS4_PD2_SEL[1:0] | | MEAS4_PD1_SEL[1:0] | |
| 0x4E | MEAS4 LEDA Current[7:0] | MEAS4_DRVA_PA[7:0] | | | | | | | |

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| 0x4F | MEAS4 LEDB Current[7:0] | MEAS4_DRVB_PA[7:0] | | | | | | | |
| **PPG MEAS5 Setup** | | | | | | | | | |
| 0x50 | MEAS5 Selects[7:0] | – | MEAS5_AMB | MEAS5_DRVB[2:0] | | | MEAS5_DRVA[2:0] | | |
| 0x51 | MEAS5 Configuration 1[7:0] | MEAS5_SINC3_SEL | MEAS5_FILT2_SEL | MEAS5_FILT_SEL | MEAS5_TINT[1:0] | | MEAS5_AVER[2:0] | | |
| 0x52 | MEAS5 Configuration 2[7:0] | – | – | MEAS5_PPG2_ADC_RGE[1:0] | | – | – | MEAS5_PPG1_ADC_RGE[1:0] | |
| 0x53 | MEAS5 Configuration 3[7:0] | MEAS5_PPG2_DACOFF[3:0] | | | | MEAS5_PPG1_DACOFF[3:0] | | | |
| 0x54 | MEAS5 Configuration 4[7:0] | MEAS5_PD_SETLNG[1:0] | | – | – | MEAS5_LED_SETLNG[1:0] | | MEAS5_LED_RGE[1:0] | |
| 0x55 | MEAS5 Configuration 5[7:0] | MEAS5_PD4_SEL[1:0] | | MEAS5_PD3_SEL[1:0] | | MEAS5_PD2_SEL[1:0] | | MEAS5_PD1_SEL[1:0] | |
| 0x56 | MEAS5 LEDA Current[7:0] | MEAS5_DRVA_PA[7:0] | | | | | | | |
| 0x57 | MEAS5 LEDB Current[7:0] | MEAS5_DRVB_PA[7:0] | | | | | | | |
| **PPG MEAS6 Setup** | | | | | | | | | |
| 0x58 | MEAS6 Selects[7:0] | – | MEAS6_AMB | MEAS6_DRVB[2:0] | | | MEAS6_DRVA[2:0] | | |
| 0x59 | MEAS6 Configuration 1[7:0] | MEAS6_SINC3_SEL | MEAS6_FILT2_SEL | MEAS6_FILT_SEL | MEAS6_TINT[1:0] | | MEAS6_AVER[2:0] | | |
| 0x5A | MEAS6 Configuration 2[7:0] | – | – | MEAS6_PPG2_ADC_RGE[1:0] | | – | – | MEAS6_PPG1_ADC_RGE[1:0] | |
| 0x5B | MEAS6 Configuration 3[7:0] | MEAS6_PPG2_DACOFF[3:0] | | | | MEAS6_PPG1_DACOFF[3:0] | | | |
| 0x5C | MEAS6 Configuration 4[7:0] | MEAS6_PD_SETLNG[1:0] | | – | – | MEAS6_LED_SETLNG[1:0] | | MEAS6_LED_RGE[1:0] | |
| 0x5D | MEAS6 Configuration 5[7:0] | MEAS6_PD4_SEL[1:0] | | MEAS6_PD3_SEL[1:0] | | MEAS6_PD2_SEL[1:0] | | MEAS6_PD1_SEL[1:0] | |
| 0x5E | MEAS6 LEDA Current[7:0] | MEAS6_DRVA_PA[7:0] | | | | | | | |
| 0x5F | MEAS6 LEDB Current[7:0] | MEAS6_DRVB_PA[7:0] | | | | | | | |
| **PPG Threshold Interrupts** | | | | | | | | | |
| 0x70 | THRESHOLD MEAS SEL[7:0] | – | THRESH2_MEAS_SEL[2:0] | | | – | THRESH1_MEAS_SEL[2:0] | | |
| 0x71 | THRESHOLD HYST[7:0] | THRESH2_PPG_SEL | THRESH1_PPG_SEL | – | TIME_HYST[1:0] | | LEVEL_HYST[2:0] | | |
| 0x72 | PPG HI THRESHOLD1[7:0] | THRESHOLD1_UPPER[7:0] | | | | | | | |
| 0x73 | PPG LO THRESHOLD1[7:0] | THRESHOLD1_LOWER[7:0] | | | | | | | |
| 0x74 | PPG HI THRESHOLD2[7:0] | THRESHOLD2_UPPER[7:0] | | | | | | | |

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| 0x75 | PPG LO THRESHOLD2[7:0] | THRESHOLD2_LOWER[7:0] | | | | | | | |
| **ECG Setup** | | | | | | | | | |
| 0x80 | ECG Configuration 1[7:0] | – | – | – | – | ECG_DEC_RATE[2:0] | | | ECG_EN |
| 0x81 | ECG Configuration 2[7:0] | ECG_IPOL | ECG_PGA_GAIN[2:0] | | | ECG_INA_RGE[1:0] | | ECG_INA_GAIN[1:0] | |
| 0x82 | ECG Configuration 3[7:0] | – | – | – | – | ECG_IMP_HI | ECG_AUTO_REC | ECG_MUX_SEL[1:0] | |
| 0x83 | ECG Configuration 4[7:0] | EN_ECG_FAST_REC[1:0] | ECG_FAST_REC_THRESHOLD[5:0] | | | | | | |
| **ECG Calibration** | | | | | | | | | |
| 0x84 | ECG CAL Configuration 1[7:0] | ECG_CAL_HIGH[10:8] | | | ECG_CAL_FREQ[2:0] | | | ECG_CAL_DUTY | ECG_CAL_EN |
| 0x85 | ECG CAL Configuration 2[7:0] | ECG_CAL_HIGH[7:0] | | | | | | | |
| 0x86 | ECG CAL Configuration 3[7:0] | ECG_OPEN_P | ECG_OPEN_N | ECG_CAL_MODE | ECG_CAL_MAG | ECG_CAL_P_SEL[1:0] | | ECG_CAL_N_SEL[1:0] | |
| **ECG Lead Detect** | | | | | | | | | |
| 0x88 | ECG Lead Detect Configuration 1[7:0] | EN_ECG_LON | EN_ECG_LOFF | – | – | ECG_LOFF_MODE | ECG_LOFF_FREQ[2:0] | | |
| 0x89 | ECG Lead Detect Configuration 2[7:0] | ECG_LOFF_IPOL | ECG_LOFF_IMAG[2:0] | | | ECG_LOFF_THRESH[3:0] | | | |
| **ECG Lead Bias** | | | | | | | | | |
| 0x90 | ECG Lead Bias Configuration 1[7:0] | – | – | – | – | ECG_RBIAS_VALUE[1:0] | | EN_ECG_RBIASP | EN_ECG_RBIASN |
| **ECG RLD and CM Amps** | | | | | | | | | |
| 0x92 | RLD Configuration 1[7:0] | RLD_EN | RLD_MODE | RLD_RBIAS | EN_RLD_OOR | ACTV_CM_P | ACTV_CM_N | RLD_GAIN[1:0] | |
| 0x93 | RLD Configuration 2[7:0] | RLD_EXT_RES | RLD_SEL_ECG | RLD_BW[1:0] | | BODY_BIAS_DAC[3:0] | | | |
| **BIOZ Setup** | | | | | | | | | |
| 0xA0 | BIOZ Configuration 1[7:0] | BIOZ_DAC_OSR[1:0] | | BIOZ_ADC_OSR[2:0] | | | ECG_BIOZ_BG_EN | BIOZ_EN[1:0] | |
| 0xA1 | BIOZ Configuration 2[7:0] | BIOZ_DHPF[1:0] | | BIOZ_DLPF[2:0] | | | – | – | EN_BIOZ_THRESH |
| 0xA2 | BIOZ Configuration 3[7:0] | BIOZ_EXT_RES | – | BIOZ_VDRV_MAG[1:0] | | BIOZ_IDRV_RGE[1:0] | | BIOZ_DRV_MODE[1:0] | |
| 0xA3 | BIOZ Configuration 4[7:0] | EN_UTIL_MODE | – | – | – | – | – | – | – |
| 0xA4 | BIOZ Configuration 5[7:0] | BIOZ_DC_CODE_SEL | BIOZ_DC_DAC_CODE[6:0] | | | | | | |
| 0xA5 | BIOZ Configuration 6[7:0] | BIOZ_AHPF[3:0] | | | | BIOZ_INA_MODE | BIOZ_DM_DIS | BIOZ_GAIN[1:0] | |

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| 0xA6 | BIOZ Configuration 7[7:0] | BIOZ_EXT_CAP | BIOZ_DC_RESTORE | BIOZ_DRV_RESET | BIOZ_DAC_RESET | BIOZ_AMP_RGE[1:0] | | BIOZ_AMP_BW[1:0] | |
| 0xA7 | BIOZ Configuration 8[7:0] | RLD_SEL_BIOZ | RLD_DRV | BIOZ_CMRES_DIS | BIOZ_STBYON | BIOZ_IPOL | BIOZ_FAST | BIOZ_INA_CHOP_EN | BIOZ_CH_FSEL |
| 0xA8 | BIOZ Low Threshold[7:0] | BIOZ_LO_THRESH[7:0] | | | | | | | |
| 0xA9 | BIOZ High Threshold[7:0] | BIOZ_HI_THRESH[7:0] | | | | | | | |
| **BIOZ Calibration** | | | | | | | | | |
| 0xAA | BIOZ Mux Configuration 1[7:0] | BMUX_RSEL[1:0] | | BMUX_BIST_EN | – | – | CONNECT_CAL_ONLY | BIOZ_MUX_EN | BIOZ_CAL_EN |
| 0xAB | BIOZ Mux Configuration 2[7:0] | BMUX_GSR_RSEL[1:0] | | GSR_LOAD_EN | – | – | – | EN_EXT_INLOAD | EN_INT_INLOAD |
| 0xAC | BIOZ Mux Configuration 3[7:0] | BIP_ASSIGN[1:0] | | BIN_ASSIGN[1:0] | | DRVP_ASSIGN[1:0] | | DRVN_ASSIGN[1:0] | |
| 0xAD | BIOZ Mux Configuration 4[7:0] | BIST_R_ERR[7:0] | | | | | | | |
| **BIOZ Lead Detect** | | | | | | | | | |
| 0xB0 | BIOZ Lead Detect Configuration 1[7:0] | EN_BIOZ_LON | EN_BIOZ_LOFF | EN_EXT_BIOZ_LOFF | EN_BIOZ_DRV_OOR | BIOZ_LOFF_IPOL | BIOZ_LOFF_IMAG[2:0] | | |
| 0xB1 | BIOZ Lead-Off Threshold[7:0] | RESP_CG_MAG_4X | RESP_CG_MAG[2:0] | | | BIOZ_LOFF_THRESH[3:0] | | | |
| **BIOZ Lead Bias** | | | | | | | | | |
| 0xB4 | BIOZ Lead Bias Configuration 1[7:0] | – | – | – | – | BIOZ_RBIAS_VALUE[1:0] | | EN_BIOZ_RBIASP | EN_BIOZ_RBIASN |
| **Respiration Setup** | | | | | | | | | |
| 0xB6 | Respiration Configuration 1[7:0] | CG_LPF_DUTY[2:0] | | | CG_CHOP_CLK[1:0] | | CG_MODE[1:0] | | RESP_EN |
| **Interrupt Enables** | | | | | | | | | |
| 0xC0 | Interrupt1 Enable 1[7:0] | A_FULL_EN1 | PPG_FRAME_RDY_EN1 | FIFO_DATA_RDY_EN1 | ALC_OVF_EN1 | EXP_OVF_EN1 | PPG_THRESH2_HILO_EN1 | PPG_THRESH1_HILO_EN1 | – |
| 0xC1 | Interrupt1 Enable 2[7:0] | INVALID_PPG_CFG_EN1 | – | LED6_COMPB_EN1 | LED5_COMPB_EN1 | LED4_COMPB_EN1 | LED3_COMPB_EN1 | LED2_COMPB_EN1 | LED1_COMPB_EN1 |
| 0xC2 | Interrupt1 Enable 3[7:0] | – | – | – | FREQ_UNLOCK_EN1 | FREQ_LOCK_EN1 | PHASE_UNLOCK_EN1 | PHASE_LOCK_EN1 | – |
| 0xC3 | Interrupt1 Enable 4[7:0] | ECG_LON_EN1 | – | ECG_FAST_REC_EN1 | RLD_OOR_EN1 | ECG_LOFF_PH_EN1 | ECG_LOFF_PL_EN1 | ECG_LOFF_NH_EN1 | ECG_LOFF_NL_EN1 |

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| 0xC4 | Interrupt1 Enable 5[7:0] | BIOZ_LON_EN1 | BIOZ_OVER_EN1 | BIOZ_UNDR_EN1 | BIOZ_DRVP_OFF_EN1 | BIOZ_LOFF_PH_EN1 | BIOZ_LOFF_PL_EN1 | BIOZ_LOFF_NH_EN1 | BIOZ_LOFF_NL_EN1 |
| 0xC5 | Interrupt2 Enable 1[7:0] | A_FULL_EN2 | PPG_FRAME_RDY_EN2 | FIFO_DATA_RDY_EN2 | ALC_OVF_EN2 | EXP_OVF_EN2 | PPG_THRESH2_HILO_EN2 | PPG_THRESH1_HILO_EN2 | – |
| 0xC6 | Interrupt2 Enable 2[7:0] | INVALID_PPG_CFG_EN2 | – | LED6_COMPB_EN2 | LED5_COMPB_EN2 | LED4_COMPB_EN2 | LED3_COMPB_EN2 | LED2_COMPB_EN2 | LED1_COMPB_EN2 |
| 0xC7 | Interrupt2 Enable 3[7:0] | – | – | – | FREQ_UNLOCK_EN2 | FREQ_LOCK_EN2 | PHASE_UNLOCK_EN2 | PHASE_LOCK_EN2 | – |
| 0xC8 | Interrupt2 Enable 4[7:0] | ECG_LON_EN2 | – | ECG_FAST_REC_EN2 | RLD_OOR_EN2 | ECG_LOFF_PH_EN2 | ECG_LOFF_PL_EN2 | ECG_LOFF_NH_EN2 | ECG_LOFF_NL_EN2 |
| 0xC9 | Interrupt2 Enable 5[7:0] | BIOZ_LON_EN2 | BIOZ_OVER_EN2 | BIOZ_UNDR_EN2 | BIOZ_DRVP_OFF_EN2 | BIOZ_LOFF_PH_EN2 | BIOZ_LOFF_PL_EN2 | BIOZ_LOFF_NH_EN2 | BIOZ_LOFF_NL_EN2 |
| **Part ID** | | | | | | | | | |
| 0xFF | Part ID[7:0] | PART_ID[7:0] | | | | | | | |

## Register Details

### Status 1 (0x00)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| **Field** | A_FULL | PPG_FRAME_RDY | FIFO_DATA_RDY | ALC_OVF | EXP_OVF | PPG_THRESH2_HILO | PPG_THRESH1_HILO | PWR_RDY |
| **Reset** | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| **Access Type** | Read Only | Read Only | Read Only | Read Only | Read Only | Read Only | Read Only | Read Only |

### A_FULL

A_FULL is set to 1 when the FIFO has reached the threshold programmed in the FIFO_A_FULL[7:0](0x0D). This is a read-only bit and it is cleared when the Status 1 Register is read. It is also cleared when the FIFO Data Register(0x0C) is read if FIFO_STAT_CLR[3](0x0E) = 1.

| A_FULL | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | Indicates that the FIFO buffer has reached the threshold set by FIFO_A_FULL[7:0](0x0D). |

### PPG_FRAME_RDY

PPG_FRAME_RDY is set to 1 when a full frame conversion has completed. A frame consists of FIFO data for all the PPG ADC conversions for the sequence programmed in the MEASx (x = 1 to 6) enable registers. This is a read-only bit and it is cleared by reading the Status 1 register. It is also cleared by reading the FIFO Data register(0x0C) if FIFO_STAT_CLR[3](0x0E) = 1.

| PPG_FRAME_RDY | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | A complete PPG frame is available in the FIFO. |

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### FIFO_DATA_RDY

FIFO_DATA_RDY bit is set to 1 when new data is available in the FIFO. This is a read-only bit and it is cleared by reading the Status 1 register. It is also cleared by reading the FIFO Data Register(0x0C) if FIFO_STAT_CLR[3](0x0E) = 1

| FIFO_DATA_RDY | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | New data is available in the FIFO |

### ALC_OVF

ALC_OVF is set to 1 when a dark current measurement is either under range or over range for any of the assigned PPG measurements. A measurement is over range if it is more than positive full-scale (524287), and under range if it is less than negative full-scale / 4 (-131072). This is a read-only bit, and it is cleared by reading the Status 1 register.

| ALC_OVF | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | The dark current data is over range or under range. |

### EXP_OVF

EXP_OVF set to 1 when an exposure measurement is either over range or under range for any of the enabled PPG measurements. A measurement is over range if it is more than positive full scale (524287) minus roughly 16384, and under range if it is lower than negative full-scale / 4 (-131072) minus roughly 16384. This is a read-only bit, and it is cleared by reading the Status 1 register.

| EXP_OVF | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | The exposure data is over range or under range. |

### PPG_THRESH2_HILO

PPG_THRESH2_HILO is set to 1 when the PPG ADC reading qualifies as above THRESHOLD2_UPPER[7:0](0x74) or below THRESHOLD2_LOWER[7:0](0x75). This is a read-only bit and it is cleared by reading the status 1 register (0x00).

See the Threshold Detect Function section for a complete explanation of how the PPG ADC is qualified as above or below threshold.

| PPG_THRESH2_HILO | DECODE |
|---|---|
| 0 | ADC reading is within the threshold 2 range. |
| 1 | ADC reading is either above the THRESHOLD2_UPPER level or below the THRESHOLD2_LOWER level. |

### PPG_THRESH1_HILO

PPG_THRESH1_HILO is set to 1 when the PPG ADC reading qualifies as above THRESHOLD1_UPPER[7:0](0x72) or below THRESHOLD1_LOWER[7:0](0x73). This is a read-only bit and it is cleared by reading the status 1 register (0x00).

See the Threshold Detect Function section for a complete explanation of how the PPG ADC is qualified as above or below threshold.

| PPG_THRESH1_HILO | DECODE |
|---|---|
| 0 | ADC reading is within the threshold 1 range. |
| 1 | ADC reading is either above the THRESHOLD1_UPPER level or below the THRESHOLD1_LOWER level. |

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### PWR_RDY

PWR_RDY is set to 1 when $V_{DVDD}$ goes below the undervoltage lockout (UVLO) threshold, which is nominally 1.38V. If this condition occurs, all registers are reset to their POR state. This bit is not triggered by a soft-reset. This is a read-only bit and it is cleared when Status 1 register is read, or by setting SHDN[1](0x11) bit to 1.

PWR_RDY is a non-maskable interrupt, so it gets asserted only on INT1.

| Value | Decode |
|---|---|
| 0 | Normal operation |
| 1 | Indicates that $V_{DVDD}$ went below the UVLO threshold. |

## Status 2 (0x01)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| **Field** | INVALID_PPG_CFG | – | LED6_COMPB | LED5_COMPB | LED4_COMPB | LED3_COMPB | LED2_COMPB | LED1_COMPB |
| **Reset** | 0 | – | 0 | 0 | 0 | 0 | 0 | 0 |
| **Access Type** | Read Only | – | Read Only | Read Only | Read Only | Read Only | Read Only | Read Only |

### INVALID_PPG_CFG

INVALID_PPG_CFG is set to 1 when the PPG frame rate set by clock divider FR_CLK_DIV[14:0](0x28, 0x29) is too fast to accommodate the programmed PPG frame sequence. This is a read-only bit and it is cleared when the Status 2 register is read.

| INVALID_PPG_CFG | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | The PPG frame rate is too fast to accommodate the programmed PPG frame sequence. |

### LED6_COMPB

LEDn_COMPB is set to 1 when the voltage at the LEDn_DRV pin is below the LED compliance voltage (n = 1 to 6). LEDn_COMPB is a read-only bit and it is cleared when the Status 2 register is read.

| LEDn_COMPB | DECODE |
|---|---|
| 0 | The LEDn_DRV pin has sufficient voltage to support the programmed current. |
| 1 | The LEDn_DRV pin is below the voltage needed to support the programmed current. Power supply rejection on LEDn is degraded and LEDn current is inaccurate. |

### LED5_COMPB

See LED6_COMPB for details.

### LED4_COMPB

See LED6_COMPB for details.

### LED3_COMPB

See LED6_COMPB for details.

### LED2_COMPB

See LED6_COMPB for details.

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### LED1_COMPB

See LED6_COMPB for details.

## Status 3 (0x02)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | – | – | FREQ_UNLOCK | FREQ_LOCK | PHASE_UNLOCK | PHASE_LOCK | – |
| Reset | – | – | – | 0 | 0 | 0 | 0 | – |
| Access Type | – | – | – | Read Only | Read Only | Read Only | Read Only | – |

### FREQ_UNLOCK

FREQ_UNLOCK is set to 1 when the PLL loses the frequency lock. This is a read-only bit and is cleared by reading the Status 3 register. If the frequency unlock state persists, the FREQ_UNLOCK bit is set again.

| FREQ_UNLOCK | DECODE |
|---|---|
| 0 | The PLL is frequency locked |
| 1 | The PLL is frequency unlocked |

### FREQ_LOCK

FREQ_LOCK is set to 1 when the PLL achieves frequency lock. This is a read-only bit and is cleared by reading the Status 3 register. If the PLL remains frequency locked, the FREQ_LOCK bit is set again.

| FREQ_LOCK | DECODE |
|---|---|
| 0 | The PLL is frequency unlocked. |
| 1 | The PLL is frequency locked. |

### PHASE_UNLOCK

PHASE_UNLOCK is set to 1 when the PLL phase was locked and then lost its phase lock. This is a read-only bit and cleared by reading the status 3 register. If the PLL remains phase unlocked, the PHASE_UNLOCK bit is set again. This bit can give inconsistent results when using the internal oscillator or a high-jitter external oscillator. If the FREQ_LOCK = 1, PHASE_LOCK = 1, and the FREQ_UNLOCK = 0, then the PLL is operating correctly.

| PHASE_UNLOCK | DECODE |
|---|---|
| 0 | The PLL is phase locked. |
| 1 | The PLL is phase unlocked. |

### PHASE_LOCK

PHASE_LOCK is set to 1 when the PLL achieves phase lock. This is a read-only bit and cleared by reading the status 3 register. If the PLL remains in phase lock, the PHASE_LOCK bit is set again.

| PHASE_LOCK | DECODE |
|---|---|
| 0 | The PLL is phase unlocked. |
| 1 | The PLL is phase locked. |

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## Status 4 (0x03)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | ECG_LON | – | ECG_FAST_REC | RLD_OOR | ECG_LOFF_PH | ECG_LOFF_PL | ECG_LOFF_NH | ECG_LOFF_NL |
| Reset | 0 | – | 0 | 0 | 0 | 0 | 0 | 0 |
| Access Type | Read Only | – | Read Only | Read Only | Read Only | Read Only | Read Only | Read Only |

### ECG_LON

ECG_LON is set to 1 when the an ECG lead-on condition is detected. This is a read-only bit and is cleared by reading the Status 4 register. If the DC lead-on state persists, the ECG_LON bit is set again.

| ECG_LON | DECODE |
|---|---|
| 0 | ECG lead-on condition has not been detected. |
| 1 | ECG lead-on condition has been detected. |

### ECG_FAST_REC

ECG_FAST_REC is set to 1 when the ECG fast recovery mode is engaged (either manually or automatically). This is a read-only bit and cleared by reading the Status 4 register. If the ECG fast recovery mode persists for the next ECG sample, the ECG_FAST_REC bit is set again.

| ECG_FAST_REC | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | ECG channel is in fast recovery mode |

### RLD_OOR

RLD_OOR is set to 1 when the right leg drive voltage has been out of range (< 0.26V or > 1.26V) for more than 128 ms if CLK_FREQ_SEL = 0, or 125ms if CLK_FREQ_SEL = 1.

The delay is bypassed if BYP_DLY[7](0x12) is set to 1.

This is a read-only bit and is cleared by reading the Status 4 register. If the right leg drive voltage out of range condition persists, the RLD_OOR bit is set again.

| RLD_OOR | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | Right leg drive voltage is out of range. |

### ECG_LOFF_PH

ECG_LOFF_PH is set to 1 when the ECGP voltage has been greater than $V_{ECG_TH_H}$ for more than 128ms if CLK_FREQ_SEL = 0, or 125ms if CLK_FREQ_SEL = 1.

The delay is bypassed if BYP_DLY[7](0x12) is set to 1.

$V_{ECG_TH_H}$ is set by ECG_LOFF_THRESH[3:0](0x89).

This is a read-only bit and it is cleared by reading the Status 4 register. If the lead-off condition for ECG_LOFF_PH persists, this bit is set again.

| ECG_LOFF_PH | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | The ECGP voltage is higher than $V_{ECG_TH_H}$ |

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### ECG_LOFF_PL

ECG_LOFF_PL is set to 1 when the ECGP voltage has been less than $V_{ECG_TH_L}$ for more than 128ms if CLK_FREQ_SEL = 0, or 125ms if CLK_FREQ_SEL = 1.

The delay is bypassed if BYP_DLY[7](0x12) is set to 1.

$V_{ECG_TH_L}$ is set by ECG_LOFF_THRESH[3:0](0x89).

This is a read-only bit and it is cleared by reading the Status 4 register. If the lead-off condition for ECG_LOFF_PL persists, this bit is set again.

| ECG_LOFF_PL | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | The ECGP voltage is lower than $V_{ECG_TH_L}$ |

### ECG_LOFF_NH

ECG_LOFF_NH is set to 1 when the ECGN voltage has been greater than $V_{ECG_TH_H}$ for more than 128ms if CLK_FREQ_SEL = 0, or 125ms if CLK_FREQ_SEL = 1.

The delay is bypassed if BYP_DLY is set to 1.

$V_{ECG_TH_H}$ is set by ECG_LOFF_THRESH[3:0](0x89).

This is a read-only bit and it is cleared by reading the Status 4 register. If the lead-off condition for ECG_LOFF_NH persists, this bit is set again.

| ECH_LOFF_NH | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | The ECGN voltage is higher than $V_{LOFF_TH+}$ |

### ECG_LOFF_NL

ECG_LOFF_NL is set to 1 when the ECGN voltage has been less than $V_{ECG_TH_L}$ for more than 128ms if CLK_FREQ_SEL = 0, or 125ms if CLK_FREQ_SEL = 1.

The delay is bypassed if BYP_DLY[7](0x12) is set to 1.

$V_{ECG_TH_L}$ is set by ECG_LOFF_THRESH[3:0](0x89).

This is a read-only bit and it is cleared by reading the Status 4 register. If the lead-off condition for ECG_LOFF_NL persists, this bit is set again.

| ECG_LOFF_NL | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | The ECGN voltage is lower than $V_{ECG_TH_L}$ |

## Status 5 (0x04)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | BIOZ_LON | BIOZ_OVER | BIOZ_UNDR | BIOZ_DRV_OOR | BIOZ_LOFF_PH | BIOZ_LOFF_PL | BIOZ_LOFF_NH | BIOZ_LOFF_NL |
| Reset | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Access Type | Read Only | Read Only | Read Only | Read Only | Read Only | Read Only | Read Only | Read Only |

### BIOZ_LON

BIOZ_LON is set to 1 when a BioZ lead-on condition is detected. This is a read-only bit and is cleared by reading the Status 5 register. If the BioZ lead-on condition persists, the BIOZ_LON bit is set again.

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| BIOZ_LON | DECODE |
|---|---|
| 0 | BioZ lead-on condition has not been detected. |
| 1 | BioZ lead-on condition has been detected. |

### BIOZ_OVER

BIOZ_OVER is set to 1 when the absolute value of the BioZ ADC reading has exceeded the BioZ high threshold set by register BIOZ_HI_THRESH[7:0](0xA9) for more than 128ms if CLK_FREQ_SEL = 0, or 125ms if CLK_FREQ_SEL = 1. This bit is cleared when the Status 5 register is read. If the BIOZ_OVER condition persists at the end of next BioZ sample, the bit is set to 1 again.

| BIOZ_OVER | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | BIOZ_HI_THRESH has been exceeded |

### BIOZ_UNDR

BIOZ_UNDR is set to 1 when the absolute value of the BioZ ADC reading has been below the BIOZ Low Threshold set by register BIOZ_LO_THRESH[7:0](0xA8) for more than 128ms if CLK_FREQ_SEL = 0, or 125ms if CLK_FREQ_SEL = 1. This bit is cleared when the Status 5 register is read. If the BIOZ_UNDR condition persists at the end of the next BioZ sample, the bit is set to 1 again.

| BIOZ_UNDER | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | BIOZ_LO_THRESH has been exceeded |

### BIOZ_DRV_OOR

BIOZ_DRV_OOR is set to 1 when the BioZ DRVN voltage peaks have been out of range (< 0.2V or > ($V_{AVDD}$ - 0.2V)) for more than 128ms if CLK_FREQ_SEL = 0, or 125ms if CLK_FREQ_SEL = 1.

The delay is bypassed if BYP_DLY[7](0x12) is set to 1.

This bit is cleared when the Status 5 register is read. If the BioZ drive out-of-range condition persists, this bit continues to remain asserted.

| BIOZ_DRV_OOR | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | DRVN is out of range |

### BIOZ_LOFF_PH

BIOZ_LOFF_PH is set to 1 when the BIP voltage has been greater than $V_{BIOZ_TH_H}$ for more than 128ms if CLK_FREQ_SEL = 0, or 125ms if CLK_FREQ_SEL = 1.

The delay is bypassed if BYP_DLY[7](0x12) is set to 1.

$V_{BIOZ_TH_H}$ is set by BIOZ_LOFF_THRESH[3:0](0xB1).

This is a read-only bit and it is cleared by reading the Status 5 register. If the lead-off condition for BIOZ_LOFF_PH persists, this bit is set again.

| BIOZ_LOFF_PH | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | The BIP voltage is greater than $V_{BIOZ_TH_H}$ |

### BIOZ_LOFF_PL

BIOZ_LOFF_PL is set to 1 when the BIP voltage has been less than $V_{BIOZ_TH_L}$ for more than 128ms if CLK_FREQ_SEL = 0, or125ms if CLK_FREQ_SEL = 1.

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The delay is bypassed if BYP_DLY[7](0x12) is set to 1.

$V_{BIOZ_TH_L}$ is set by BIOZ_LOFF_THRESH[3:0](0xB1).

This is a read-only bit and it is cleared by reading the Status 5 register. If the lead-off condition for BIOZ_LOFF_PL persists, this bit is set again.

| BIOZ_LOFF_PL | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | The BIP voltage is lower than $V_{BIOZ_TH_L}$ |

### BIOZ_LOFF_NH

BIOZ_LOFF_NH is set to 1 when the BIN voltage has been greater than $V_{BIOZ_TH_H}$ for more than 128ms if CLK_FREQ_SEL = 0, or 125ms if CLK_FREQ_SEL = 1.

The delay is bypassed if BYP_DLY[7](0x12) is set to 1.

$V_{BIOZ_TH_H}$ is set by BIOZ_LOFF_THRESH[3:0](0xB1).

This is a read-only bit and it is cleared by reading the Status 5 register. If the lead-off condition for BIOZ_LOFF_NH persists, this bit is set again.

| BIOZ_LOFF_NH | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | The BIN voltage is higher than $V_{BIOZ_TH_H}$ |

### BIOZ_LOFF_NL

BIOZ_LOFF_NL is set to 1 when the BIN voltage has been less than $V_{BIOZ_TH_L}$ for more than 128ms if CLK_FREQ_SEL = 0, or 125ms if CLK_FREQ_SEL = 1.

The delay is bypassed if BYP_DLY[7](0x12) is set to 1.

$V_{BIOZ_TH_L}$ is set by BIOZ_LOFF_THRESH[3:0](0xB1).

This is a read-only bit and it is cleared by reading the Status 5 register. If the lead-off condition for BIOZ_LOFF_NL persists, this bit is set again.

| BIOZ_LOFF_NL | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | The BIN voltage is lower than $V_{BIOZ_TH_L}$ |

## FIFO Write Pointer (0x08)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | FIFO_WR_PTR[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Read Only | | | | | | | |

### FIFO_WR_PTR

FIFO_WR_PTR points to the FIFO location where the next sample will be written. This pointer advances for each sample pushed on to the circular FIFO. The write pointer wraps around to count 0x00 as the next FIFO location after count 0xFF.

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### FIFO Read Pointer (0x09)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | FIFO_RD_PTR[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read, Dual | | | | | | | |

**FIFO_RD_PTR**

FIFO_RD_PTR points to the FIFO location from which the next sample is to be read through the serial interface. This pointer advances each time a sample is read from the circular FIFO. The read pointer can be both read and written to. This allows rereading (or retrying) samples from the FIFO. However, writing to FIFO_RD_PTR can have adverse effects if it results in the FIFO being almost full. The read pointer wraps around to count 0x00 after count 0xFF.

### FIFO Counter 1 (0x0A)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | FIFO_DATA_COUNT[8] | OVF_COUNTER[6:0] | | | | | | |
| Reset | 0x0 | 0x00 | | | | | | |
| Access Type | Read Only | Read Only | | | | | | |

**FIFO_DATA_COUNT**

FIFO_DATA_COUNT[8](0x0A) is a read-only bit which holds the most significant bit of the number of items available in the FIFO for the host to read. The lower 8 bits are in the FIFO_DATA_COUNT[7:0](0x0B) register. FIFO_DATA_COUNT[8:0] increments when a new item is pushed to the FIFO, and decrements when the host reads an item from the FIFO.

FIFO_DATA_COUNT[8:0] is useful for debug.

**OVF_COUNTER**

The overflow counter OVF_COUNTER logs the number of samples lost if the FIFO is not read in a timely fashion. When FIFO is full any new samples results in either new or old samples getting lost depending on the FIFO_RO[1](0x0E) setting.

This is a read-only register. When a complete sample is read from FIFO and the read pointer advances, the OVF_COUNTER is reset to zero. It should be read immediately before reading the FIFO in order to check if an overflow condition has occured. This counter saturates at count value 0x7F.

### FIFO Counter 2 (0x0B)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | FIFO_DATA_COUNT[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Read Only | | | | | | | |

**FIFO_DATA_COUNT**

FIFO_DATA_COUNT[7:0] is a read-only register that holds the lower 8 bits of the number of items available in the FIFO for the host to read.

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See the FIFO_DATA_COUNT[8](0x0A) description for details.

### FIFO Data Register (0x0C)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | FIFO_DATA[7:0] | | | | | | | |
| Reset | 0xFF | | | | | | | |
| Access Type | Read Only | | | | | | | |

#### FIFO_DATA

FIFO_DATA is used to get data from the FIFO using burst reads only. When burst reading from this register, the register address pointer does not auto-increment, and the FIFO_RD_PTR[7:0](0x09) advances to provide subsequent samples. Each sample is three bytes, so burst reading three bytes in the FIFO_DATA register through the serial interface advances the FIFO_RD_PTR by one count. The format and data type of the data stored in the FIFO is determined by the tag associated with the data. For details and examples of various data types in some use cases, see the FIFO Description section. This is a read-only register.

### FIFO Configuration 1 (0x0D)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | FIFO_A_FULL[7:0] | | | | | | | |
| Reset | 0x7F | | | | | | | |
| Access Type | Write, Read | | | | | | | |

#### FIFO_A_FULL

FIFO_A_FULL sets the high-water mark for the FIFO and determines when status bit A_FULL[7](0x00) is asserted. The A_FULL bit is asserted when the FIFO holds (256 - FIFO_A_FULL) samples. For example, if set to 0x0F, A_FULL is asserted when there are 15 empty spaces left (241 samples in FIFO). If A_FULL_EN1[7](0xC0) or A_FULL_EN2[7](0xC5) is set to 1, then A_FULL being asserted results in an interrupt on the corresponding interrupt pin (INT1 or INT2). This condition should prompt the processor to read samples from the FIFO before it fills and overflows. The A_FULL bit is cleared and the interrupt is deasserted when the Status 1 register (0x00) is read.

| FIFO_A_FULL | Free Spaces Before Interrupt is Asserted | Number of Samples in FIFO |
|---|---|---|
| 0x00 | 0 | 256 |
| 0x01 | 1 | 255 |
| 0x02 | 2 | 254 |
| 0x03 | 3 | 253 |
| ---- | ---- | ---- |
| 0xFE | 254 | 2 |
| 0xFF | 255 | 1 |

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## FIFO Configuration 2 (0x0E)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | – | FIFO_MARK | FLUSH_FIFO | FIFO_STAT_CLR | A_FULL_TYPE | FIFO_RO | – |
| Reset | – | – | 0 | 0 | 1 | 0 | 0 | – |
| Access Type | – | – | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | – |

### FIFO_MARK

When FIFO_MARK is set to 1, a marker tag is pushed to the FIFO. FIFO_MARK is a self clearing bit. The marker tag is useful for differentiating the data in the FIFO before and after the tag.

See the FIFO Description section for the marker tag information.

| Value | Decode |
|---|---|
| 0 | Normal data saved in FIFO |
| 1 | Marker saved in FIFO |

### FLUSH_FIFO

When the FLUSH_FIFO bit is set to 1, the FIFO is flushed, and FIFO_WR_PTR[7:0](0x08), FIFO_RD_PTR[7:0](0x09), FIFO_DATA_COUNT[8:0](0x0A, 0x0B) and OVF_COUNTER[6:0](0x0A) are reset to zero. The contents of the FIFO are lost. FLUSH_FIFO is a self-clearing bit.

| Value | Decode |
|---|---|
| 0 | Normal mode |
| 1 | FIFO is flushed |

### FIFO_STAT_CLR

FIFO_STAT_CLR determines if a FIFO_DATA[7:0](0x0C) register read clears the status bits A_FULL[7](0x00), PPG_FRAME_RDY[6](0x00), FIFO_DATA_RDY[5](0x00), and their corresponding interrupts.

| Value | Decode |
|---|---|
| 0 | A_FULL, PPG_FRAME_RDY, and FIFO_DATA_RDY status and interrupts do not get cleared by a FIFO_DATA[7:0](0x0C) register read. They get cleared by a Status 1 register read. |
| 1 | A_FULL, PPG_FRAME_RDY, and FIFO_DATA_RDY status and interrupts get cleared by a FIFO_DATA[7:0](0x0C) register read or a Status 1 register read. |

### A_FULL_TYPE

A_FULL_TYPE defines the behavior of status bit A_FULL[7](0x00) and its corresponding interrupt. When A_FULL_TYPE is set to 0, A_FULL is asserted every time the FIFO almost-full condition is detected. When A_FULL_TYPE is set to 1, A_FULL is asserted only for any new almost full condition.

| Value | Decode |
|---|---|
| 0 | A_FULL interrupt gets asserted when the almost full condition is detected. It is cleared by a Status 1 register read, but reasserts for every sample if the almost-full condition persists. |
| 1 | A_FULL interrupt gets asserted when the almost-full condition is detected. The interrupt gets cleared by a Status 1 register read, and does not reassert until the FIFO is read and then a new almost-full condition is detected. |

### FIFO_RO

FIFO_RO bit controls the behavior of the FIFO when the FIFO becomes completely filled with data. Push to FIFO is enabled when FIFO is full if FIFO_RO is set to 1 and old samples are lost. Both FIFO Write Pointer (0x08) and FIFO Read Pointer (0x09) increment for each sample after the FIFO is full. If FIFO_RO is set to 0, new samples are lost and

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the FIFO is not updated. FIFO Write Pointer and FIFO Read Pointer do not increment until a sample is read from the FIFO.

| Value | Decode |
|---|---|
| 0 | The FIFO stops on full. |
| 1 | The FIFO automatically rolls over on full. |

## System Sync (0x10)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| **Field** | TIMING_SYS_RESET | – | – | – | – | – | – | – |
| **Reset** | 0 | – | – | – | – | – | – | – |
| **Access Type** | Write, Read | – | – | – | – | – | – | – |

### TIMING_SYS_RESET

TIMING_SYS_RESET works together with the TRIG_FCFG[7:5](0x13) bit to synchronize the timing subsystems of multiple AFEs.

Setting the TIMING_SYS_RESET to 1 resets the BIOZ_NDIV[7:6](0x1A), ECG_NDIV[10:0](0x1B, 0x1C), and ECG_FDIV[2:0](0x1B) dividers.

See the PLL Synchronization section for a description of how to configure the PLL controller-target operation.

TIMING_SYS_RESET is a self clearing bit.

| TIMING_SYS_RESET | DECODE |
|---|---|
| 0 | Normal mode |
| 1 | Reset pulse is sent on TRIG if PLL is locked, and TRIG_FCFG = 0x2. |

## System Configuration 1 (0x11)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| **Field** | – | DISABLE_I2C | ECG_PPG_TIMING_DATA | BIOZ_PPG_TIMING_DATA | ECG_BIOZ_TIMING_DATA | – | SHDN | RESET |
| **Reset** | – | 0 | 0 | 0 | 0 | – | 0 | 0 |
| **Access Type** | – | Write, Read | Write, Read | Write, Read | Write, Read | – | Write, Read | Write, Read |

### DISABLE_I2C

When DISABLE_I2C is set to 0 (default), the part uses the $I^2C$ or SPI serial interface depending on the state of the CSB/I2C_SEL pin. When DISABLE_I2C is set to 1, the part uses the SPI serial interface.

For SPI interface, users must set this DISABLE_I2C to 1 during initialization after powerup. See the Digital Interface section for more information.

| DISABLE_I2C | DECODE |
|---|---|
| 0 | CSB/I2C_SEL pin selects interface |
| 1 | Part uses SPI interface only |

### ECG_PPG_TIMING_DATA

ECG_PPG_TIMING_DATA enables saving the time delay between the ECG sample pushed to the FIFO and the start

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of the next PPG measurement. ECG-to-PPG timing data is identified by a specific tag. For information on PPG timing data format and tag, see the FIFO Description section. For details on ECG-to-PPG timing data and some examples, see the ECG-to-PPG Timing Data section.

When both ECG and PPG are enabled and ECG_PPG_TIMING_DATA is set to 1, the PPG frame rate uses the ECG ADC clock. Since ECG_PPG_TIMING_DATA changes the PPG frame-rate clock selection, this bit should be modified before enabling PPG using the MEASx_EN (x = 1 to 6) registers.

If ECG is not enabled, the ECG-to-PPG timing data is not saved in the FIFO.

| ECG_PPG_TIMING_DATA | DECODE |
|---|---|
| 0 | ECG-to-PPG timing data is not saved in the FIFO. |
| 1 | ECG-to-PPG timing data is saved in FIFO if both PPG and ECG are enabled. |

### BIOZ_PPG_TIMING_DATA

BIOZ_PPG_TIMING_DATA enables saving timing delay between the BioZ sample pushed to the FIFO and the start of the next PPG measurement. BioZ-to-PPG timing data is identified by a specific tag. For information on data format and tag, see the FIFO Description section. For details on BioZ-to-PPG timing data and some examples, see the BioZ-to-PPG Timing Data section.

When both BioZ and PPG are enabled and BIOZ_PPG_TIMING_DATA is set to 1, the PPG frame rate uses the BioZ-ADC clock. Since BIOZ_PPG_TIMING_DATA changes the PPG frame-rate clock selection, this bit should be modified before enabling PPG using the MEASx_EN (x = 1 to 6) registers. If BioZ is not enabled, the BioZ-to-PPG timing data is not saved in the FIFO.

When ECG, BioZ, and PPG are all enabled and both ECG_PPG_TIMING_DATA and BIOZ_PPG_TIMING_DATA are set to 1, the PPG frame rate uses the ECG-ADC clock.

| BIOZ_PPG_TIMING_DATA | DECODE |
|---|---|
| 0 | BioZ-to-PPG timing data is not saved in the FIFO. |
| 1 | BioZ-to-PPG timing data is saved in the FIFO if both PPG and BioZ are enabled. |

### ECG_BIOZ_TIMING_DATA

ECG_BIOZ_TIMING_DATA enables saving the time delay between the ECG sample being pushed to the FIFO and the next BioZ sample being pushed to the FIFO. ECG to BioZ timing data is identified by a specific tag. For information on data format and tag, see FIFO Description. For details on ECG to BioZ timing data and some examples, see ECG to BIOZ Timing Data.

If ECG or BioZ is not enabled, the ECG to BioZ timing data is not saved in the FIFO.

| ECG_BIOZ_TIMING_DATA | DECODE |
|---|---|
| 0 | ECG to BioZ timing data is not saved in FIFO |
| 1 | ECG to BioZ timing data is saved in FIFO if both ECG and BioZ are enabled. |

### SHDN

Setting SHDN to 1 puts the MAX86178 into shutdown mode. While in shutdown mode, all configuration registers retain their values and write/read operations function normally. All interrupts are cleared to zero in this mode. Also in this mode, the oscillator is shut down and the part draws minimum current. If this bit is asserted during an active conversion, then the conversion is aborted. Set SHDN to 0 to put the part back in normal mode. See the Shutdown Sequence section for more details.

| SHDN | DECODE |
|---|---|
| 0 | Normal mode |
| 1 | Shutdown mode |

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### RESET

Setting RESET to 1 resets all registers to their power-on-reset state. This is a self-clearing bit and resets to 0 after the reset sequence is completed.

| RESET | DECODE |
|---|---|
| 0 | Normal mode |
| 1 | All registers restored to power-on-reset state |

## System Configuration 2 (0x12)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | BYP_DLY | – | – | ECG_SAMP_SYNC_FREQ[4:0] | | | | |
| Reset | 0 | – | – | 0x00 | | | | |
| Access Type | Write, Read | – | – | Write, Read | | | | |

### BYP_DLY

BYP_DLY bypasses the delays associated with the status bits in the Status 4 (0x03) and Status 5 (0x04) registers. These status bits are the threshold, lead-on, and lead-off features of the ECG and BioZ channels.

| BYP_DLY | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | ECG and BioZ status delays bypassed |

### ECG_SAMP_SYNC_FREQ

ECG_SAMP_SYNC_FREQ divides the ECG sample rate by the ratio shown in the decode table and outputs a sample sync pulse on TRIG when TRIG_FCFG = 0x3. The sample sync pulse width is equal to a quarter of the original ECG sample-rate period.

| ECG_SAMP_SYNC_FREQ | DIVIDER |
|---|---|
| 0x00 | 1 |
| 0x01 | 2 |
| 0x02 | 3 |
| ... | ... |
| 0x1C | 29 |
| 0x1D | 30 |
| 0x1E | 31 |
| 0x1F | 32 |

## Pin Functional Configuration (0x13)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | TRIG_FCFG[2:0] | | | TRIG_ICFG | INT2_FCFG[1:0] | | INT1_FCFG[1:0] | |
| Reset | 0x0 | | | 0 | 0x0 | | 0x1 | |
| Access Type | Write, Read | | | Write, Read | Write, Read | | Write, Read | |

### TRIG_FCFG

TRIG_FCFG controls the function of the TRIG pin. (See the PLL Synchronization Using TRIG Pin section in the detailed

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description.)

When TRIG_FCFG is set to 0x0 and PPG_SYNC_MODE[5](0x21) = 1, the TRIG input pin is used for external PPG frame sync. In this mode the device is in one-shot mode for PPG measurements. A frame cycle begins upon receipt of an active edge on the TRIG input (see TRIG_ICFG[1](0x13)).

When TRIG_FCFG is set to 0x1, the device is a target for PLL synchronization. When PLL_EN is set to 1, the TRIG signal is used to receive a timing-system reset pulse to synchronize the PLLs on one or more AFEs. The controller and target reset the ECG_NDIV, ECG_FDIV, and BIOZ_NDIV counters on the next rising edge of FCLK after the reset pulse rising edge.

When TRIG_FCFG is set to 0x2, the device is a controller for PLL synchronization and together with the TIMING_SYS_RESET bit it is used to synchronize the timing subsystems of multiple AFEs.

When TRIG_FCFG is set to 0x3, the device outputs the ECG Sample Sync pulse on the TRIG pin at the frequency programmed in ECG_SAMP_SYNC_FREQ. These pulses can be used for synchronizing other devices to the ECG sample rate. The width of the sync pulse is 1/4 of the ECG sample period.

When TRIG_FCFG is set to 0x4, the device outputs the LED_TX pulse on the TRIG pin. LED_TX is asserted 500ns before any of the six LED Driver pins, LEDn_DRV (n = 1 to 6) gets asserted. This feature is useful in combination with the MAX20345 boost converter to switch it from a fully "off" state to a fully "on" state; thus, reducing the boost quiescent power and the effects of boost ripple on the MAX86178.

If the broadcast feature is used for synchronizing timing subsystems of multiple AFEs, it is not necessary to use TRIG_FCFG = 0x1 or 0x2, and the TRIG pin can be used for other purposes shown in the table.

| TRIG_FCFG[2:0] | Pin Direction | Description |
|---|---|---|
| 0b000 | Input | The TRIG pin is used for external frame synchronization for PPG. See the PPG_SYNC_MODE description |
| 0b001 | Input | The TRIG pin is used for PLL synchronization in the target device |
| 0b010 | Output | The TRIG pin is used for PLL synchronization in the controller device |
| 0b011 | Output | The TRIG pin is used for ECG sample-sync pulses |
| 0b100 | Output | The TRIG pin outputs the LED_TX pulse |

### TRIG_ICFG

TRIG_ICFG sets the input active edge of the TRIG pin.

| TRIG_ICFG | DECODE |
|---|---|
| 0 | The TRIG pin active edge is falling. |
| 1 | The TRIG pin active edge is rising. |

### INT2_FCFG

INTx_FCFG (x = 1, 2) controls the function of the corresponding INTx pin.

| INTx_FCFG (x = 1, 2) | DECODE |
|---|---|
| 0x0 | INTx is disabled |
| 0x1 | INTx is enabled and is cleared by reading the corresponding status register or FIFO as applicable. |
| 0x2 | INTx is enabled and gets cleared automatically after 30.5µs, or by reading the corresponding status register or FIFO as applicable. |
| 0x3 | INTx is enabled and gets cleared automatically after 244µs, or by reading the corresponding status register or FIFO as applicable. |

### INT1_FCFG

See INT2_FCFG for details.

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## Output Pin Configuration (0x14)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | TRIG_OCFG[1:0] | | – | – | INT2_OCFG[1:0] | | INT1_OCFG[1:0] | |
| Reset | 0x0 | | – | – | 0x0 | | 0x0 | |
| Access Type | Write, Read | | – | – | Write, Read | | Write, Read | |

### TRIG_OCFG

TRIG_OCFG selects the output drive type for the TRIG pin.

| TRIG_OCFG | TRIG OUTPUT DRIVE TYPE |
|---|---|
| 0x0 | Open-drain, active-low output. |
| 0x1 | Active drive to IOVDD and DGND; the active level is a high output. |
| 0x2 | Active drive to IOVDD and DGND; the active level is a low output. |
| 0x3 | Do not use. |

### INT2_OCFG

INTx_OCFG (x = 1, 2) selects the output drive type for the corresponding INTx pin.

| INTx_OCFG (x = 1, 2) | INTx OUTPUT DRIVE TYPE |
|---|---|
| 0x0 | Open-drain, active-low output. |
| 0x1 | Active drive to IOVDD and DGND; the active level is a high output. |
| 0x2 | Active drive to IOVDD and DGND; the active level is a low output. |
| 0x3 | Do not use. |

### INT1_OCFG

See INT2_OCFG for details.

## I2C Broadcast Address (0x15)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | I2C_BCAST_ADDR[6:0] | | | | | | | I2C_BCAST_EN |
| Reset | 0x00 | | | | | | | 0 |
| Access Type | Write, Read | | | | | | | Write, Read |

### I2C_BCAST_ADDR

I2C_BCAST_ADDR is used to define the upper 7 bits of the $I^2C$ address in $I^2C$ broadcast mode (I2C_BCAST_EN = 1) when writing to multiple devices simultaneously using the $I^2C$ serial interface. I2C_BCAST_ADDR is ignored in SPI mode.

See the $I^2C$ Broadcast section for more details.

### I2C_BCAST_EN

I2C_BCAST_EN enables write transactions to multiple devices using the broadcast address programmed in I2C_BCAST_ADDR in $I^2C$ mode. $I^2C$ read transactions are not supported when I2C_BCAST_ADDR is used.

Note that for devices using SPI for serial interface, broadcast write transactions can be achieved by driving the CSB

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pins low on multiple devices at the same time.

| I2C_BCAST_EN | DECODE |
|---|---|
| 0 | Normal mode. I2C transactions are for one device only. |
| 1 | I2C broadcast mode. Write transactions to multiple devices are enabled. |

## PLL Configuration 1 (0x18)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MDIV[9:8] | | – | – | – | – | PLL_LOCK_WNDW | PLL_EN |
| Reset | 0x1 | | – | – | – | – | 0 | 0 |
| Access Type | Write, Read | | – | – | – | – | Write, Read | Write, Read |

### MDIV

MDIV[9:0] multiplies the REF_CLK by MDIV + 1 to set the frequency of the PLL. MDIV[9:8] are the 2 MSBs of MDIV[9:0]. The lower 8 bits are in MDIV[7:0].

MDIV must be set such that PLL_CLK is between 4.0MHz and 28.0MHz.

For information on how to set MDIV[9:0] see the Timing Subsystem section.

### PLL_LOCK_WNDW

PLL_LOCK_WNDW selects the time window for the PLL phase lock detector. The PLL lock detector compares the rising edges of FCLK and the output of the M divider, and determines the PLL to be locked if the difference between the two is less than PLL_LOCK_WNDW. Setting PLL_LOCK_WNDW = 1 helps to avoid false PHASE_UNLOCK interrupts when the FCLK reference has high jitter.

| PLL_LOCK_WNDW | PLL PHASE LOCK WINDOW |
|---|---|
| 0 | 1 PLL clock period |
| 1 | 2 PLL clock periods (recommended when using high-jitter FCLK input) |

### PLL_EN

PLL_EN enables the internal PLL, which multiplies the reference clock to a frequency between 4MHz and 28MHz. For details on the timing subsystem with PLL enabled, see the Timing Subsystem section. PLL_EN must be set to 1 before enabling ECG or BioZ measurements.

| PLL_EN | DECODE |
|---|---|
| 0 | PLL is disabled |
| 1 | PLL is enabled |

## PLL Configuration 2 (0x19)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MDIV[7:0] | | | | | | | |
| Reset | 0xBB | | | | | | | |
| Access Type | Write, Read | | | | | | | |

### MDIV

MDIV[7:0] are the 8 LSBs of MDIV[9:0]

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See the MDIV[9:8] register for details.

## PLL Configuration 3 (0x1A)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | BIOZ_NDIV[1:0] | | – | – | BIOZ_KDIV[3:0] | | | |
| Reset | 0x0 | | – | – | 0x0 | | | |
| Access Type | Write, Read | | – | – | Write, Read | | | |

### BIOZ_NDIV

BIOZ_NDIV divides down the PLL clock as shown in the table below and sets the clock for the BioZ ADC.

For information on how to set BIOZ_NDIV see the Timing Subsystem section.

| BIOZ_NDIV | BioZ N-Divider |
|---|---|
| 0x0 | 256 |
| 0x1 | 512 |
| 0x2 | 1024 |
| 0x3 | 1024 |

### BIOZ_KDIV

BIOZ_KDIV divides down the PLL clock as shown in the table below and sets the clock for the DDS DAC.

For information on how to set BIOZ_KDIV see the Timing Subsystem section.

| BIOZ_KDIV | BIOZ K DIVIDER |
|---|---|
| 0x0 | 1 |
| 0x1 | 2 |
| 0x2 | 4 (See Note) |
| 0x3 | 8 |
| 0x4 | 16 |
| 0x5 | 32 |
| 0x6 | 64 |
| 0x7 | 128 |
| 0x8 | 256 |
| 0x9 | 512 |
| 0xA | 1024 |
| 0xB | 2048 |
| 0xC | 4096 |
| 0xD | 8192 |
| 0xE | 8192 |
| 0xF | 8192 |

Note: this setting results in a larger offset in the receive channel. If a calibration is applied to the measurement, this effect is negated by the offset measurement and subtraction. If no calibration is applied and accurate impedance is desired, it is recommended to avoid using this setting, and instead adjusting MDIV by a factor of two to achieve the desired frequency with a different BIOZ_KDIV setting.

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## PLL Configuration 4 (0x1B)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | ECG_NDIV[10:8] | | | – | – | ECG_FDIV[2:0] | | |
| Reset | 0x1 | | | – | – | 0x0 | | |
| Access Type | Write, Read, Dual | | | – | – | Write, Read | | |

### ECG_NDIV

ECG_NDIV[10:0] divides the PLL clock according to the table below, and together with ECG_FDIV[2:0] sets the clock for the ECG ADC.

ECG_NDIV[10:8] are the most significant bits of 11-bit ECG_NDIV[10:0]. The lower 8 bits are in ECG_NDIV[7:0](0x1C).

ECG_NDIV[10:0] can be set to any value between 16 and 2047. If a value smaller than 16 is written, it is automatically overwritten to 16.

For information on how to set ECG_NDIV[10:0] see the Timing Subsystem section.

| ECG_NDIV[10:0] | ECG N DIVIDER |
|---|---|
| 0x000 to 0x00F | 16 |
| 0x010 | 16 |
| 0x011 | 17 |
| 0x012 | 18 |
| ... | ... |
| 0x7FE | 2046 |
| 0x7FF | 2047 |

### ECG_FDIV

ECG_FDIV along with ECG_NDIV divide the PLL clock to provide an ECG ADC clock. When respiration mode is enabled (RESP_EN = 1), ECG_FDIV also divides the respiration sampling rate.

When ECG_EN = 1 or RESP_EN = 1, ECG_FDIV must be non-zero.

For information on how to set ECG_FDIV see the Timing Subsystem section.

| ECG_FDIV | ECG F DIVIDER |
|---|---|
| 0x0 | ECG_ADC_CLK disabled |
| 0x1 | 1 |
| 0x2 | 2 |
| 0x3 | 4 |
| 0x4 | 8 |
| 0x5 | 16 |
| 0x6 | 16 |
| 0x7 | 16 |

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## PLL Configuration 5 (0x1C)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | ECG_NDIV[7:0] | | | | | | | |
| Reset | 0x3F | | | | | | | |
| Access Type | Write, Read, Dual | | | | | | | |

### ECG_NDIV

ECG_NDIV[7:0] has the lower 8 bits of 11-bit ECG_NDIV[10:0].

See ECG_NDIV[10:8] for details.

## PLL Configuration 6 (0x1D)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | REF_CLK_SEL | CLK_FREQ_SEL | CLK_FINE_TUNE[4:0] | | | | |
| Reset | – | 0 | 1 | 0x00 | | | | |
| Access Type | – | Write, Read | Write, Read | Write, Read | | | | |

### REF_CLK_SEL

REF_CLK_SEL selects the reference clock (REF_CLK) for the PLL. The reference clock is either external on FCLK (REF_CLK_SEL = 1) or an internal oscillator (REF_CLK_SEL = 0).

| REF_CLK_SEL | DECODE |
|---|---|
| 0 | Internal 32.0kHz or 32.768kHz oscillator used for REF_CLK |
| 1 | External oscillator used for REF_CLK |

### CLK_FREQ_SEL

CLK_FREQ_SEL selects the PLL reference-clock frequency. When using the internal oscillator (REF_CLK_SEL = 0), this bit sets the frequency of the internal oscillator. When using an external clock on the FCLK pin (REF_CLK_SEL = 1), this bit must match the frequency of the external clock. This bit sets the internal timing durations according to the clock frequency.

| CLK_FREQ_SEL | DECODE |
|---|---|
| 0 | PLL reference clock is 32.0kHz |
| 1 | PLL reference clock is 32.768kHz |

### CLK_FINE_TUNE

CLK_FINE_TUNE is used to fine-tune the internal slow oscillator. This is accomplished by measuring the time between interrupts using a microcontroller, crystal-based real-time oscillator as a reference and computing the error in the time between interrupts. CLK_FINE_TUNE is a 2's complement code with a resolution of 0.2% per LSB. The total range is +3.0% to -3.2% around the factory trimmed value.

| CLK_FINE_TUNE | SHIFT IN FREQUENCY (%) | CLK_FINE_TUNE | SHIFT IN FREQUENCY(%) |
|---|---|---|---|
| 0x00 | 0.0 | 0x10 | -3.2 |
| 0x01 | 0.2 | 0x11 | -3.0 |
| 0x02 | 0.4 | 0x12 | -2.8 |
| 0x03 | 0.6 | 0x13 | -2.6 |

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| CLK_FINE_TUNE | SHIFT IN FREQUENCY (%) | CLK_FINE_TUNE | SHIFT IN FREQUENCY(%) |
|---|---|---|---|
| 0x04 | 0.8 | 0x14 | -2.4 |
| 0x05 | 1.0 | 0x15 | -2.2 |
| 0x06 | 1.2 | 0x16 | -2.0 |
| 0x07 | 1.4 | 0x17 | -1.8 |
| 0x08 | 1.6 | 0x18 | -1.6 |
| 0x09 | 1.8 | 0x19 | -1.4 |
| 0x0A | 2.0 | 0x1A | -1.2 |
| 0x0B | 2.2 | 0x1B | -1.0 |
| 0x0C | 2.4 | 0x1C | -0.8 |
| 0x0D | 2.6 | 0x1D | -0.6 |
| 0x0E | 2.8 | 0x1E | -0.4 |
| 0x0F | 3.0 | 0x1F | -0.2 |

## PPG Configuration 1 (0x20)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | – | MEAS6_EN | MEAS5_EN | MEAS4_EN | MEAS3_EN | MEAS2_EN | MEAS1_EN |
| Reset | – | – | 0 | 0 | 0 | 0 | 0 | 0 |
| Access Type | – | – | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read |

### MEAS6_EN

See MEAS1_EN for details.

### MEAS5_EN

See MEAS1_EN for details.

### MEAS4_EN

See MEAS1_EN for details.

### MEAS3_EN

See MEAS1_EN for details.

### MEAS2_EN

See MEAS1_EN for details.

### MEAS1_EN

MEASx_EN (x = 1 to 6) enables or disables the PPG measurement programmed in the corresponding PPG MEASx Setup registers.

| MEASx_EN (x = 1 to 6) | Decode |
|---|---|
| 0 | Measurement x is disabled |
| 1 | Measurement x is enabled |

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## PPG Configuration 2 (0x21)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | – | PPG_SYNC_MODE | – | PPG2_PWRDN | PPG1_PWRDN | – | – |
| Reset | – | – | 0 | – | 0 | 0 | – | – |
| Access Type | – | – | Write, Read | – | Write, Read | Write, Read | – | – |

### PPG_SYNC_MODE

PPG_SYNC_MODE selects the PPG frame synchronization mode.

| PPG_SYNC_MODE | Operating Modes | Description |
|---|---|---|
| 0 | Internal frame sync | The device is in free-running mode. Frame-sync pulses are generated internally by PPG_FR_CLK and the frame rate divider set by FR_CLK_DIV[14:0](0x28, 0x29). |
| 1 | External frame sync | The device is in one-shot mode for PPG measurements. A frame cycle begins upon receipt of an active edge on the TRIG input. A frame cycle consists of a powerup cycle and the execution of each enabled measurement. In this mode,TRIG_FCFG[7:5](0x13) must be set to 0x0. |

### PPG2_PWRDN

PPG2_PWRDN enables or disables PPG channel 2.

| PPG2_PWRDN | Decode |
|---|---|
| 0 | PPG channel 2 is enabled. |
| 1 | PPG channel 2 is powered down. |

### PPG1_PWRDN

PPG1_PWRDN enables or disables PPG channel 1.

| PPG1_PWRDN | Decode |
|---|---|
| 0 | PPG channel 1 is enabled |
| 1 | PPG channel 1 is powered down |

## PPG Configuration 3 (0x22)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | SMP_AVE[2:0] | | | ALC_DISABLE | – | COLLECT_RAW_DATA | MEAS1_CONFIG_SEL |
| Reset | – | 0x0 | | | 0 | – | 0 | 0 |
| Access Type | – | Write, Read | | | Write, Read | – | Write, Read | Write, Read |

### SMP_AVE

SMP_AVE sets the number of adjacent samples from each individual PPG channel that are averaged on-chip before being written to the FIFO. SMP_AVE must be programmed to 0 when Threshold Interrupts are enabled or when External Frame Sync is used.

| SMP_AVE | Number of Samples Averaged |
|---|---|
| 0x0 | 1 (No averaging) |

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| SMP_AVE | Number of Samples Averaged |
|---|---|
| 0x1 | 2 |
| 0x2 | 4 |
| 0x3 | 8 |
| 0x4 to 0x7 | 16 |

### ALC_DISABLE

ALC_DISABLE disables the front-end analog ambient-light cancelation circuit for PPG measurements. This bit does not alter the digital ambient-light cancelation.

| ALC_DISABLE | Decode |
|---|---|
| 0 | Normal operation |
| 1 | Front-end analog ambient-light cancelation is disabled. |

### COLLECT_RAW_DATA

COLLECT_RAW_DATA pushes each ambient conversion and exposure conversion within a PPG measurement to the FIFO, separately. Setting COLLECT_RAW_DATA to 1 inhibits the digital ambient cancelation. This allows for a customized ambient rejection algorithm to be run in a host processor.

When COLLECT_RAW_DATA is set to 1, PROX_AUTO[3](0x23), THRESH1_MEAS_SEL[3:0](0x70), and THRESH2_MEAS_SEL[7:4](0x70) should be set to zero.

| COLLECT_RAW_DATA | Decode |
|---|---|
| 0 | Computed data for each measurement is saved in the FIFO. |
| 1 | Raw data for all ambient conversions and LED conversions in each measurement is saved in the FIFO. |

### MEAS1_CONFIG_SEL

MEAS1_CONFIG_SEL selects whether all enabled PPG measurements use a unique configuration or use the configuration settings defined in the MEAS1 setup registers (0x20 to 0x26). The unique configuration for each measurement is defined in each MEASx setup register. Setting MEAS1_CONFIG_SEL = 1 allows for reduced setup configuration writes. When MEAS1_CONFIG_SEL = 1, the configuration settings used by all enabled measurements are as follows:

MEAS1_SINC3_SEL

MEAS1_FILT2_SEL

MEAS1_FILT_SEL

MEAS1_TINT

MEAS1_AVER

MEAS1_PPG1_ADC_RGE

MEAS1_PPG2_ADC_RGE

MEAS1_PD_SETLNG

MEAS1_LED_SETLNG

MEAS1_LED_RGE

MEAS1_PD1_SEL

MEAS1_PD2_SEL

MEAS1_PD3_SEL

MEAS1_PD4_SEL

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| MEAS1_CONFIG_SEL | Decode |
|---|---|
| 0 | Specific configuration defined in each measurement setup registers is used. |
| 1 | MEAS1 configuration is used for all enabled measurements. |

## PPG Configuration 4 (0x23)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | – | – | PROX_DATA_EN | PROX_AUTO | – | – | – |
| Reset | – | – | – | 0 | 0 | – | – | – |
| Access Type | – | – | – | Write, Read | Write, Read | – | – | – |

### PROX_DATA_EN

PROX_DATA_EN enables MEAS6 data to be saved in the FIFO when PROX_AUTO = 1. If PROX_AUTO is set to 0, PROX_DATA_EN is ignored.

| PROX_DATA_EN | Decode |
|---|---|
| 0 | MEAS6 data is not saved in FIFO when PROX_AUTO = 1 |
| 1 | MEAS6 data is saved in FIFO when PROX_AUTO = 1 |

### PROX_AUTO

PROX_AUTO enables automatic proximity detect mode. For details, see the Automatic Proximity-Detect Mode section.

| PROX_AUTO | Decode |
|---|---|
| 0 | Normal mode |
| 1 | Automatic proximity detect mode. |

## Photodiode Bias (0x24)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | PD4_BIAS[1:0] | | PD3_BIAS[1:0] | | PD2_BIAS[1:0] | | PD1_BIAS[1:0] | |
| Reset | 0x1 | | 0x1 | | 0x1 | | 0x1 | |
| Access Type | Write, Read | | Write, Read | | Write, Read | | Write, Read | |

### PD4_BIAS

PDm_BIAS (m = 1 to 4) selects the photodiode-bias current for the photodiode connected to the corresponding PDm_IN pin. PDm_BIAS should be adjusted according to the photodiode capacitance on the PDm_IN pin. See the Photodiode Biasing section for more information.

| PDm_BIAS (m = 1 to 4) | Range of Photodiode Capacitance (pF) |
|---|---|
| 0x0 | Not recommended |
| 0x1 | 0 to 125 |
| 0x2 | 125 to 250 |
| 0x3 | 250 to 500 |

### PD3_BIAS

See PD4_BIAS for details.

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### PD2_BIAS

See PD4_BIAS for details.

### PD1_BIAS

See PD4_BIAS for details.

## FR Clock Divider MSB (0x28)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | FR_CLK_DIV[14:8] | | | | | | |
| Reset | – | 0x01 | | | | | | |
| Access Type | – | Write, Read | | | | | | |

### FR_CLK_DIV

FR_CLK_DIV[14:8] has the upper 7 bits of the 15-bit FR_CLK_DIV[14:0] clock divider, which defines the PPG frame rate.

For more information on PPG frame rate see the Timing Subsystem section.

| FR_CLK_DIV[14:0] | Frame Rate with REF_CLK_SEL = 0 (fps) | Frame Rate with RED_CLK_SEL = 1 (fps) |
|---|---|---|
| 0x7FFF | Reserved | Reserved |
| 0x7FFE | 0.976622 | 1.000061 |
| 0x7FFD | 0.976652 | 1.000092 |
| .... | | |
| 0x0100 | 125 | 128 (default) |
| …. | | |
| 0x0012 | 1777.78 | 1820.44 |
| 0x0011 | 1882.35 | 1927.53 |
| 0x0010 - 0x0000 | 1882.35 | 1927.53 |

## FR Clock Divider LSB (0x29)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | FR_CLK_DIV[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

### FR_CLK_DIV

FR_CLK_DIV[7:0] has the lower byte of the 15-bit FR_CLK_DIV clock divider that defines the PPG frame rate.

See FR_CLK_DIV[14:8] for more details.

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## MEAS1 Selects (0x30)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | MEAS1_AMB | MEAS1_DRVB[2:0] | | | MEAS1_DRVA[2:0] | | |
| Reset | – | 0 | 0x0 | | | 0x0 | | |
| Access Type | – | Write, Read | Write, Read | | | Write, Read | | |

### MEAS1_AMB

MEASx_AMB (x = 1 to 6) enables or disables direct ambient measurement. When MEASx_AMB is set to 1, MEASx_DRVA and MEASx_DRVB are ignored.

The direct ambient measurement should always be the last enabled measurement in the frame.

| MEASx_AMB (x = 1 to 6) | Decode |
|---|---|
| 0 | Normal mode |
| 1 | Direct ambient conversion |

### MEAS1_DRVB

MEASx_DRVB (x = 1 to 6) selects the LEDn_DRV pin (n = 1 to 6) driven by LED driver B.

| MEASx_DRVB (x = 1 to 6) | Pin driven by LED driver B |
|---|---|
| 0x0 | LED1_DRV |
| 0x1 | LED2_DRV |
| 0x2 | LED3_DRV |
| 0x3 | LED4_DRV |
| 0x4 | LED5_DRV |
| 0x5 | LED6_DRV |
| 0x6 | Do not use |
| 0x7 | Do not use |

### MEAS1_DRVA

MEASx_DRVA (x = 1 to 6) selects the LEDn_DRV pin (n = 1 to 6) driven by LED driver A.

| MEASx_DRVA (x = 1 to 6) | Pin driven by LED driver A |
|---|---|
| 0x0 | LED1_DRV |
| 0x1 | LED2_DRV |
| 0x2 | LED3_DRV |
| 0x3 | LED4_DRV |
| 0x4 | LED5_DRV |
| 0x5 | LED6_DRV |
| 0x6 | Do not use |
| 0x7 | Do not use |

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## MEAS1 Configuration 1 (0x31)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS1_SINC3_SEL | MEAS1_FILT2_SEL | MEAS1_FILT_SEL | MEAS1_TINT[1:0] | | MEAS1_AVER[2:0] | | |
| Reset | 0 | 1 | 0 | 0x3 | | 0x0 | | |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | | Write, Read | | |

### MEAS1_SINC3_SEL

MEASx_SINC3_SEL (x = 1 to 6) enables the SINC3 decimation filter for the PPG ADC. If MEASx_SINC3_SEL is set to 1, MEASx_TINT must be set to 3 and MEASx_FILT2_SEL must be set to 0.

| MEASx_SINC3_SEL (x = 1 to 6) | Decode |
|---|---|
| 0 | SINC3 filter is not used. Either a second-order or third-order decimation filter is used depending on the MEASx_FILT2_SEL, MEASx_FILT_SEL, and MEASx_TINT settings. |
| 1 | SINC3 decimation filter is used (if MEASx_TINT = 3 (117.1µs)) |

### MEAS1_FILT2_SEL

MEASx_FILT2_SEL enables either a second-order decimation filter or a third order decimation filter. When MEASx_FILT2_SEL = 1 (second-order decimation filter), MEASx_SINC3_SEL must be set to 0 and MEASx_TINT must be 0x3.

| MEASx_FILT2_SEL | Decode |
|---|---|
| 0 | 3rd order decimation filter is used. |
| 1 | 2nd order decimation filter is used if MEASx_SINC3_SEL = 0 and MEASx_TINT = 3 |

### MEAS1_FILT_SEL

MEASx_FILT_SEL (x = 1 to 6) selects the digital ambient light rejection method to be used. See the Ambient Rejection section for details.

| MEASx_FILT_SEL (x = 1 to 6) | Decode |
|---|---|
| 0 | Central difference method (CDM) is enabled. |
| 1 | Forward difference method (FDM) is enabled. |

### MEAS1_TINT

MEASx_TINT (x = 1 to 6) selects the integration time of PPG ADCs.

| MEASx_TINT (x = 1 to 6) | Integration Time with 3rd-Order Decimation Filter (µs) | Integration Time with 2nd-Order Decimation Filter (µs) |
|---|---|---|
| 0x0 | 14.6 | Not applicable |
| 0x1 | 29.2 | Not applicable |
| 0x2 | 58.6 | Not applicable |
| 0x3 | 117.0 | 118.2 |

### MEAS1_AVER

MEASx_AVER (x = 1 to 6) selects the number of exposures to be averaged in order to get one measurement sample in the FIFO. When setting MEASx_AVER to any value other than 0, MEASx_FILT_SEL must be 0. With MEASx_AVER, each sample in FIFO is a computed average of (2 x $2^{MEASx_AVER}$ + 1) ADC conversions of interleaved ambient and

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exposure measurements. When MEASx_FILT_SEL = 1, MEASx_AVER is ignored.

| MEASx_AVER (x = 1 to 6) | Number of LED Pulses |
|---|---|
| 0x0 | 1 |
| 0x1 | 2 |
| 0x2 | 4 |
| 0x3 | 8 |
| 0x4 | 16 |
| 0x5 | 32 |
| 0x6 | 64 |
| 0x7 | 128 |

## MEAS1 Configuration 2 (0x32)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | – | MEAS1_PPG2_ADC_RGE [1:0] | | – | – | MEAS1_PPG1_ADC_RGE [1:0] | |
| Reset | – | – | 0x2 | | – | – | 0x2 | |
| Access Type | – | – | Write, Read | | – | – | Write, Read | |

### MEAS1_PPG2_ADC_RGE

MEASx_PPG2_ADC_RGE (x = 1 to 6) selects the positive full-scale range of the PPG ADC on channel 2 for measurement x.

| MEASx_PPG2_ADC_RGE (x = 1 to 6) | Full-Scale Range for ADC2 (µA) |
|---|---|
| 0x0 | 4.0 |
| 0x1 | 8.0 |
| 0x2 | 16.0 |
| 0x3 | 32.0 |

### MEAS1_PPG1_ADC_RGE

MEASx_PPG1_ADC_RGE (x = 1 to 6) selects the positive full-scale range of the PPG ADC on channel 1 for measurement x.

| MEASx_PPG1_ADC_RGE (x = 1 to 6) | Full-Scale Range for ADC1 (µA) |
|---|---|
| 0x0 | 4.0 |
| 0x1 | 8.0 |
| 0x2 | 16.0 |
| 0x3 | 32.0 |

## MEAS1 Configuration 3 (0x33)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS1_PPG2_DACOFF[3:0] | | | | MEAS1_PPG1_DACOFF[3:0] | | | |
| Reset | 0x0 | | | | 0x0 | | | |
| Access Type | Write, Read | | | | Write, Read | | | |

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### MEAS1_PPG2_DACOFF

MEASx_PPG2_DACOFF (x = 1 to 6) selects the offset DAC current added to the ADC on PPG channel 2 during the exposure interval. This allows for a larger convertible exposure range for ADC2 by sourcing some of the photodiode DC exposure current from the offset DAC.

| MEASx_PPG2_DACOFF (x = 1 to 6) | Injected Offset Current to ADC2 (µA) |
|---|---|
| 0x0 | 0 |
| 0x1 | 2 |
| 0x2 | 4 |
| 0x3 | 6 |
| 0x4 | 8 |
| 0x5 | 10 |
| 0x6 | 12 |
| 0x7 | 14 |
| 0x8 | 16 |
| 0x9 | 18 |
| 0xA | 20 |
| 0xB | 22 |
| 0xC | 24 |
| 0xD | 26 |
| 0xE | 28 |
| 0XF | 30 |

### MEAS1_PPG1_DACOFF

MEASx_PPG1_DACOFF (x = 1 to 6) selects the offset DAC current added to ADC on PPG channel 1 during the exposure interval. This allow for a larger convertible exposure range for ADC1 by sourcing some of the photodiode DC exposure current from the offset DAC.

| MEASx_PPG1_DACOFF (x = 1 to 6) | Injected Offset Current to ADC1 (µA) |
|---|---|
| 0x0 | 0 |
| 0x1 | 2 |
| 0x2 | 4 |
| 0x3 | 6 |
| 0x4 | 8 |
| 0x5 | 10 |
| 0x6 | 12 |
| 0x7 | 14 |
| 0x8 | 16 |
| 0x9 | 18 |
| 0xA | 20 |
| 0xB | 22 |
| 0xC | 24 |
| 0xD | 26 |
| 0xE | 28 |
| 0xF | 30 |

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## MEAS1 Configuration 4 (0x34)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS1_PD_SETLNG[1:0] | | – | – | MEAS1_LED_SETLNG[1:0] | | MEAS1_LED_RGE[1:0] | |
| Reset | 0x1 | | – | – | 0x1 | | 0x3 | |
| Access Type | Write, Read | | – | – | Write, Read | | Write, Read | |

### MEAS1_PD_SETLNG

MEASx_PD_SETLNG (x = 1 to 6) selects the time between dark and exposure samples for measurement x. This accomodates photodiodes with longer settling time.

PD settling time should always be more than LED settling time selected in MEASx_LED_SETLNG. Note that for the same setting of both MEASx_PD_SETLNG and MEASx_LED_SETLNG, the photodiode settling time is 0.1µs higher than the LED settling time, and thus, satisfies the requirement of the higher PD settling time.

| MEASx_PD_SETLNG (x = 1 to 6) | Photodiode Settling Time (µs) |
|---|---|
| 0x0 | 7.8 |
| 0x1 | 11.8 |
| 0x2 | 15.8 |
| 0x3 | 23.8 |

### MEAS1_LED_SETLNG

MEASx_LED_SETLNG (x = 1 to 6) selects the delay from the rising edge of LED to the start of the exposure ADC integration. This allows for the LED current to settle before the start of ADC integration. LED settling time for a measurement must always be less than the photodiode settling time for the same measurement.

| MEASx_LED_SETLNG (x = 1 to 6) | LED Settling Time (µs) |
|---|---|
| 0x0 | 7.7 |
| 0x1 | 11.7 |
| 0x2 | 15.7 |
| 0x3 | 23.7 |

### MEAS1_LED_RGE

MEASx_LED_RGE (x = 1 to 6) selects the drive current range for both LED current drivers, DRVA and DRVB, for measurement x.

| MEASx_LED_RGE (x = 1 to 6) | LED Full-Scale Range (mA) |
|---|---|
| 0x0 | 32 |
| 0x1 | 64 |
| 0x2 | 96 |
| 0x3 | 128 |

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## MEAS1 Configuration 5 (0x35)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS1_PD4_SEL[1:0] | | MEAS1_PD3_SEL[1:0] | | MEAS1_PD2_SEL[1:0] | | MEAS1_PD1_SEL[1:0] | |
| Reset | 0x0 | | 0x0 | | 0x3 | | 0x2 | |
| Access Type | Write, Read | | Write, Read | | Write, Read | | Write, Read | |

### MEAS1_PD4_SEL

MEASx_PD4_SEL (x = 1 to 6) connects the PD4 input to the selected optical channel.

| MEASx_PD4_SEL (x = 1 to 6) | Decode |
|---|---|
| 0x0, 0x1 | PD4 not selected |
| 0x2 | PD4 is connected to PPG channel 1 |
| 0x3 | PD4 is connected to PPG channel 2 |

### MEAS1_PD3_SEL

MEASx_PD3_SEL (x = 1 to 6) selects the PD3 input to each of the optical channels.

| MEASx_PD3_SEL (x = 1 to 6) | Decode |
|---|---|
| 0x0, 0x1 | PD3 not selected |
| 0x2 | PD3 is connected to PPG channel 1 |
| 0x3 | PD3 is connected to PPG channel 2 |

### MEAS1_PD2_SEL

MEASx_PD2_SEL (x = 1 to 6) selects the PD2 input to each of the optical channels.

| MEASx_PD2_SEL (x = 1 to 6) | Decode |
|---|---|
| 0x0, 0x1 | PD2 not selected |
| 0x2 | PD2 is connected to PPG channel 1 |
| 0x3 | PD2 is connected to PPG channel 2 |

### MEAS1_PD1_SEL

MEASx_PD1_SEL (x = 1 to 6) selects the PD1 input to each of the optical channels.

| MEASx_PD1_SEL (x = 1 to 6) | Decode |
|---|---|
| 0x0, 0x1 | PD1 not selected |
| 0x2 | PD1 is connected to PPG channel 1 |
| 0x3 | PD1 is connected to PPG channel 2 |

## MEAS1 LEDA Current (0x36)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS1_DRVA_PA[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

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### MEAS1_DRVA_PA

MEASx_DRVA_PA (x = 1 to 6) selects the LED drive current on LED driver A for measurement x. If MEASn_DRVA_PA is set to 0x00, LED Driver A is disabled for measurement x. The full-scale range selected by MEASx_LED_RGE[4:5](0x22) determines the LED current for each LSB of MEASx_DRVA_PA setting. For example, when MEASx_LED_RGE = 0, one LSB of MEASx_DRVA_PA is 0.125mA of the LED driver current on DRVA, but when MEASx_LED_RGE = 3, each LSB of MEASx_DRVA_PA setting is 0.5mA of the LED driver current on DRVA.

| MEASx_LED_RGE (x = 1 to 6) | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| **MEASx_DRVA_PA (x = 1 to 6)** | **LED Current (mA)** | **LED Current (mA)** | **LED Current (mA)** | **LED Current (mA)** |
| 0x00 | 0.000 | 0.000 | 0.000 | 0.000 |
| 0x01 | 0.125 | 0.250 | 0.375 | 0.500 |
| 0x02 | 0.250 | 0.500 | 0.750 | 1.000 |
| 0x03 | 0.375 | 0.750 | 1.125 | 1.500 |
| ..... | | | | |
| 0xFC | 31.500 | 63.000 | 94.500 | 126.000 |
| 0xFD | 31.625 | 63.250 | 94.875 | 126.500 |
| 0xFE | 31.750 | 63.500 | 95.250 | 127.000 |
| 0xFF | 31.875 | 63.750 | 95.625 | 127.500 |
| LSB | 0.125 | 0.250 | 0.375 | 0.500 |

## MEAS1 LEDB Current (0x37)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| **Field** | MEAS1_DRVB_PA[7:0] | | | | | | | |
| **Reset** | 0x00 | | | | | | | |
| **Access Type** | Write, Read | | | | | | | |

### MEAS1_DRVB_PA

MEASx_DRVB_PA (x = 1 to 6) selects the LED drive current on LED driver B for measurement x. If MEASx_DRVB_PA is set to 0x00, LED Driver B is disabled for measurement x. The full-scale range selected by MEASx_LED_RGE[4:5](0x22) determines the LED current for each LSB of the MEASx_DRVB_PA setting. For example, when MEASx_LED_RGE = 0, one LSB of MEASx_DRVB_PA is 0.125mA of the LED driver current on DRVB, but when MEASx_LED_RGE = 3, each LSB of MEASx_DRVB_PA setting is 0.5mA of the LED driver current on DRVB.

| MEASx_LED_RGE (x = 1 to 6) | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| **MEASx_DRVB_PA (x= 1 to 6)** | **LED Current (mA)** | **LED Current (mA)** | **LED Current (mA)** | **LED Current (mA)** |
| 0x00 | 0.000 | 0.000 | 0.000 | 0.000 |
| 0x01 | 0.125 | 0.250 | 0.375 | 0.500 |
| 0x02 | 0.250 | 0.500 | 0.750 | 1.000 |
| 0x03 | 0.375 | 0.750 | 1.125 | 1.500 |
| ..... | | | | |
| 0xFC | 31.500 | 63.000 | 94.500 | 126.000 |
| 0xFD | 31.625 | 63.250 | 94.875 | 126.500 |
| 0xFE | 31.750 | 63.500 | 95.250 | 127.000 |

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| MEASx_LED_RGE (x = 1 to 6) | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0xFF | 31.875 | 63.750 | 95.625 | 127.500 |
| Each LSB | 0.125 | 0.250 | 0.375 | 0.500 |

## MEAS2 Selects (0x38)

(

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | MEAS2_AMB | MEAS2_DRVB[2:0] | | | MEAS2_DRVA[2:0] | | |
| Reset | – | 0 | 0x0 | | | 0x0 | | |
| Access Type | – | Write, Read | Write, Read | | | Write, Read | | |

### MEAS2_AMB

See MEAS1_AMB[6](0x30) for details.

### MEAS2_DRVB

See MEAS1_DRVB[3:5](0x30) for details.

### MEAS2_DRVA

See MEAS1_DRVA[2:0](0x30) for details.

## MEAS2 Configuration 1 (0x39)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS2_SINC3_SEL | MEAS2_FILT2_SEL | MEAS2_FILT_SEL | MEAS2_TINT[1:0] | | MEAS2_AVER[2:0] | | |
| Reset | 0 | 1 | 0 | 0x3 | | 0x0 | | |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | | Write, Read | | |

### MEAS2_SINC3_SEL

See MEAS1_SINC3_SEL[7](0x31) for details

### MEAS2_FILT2_SEL

See MEAS1_FILT2_SEL[6](0x31) for details.

### MEAS2_FILT_SEL

See MEAS1_FILT_SEL[5](0x31) for details.

### MEAS2_TINT

See MEAS1_TINT[4:3](0x31) for details.

### MEAS2_AVER

See MEAS1_AVER[2:0](0x31) for details.

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### MEAS2 Configuration 2 (0x3A)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | – | MEAS2_PPG2_ADC_RGE[1:0] | | – | – | MEAS2_PPG1_ADC_RGE[1:0] | |
| Reset | – | – | 0x2 | | – | – | 0x2 | |
| Access Type | – | – | Write, Read | | – | – | Write, Read | |

**MEAS2_PPG2_ADC_RGE**

See MEAS1_PPG2_ADC_RGE[5:4](0x32) for details.

**MEAS2_PPG1_ADC_RGE**

See MEAS1_PPG1_ADC_RGE[1:0](0x32) for details.

### MEAS2 Configuration 3 (0x3B)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS2_PPG2_DACOFF[3:0] | | | | MEAS2_PPG1_DACOFF[3:0] | | | |
| Reset | 0x0 | | | | 0x0 | | | |
| Access Type | Write, Read | | | | Write, Read | | | |

**MEAS2_PPG2_DACOFF**

See MEAS1_PPG2_DACOFF[7:4](0x33) for details.

**MEAS2_PPG1_DACOFF**

See MEAS1_PPG1_DACOFF[3:0](0x33) for details.

### MEAS2 Configuration 4 (0x3C)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS2_PD_SETLNG[1:0] | | – | – | MEAS2_LED_SETLNG[1:0] | | MEAS2_LED_RGE[1:0] | |
| Reset | 0x1 | | – | – | 0x1 | | 0x3 | |
| Access Type | Write, Read | | – | – | Write, Read | | Write, Read | |

**MEAS2_PD_SETLNG**

See MEAS1_PD_SETLNG[7:6](0x34) for details.

**MEAS2_LED_SETLNG**

See MEAS1_LED_SETLNG[3:2](0x34) for details.

**MEAS2_LED_RGE**

See MEAS1_LED_RGE[1:0](0x34) for details.

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## MEAS2 Configuration 5 (0x3D)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS2_PD4_SEL[1:0] | | MEAS2_PD3_SEL[1:0] | | MEAS2_PD2_SEL[1:0] | | MEAS2_PD1_SEL[1:0] | |
| Reset | 0x0 | | 0x0 | | 0x3 | | 0x2 | |
| Access Type | Write, Read | | Write, Read | | Write, Read | | Write, Read | |

**MEAS2_PD4_SEL**

See MEAS1_PD4_SEL[7:6](0x35) for details.

**MEAS2_PD3_SEL**

See MEAS1_PD3_SEL[5:4](0x35) for details.

**MEAS2_PD2_SEL**

See MEAS1_PD2_SEL[3:2](0x35) for details.

**MEAS2_PD1_SEL**

See MEAS1_PD1_SEL[1:0](0x35) for details.

## MEAS2 LEDA Current (0x3E)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS2_DRVA_PA[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

**MEAS2_DRVA_PA**

See MEAS1_DRVA_PA[7:0](0x36) for details.

## MEAS2 LEDB Current (0x3F)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS2_DRVB_PA[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

**MEAS2_DRVB_PA**

See MEAS1_DRVB_PA[7:0](0x37) for details.

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## MEAS3 Selects (0x40)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | MEAS3_AMB | MEAS3_DRVB[2:0] | | | MEAS3_DRVA[2:0] | | |
| Reset | – | 0 | 0x0 | | | 0x0 | | |
| Access Type | – | Write, Read | Write, Read | | | Write, Read | | |

### MEAS3_AMB

See MEAS1_AMB[6](0x30) for details.

### MEAS3_DRVB

See MEAS1_DRVB[3:5](0x30) for details.

### MEAS3_DRVA

See MEAS1_DRVA[2:0](0x30) for details.

## MEAS3 Configuration 1 (0x41)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS3_SINC3_SEL | MEAS3_FILT2_SEL | MEAS3_FILT_SEL | MEAS3_TINT[1:0] | | MEAS3_AVER[2:0] | | |
| Reset | 0 | 1 | 0 | 0x3 | | 0x0 | | |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | | Write, Read | | |

### MEAS3_SINC3_SEL

See MEAS1_SINC3_SEL[7](0x30) for details

### MEAS3_FILT2_SEL

See MEAS1_FILT2_SEL for details.

### MEAS3_FILT_SEL

See MEAS1_FILT_SEL[5](0x31) for details.

### MEAS3_TINT

See MEAS1_TINT[4:3](0x31) for details.

### MEAS3_AVER

See MEAS1_AVER[2:0](0x31) for details.

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### MEAS3 Configuration 2 (0x42)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | – | MEAS3_PPG2_ADC_RGE[1:0] | | – | – | MEAS3_PPG1_ADC_RGE[1:0] | |
| Reset | – | – | 0x2 | | – | – | 0x2 | |
| Access Type | – | – | Write, Read | | – | – | Write, Read | |

**MEAS3_PPG2_ADC_RGE**

See MEAS1_PPG2_ADC_RGE[5:4](0x32) for details.

**MEAS3_PPG1_ADC_RGE**

See MEAS1_PPG1_ADC_RGE[1:0](0x32) for details.

### MEAS3 Configuration 3 (0x43)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS3_PPG2_DACOFF[3:0] | | | | MEAS3_PPG1_DACOFF[3:0] | | | |
| Reset | 0x0 | | | | 0x0 | | | |
| Access Type | Write, Read | | | | Write, Read | | | |

**MEAS3_PPG2_DACOFF**

See MEAS1_PPG2_DACOFF[7:4](0x33) for details.

**MEAS3_PPG1_DACOFF**

See MEAS1_PPG1_DACOFF[3:0](0x33) for details.

### MEAS3 Configuration 4 (0x44)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS3_PD_SETLNG[1:0] | | – | – | MEAS3_LED_SETLNG[1:0] | | MEAS3_LED_RGE[1:0] | |
| Reset | 0x1 | | – | – | 0x1 | | 0x3 | |
| Access Type | Write, Read | | – | – | Write, Read | | Write, Read | |

**MEAS3_PD_SETLNG**

See MEAS1_PD_SETLNG[7:6](0x34) for details.

**MEAS3_LED_SETLNG**

See MEAS1_LED_SETLNG[3:2](0x34) for details.

**MEAS3_LED_RGE**

See MEAS1_LED_RGE[1:0](0x34) for details.

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### MEAS3 Configuration 5 (0x45)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS3_PD4_SEL[1:0] | | MEAS3_PD3_SEL[1:0] | | MEAS3_PD2_SEL[1:0] | | MEAS3_PD1_SEL[1:0] | |
| Reset | 0x0 | | 0x0 | | 0x3 | | 0x2 | |
| Access Type | Write, Read | | Write, Read | | Write, Read | | Write, Read | |

**MEAS3_PD4_SEL**

See MEAS1_PD4_SEL[7:6](0x35) for details.

**MEAS3_PD3_SEL**

See MEAS1_PD3_SEL[5:4](0x35) for details.

**MEAS3_PD2_SEL**

See MEAS1_PD2_SEL[3:2](0x35) for details.

**MEAS3_PD1_SEL**

See MEAS1_PD1_SEL[1:0](0x35) for details.

### MEAS3 LEDA Current (0x46)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS3_DRVA_PA[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

**MEAS3_DRVA_PA**

See MEAS1_DRVA_PA[7:0](0x36) for details.

### MEAS3 LEDB Current (0x47)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS3_DRVB_PA[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

**MEAS3_DRVB_PA**

See MEAS1_DRVB_PA[7:0](0x37) for details.

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## MEAS4 Selects (0x48)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | MEAS4_AMB | MEAS4_DRVB[2:0] | | | MEAS4_DRVA[2:0] | | |
| Reset | – | 0 | 0x0 | | | 0x0 | | |
| Access Type | – | Write, Read | Write, Read | | | Write, Read | | |

**MEAS4_AMB**

See MEAS1_AMB[6](0x30) for details.

**MEAS4_DRVB**

See MEAS1_DRVB[3:5](0x30) for details.

**MEAS4_DRVA**

See MEAS1_DRVA[2:0](0x30) for details.

## MEAS4 Configuration 1 (0x49)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS4_SINC3_SEL | MEAS4_FILT2_SEL | MEAS4_FILT_SEL | MEAS4_TINT[1:0] | | MEAS4_AVER[2:0] | | |
| Reset | 0 | 1 | 0 | 0x3 | | 0x0 | | |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | | Write, Read | | |

**MEAS4_SINC3_SEL**

See MEAS1_SINC3_SEL[7](0x31) for details.

**MEAS4_FILT2_SEL**

See MEAS1_FILT2_SEL[6](0x31) for details.

**MEAS4_FILT_SEL**

See MEAS1_FILT_SEL[5](0x31) for details.

**MEAS4_TINT**

See MEAS1_TINT[4:3](0x31) for details.

**MEAS4_AVER**

See MEAS1_AVER[2:0](0x31) for details.

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### MEAS4 Configuration 2 (0x4A)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | – | MEAS4_PPG2_ADC_RGE[1:0] | | – | – | MEAS4_PPG1_ADC_RGE[1:0] | |
| Reset | – | – | 0x2 | | – | – | 0x2 | |
| Access Type | – | – | Write, Read | | – | – | Write, Read | |

**MEAS4_PPG2_ADC_RGE**

See MEAS1_PPG2_ADC_RGE[5:4](0x32) for details.

**MEAS4_PPG1_ADC_RGE**

See MEAS1_PPG1_ADC_RGE[1:0](0x32) for details.

### MEAS4 Configuration 3 (0x4B)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS4_PPG2_DACOFF[3:0] | | | | MEAS4_PPG1_DACOFF[3:0] | | | |
| Reset | 0x0 | | | | 0x0 | | | |
| Access Type | Write, Read | | | | Write, Read | | | |

**MEAS4_PPG2_DACOFF**

See MEAS1_PPG2_DACOFF[7:4](0x33) for details.

**MEAS4_PPG1_DACOFF**

See MEAS1_PPG1_DACOFF[3:0](0x33) for details.

### MEAS4 Configuration 4 (0x4C)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS4_PD_SETLNG[1:0] | | – | – | MEAS4_LED_SETLNG[1:0] | | MEAS4_LED_RGE[1:0] | |
| Reset | 0x1 | | – | – | 0x1 | | 0x3 | |
| Access Type | Write, Read | | – | – | Write, Read | | Write, Read | |

**MEAS4_PD_SETLNG**

See MEAS1_PD_SETLNG[7:6](0x34) for details.

**MEAS4_LED_SETLNG**

See MEAS1_LED_SETLNG[3:2](0x34) for details.

**MEAS4_LED_RGE**

See MEAS1_LED_RGE[1:0](0x34) for details.

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### MEAS4 Configuration 5 (0x4D)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS4_PD4_SEL[1:0] | | MEAS4_PD3_SEL[1:0] | | MEAS4_PD2_SEL[1:0] | | MEAS4_PD1_SEL[1:0] | |
| Reset | 0x0 | | 0x0 | | 0x3 | | 0x2 | |
| Access Type | Write, Read | | Write, Read | | Write, Read | | Write, Read | |

**MEAS4_PD4_SEL**

See MEAS1_PD4_SEL[7:6](0x35) for details.

**MEAS4_PD3_SEL**

See MEAS1_PD3_SEL[5:4](0x35) for details.

**MEAS4_PD2_SEL**

See MEAS1_PD2_SEL[3:2](0x35) for details.

**MEAS4_PD1_SEL**

See MEAS1_PD1_SEL[1:0](0x35) for details.

### MEAS4 LEDA Current (0x4E)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS4_DRVA_PA[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

**MEAS4_DRVA_PA**

See MEAS1_DRVA_PA[7:0](0x36) for details.

### MEAS4 LEDB Current (0x4F)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS4_DRVB_PA[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

**MEAS4_DRVB_PA**

See MEAS1_DRVB_PA[7:0](0x37) for details.

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## MEAS5 Selects (0x50)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | MEAS5_AMB | MEAS5_DRVB[2:0] | | | MEAS5_DRVA[2:0] | | |
| Reset | – | 0 | 0x0 | | | 0x0 | | |
| Access Type | – | Write, Read | Write, Read | | | Write, Read | | |

**MEAS5_AMB**

See MEAS1_AMB[6](0x30) for details.

**MEAS5_DRVB**

See MEAS1_DRVB[3:5](0x30) for details.

**MEAS5_DRVA**

See MEAS1_DRVA[2:0](0x30) for details.

## MEAS5 Configuration 1 (0x51)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS5_SINC3_SEL | MEAS5_FILT2_SEL | MEAS5_FILT_SEL | MEAS5_TINT[1:0] | | MEAS5_AVER[2:0] | | |
| Reset | 0 | 1 | 0 | 0x3 | | 0x0 | | |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | | Write, Read | | |

**MEAS5_SINC3_SEL**

See MEAS1_SINC3_SEL[7](0x31) for details.

**MEAS5_FILT2_SEL**

See MEAS1_FILT2_SEL[6](0x31) for details.

**MEAS5_FILT_SEL**

See MEAS1_FILT_SEL[5](0x31) for details.

**MEAS5_TINT**

See MEAS1_TINT[4:3](0x31) for details.

**MEAS5_AVER**

See MEAS1_AVER[2:0](0x31) for details.

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### MEAS5 Configuration 2 (0x52)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | – | MEAS5_PPG2_ADC_RGE[1:0] | | – | – | MEAS5_PPG1_ADC_RGE[1:0] | |
| Reset | – | – | 0x2 | | – | – | 0x2 | |
| Access Type | – | – | Write, Read | | – | – | Write, Read | |

**MEAS5_PPG2_ADC_RGE**

See MEAS1_PPG2_ADC_RGE[5:4](0x32) for details.

**MEAS5_PPG1_ADC_RGE**

See MEAS1_PPG1_ADC_RGE[1:0](0x32) for details.

### MEAS5 Configuration 3 (0x53)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS5_PPG2_DACOFF[3:0] | | | | MEAS5_PPG1_DACOFF[3:0] | | | |
| Reset | 0x0 | | | | 0x0 | | | |
| Access Type | Write, Read | | | | Write, Read | | | |

**MEAS5_PPG2_DACOFF**

See MEAS1_PPG2_DACOFF[7:4](0x33) for details.

**MEAS5_PPG1_DACOFF**

See MEAS1_PPG1_DACOFF[3:0](0x33) for details.

### MEAS5 Configuration 4 (0x54)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS5_PD_SETLNG[1:0] | | – | – | MEAS5_LED_SETLNG[1:0] | | MEAS5_LED_RGE[1:0] | |
| Reset | 0b01 | | – | – | 0x1 | | 0x3 | |
| Access Type | Write, Read | | – | – | Write, Read | | Write, Read | |

**MEAS5_PD_SETLNG**

See MEAS1_PD_SETLNG[7:6](0x34) for details.

**MEAS5_LED_SETLNG**

See MEAS1_LED_SETLNG[3:2](0x34) for details.

**MEAS5_LED_RGE**

See MEAS1_LED_RGE[1:0](0x34) for details.

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### MEAS5 Configuration 5 (0x55)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS5_PD4_SEL[1:0] | | MEAS5_PD3_SEL[1:0] | | MEAS5_PD2_SEL[1:0] | | MEAS5_PD1_SEL[1:0] | |
| Reset | 0x0 | | 0x0 | | 0x3 | | 0x2 | |
| Access Type | Write, Read | | Write, Read | | Write, Read | | Write, Read | |

**MEAS5_PD4_SEL**

See MEAS1_PD4_SEL[7:6](0x35) for details.

**MEAS5_PD3_SEL**

See MEAS1_PD3_SEL[5:4](0x35) for details.

**MEAS5_PD2_SEL**

See MEAS1_PD2_SEL[3:2](0x35) for details.

**MEAS5_PD1_SEL**

See MEAS1_PD1_SEL[1:0](0x35) for details.

### MEAS5 LEDA Current (0x56)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS5_DRVA_PA[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

**MEAS5_DRVA_PA**

See MEAS1_DRVA_PA[7:0](0x36) for details.

### MEAS5 LEDB Current (0x57)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS5_DRVB_PA[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

**MEAS5_DRVB_PA**

See MEAS1_DRVB_PA[7:0](0x37) for details.

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## MEAS6 Selects (0x58)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | MEAS6_AMB | MEAS6_DRVB[2:0] | | | MEAS6_DRVA[2:0] | | |
| Reset | – | 0 | 0x0 | | | 0x0 | | |
| Access Type | – | Write, Read | Write, Read | | | Write, Read | | |

**MEAS6_AMB**

See MEAS1_AMB[6](0x30) for details.

**MEAS6_DRVB**

See MEAS1_DRVB[3:5](0x30) for details.

**MEAS6_DRVA**

See MEAS1_DRVA[2:0](0x30) for details.

## MEAS6 Configuration 1 (0x59)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS6_SINC3_SEL | MEAS6_FILT2_SEL | MEAS6_FILT_SEL | MEAS6_TINT[1:0] | | MEAS6_AVER[2:0] | | |
| Reset | 0 | 1 | 0 | 0x3 | | 0x0 | | |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | | Write, Read | | |

**MEAS6_SINC3_SEL**

See MEAS1_SINC3_SEL[7](0x31) for details.

**MEAS6_FILT2_SEL**

See MEAS1_FILT2_SEL[6](0x31) for details.

**MEAS6_FILT_SEL**

See MEAS1_FILT_SEL[5](0x31) for details.

**MEAS6_TINT**

See MEAS1_TINT[4:3](0x31) for details.

**MEAS6_AVER**

See MEAS1_AVER[2:0](0x31) for details.

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### MEAS6 Configuration 2 (0x5A)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | – | MEAS6_PPG2_ADC_RGE[1:0] | | – | – | MEAS6_PPG1_ADC_RGE[1:0] | |
| Reset | – | – | 0x2 | | – | – | 0x2 | |
| Access Type | – | – | Write, Read | | – | – | Write, Read | |

**MEAS6_PPG2_ADC_RGE**

See MEAS1_PPG2_ADC_RGE[5:4](0x32) for details.

**MEAS6_PPG1_ADC_RGE**

See MEAS1_PPG1_ADC_RGE[1:0](0x32) for details.

### MEAS6 Configuration 3 (0x5B)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS6_PPG2_DACOFF[3:0] | | | | MEAS6_PPG1_DACOFF[3:0] | | | |
| Reset | 0x0 | | | | 0x0 | | | |
| Access Type | Write, Read | | | | Write, Read | | | |

**MEAS6_PPG2_DACOFF**

See MEAS1_PPG2_DACOFF[7:4](0x33) for details.

**MEAS6_PPG1_DACOFF**

See MEAS1_PPG1_DACOFF[3:0](0x33) for details.

### MEAS6 Configuration 4 (0x5C)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS6_PD_SETLNG[1:0] | | – | – | MEAS6_LED_SETLNG[1:0] | | MEAS6_LED_RGE[1:0] | |
| Reset | 0b01 | | – | – | 0x1 | | 0x3 | |
| Access Type | Write, Read | | – | – | Write, Read | | Write, Read | |

**MEAS6_PD_SETLNG**

See MEAS1_PD_SETLNG[7:6](0x34) for details.

**MEAS6_LED_SETLNG**

See MEAS1_LED_SETLNG[3:2](0x34) for details.

**MEAS6_LED_RGE**

See MEAS1_LED_RGE[1:0](0x34) for details.

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### MEAS6 Configuration 5 (0x5D)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS6_PD4_SEL[1:0] | | MEAS6_PD3_SEL[1:0] | | MEAS6_PD2_SEL[1:0] | | MEAS6_PD1_SEL[1:0] | |
| Reset | 0x0 | | 0x0 | | 0x3 | | 0x2 | |
| Access Type | Write, Read | | Write, Read | | Write, Read | | Write, Read | |

**MEAS6_PD4_SEL**

See MEAS1_PD4_SEL[7:6](0x35) for details.

**MEAS6_PD3_SEL**

See MEAS1_PD3_SEL[5:4](0x35) for details.

**MEAS6_PD2_SEL**

See MEAS1_PD2_SEL[3:2](0x35) for details.

**MEAS6_PD1_SEL**

See MEAS1_PD1_SEL[1:0](0x35) for details.

### MEAS6 LEDA Current (0x5E)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS6_DRVA_PA[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

**MEAS6_DRVA_PA**

See MEAS1_DRVA_PA[7:0](0x36) for details.

### MEAS6 LEDB Current (0x5F)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | MEAS6_DRVB_PA[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

**MEAS6_DRVB_PA**

See MEAS1_DRVB_PA[7:0](0x37) for details.

### THRESHOLD MEAS SEL (0x70)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | THRESH2_MEAS_SEL[2:0] | | | – | THRESH1_MEAS_SEL[2:0] | | |
| Reset | – | 0x0 | | | – | 0x0 | | |
| Access Type | – | Write, Read | | | – | Write, Read | | |

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### THRESH2_MEAS_SEL

THRESH2_MEAS_SEL enables the threshold detect function and selects the PPG measurement for the second instance of the threshold function. For details see the Threshold Detect Function section.

If the threshold detect function is enabled, COLLECT_RAW_DATA[1](0x22) and SMP_AVE[5:3](0x22) must be set to zero.

| THRESH2_MEAS_SEL | Measurement Selected for Threshold 2 |
|---|---|
| 0x0 | Threshold 2 is disabled |
| 0x1 | MEAS1 |
| 0x2 | MEAS2 |
| 0x3 | MEAS3 |
| 0x4 | MEAS4 |
| 0x5 | MEAS5 |
| 0x6 | MEAS6 |
| 0x7 | Do not use |

### THRESH1_MEAS_SEL

THRESH1_MEAS_SEL enables the threshold detect function and selects the PPG measurement for the first instance of the threshold function. For details see the Threshold Detect Function section.

If the threshold detect function is enabled, COLLECT_RAW_DATA[1](0x22) and SMP_AVE[5:3](0x22) must be set to zero.

| THRESH1_MEAS_SEL | Measurement Selected for Threshold 1 |
|---|---|
| 0x0 | Threshold 1 is disabled |
| 0x1 | MEAS1 |
| 0x2 | MEAS2 |
| 0x3 | MEAS3 |
| 0x4 | MEAS4 |
| 0x5 | MEAS5 |
| 0x6 | MEAS6 |
| 0x7 | Do not use |

## THRESHOLD HYST (0x71)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | THRESH2_PPG_SEL | THRESH1_PPG_SEL | – | TIME_HYST[1:0] | | LEVEL_HYST[2:0] | | |
| Reset | 0 | 0 | – | 0x0 | | 0x0 | | |
| Access Type | Write, Read | Write, Read | – | Write, Read | | Write, Read | | |

### THRESH2_PPG_SEL

THRESH2_PPG_SEL selects the optical channel for THRESHOLD 2.

| THRESH2_PPG_SEL | Optical Channel for Threshold 2 |
|---|---|
| 0 | PPG1 |
| 1 | PPG2 |

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### THRESH1_PPG_SEL

THRESH1_PPG_SEL selects the optical channel for THRESHOLD 1.

| THRESH1_PPG_SEL | Optical Channel for Threshold 1 |
|---|---|
| 0 | PPG1 |
| 1 | PPG2 |

### TIME_HYST

Time hysteresis selects the number of consecutive samples outside the limits defined by THRESHOLDx_UPPER and THRESHOLDx_LOWER in order to trigger the threshold interrupt THRESHx_HILO. TIME_HYST applies to both instances of threshold interrupts. For details, see the Threshold Detect Function section.

Note: When using External Frame Trigger Input mode for synchronization, the Time Hysteresis field must be set to Zero.

When PPG_SYNC_MODE[5](0x21) = 0x1, write TIME_HYST[4:3](0x71) = 0b00.

| TIME_HYST | Number of Samples before Interrupt is Set |
|---|---|
| 0x0 | Time hysteresis is disabled |
| 0x1 | 2 |
| 0x2 | 4 |
| 0x3 | 8 |

### LEVEL_HYST

LEVEL_HYST sets the variation in ADC counts permitted when the THRESHx_HILO interrupt is triggered. This value is in ADC counts and is applied at ±0.5 x LEVEL_HYST around the THRESHOLDx_UPPER and THRESHOLDx_LOWER. LEVEL_HYST applies to both instances of threshold interrupts. For details, see the Threshold Detect Function section.

| Level_HYST | Magnitude of Hysteresis (LSBs) |
|---|---|
| 0x0 | Level Hysteresis is disabled |
| 0x1 | 2 |
| 0x2 | 4 |
| 0x3 | 8 |
| 0x4 | 16 |
| 0x5 | 32 |
| 0x6 | 64 |
| 0x7 | 128 |

## PPG HI THRESHOLD1 (0x72)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | THRESHOLD1_UPPER[7:0] | | | | | | | |
| Reset | 0xFF | | | | | | | |
| Access Type | Write, Read | | | | | | | |

### THRESHOLD1_UPPER

Defines the upper threshold limit for THRESHOLD x (x = 1, 2). Each LSB of THRESHOLDx_UPPER represents 2048 LSBs of the corresponding selected measurement (THRESHx_MEAS_SEL) ADC code.

THRESHOLDx_UPPER must be programmed to be greater than THRESHOLDx_LOWER; otherwise, the interrupt behavior is undefined.

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| THRESHOLDx_UPPER (x = 1, 2) | Upper Limit for THRESHOLD x |
|---|---|
| 0x00 | 0, upper threshold is disabled |
| 0x01 | 2048 |
| 0x02 | 4096 |
| 0x03 | 6144 |
| • | • |
| • | • |
| • | • |
| 0xFD | 518144 |
| 0xFE | 520192 |
| 0xFF | 522240 |

## PPG LO THRESHOLD1 (0x73)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | THRESHOLD1_LOWER[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

### THRESHOLD1_LOWER

Defines the lower threshold limit for THRESHOLD x (x = 1, 2). Each LSB of THRESHOLDx_LOWER represents 2048 LSBs of the selected measurement (THRESHx_MEAS_SEL) ADC code.

| THRESHOLDx_LOWER | Lower Limit for THRESHOLDx |
|---|---|
| 0x00 | 0 |
| 0x01 | 2048 |
| 0x02 | 4096 |
| 0x03 | 6144 |
| • | • |
| • | • |
| • | • |
| 0xFD | 518144 |
| 0xFE | 520192 |
| 0xFF | 522240 |

## PPG HI THRESHOLD2 (0x74)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | THRESHOLD2_UPPER[7:0] | | | | | | | |
| Reset | 0xFF | | | | | | | |
| Access Type | Write, Read | | | | | | | |

### THRESHOLD2_UPPER

See THRESHOLD1_UPPER for details.

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## PPG LO THRESHOLD2 (0x75)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | THRESHOLD2_LOWER[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

### THRESHOLD2_LOWER

See THRESHOLD1_LOWER for details.

## ECG Configuration 1 (0x80)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | – | – | – | ECG_DEC_RATE[2:0] | | | ECG_EN |
| Reset | – | – | – | – | 0x0 | | | 0 |
| Access Type | – | – | – | – | Write, Read | | | Write, Read |

### ECG_DEC_RATE

ECG_DEC_RATE sets the decimation ratio for the ECG_ADC as detailed in the table below.

For information on how to set ECG_DEC_RATE see the Timing Subsystem section

| ECG_DEC_RATE | DECIMATION RATIO |
|---|---|
| 0x0 | 16 |
| 0x1 | 32 |
| 0x2 | 64 |
| 0x3 | 128 |
| 0x4 | 256 |
| 0x5 | 512 |
| 0x6 | 512 |
| 0x7 | 512 |

### ECG_EN

ECG_EN enables the ECG channel and ECG data conversions.

| ECG_EN | DECODE |
|---|---|
| 0 | ECG is disabled. |
| 1 | ECG is enabled. |

## ECG Configuration 2 (0x81)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | ECG_IPOL | ECG_PGA_GAIN[2:0] | | | ECG_INA_RGE[1:0] | | ECG_INA_GAIN[1:0] | |
| Reset | 0 | 0x0 | | | 0x0 | | 0x1 | |
| Access Type | Write, Read | Write, Read | | | Write, Read | | Write, Read | |

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### ECG_IPOL

ECG_IPOL selects the ECG input polarity.

| ECG_IPOL | ECG INPUT POLARITY |
|---|---|
| 0 | Non-Inverted |
| 1 | Inverted |

### ECG_PGA_GAIN

ECG_PGA_GAIN selects the gain for the PGA in the ECG channel. The total ECG channel gain is the product of INA gain and PGA gain.

| ECG_PGA_GAIN | PGA GAIN (V/V) |
|---|---|
| 0x0 | 1 |
| 0x1 | 2 |
| 0x2 | 4 |
| 0x3 | 8 |
| 0x4 | Do not use |
| 0x5 | Do not use |
| 0x6 | Do not use |
| 0x7 | 16 |

### ECG_INA_RGE

ECG_INA_RGE selects the gain range of the ECG input amplifier.

See ECG_INA_GAIN for details.

### ECG_INA_GAIN

ECG_INA_GAIN selects the gain of the ECG input amplifier. See the ECG Noise Measurements section for typical noise performance at each gain setting.

| | INA GAIN (V/V) | | | |
|---|---|---|---|---|
| **ECG_INA_GAIN** | **ECG_INA_RGE = 0x0** | ECG_INA_RGE = **0x1** | ECG_INA_RGE = **0x2** | ECG_INA_RGE = **0x3** |
| 0x0 | 10 | 7.5 | 5 | 2.5 |
| 0x1 | 20 | 15 | 10 | 5 |
| 0x2 | 40 | 30 | 20 | 10 |
| 0x3 | 60 | 45 | 30 | 15 |

## ECG Configuration 3 (0x82)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| **Field** | – | – | – | – | ECG_IMP_HI | ECG_AUTO_REC | ECG_MUX_SEL[1:0] | |
| **Reset** | – | – | – | – | 0 | 0 | 0x0 | |
| **Access Type** | – | – | – | – | Write, Read | Write, Read | Write, Read | |

### ECG_IMP_HI

ECG_IMP_HI selects the combined output impedance of CAPP and CAPN. This impedance together with the value of the external capacitor connected between CAPP and CAPN sets the HPF corner frequency of the ECG channel.

If ECG_IMP_HI is set to 0 then the input impedance is independent of the ECG_INA_RGE setting.

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If ECG_IMP_HI is set to 1 then the input impedance is dependent on the ECG_INA_RGE setting.

| | CAPP AND CAPN COMBINED OUTPUT IMPEDANCE (kΩ) | |
|---|---|---|
| **ECG_INA_RGE** | **ECG_IMP_HI = 0** | ECG_IMP_HI = **1** |
| 0x0 | 400 | 400 |
| 0x1 | 400 | 533 |
| 0x2 | 400 | 800 |
| 0x3 | 400 | 1600 |

### ECG_AUTO_REC

ECG_AUTO_REC enables analog automatic recovery mode in the ECG INA. When enabled, the INA automatically enables the fast recovery buffers when the INA is saturated.

| ECG_AUTO_REC | DECODE |
|---|---|
| 0 | Analog automatic fast recovery disabled |
| 1 | Analog automatic fast recovery enabled |

### ECG_MUX_SEL

ECG_MUX_SEL[1:0] defines how ECGP/ECGN inputs and RLD output are routed to the ECG_EL1/2/3 pins. The selection table is shown below. The pins are disconnected from the ECG/RLD circuit blocks by default. ECG_OPEN_P and ECG_OPEN_N must also be set to 0 to connect the ECG inputs.

ECGP and ECGN connections can be swapped by the ECG_IPOL bit. For example, ECG_EL1 is routed to the ECGP input, and ECG_EL2 is routed to the ECGN input when ECG_RLD_MUX_SEL = 0x1 if ECG_IPOL = 0. ECG_EL1 is routed to ECGN input, and vice versa, if ECG_IPOL = 1.

| ECG_MUX_SEL | ECGP | ECGN | RLD |
|---|---|---|---|
| 0x0 | N/C | N/C | N/C |
| 0x1 | ECG_EL1 | ECG_EL2 | ECG_EL3 |
| 0x2 | ECG_EL3 | ECG_EL1 | ECG_EL2 |
| 0x3 | ECG_EL2 | ECG_EL3 | ECG_EL1 |

## ECG Configuration 4 (0x83)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| **Field** | EN_ECG_FAST_REC[1:0] | | ECG_FAST_REC_THRESHOLD[5:0] | | | | | |
| **Reset** | 0x0 | | 0x3F | | | | | |
| **Access Type** | Write, Read | | Write, Read | | | | | |

### EN_ECG_FAST_REC

EN_ECG_FAST_REC enables digital automatic fast recovery mode or manual mode in the ECG INA. Manual fast recovery mode remains active once it is enabled until it is manually disabled by setting EN_ECG_FAST_REC to 0x0. Automatic fast recovery mode is activated when the ECG ADC count is outside of the threshold set by ECG_FAST_REC_THRESHOLD for approximately 125ms, and remains active for approximately 500ms.

| EN_ECG_FAST_REC | DECODE |
|---|---|
| 0x0 | Normal mode |
| 0x1 | Manual fast recovery mode enabled |
| 0x2 | Digital automatic fast recovery mode enabled |
| 0x3 | Do not use |

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### ECG_FAST_REC_THRESHOLD

If EN_ECG_FAST_REC is set to 0x2 (digital automatic fast-recovery mode) and the output of an ECG measurement exceeds the symmetric thresholds defined by ±(2048 x ECG_FAST_REC_THRESHOLD) for more than 125ms, the fast recovery mode is automatically engaged and remains active for 500ms. The default value (ECG_FAST_REC_THRESHOLD = 0x3F) corresponds to an ECG output upper threshold of 0x1F800 and an ECG output lower threshold of 0x20800, or ±98.4% of full-scale.

## ECG CAL Configuration 1 (0x84)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | ECG_CAL_HIGH[10:8] | | | ECG_CAL_FREQ[2:0] | | | ECG_CAL_DUTY | ECG_CAL_EN |
| Reset | 0x0 | | | 0x0 | | | 0 | 0 |
| Access Type | Write, Read | | | Write, Read | | | Write, Read | Write, Read |

### ECG_CAL_HIGH

ECG_CAL_HIGH[10:8] are the most significant 3 bits of ECG_CAL_HIGH[10:0]. ECG_CAL_HIGH[10:0] determines the time high (or duty cycle) for the calibration source when ECG_CAL_DUTY is set to 0. Time high is calculated by the equation

$t_{HIGH}$ = ECG_CAL_HIGH[10:0] x 1 / ECG_ADC_CLK, where ECG_ADC_CLK is between 19kHz and 32.768kHz (see the Timing Subsystem section).

$t_{HIGH}$ should be lower than the calibration frequency period set by ECG_CAL_FREQ.

### ECG_CAL_FREQ

ECG_CAL_FREQ selects the frequency of the calibration source ($f_{CAL}$), relative to the ECG ADC clock.

| CAL_FREQ | CALIBRATION SIGNAL FREQUENCY | CALIBRATION SIGNAL FREQUENCY WITH ECG_ADC_CLK = 32.768kHz (Hz) |
|---|---|---|
| 0x0 | ECG_ADC_CLK / 128 | 256 |
| 0x1 | ECG_ADC_CLK / 512 | 64 |
| 0x2 | ECG_ADC_CLK / 2048 | 16 |
| 0x3 | ECG_ADC_CLK / 8192 | 4 |
| 0x4 | ECG_ADC_CLK / $2^{15}$ | 1 |
| 0x5 | ECG_ADC_CLK / $2^{17}$ | 1/4 |
| 0x6 | ECG_ADC_CLK / $2^{19}$ | 1/16 |
| 0x7 | ECG_ADC_CLK / $2^{21}$ | 1/64 |

### ECG_CAL_DUTY

ECG_CAL_DUTY selects between time-high and 50% duty modes of the calibration source.

| CAL_DUTY | DECODE |
|---|---|
| 0 | Use CAL_HIGH to select time high. |
| 1 | Duty cycle is 50% |

### ECG_CAL_EN

When ECG_EN is set to 1, ECG_CAL_EN enables ECG calibration sources VCALP and VCALN. Before enabling the calibration voltage sources, ensure that the input switches are disconnected (OPEN_P and OPEN_N are set to 1 in register 0xA2).

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| ECG_CAL_EN | DECODE |
|---|---|
| 0 | ECG calibration sources and modes disabled |
| 1 | ECG calibration sources and modes enabled |

## ECG CAL Configuration 2 (0x85)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | ECG_CAL_HIGH[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

### ECG_CAL_HIGH

ECG_CAL_HIGH[7:0] are the least significant 8 bits of ECG_CAL_HIGH[10:0]. See ECG_CAL_HIGH[10:8] for details.

## ECG CAL Configuration 3 (0x86)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | ECG_OPEN_P | ECG_OPEN_N | ECG_CAL_MODE | ECG_CAL_MAG | ECG_CAL_P_SEL[1:0] | | ECG_CAL_N_SEL[1:0] | |
| Reset | 1 | 1 | 0 | 0 | 0x0 | | 0x0 | |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | | Write, Read | |

### ECG_OPEN_P

ECG_OPEN_P controls the ECGP input switch. This switch must be connected (ECG_OPEN_P = 0) to measure a patient ECG signal, and must be disconnected (ECG_OPEN_P = 1) before enabling the calibration voltage sources.

| ECG_OPEN_P | DECODE |
|---|---|
| 0 | The ECGP pin is internally connected to the ECG channel |
| 1 | The ECGP pin is internally isolated from the ECG channel |

### ECG_OPEN_N

ECG_OPEN_N controls the ECGN input switch. This switch must be connected (ECG_OPEN_N = 0) to measure a patient ECG signal, and must be disconnected (ECG_OPEN_N = 1) before enabling the calibration voltage sources.

| ECG_OPEN_N | DECODE |
|---|---|
| 0 | The ECGN pin is internally connected to the ECG channel |
| 1 | The ECGN pin is internally isolated from the ECG channel |

### ECG_CAL_MODE

ECG_CAL_MODE is used to select the mode of calibration source.

| ECG_CAL_MODE | DECODE |
|---|---|
| 0 | Unipolar, sources swing between ($V_{MID_ECG}$ ± $V_{CAL_MAG}$) and ($V_{MID_ECG}$) |
| 1 | Bipolar, sources swing between ($V_{MID_ECG}$ + $V_{CAL_MAG}$) and ($V_{MID_ECG}$ - $V_{CAL_MAG}$) |

### ECG_CAL_MAG

ECG_CAL_MAG selects the magnitude of the calibration sources.

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| ECG_CAL_MAG | DECODE |
|---|---|
| 0 | $V_{CAL_MAG}$ = 0.5mV |
| 1 | $V_{CAL_MAG}$ = 1.0mV |

### ECG_CAL_P_SEL

ECG_ECG_CAL_P_SEL selects which calibration voltage source is connected to the ECGP input.

| ECG_CAL_P_SEL | DECODE |
|---|---|
| 0x0 | No calibration signal applied |
| 0x1 | ECGP is connected to $V_{MID_ECG}$ |
| 0x2 | ECGP is connected to VCALP (only available if ECG_CAL_EN = 1) |
| 0x3 | ECGP is connected to VCALN (only available if ECG_CAL_EN = 1) |

### ECG_CAL_N_SEL

ECG_CAL_N_SEL selects which calibration voltage source is connected to the ECGN input.

| ECG_CAL_N_SEL | DECODE |
|---|---|
| 0x0 | No calibration signal applied |
| 0x1 | ECGN is connected to $V_{MID_ECG}$ |
| 0x2 | ECGN is connected to VCALP (only available if ECG_CAL_EN = 1) |
| 0x3 | ECGN is connected to VCALN (only available if ECG_CAL_EN = 1) |

## ECG Lead Detect Configuration 1 (0x88)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | EN_ECG_LON | EN_ECG_LOFF | – | – | ECG_LOFF_MODE | ECG_LOFF_FREQ[2:0] | | |
| Reset | 0 | 0 | – | – | 0 | 0x0 | | |
| Access Type | Write, Read | Write, Read | – | – | Write, Read | Write, Read | | |

### EN_ECG_LON

EN_ECG_LON enables ultra-low-power (ULP) DC lead-on detection. ECG lead-on detection only functions when ECG is disabled (ECG_EN[0](0x80) = 0).

| EN_ECG_LON | DECODE |
|---|---|
| 0 | ECG lead-on detection is disabled |
| 1 | ECG lead-on detection is enabled |

### EN_ECG_LOFF

EN_ECG_LOFF enables lead-off detection, which is either DC lead-off detection or AC lead-off detection depending on ECG_LOFF_MODE.

If ECG is not enabled, EN_ECG_LOFF is ignored and lead-off detection is disabled.

### ECG_LOFF_MODE

ECG_LOFF_MODE selects between DC and AC lead-off detection.

| ECG_LOFF_MODE | DECODE |
|---|---|
| 0 | DC lead-off detection mode |

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| ECG_LOFF_MODE | DECODE |
|---|---|
| 1 | AC lead-off detection mode |

### ECG_LOFF_FREQ

ECG_LOFF_FREQ selects the frequency divider of the square-wave stimulus frequency for AC lead-off, by dividing the ECG_ADC_CLK by N.

For information on how to set ECG_LOFF_FREQ see the Timing Subsystem section.

| ECG_LOFF_FREQ | N DIVIDER |
|---|---|
| 0x0 | Stimulus disabled (DC) |
| 0x1 | 4 |
| 0x2 | 8 |
| 0x3 | 16 |
| 0x4 | 32 |
| 0x5 | 64 |
| 0x6 | 128 |
| 0x7 | 256 |

## ECG Lead Detect Configuration 2 (0x89)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | ECG_LOFF_IPOL | ECG_LOFF_IMAG[2:0] | | | ECG_LOFF_THRESH[3:0] | | | |
| Reset | 0 | 0x0 | | | 0x0 | | | |
| Access Type | Write, Read | Write, Read | | | Write, Read | | | |

### ECG_LOFF_IPOL

ECG_LOFF_IPOL selects the current polarity for ECG DC lead-off detection.

| ECG_LOFF_IPOL | POLARITY |
|---|---|
| 0 | Non-inverted. ECGP sources current, ECGN sinks current. |
| 1 | Inverted. ECGP sinks current, ECGN sources current. |

### ECG_LOFF_IMAG

ECG_LOFF_IMAG selects the DC/AC lead-off current amplitude.

| ECG_LOFF_IMAG | ECG DC/AC LEAD-OFF CURRENT ($nA/nA_{PK}$) |
|---|---|
| 0x0 | 0 (sources are disabled and disconnected) |
| 0x1 | 5 |
| 0x2 | 10 |
| 0x3 | 20 |
| 0x4 | 50 |
| 0x5 | 100 |
| 0x6 | 200 |
| 0x7 | 400 |

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### ECG_LOFF_THRESH

ECG_LOFF_THRESH selects the voltage threshold for the DC or AC lead-off window comparators, which are centered at $V_{MID_ECG}$. If the voltage of either ECGP or ECGN goes above the high threshold or below the low threshold, the corresponding ECG_DC_LOFF status bit is set to 1 in register 0x03.

| ECG_DC_LOFF_THRESH | ECG DC/AC LEAD-OFF THRESHOLD |
|---|---|
| 0x0 | $V_{MID_ECG}$ ± 25mV |
| 0x1 | $V_{MID_ECG}$ ± 50mV |
| 0x2 | $V_{MID_ECG}$ ± 75mV |
| ... | |
| 0xE | $V_{MID_ECG}$ ± 375mV |
| 0xF | $V_{MID_ECG}$ ± 400mV |

## ECG Lead Bias Configuration 1 (0x90)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| **Field** | – | – | – | – | ECG_RBIAS_VALUE[1:0] | | EN_ECG_RBIASP | EN_ECG_RBIASN |
| **Reset** | – | – | – | – | 0x0 | | 0 | 0 |
| **Access Type** | – | – | – | – | Write, Read | | Write, Read | Write, Read |

### ECG_RBIAS_VALUE

ECG_RBIAS_VALUE selects the ECG lead bias resistance, which is between ECGP and $V_{MID_ECG}$ or $V_{RLD}$(EN_ECG_RBIASP = 1), and ECGN and $V_{MID_ECG}$ or $V_{RLD}$(EN_ECG_RBIASN = 1).

| ECG_RBIAS_VALUE | BIAS RESISTANCE (MΩ) |
|---|---|
| 0x0 | 50 |
| 0x1 | 100 |
| 0x2 | 200 |
| 0x3 | Do not use |

### EN_ECG_RBIASP

EN_ECG_RBIASP enables the ECG lead bias between ECGP and $V_{MID_ECG}$ or $V_{RLD}$ with the value selected by ECG_RIAS_VALUE.

| EN_ECG_RBIASP | DECODE |
|---|---|
| 0 | ECGP is not resistively connected to $V_{MID_ECG}$ or $V_{RLD}$. |
| 1 | ECGP is connected to $V_{MID_ECG}$ or $V_{RLD}$ using a resistor selected by ECG_RBIAS_VALUE. |

### EN_ECG_RBIASN

EN_ECG_RBIASN enables ECG lead bias between ECGN and $V_{MID_ECG}$ or $V_{RLD}$ with the value selected by ECG_RIAS_VALUE.

| EN_ECG_RBIAS_N | DECODE |
|---|---|
| 0 | ECGN is not resistively connected to $V_{MID_ECG}$ or $V_{RLD}$. |
| 1 | ECGN is connected to $V_{MID_ECG}$ or $V_{RLD}$ using a resistor selected by ECG_RBIAS_VALUE. |

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## RLD Configuration 1 (0x92)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | RLD_EN | RLD_MODE | RLD_RBIAS | EN_RLD_OOR | ACTV_CM_P | ACTV_CM_N | RLD_GAIN[1:0] | |
| Reset | 0 | 0 | 0 | 0 | 0 | 0 | 0x0 | |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | |

### RLD_EN

RLD_EN enables the right leg drive circuit. RLD_EN is not controlled by ECG_EN. Write both ECG_EN and RLD_EN to 0 when disabling ECG block.

| RLD_EN | DECODE |
|---|---|
| 0 | RLD is disabled and the RLD pin is high-impedance. |
| 1 | RLD is enabled. |

### RLD_MODE

RLD_MODE controls the RLD amplifier feedback switch used to control the right leg drive AC feedback loop.

| RLD_MODE | DECODE |
|---|---|
| 0 | Open-loop body-bias mode. The feedback network is shorted and the RLD amplifier acts as a DC buffer to bias the body to a voltage selected by BODY_BIAS_DAC. |
| 1 | Closed-loop right leg drive mode. The RLD amplifier applies inverting gain to the AC common-mode input signal, forming a feedback loop through the body to bring the ECGP and ECGN inputs to the voltage selected by BODY_BIAS_DAC. |

### RLD_RBIAS

RLD_RBIAS selects the lead-bias voltage for the ECG or BioZ inputs, which is either the $V_{MID_ECG}$ reference or the output of the right leg drive common mode averager ($V_{RLD}$).

| RLD_RBIAS | DECODE |
|---|---|
| 0 | $V_{MID_ECG}$ is used as the lead-bias voltage |
| 1 | $V_{RLD}$ is used as the lead-bias voltage |

### EN_RLD_OOR

EN_RLD_OOR enabled the RLD out or range comparator, which sets the RLD_OOR[3](0x04) status bit if the RLD amplifier output voltage is below 0.127 x $V_{AVDD}$ or above 0.870 x $V_{AVDD}$.

| EN_RLD_OOR | DECODE |
|---|---|
| 0 | OOR comparator is disabled |
| 1 | OOR comparator is enabled |

### ACTV_CM_P

ACTV_CM_P enables the positive input to the common-mode averager. The positive input is taken from either ECGP or CAPP, selected by RLD_SEL_ECG[6](0x93). Enable both inputs for closed-loop right leg applications.

| ACTV_CM_P | DECODE |
|---|---|
| 0 | Positive input is disabled. The input buffer is disabled and its output is high-impedance. |
| 1 | Positive input is enabled |

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### ACTV_CM_N

ACTV_CM_N enables the negative input to the common-mode averager. The positive input is taken from either ECGN or CAPN, selected by RLD_SEL_ECG[6](0x93). Enable both inputs for closed-loop right leg drive applications.

| ACTV_CM_N | DECODE |
|---|---|
| 0 | Negative input is disabled. The input buffer is disabled and its output is high-impedance. |
| 1 | Negative input is enabled |

### RLD_GAIN

RLD_GAIN selects the internal RLD gain when RLD_EXT_RES is set to 0 and RLD_MODE is set to 1.

| RLD_GAIN | GAIN WITH BOTH INPUTS ENABLES (V/V) |
|---|---|
| 0x0 | 12 |
| 0x1 | 24 |
| 0x2 | 48 |
| 0x3 | 97 |

## RLD Configuration 2 (0x93)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | RLD_EXT_RES | RLD_SEL_ECG | RLD_BW[1:0] | | BODY_BIAS_DAC[3:0] | | | |
| Reset | 0 | 0 | 0x0 | | 0x0 | | | |
| Access Type | Write, Read | Write, Read | Write, Read | | Write, Read | | | |

### RLD_EXT_RES

RLD_EXT_RES disconnects the internal feedback resistor to use an external gain-setting resistor. RLD_EXT_RES is effective only when RLD_MODE[6](0x92) is set to 1. The gain setting resistor $R_{RLDFB}$ must be connected between the RLD and RLD_INV pins, which each have an internal 50kΩ series resistor. The resulting gain is (100kΩ + $R_{RLDFB}$) / 150kΩ when both common-mode averager inputs are enabled, and (100kΩ + $R_{RLDFB}$) / 250kΩ when only one is enabled.

| RLD_EXT_RES | DECODE |
|---|---|
| 0 | Internal RLD feedback resistor is connected. If an external feedback resistor is connected, the two resistances are in parallel. |
| 1 | Internal RLD feedback resistor is disconnected. Gain is set by the external feedback resistor. |

### RLD_SEL_ECG

RLD_SEL_ECG selects ECGP/ECGN or CAPP/CAPN as the inputs to the RLD common-mode averager. The common-mode averager input can be switched to the BioZ inputs by setting RLD_SEL_BIOZ[7](0xA7) to 1.

The voltages at CAPP/CAPN are buffered by the ECG INA, so their common-mode voltage lags slightly behind the ECGP/ECGN common-mode voltage. This introduces phase lag into the RLD feedback loop resulting in less common-mode signal attenuation.

| RLD_SEL_BIOZ | RLD_SEL_ECG | RLD INPUT SELECTION |
|---|---|---|
| 0 | 0 | CAPP/CAPN |
| 0 | 1 | ECGP/ECGN |
| 1 | x | BIP/BIN |

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### RLD_BW

RLD_BW selects the bandwidth for the right leg drive amplifier.

| | RLD BANDWIDTH (Hz) | | | |
|---|---|---|---|---|
| **RLD_BW** | **RLD_GAIN = 12** | **RLD_GAIN = 24** | **RLD_GAIN = 48** | **RLD_GAIN = 97** |
| **0** | 917 | 697 | 647 | 555 |
| **1** | 937 | 724 | 698 | 648 |
| **2** | 944 | 733 | 716 | 681 |
| **3** | 948 | 736 | 724 | 696 |

### BODY_BIAS_DAC

BODY_BIAS_DAC sets the voltage at the noninverting terminal of the RLD amplifier. The voltage is $V_{MID_ECG}$ - (BODY_BIAS_DAC x 0.04V), where $V_{MID_ECG}$ is 0.76V (typ), and BODY_BIAS_DAC is a 2's complement representation.

| BODY_BIAS_DAC | DECODE |
|---|---|
| 0x0 | $V_{MID_ECG}$ = 0.76V (mid) |
| 0x1 | $V_{MID_ECG}$ - 1 x 0.04V = 0.72V |
| 0x2 | $V_{MID_ECG}$ - 2 x 0.04V = 0.68V |
| ... | ... |
| 0x7 | $V_{MID_ECG}$ - 7 x 0.04V = 0.48V (min) |
| 0x8 | $V_{MID_ECG}$ + 8 x 0.04V = 1.08V (max) |
| ... | |
| 0xE | $V_{MID_ECG}$ + 2 x 0.04V = 0.84V |
| 0xF | $V_{MID_ECG}$ + 1 x 0.04V = 0.80V |

## BIOZ Configuration 1 (0xA0)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| **Field** | BIOZ_DAC_OSR[1:0] | | BIOZ_ADC_OSR[2:0] | | | ECG_BIOZ_BG_EN | BIOZ_EN[1:0] | |
| **Reset** | 0x0 | | 0x0 | | | 0 | 0x0 | |
| **Access Type** | Write, Read | | Write, Read | | | Write, Read | Write, Read | |

### BIOZ_DAC_OSR

BIOZ_DAC_OSR[1:0] sets the oversample ratio of the BioZ DDS DAC. For information on how to set BIOZ_DAC_OSR[1:0], see the Timing Subsystem system.

| BIOZ_DAC_OSR | DAC OVER SAMPLING RATIO |
|---|---|
| 0x0 | 32 |
| 0x1 | 64 |
| 0x2 | 128 |
| 0x3 | 256 |

### BIOZ_ADC_OSR

BIOZ_ADC_OSR[2:0] sets the oversample ratio of the BioZ ADC. For information on how to set BIOZ_ADC_OSR[2:0], see the Timing Subsystem section.

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| BIOZ_ADC_OSR | ADC OVER SAMPLING RATIO |
|---|---|
| 0x0 | 8 |
| 0x1 | 16 |
| 0x2 | 32 |
| 0x3 | 64 |
| 0x4 | 128 |
| 0x5 | 256 |
| 0x6 | 512 |
| 0x7 | 1024 |

### ECG_BIOZ_BG_EN

ECG_BIOZ_BG_EN enables the shared ECG and BioZ bandgap bias, which is required for all functions except the ULP LON. The bias power-up time is approximately 512ms and should be kept on between subsequent measurements. The bandgap bias also automatically enables if the ECG or BioZ channels are enabled, but it is recommended to set ECG_BIOZ_BG_EN = 1 before starting measurements.

| ECG_BIOZ_BG_EN | DECODE |
|---|---|
| 0 | ECG and BioZ bandgap bias disabled |
| 1 | ECG and BioZ bandgap bias enabled |

### BIOZ_EN

BIOZ_EN[1:0] enables the bioimpedance drive and receive channels for the in-phase (I channel) or quadrature-phase (Q channel) components when enabled when EN_UTIL_MODE is set to 0. This channel is disabled and is in a low-power state when BIOZ_EN[1:0] is set to 0x0 or 0x3. Enabling BioZ I or Q measurement also enables PLL and BG. It is recommended to set PLL_ENABLE and BG_ENABLE = 1 before enabling BioZ I or Q.

See the FIFO Description for the Tags used for the I channel and Q channel data.

| EN_UTIL_MODE | BIOZ_EN[1:0] | DECODE |
|---|---|---|
| 0 | 0x0 | Disable all BioZ measurements |
| 0 | 0x1 | Enable BioZ I measurement |
| 0 | 0x2 | Enable BioZ Q measurement |
| 0 | 0x3 | Disable all BioZ measurements |
| 1 | 0xX | BioZ ADC used for Utility (see the description for EN_UTIL_MODE) |

## BIOZ Configuration 2 (0xA1)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | BIOZ_DHPF[1:0] | | BIOZ_DLPF[2:0] | | | – | – | EN_BIOZ_THRESH |
| Reset | 0x0 | | 0x0 | | | – | – | 0 |
| Access Type | Write, Read | | Write, Read | | | – | – | Write, Read |

### BIOZ_DHPF

BIOZ_DHPF sets the BioZ channel digital high-pass filter cutoff frequency.

| BIOZ_DHPF[1:0] | CUTOFF FREQUENCY (Hz) |
|---|---|
| 0x0 | Bypass |
| 0x1 | 0.00025 x SR_BIOZ |

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| BIOZ_DHPF[1:0] | CUTOFF FREQUENCY (Hz) |
|---|---|
| 0x2 | 0.002 x SR_BIOZ |
| 0x3 | 0.002 x SR_BIOZ |

### BIOZ_DLPF

BIOZ_DLPF sets the BioZ channel digital low-pass filter cutoff frequency.

| BIOZ_DLPF[2:0] | CUTOFF FREQUENCY (Hz) |
|---|---|
| 0x0 | Bypass |
| 0x1 | 0.005 x SR_BIOZ |
| 0x2 | 0.02 x SR_BIOZ |
| 0x3 | 0.08 x SR_BIOZ |
| 0x4 to 0x7 | 0.25 x SR_BIOZ |

### EN_BIOZ_THRESH

When EN_BIOZ_THRESH is set to 1, the BioZ I or Q data is compared with the BIOZ_LO_THRESH[7:0](0xA8) and BIOZ_HI_THRESH[7:0](0xA9) to detect a lead-off condition. The I or Q data is compared to the thresholds and is determined by BIOZ_EN[1:0](0xA0). The status is reflected in BIOZ_OVER[6](0x04) and BIOZ_UNDR[5](0x04).

When EN_BIOZ_THRESH is set to 0, the threshold compare function is disabled.

## BIOZ Configuration 3 (0xA2)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | BIOZ_EXT_RES | – | BIOZ_VDRV_MAG[1:0] | | BIOZ_IDRV_RGE[1:0] | | BIOZ_DRV_MODE[1:0] | |
| Reset | 0 | – | 0x0 | | 0x0 | | 0x0 | |
| Access Type | Write, Read | – | Write, Read, Dual | | Write, Read, Dual | | Write, Read | |

### BIOZ_EXT_RES

BIOZ_EXT_RES selects the external $R_{EXT}$ resistor or the internal range settings resistors.

When BIOZ_EXT_RES is set to 0, the internal range resistors are used and the current magnitude is set by both BIOZ_VDRV_MAG[5:4](0xA2) and BIOZ_IDRV_RGE[3:2](0xA2). BIOZ_VDRV_MAG and BIOZ_IDRV_RGE can be automatically overwritten depending on the stimulus frequency according to patient safety requirements. Note that $R_{EXT}$ is not disconnected, so the DRVXR pin should be unconnected. If DRVXR is connected, $R_{EXT}$ is connected in parallel with the internal range resistor, which results in a larger current magnitude.

When BIOZ_EXT_RES is set to 1, BIOZ_VDRV_MAG and the value of $R_{EXT}$ set the current magnitude. BIOZ_VDRV_MAG is not automatically overwritten.

### BIOZ_VDRV_MAG

In voltage drive mode, BIOZ_VDRV_MAG sets the voltage amplitude at DRVR, which is connected to BIOZ_EL1. BIOZ_IDRV_RGE is ignored.

In current drive mode, BIOZ_VDRV_MAG and BIOZ_IDRV_RGE set the current magnitude. See BIOZ_IDRV_RGE for details.

| BIOZ_VDRV_MAG[1:0] | VOLTAGE MAGNITUDE ($mV_{PK}$) | VOLTAGE MAGNITUDE ($mV_{RMS}$) |
|---|---|---|
| 0x0 | 50 | 35.4 |
| 0x1 | 100 | 70.7 |
| 0x2 | 250 | 177 |

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| BIOZ_VDRV_MAG[1:0] | VOLTAGE MAGNITUDE ($mV_{PK}$) | VOLTAGE MAGNITUDE ($mV_{RMS}$) |
|---|---|---|
| 0x3 | 500 | 354 |

### BIOZ_IDRV_RGE

BIOZ_IDRV_RGE[1:0] sets the value of the internal current range resistor, which determines the current magnitude when BIOZ_EXT_RES = 0.

| BIOZ_IDRV_RGE[1:0] | INTERNAL RANGE RESISTOR VALUE |
|---|---|
| 0x0 | 552.5kΩ (VDRV magnitude reduced by 4x) |
| 0x1 | 110.5kΩ |
| 0x2 | 5.525kΩ |
| 0x3 | 276.25Ω |

When BIOZ_EXT_RES = 1, the external resistor connected between DRVXR and DRVXC, and determines the drive-current amplitude: $I_{MAG} = V_{DRVR} / R_{EXT}$.

When BIOZ_EXT_RES = 0, BIOZ_VDRV_MAG and BIOZ_IDRV_RGE together select the magnitude of the stimulus current. When BIOZ_IDRV_RGE = 0x0 in current mode, the drive voltage at DRVR is reduced by a factor of four to support smaller current magnitudes.

| BIOZ_IDRV_RGE[1:0] | BIOZ_VDRV_MAG[1:0] | AMPLITUDE OF $V_{DRVR}$ ($mV_{PK}$) | AMPLITUDE OF $V_{DRVR}$ ($mV_{RMS}$) | AMPLITUDE OF CURRENT (PEAK) | AMPLITUDE OF CURRENT (RMS) |
|---|---|---|---|---|---|
| 0x0 | 0x0 | 12.5 | 8.8 | 23nA | 16nA |
| 0x0 | 0x1 | 25 | 17.7 | 45nA | 32nA |
| 0x0 | 0x2 | 62.5 | 44.2 | 113nA | 80nA |
| 0x0 | 0x3 | 125 | 88.4 | 226nA | 160nA |
| 0x1 | 0x0 | 50 | 35.4 | 452nA | 320nA |
| 0x1 | 0x1 | 100 | 70.7 | 905nA | 640nA |
| 0x1 | 0x2 | 250 | 177 | 2.262µA | 1.6µA |
| 0x1 | 0x3 | 500 | 354 | 4.525µA | 3.2µA |
| 0x2 | 0x0 | 50 | 35.4 | 9.05µA | 6.4µA |
| 0x2 | 0x1 | 100 | 70.7 | 18.10µA | 12.8µA |
| 0x2 | 0x2 | 250 | 177 | 45.25µA | 32µA |
| 0x2 | 0x3 | 500 | 354 | 90.50µA | 64µA |
| 0x3 | 0x0 | 50 | 35.4 | 181µA | 128µA |
| 0x3 | 0x1 | 100 | 70.7 | 362µA | 256µA |
| 0x3 | 0x2 | 250 | 177 | 905µA | 640µA |
| 0x3 | 0x3 | 500 | 354 | 1.81mA | 1.28mA |

### BIOZ_DRV_MODE

BIOZ_DRV_MODE selects the stimulus type of the BioZ transmit channel.

When RESP_EN[0](0xB6) = 1, BIOZ_DRV_MODE is ignored.

| BIOZ_DRV_MODE[1:0] | DRIVE TYPE |
|---|---|
| 0x0 | Current Drive. A sine-wave current is driven into the body through selectable electrode pins. |
| 0x1 | Voltage Drive. A sine-wave voltage is applied to BIOZ_EL1 while BIOZ_EL4 is driven to $V_{MID_TX}$. |
| 0x2 | H-Bridge Drive. BIOZ_EL1 and BIOZ_EL4 are alternately switched between AVDD and AGND. |
| 0x3 | Standby. The transmit channel is reset and held in a low-power state, driving the electrodes to $V_{MID_TX}$. |

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## BIOZ Configuration 4 (0xA3)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| **Field** | EN_UTIL_MODE | – | – | – | – | – | – | – |
| **Reset** | 0 | – | – | – | – | – | – | – |
| **Access Type** | Write, Read | – | – | – | – | – | – | – |

### EN_UTIL_MODE

When EN_UTIL_MODE is set to 0, BIOZ_EN[1:0] selects the I or Q channel.

When EN_UTIL_MODE is set to 1, the BioZ ADC is used as a utility ADC to sense a differential ECG-related voltage selected by BIOZ_EN[1:0].

See the FIFO Description section for the Tags used in this mode.

| EN_UTIL_MODE | BIOZ_EN | UTILITY FUNCTION |
|---|---|---|
| 0 | 0x0 to 0x3 | Utility mode is disabled. |
| 1 | 0x0 | $V_{ECGP}$ - $V_{ECGN}$ differential input is connected to the BioZ channel. |
| 1 | 0x1 | $V_{ECGP}$ - $V_{MID_ECG}$ differential input is connected to the BioZ channel. |
| 1 | 0x2 | $V_{ECGN}$ - $V_{MID_ECG}$ differential input is connected to the BioZ channel. |
| 1 | 0x3 | $V_{CAPP}$ - $V_{CAPN}$ differential input is connected to the BioZ channel. |

## BIOZ Configuration 5 (0xA4)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| **Field** | BIOZ_DC_CODE_SEL | BIOZ_DC_DAC_CODE[6:0] | | | | | | |
| **Reset** | 0 | 0x00 | | | | | | |
| **Access Type** | Write, Read | Write, Read | | | | | | |

### BIOZ_DC_CODE_SEL

When BIOZ_DC_CODE_SEL is set to 0, the DDS DAC code (sine-wave sample) is selected for the 10-bit BioZ DAC. When BIOZ_DC_CODE_SEL is set to 1, BIOZ_DC_DAC_CODE[6:0] is selected for the upper 7 bits of the BioZ DAC, and the lower 3 bits are set to 0.

### BIOZ_DC_DAC_CODE

BIOZ_DC_DAC_CODE[6:0] are the upper 7 bits of the 10-bit DC code for the BioZ DAC when BIOZ_DC_CODE_SEL is set to 1. The lower 3 bits for the DAC code are set to 0.

## BIOZ Configuration 6 (0xA5)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| **Field** | BIOZ_AHPF[3:0] | | | | BIOZ_INA_MODE | BIOZ_DM_DIS | BIOZ_GAIN[1:0] | |
| **Reset** | 0x0 | | | | 0 | 0 | 0x0 | |
| **Access Type** | Write, Read | | | | Write, Read | Write, Read | Write, Read | |

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### BIOZ_AHPF

BIOZ_AHPF sets the corner frequency of the internal analog high-pass filter, or sets the filter resistance when using external capacitors on BIP and BIN. The resistance is center tapped with the midpoint connected to $V_{MID_RX}$ through a 100MΩ resistor. The 100MΩ midpoint resistor can be bypassed by settings BIOZ_CMRES_DIS[5](0xA7) to 1. When using external capacitors, the analog HPF corner frequency is set by the series capacitance and the selected common-mode resistance according to the following equation:

- $f_{-3dB} = 1/(2 \times \pi \times R_{AHPF} \times C_{SERIES})$

where $C_{SERIES}$ is the series combination of the external capacitors on BIP and BIN:

- $C_{SERIES} = (C_{BIP} \times C_{BIN}) / (C_{BIP} + C_{BIN})$

| BIOZ_AHPF[3:0] | DECODE |
|---|---|
| 0x0 | 100Hz |
| 0x1 | 200Hz |
| 0x2 | 500Hz |
| 0x3 | 1,000Hz |
| 0x4 | 2,000Hz |
| 0x5 | 5,000Hz |
| 0x6 | 10,000Hz |
| 0x7 | Resistor opened, internal capacitors shorted (AHPF bypassed) |
| 0x8 | 42.4MΩ, internal capacitors shorted |
| 0x9 | 21.2MΩ, internal capacitors shorted |
| 0xA | 8.4MΩ, internal capacitors shorted |
| 0xB | 4.2MΩ, internal capacitors shorted |
| 0xC | 2.2MΩ, internal capacitors shorted |
| 0xD | 848kΩ, internal capacitors shorted |
| 0xE | 848kΩ, internal capacitors shorted |
| 0xF | Resistor opened, internal capacitor shorted (AHPF bypassed) |

### BIOZ_INA_MODE

BIOZ_INA_MODE sets the BioZ receive channel instrumentation amplifier (INA) power mode.

| BIOZ_INA_MODE | DECODE |
|---|---|
| 0 | BioZ INA is in high-power mode (low noise mode) |
| 1 | BioZ INA is in low-power mode |

### BIOZ_DM_DIS

BIOZ_DM_DIS disables the BioZ receive channel demodulator to allow a direct conversion of the differential input voltage across BIP and BIN.

| BIOZ_DM_DIS | DECODE |
|---|---|
| 0 | BioZ demodulation clock enabled |
| 1 | BioZ demodulation clock disabled |

### BIOZ_GAIN

BIOZ_GAIN sets the combined gain of the BioZ receive channel INA and PGA.

| BIOZ_GAIN[1:0] | TOTAL GAIN (V/V) | INA GAIN (V/V) | PGA GAIN (V/V) |
|---|---|---|---|
| 0x0 | 1 | 1 | 1 |

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| BIOZ_GAIN[1:0] | TOTAL GAIN (V/V) | INA GAIN (V/V) | PGA GAIN (V/V) |
|---|---|---|---|
| 0x1 | 2 | 2 | 1 |
| 0x2 | 5 | 2 | 2.5 |
| 0x3 | 10 | 2 | 5 |

## BIOZ Configuration 7 (0xA6)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | BIOZ_EXT_CAP | BIOZ_DC_RESTORE | BIOZ_DRV_RESET | BIOZ_DAC_RESET | BIOZ_AMP_RGE[1:0] | | BIOZ_AMP_BW[1:0] | |
| Reset | 1 | 0 | 0 | 0 | 0x0 | | 0x0 | |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | | Write, Read | |

### BIOZ_EXT_CAP

BIOZ_EXT_CAP selects the external capacitor $C_{EXT}$ connected between DRVXC and DRVSJ, which AC-couples the stimulus current in current mode; thus, preventing DC current from passing into the patient stimulus electrodes. When not using an AC-coupling external capacitor, it is recommended to short DRVXC and DRVSJ since the internal switch on-resistance is approximately 35Ω.

| BIOZ_EXT_CAP | DECODE |
|---|---|
| 0 | No external capacitor used. The internal switch shorts the DRVXC and DRVSJ pins together with an on-resistance of approximately 35Ω. |
| 1 | External capacitor used. The internal switch is open, and AC current is coupled through the external capacitor. |

### BIOZ_DC_RESTORE

BIOZ_DC_RESTORE closes the DC_RESTORE switch in the current-generator amplifier circuit, which applies a feedback resistance of approximately 10M

Ω to the current-drive amplifier. This maintains the DC bias of the drive electrodes during a lead-off event, which reduces the amplifier setting time when the lead is reconnected.

When using external AC coupling capacitors on the pins assigned to DRVP or DRVN, it is recommended to set DC_RESTORE to 1 to absorb any DC offset currents and prevent amplifier saturation.

| BIOZ_DC_RESTORE | DECODE |
|---|---|
| 0 | DC_RESTORE switch is open. No feedback resistance is applied to the current drive amplifier. |
| 1 | DC_RESTORE switch is closed. A 10MΩ feedback resistance is applied to the current-drive amplifier. |

### BIOZ_DRV_RESET

BIOZ_DRV_RESET places the BioZ transmit channel in a reset state by disabling the DDS DAC and closing the RESET switch of the current-drive amplifier. This shorts the feedback network of the amplifier, configuring it as a unity gain buffer and driving both drive electrodes to $V_{MID_TX}$.

| BIOZ_DRV_RESET | DECODE |
|---|---|
| 0 | Normal Operation. The RESET switch is open. |
| 1 | Reset Condition. The DDC DAC is disabled and the RESET switch is closed, shorting the current drive amplifier feedback. |

### BIOZ_DAC_RESET

BIOZ_DAC_RESET forces the DDS DAC output to zero. The human body load is driven by the reference voltage $V_{MID_TX}$, and the AC current going through the human body load is zero.

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### BIOZ_AMP_RGE

BIOZ_AMP_RGE selects the output stage option for the voltage-drive amplifier and current-drive amplifier within the BioZ transmit channel. Higher strength is recommended for higher output current loading. Higher settings increase supply-current consumption.

It is recommended to match this settings with the BIOZ_IDRV_RGE setting.

| BIOZ_AMP_RGE[1:0] | BIOZ AMPLIFIER RANGE |
|---|---|
| 0x0 | Low |
| 0x1 | Medium-Low |
| 0x2 | Medium-High |
| 0x3 | High |

### BIOZ_AMP_BW

BIOZ_AMP_BW sets the gain bandwidth of the voltage-drive amplifier and current-drive amplifier in the BioZ transmit channel. Higher bandwidth is recommended for high-frequency applications including bioimpedance analysis and impedance cardiography. Low bandwidth is recommended for low-frequency applications including galvanic skin response to reduce power consumption.

| BIOZ_AMP_BW[1:0] | BIOZ AMPLIFIER BANDWIDTH |
|---|---|
| 0x0 | Low |
| 0x1 | Medium-Low |
| 0x2 | Medium-High |
| 0x3 | High |

## BIOZ Configuration 8 (0xA7)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | RLD_SEL_BIOZ | RLD_DRV | BIOZ_CMRES_DIS | BIOZ_STBYON | BIOZ_IPOL | BIOZ_FAST | BIOZ_INA_CHOP_EN | BIOZ_CH_FSEL |
| Reset | 0 | 0 | 0 | 0 | 0 | | 0 | |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read |

### RLD_SEL_BIOZ

RLD_SEL_BIOZ selects the BIP and BIN voltages as the inputs for the RLD common mode averager, and overrides RLD_SEL_ECG.

| RLD_SEL_BIOZ | RLD_SEL_ECG | RLD INPUT SELECTION |
|---|---|---|
| 0 | 0 | CAPP and CAPN |
| 0 | 1 | ECGP and ECGN |
| 1 | x | BIP and BIN |

### RLD_DRV

RLD_DRV selects between $V_{MID_TX}$ and $V_{RLD}$ for the BioZ transmit channel common-mode voltage, applied to the VDRV and IDRV amplifiers' noninverting terminals. This enables the DRVN pin to function as right leg drive during BioZ measurements.

| RLD_DRV | DECODE |
|---|---|
| 0 | $V_{MID_TX}$ selected as the BioZ common-mode voltage. |
| 1 | $V_{RLD}$ selected as the BioZ common-mode voltage. |

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### BIOZ_CMRES_DIS

BIOZ_CMRES_DIS controls the resistance between the analog HPF midpoint and $V_{MID_RX}$, which affects the common-mode input impedance of the BioZ receive channel when the analog HPF is enabled.

| BIOZ_CMRES_DIS | DECODE |
|---|---|
| 0 | 100MΩ |
| 1 | 0Ω |

### BIOZ_STBYON

BIOZ_STBYON selects the BioZ receive channel behavior when the transmit channel is in standby mode (BIOZ_DRV_MODE = 0x3). When in standby mode, the transmit channel does not generate a stimulus, so the receive channel can be powered down in most cases.

| BIOZ_STBYON | RECEIVE CHANNEL BEHAVIOR IN STANDBY MODE |
|---|---|
| 0 | Disabled. The receive channel amplifiers, demodulator, and ADC are disabled. |
| 1 | Enabled. The BioZ receive channel amplifiers, demodulator, and ADC remain enabled. |

### BIOZ_IPOL

BIOZ_IPOL selects the polarity of the BioZ receive channel by inverting the demodulator clock phase. This is useful if either the drive electrodes or the receive electrodes are connected with inverted polarity.

| BIOZ_IPOL | DECODE |
|---|---|
| 0 | Non-inverted |
| 1 | Inverted |

### BIOZ_FAST

When BIOZ_FAST_START is set to 1, the FAST_START function is enabled until the register bit is set to 0.

The FAST START function is to connect the BioZ inputs to $V_{MID_RX}$ through 10kΩ resistors. This helps to quickly establish the DC bias of the BioZ inputs, especially when the inputs have high capacitance.

This function can be applied any time after ECG_BIOZ_BG_EN is asserted, and is recommended to be used before BioZ measurements begin.

| BIOZ_FAST | DECODE |
|---|---|
| 0 | Normal operation |
| 1 | BIP and BIN are resistively connected to $V_{MID_RX}$ through 10kΩ resistors. |

### BIOZ_INA_CHOP_EN

BIOZ_INA_CHOP_EN enables 16kHz chopping in the BioZ receive channel instrumentation amplifier.

| BIOZ_INA_CHOP_EN | DECODE |
|---|---|
| 0 | 16kHz chopping disabled |
| 1 | 16kHz chopping enabled |

### BIOZ_CH_FSEL

BIOZ_CH_FSEL selects the chopping frequency of the BioZ receive channel PGA and AAF.

Set to 0 if the F_BIOZ is not equal to (BIOZ_ADC_CLK/16)

Set to 1 if the F_BIOZ is equal to (BIOZ_ADC_CLK/16)

Note: The synthesis frequency CANNOT equal the PGA chopping frequency or the correlator will not work. Use BIOZ_CH_FSEL to ensure they are not equal.

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| BIOZ_CH_FSEL | DECODE |
|---|---|
| 0 | BioZ PGA chopping frequency is ($f_{BIOZ_ADC_CLK}$/16) |
| 1 | BioZ PGA chopping frequency is ($f_{BIOZ_ADC_CLK}$/8) |

## BIOZ Low Threshold (0xA8)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | BIOZ_LO_THRESH[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Write, Read | | | | | | | |

### BIOZ_LO_THRESH

BIOZ_LO_THRESH sets the BioZ underrange threshold.

If the BioZ measurement is within the symmetric thresholds defined by ±32 x BIOZ_LO_THRESH for longer than approximately 128ms, the BIOZ_UNDR status bit is asserted.

## BIOZ High Threshold (0xA9)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | BIOZ_HI_THRESH[7:0] | | | | | | | |
| Reset | 0xFF | | | | | | | |
| Access Type | Write, Read | | | | | | | |

### BIOZ_HI_THRESH

BIOZ_HI_THRESH sets the BioZ overrange threshold.

If the BioZ measurement is outside of the symmetric thresholds defined by ±2048 x BIOZ_HI_THRESH for longer than approximately 128ms, the BIOZ_OVER status bit is asserted.

The default value (BIOZ_HI_THRESH = 0xFF) corresponds to a BioZ output upper threshold of 0x7F800, or about 99.6% of the full-scale range.

## BIOZ Mux Configuration 1 (0xAA)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | BMUX_RSEL[1:0] | | BMUX_BIST_EN | – | – | CONNECT_CAL_ONLY | BIOZ_MUX_EN | BIOZ_CAL_EN |
| Reset | 0x0 | | 0 | – | – | 0 | 0 | 0 |
| Access Type | Write, Read | | Write, Read | – | – | Write, Read | Write, Read | Write, Read |

### BMUX_RSEL

BMUX_RSEL selects the value of the resistive calibration load applied across DRVP and BIP, and DRVN and BIN for non-GSR applications. This load is only applied when BMUX_BIST_EN = 1. The resistor error is measured during factory test, and the error value is saved in BIST_R_ERR[7:0](0xAD).

| BMUX_RSEL[1:0] | CALIBRATION RESISTANCE (Ω) |
|---|---|
| 0x0 | 5000 |
| 0x1 | 800 |

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| BMUX_RSEL[1:0] | CALIBRATION RESISTANCE (Ω) |
|---|---|
| 0x2 | 500 |
| 0x3 | 200 |

### BMUX_BIST_EN

BMUX_BIST_EN enables the built-in self-test resistor between DRVP and BIP, and DRVN and BIN for non-GSR applications. To avoid external interference, BIOZ_MUX_EN and BIOZ_CAL_EN should both be disabled when using BMUX_BIST_EN. The resistive value is selected by BMUX_RSEL.

| BMUX_BIST_EN | DECODE |
|---|---|
| 0 | Disabled. The internal resistive load is disconnected. |
| 1 | Enabled. The internal resistive load is connected between DRVP and BIP, and DRVN and BIN. |

### CONNECT_CAL_ONLY

When both BIOZ_MUX_EN and BIOZ_CAL_EN are set to 1, enabling CONNECT_CAL_ONLY connects only the CAL1-CAL4 pins and disconnects the BIOZ_ELx pins. See BIOZ_MUX_EN for details.

### BIOZ_MUX_EN

BIOZ_MUX_EN enables the BioZ input/output MUX connections to the BIOZ_ELx pins and/or CALx pins selected by DRVP_ASSIGN, DRVN_ASSIGN, BIP_ASSIGN, BIN_ASSIGN, BIOZ_CAL_EN, and CONNECT_CAL_ONLY.

| BIOZ_MUX_EN | BIOZ_CAL_EN | CONNECT_CAL_ONLY | DECODE |
|---|---|---|---|
| 0 | X | X | MUX Disabled. All BIOZ_ELx and CALx pins are disconnected. |
| 1 | 0 | X | BIOZ_ELx pins are connected according to BIP_ASSIGN, BIN_ASSIGN, DRVP_ASSIGN, and DRVN_ASSIGN. CALx pins are disconnected. |
| 1 | 1 | 0 | BIOZ_ELx pins are connected according to BIP_ASSIGN, BIN_ASSIGN, DRVP_ASSIGN, and DRVN_ASSIGN. CALx pins are also connected. |
| 1 | 1 | 1 | BIOZ_ELx pins are disconnected. CALx pins are connected only. |

### BIOZ_CAL_EN

BIOZ_CAL_EN connects the calibration pins (CAL1–CAL4) of the BioZ input/output MUX to measure the external calibration resistor when BIOZ_MUX_EN = 1. See BIOZ_MUX_EN for details.

## BIOZ Mux Configuration 2 (0xAB)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | BMUX_GSR_RSEL[1:0] | | GSR_LOAD_EN | – | – | – | EN_EXT_INLOAD | EN_INT_INLOAD |
| Reset | 0x0 | | 0 | – | – | – | 0 | 0 |
| Access Type | Write, Read | | Write, Read | – | – | – | Write, Read | Write, Read |

### BMUX_GSR_RSEL

BMUX_GSR_RSEL selects the value of the resistive calibration load applied across DRVP and BIP, and DRVN and BIN for GSR applications. This load is only applied when GSR_LOAD_EN = 1.

| BMUX_GSR_RSEL[1:0] | LOAD RESISTANCE (kΩ) |
|---|---|
| 0x0 | 25 |
| 0x1 | 100 |

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| BMUX_GSR_RSEL[1:0] | LOAD RESISTANCE (kΩ) |
|---|---|
| 0x2 | 500 |
| 0x3 | 1000 |

### GSR_LOAD_EN

GSR_LOAD_EN enables the built-in GSR load resistor between DRVP and BIP, and DRVN and BIN. To avoid external interference, BIOZ_MUX_EN and BIOZ_CAL_EN should both be disabled when using GSR_LOAD_EN. The resistive value is selected by BMUX_GSR_RSEL.

| GSR_LOAD_EN | DECODE |
|---|---|
| 0 | Disabled. The internal resistive load is disconnected. |
| 1 | Enabled. The internal resistive load is connected between DRVP and BIP, and DRVN and BIN. |

### EN_EXT_INLOAD

EN_EXT_INLOAD enables the external guard-trace-drive circuit, which outputs the buffered voltage from BIP and BIN onto the BIOZ_EL2 and BIOZ_EL3 pins.

| EN_EXT_INLOAD | DECODE |
|---|---|
| 0 | External guard-drive circuit disabled. |
| 1 | External guard-drive circuit enabled. |

### EN_INT_INLOAD

EN_INT_INLOAD enables the circuit that compensates for input capacitive loading on BIN and BIP.

| EN_INT_INLOAD | DECODE |
|---|---|
| 0 | Input capacitive loading compensation circuit disabled. |
| 1 | Input capacitive loading compensation circuit enabled. |

## BIOZ Mux Configuration 3 (0xAC)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | BIP_ASSIGN[1:0] | | BIN_ASSIGN[1:0] | | DRVP_ASSIGN[1:0] | | DRVN_ASSIGN[1:0] | |
| Reset | 0x0 | | 0x0 | | 0x0 | | 0x0 | |
| Access Type | Write, Read | | Write, Read | | Write, Read | | Write, Read | |

### BIP_ASSIGN

BIP_ASSIGN selects the electrode pin used for the BioZ positive input (BIP) when BIOZ_MUX_EN = 1. The MUX is also used to select the Utility ADC signal (see the Utility Mode Measurement section).

| BIP_ASSIGN[1:0] | ASSIGNED ELECTRODE PIN |
|---|---|
| 0x0 | BIOZ_EL1 |
| 0x1 | BIOZ_EL2A |
| 0x2 | BIOZ_EL2B |
| 0x3 | Utility ADC Mode |

### BIN_ASSIGN

BIN_ASSIGN selects the electrode pin used for the BioZ negative input (BIN) when BIOZ_MUX_EN = 1. The MUX is also used to select the utility ADC signal (see the Utility Mode Measurement section).

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| BIN_ASSIGN[1:0] | ASSIGNED ELECTRODE PIN |
|---|---|
| 0x0 | BIOZ_EL4 |
| 0x1 | BIOZ_EL3A |
| 0x2 | BIOZ_EL3B |
| 0x3 | Utility ADC Mode |

### DRVP_ASSIGN

DRVP_ASSIGN selects the electrode pin used for the BioZ positive drive (DRVP) in current mode when BIOZ_MUX_EN = 1.

In voltage mode or H-bridge mode, the BioZ positive drive only connects to BIOZ_EL1.

The BIOZ_EL1 MUX switch has lower on-resistance than BIOZ_EL2A and BIOZ_EL2B, so it is recommended to assign DRVP to BIOZ_EL1 when BIOZ_IDRV_RGE = 0x3.

| BIOZ_DRV_MODE[1:0] | DRVP_ASSIGN[1:0] | ASSIGNED ELECTRODE PIN |
|---|---|---|
| 0x0 or 0x3 | 0x0 | BIOZ_EL1 (low resistance) |
| 0x0 or 0x3 | 0x1 | BIOZ_EL2A |
| 0x0 or 0x3 | 0x2 | BIOZ_EL2B |
| 0x0 or 0x3 | 0x3 | No Connection |
| 0x1 | X | BIOZ_EL1 |
| 0x2 | X | BIOZ_EL1 |

### DRVN_ASSIGN

DRVN_ASSIGN selects the electrode pin used for the BioZ negative drive (DRVN) in current mode when BIOZ_MUX_EN = 1.

In voltage mode or H-bridge mode, the BioZ negative drive only connects to BIOZ_EL4.

The BIOZ_EL4 MUX switch has lower on-resistance than BIOZ_EL3A and BIOZ_EL3B, so it is recommended to assign DRVN to BIOZ_EL4 when BIOZ_IDRV_RGE = 0x3.

| BIOZ_DRV_MODE[1:0] | DRVN_ASSIGN[1:0] | ASSIGNED ELECTRODE PIN |
|---|---|---|
| 0x0 or 0x3 | 0x0 | BIOZ_EL4 (low resistance) |
| 0x0 or 0x3 | 0x1 | BIOZ_EL3A |
| 0x0 or 0x3 | 0x2 | BIOZ_EL3B |
| 0x0 or 0x3 | 0x3 | No Connection |
| 0x1 | X | BIOZ_EL4 |
| 0x2 | X | BIOZ_EL4 |

## BIOZ Mux Configuration 4 (0xAD)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | BIST_R_ERR[7:0] | | | | | | | |
| Reset | 0x00 | | | | | | | |
| Access Type | Read Only | | | | | | | |

### BIST_R_ERR

The on-chip built-in self-test (BIST) resistors are available for calibrating the BioZ channel magnitude/phase error. 5kΩ, 800Ω, 500Ω, or 200Ω of resistors can be chosen by BMUX_RSEL. Due to process variations, the actual resistances vary by up to ±25%. During factory test, the actual resistance of the 800Ω resistor is measured, and the error stored in

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the BIST_R_ERR field. The actual resistance can be calculated with the following equation, where BIST_R_ERR is a 2's complement representation. The other values (5kΩ, 500Ω, and 200Ω) are process-matched to the 800Ω resistor with approximately 2% precision.

$R_{ACTUAL} = R_{NOMINAL}$ x (1 + BIST_R_ERR / 512)

For example, when BIST_R_ERR = 64 and BMUX_RSEL = 0x1, $R_{ACTUAL}$ = 800Ω x (1 + 64 / 512) = 900Ω.

BIST_R_ERR is a read-only register.

## BIOZ Lead Detect Configuration 1 (0xB0)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | EN_BIOZ_LON | EN_BIOZ_LOFF | EN_EXT_BIOZ_LOFF | EN_BIOZ_DRV_OOR | BIOZ_LOFF_IPOL | BIOZ_LOFF_IMAG[2:0] | | |
| Reset | 0 | 0 | 0 | 0 | 0 | 0x0 | | |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | | |

### EN_BIOZ_LON

EN_BIOZ_LON enables ultra-low-power (ULP) DC lead-on detection on the BIP and BIN inputs. ULP mode only functions when BioZ is not enabled (BIOZ_EN = 0x0).

| EN_BIOZ_LON | DECODE |
|---|---|
| 0 | ULP lead-on detection is disabled |
| 1 | ULP lead-on detection is enabled |

### EN_BIOZ_LOFF

EN_BIOZ_LOFF enables DC lead-off detection on the BIP and BIN inputs, and only functions when BioZ is enabled (BIOZ_EN = 0x1 or 0x2). When enabled, the lead-off status is reported by the BIOZ_LOFF_PH, BIOZ_LOFF_PL, BIOZ_LOFF_NH, and BIOZ_LOFF_NL status bits and interrupts.

| EN_BIOZ_LOFF | DECODE |
|---|---|
| 0 | BioZ DC lead-off detection is disabled. |
| 1 | BioZ DC lead-off detection is enabled. |

### EN_EXT_BIOZ_LOFF

EN_EXT_BIOZ_LOFF enables DC lead-off detection on the BIOZ_EL2B and BIOZ_EL3B pins when the BIOZ_EL2A and BIOZ_EL3A pins are being used as AC-coupled BioZ inputs. BIOZ_EL2B and BIOZ_EL3B must be connected externally to the electrode side of the AC-coupling capacitors for this feature to function.

| EN_EXT_BIOZ_LOFF | DECODE |
|---|---|
| 0 | DC lead-off detection is applied to the internal BIP and BIN nodes. |
| 1 | DC lead-off detection is applied externally through BIOZ_EL2B and BIOZ_EL3B for AC-coupled applications. |

### EN_BIOZ_DRV_OOR

EN_BIOZ_DRV_OOR enables the voltage monitor on DRVN to detect the drive electrode lead-off conditions. If the total impedance between DRVP and DRVN is high, due to either the DRVP or DRVN electrode being disconnected, the AC voltage at DRVN is large and trigger a BIOZ_DRV_OOR status and interrupt.

| EN_BIOZ_DRV_OOR | DECODE |
|---|---|
| 0 | Drive voltage out-of-range detection is disabled. |
| 1 | Drive voltage out-of-range detection is enabled. |

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### BIOZ_LOFF_IPOL

BIOZ_LOFF_IPOL sets the polarity of the matched DC current sources used for DC lead-off detection.

| BIOZ_LOFF_IPOL | LEAD-OFF CURRENT POLARITY |
|---|---|
| 0 | Non-inverted. BIP sources current; BIN sinks current. |
| 1 | Inverted. BIP sinks current; BIN sources current. |

### BIOZ_LOFF_IMAG

BIOZ_LOFF_IMAG selects the DC lead-off current magnitude.

| BIOZ_LOFF_IMAG[2:0] | DC CURRENT MAGNITUDE (nA) |
|---|---|
| 0x0 | 0 (current sources disabled) |
| 0x1 | 5 |
| 0x2 | 10 |
| 0x3 | 20 |
| 0x4 | 50 |
| 0x5 | 100 |
| 0x6 | 100 |
| 0x7 | 100 |

## BIOZ Lead-Off Threshold (0xB1)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | RESP_CG_MAG_4X | RESP_CG_MAG[2:0] | | | BIOZ_LOFF_THRESH[3:0] | | | |
| Reset | 0 | 0x0 | | | 0x0 | | | |
| Access Type | Write, Read | Write, Read | | | Write, Read | | | |

### RESP_CG_MAG_4X

RESP_CG_MAG_4x selects the amplitude of the respiration current along with RESP_CG_MAG.

| RESP_CG_MAG_4X | DECODE |
|---|---|
| 0 | RESP_CG_MAG current multiplied by 1x |
| 1 | RESP_CG_MAG current multiplied by 4x |

### RESP_CG_MAG

RESP_CG_MAG together with RESP_CG_MAG_4X selects the amplitude of the respiration drive current.

| CG_MAG[2:0] | OUTPUT CURRENT WITH RESP_CG_MAG_4X = 0 ($\mu A_{PK}$) | OUTPUT CURRENT WITH RESP_CG_MAG_4X = 1 ($\mu A_{PK}$) |
|---|---|---|
| 0x0 | 0 | 0 |
| 0x1 | 8 | 32 |
| 0x2 | 16 | 64 |
| 0x3 | 32 | 128 |
| 0x4 | 48 | 192 |
| 0x5 | 64 | 256 |
| 0x6 | 80 | 320 |
| 0x7 | 96 | 384 |

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### BIOZ_LOFF_THRESH

BIOZ_LOFF_THRESH selects the voltage threshold for the DC lead-off window comparators, which are centered at $V_{MID_RX}$. If the voltage of either BIP or BIN goes above the high threshold or below the low threshold for approximately 128ms, the corresponding BIOZ_LOFF status bit is set to 1 in register 0x04.

| BIOZ_LOFF_THRESH[3:0] | BIOZ DC LEAD-OFF THRESHOLD |
|---|---|
| 0x0 | $V_{MID_RX}$ ±200mV |
| 0x1 | $V_{MID_RX}$ ±225mV |
| 0x2 | $V_{MID_RX}$ ±250mV |
| ... | ... |
| 0xE | $V_{MID_RX}$ ±550mV |
| 0xF | $V_{MID_RX}$ ±575mV |

## BIOZ Lead Bias Configuration 1 (0xB4)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | – | – | – | BIOZ_RBIAS_VALUE[1:0] | | EN_BIOZ_RBIASP | EN_BIOZ_RBIASN |
| Reset | – | – | – | – | 0x0 | | 0 | 0 |
| Access Type | – | – | – | – | Write, Read | | Write, Read | Write, Read |

### BIOZ_RBIAS_VALUE

BIOZ_RBIAS_VALUE selects the BioZ input lead bias resistance, which is between BIP and $V_{MID_RX}$ (EN_BIOZ_RBIASP = 1), and BIN and $V_{MID_RX}$ (EN_BIOZ_RBIASN = 1).

| BIOZ_RBIAS_VALUE[1:0] | BIAS RESISTANCE (MΩ) |
|---|---|
| 0x0 | 50 |
| 0x1 | 100 |
| 0x2 | 200 |
| 0x3 | Do not use |

### EN_BIOZ_RBIASP

EN_BIOZ_RBIASP enables the BioZ lead bias on BIP. The resistor connecting BIP to $V_{MID_RX}$ is selected in BIOZ_RBIAS_VALUE.

| EN_BIOZ_RBIASP | DECODE |
|---|---|
| 0 | BIP is not resistively connected to $V_{MID_RX}$. |
| 1 | BIP is connected to $V_{MID_RX}$ using a resistor (selected by BIOZ_RBIAS_VALUE). |

### EN_BIOZ_RBIASN

EN_BIOZ_RBIASN enables BioZ lead bias on BIN. The resistor connecting BIN to $V_{MID_RX}$ is selected in BIOZ_RBIAS_VALUE.

| EN_BIOZ_RBIASN | DECODE |
|---|---|
| 0 | BIN is not resistively connected to $V_{MID_RX}$. |
| 1 | BIN is connected to $V_{MID_RX}$ using a resistor (selected by BIOZ_RBIAS_VALUE). |

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## Respiration Configuration 1 (0xB6)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | CG_LPF_DUTY[2:0] | | | CG_CHOP_CLK[1:0] | | CG_MODE[1:0] | | RESP_EN |
| Reset | 0x0 | | | 0x0 | | 0x0 | | 0 |
| Access Type | Write, Read | | | Write, Read | | Write, Read | | Write, Read |

### CG_LPF_DUTY

CG_LPF_DUTY sets the duty cycle of the respiration current generator common-mode feedback low-pass filter. Doubling the duty cycle doubles the common-mode-feedback bandwidth. Higher bandwidths result in faster current source settling, while lower bandwidths maintain higher current source output impedance.

| CG_LPF_DUTY[2:0] | TIME HIGH (BIOZ_SYNTH_CLK CYCLES) | APPROXIMATE BANDWIDTH (Hz) |
|---|---|---|
| 0x0 | 1 | 0.98 |
| 0x1 | 2 | 1.95 |
| 0x2 | 4 | 3.90 |
| 0x3 | 8 | 7.79 |
| 0x4 | 16 | 15.54 |
| 0x5 | 32 | 30.89 |
| 0x6 | 64 | 61.08 |
| 0x7 | Always On | N/A |

### CG_CHOP_CLK

CG_CHOP_CLK selects the respiration current source chopping clock divider ratio from BIOZ_SYNTH_CLK. The respiration current generators are chopped at a frequency of BIOZ_SYNTH_CLK / CG_CHOP_CLK.

| CG_CHOP_CLK[1:0] | DIVIDER RATIO |
|---|---|
| 0x0 | 256 |
| 0x1 | 512 |
| 0x2 | 1024 |
| 0x3 | 2048 |

### CG_MODE

CG_MODE selects a current source dynamic matching and common-mode feedback scheme for the respiration-current generators. These schemes can help mitigate flicker noise in the respiration current generators.

| CG_MODE[1:0] | CURRENT SOURCE ROTATION SCHEME |
|---|---|
| 0x0 | Dynamic matching disabled with analog low-pass filter |
| 0x1 | Dynamic matching enabled without analog low-pass filter |
| 0x2 | Dynamic matching enabled with analog low-pass filter |
| 0x3 | Dynamic matching enabled with internal resistive-common mode load |

### RESP_EN

When the BioZ channel is enabled, RESP_EN switches the stimulus source from the transmit channel to the balanced respiration current generators. BIOZ_BG_EN must be set to 1 and BIOZ_EN must be set to 0x1 or 0x2.

| RESP_EN | DECODE |
|---|---|
| 0 | Respiration mode is disabled, and the BioZ transmit channel is used. |

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| RESP_EN | DECODE |
|---|---|
| 1 | Respiration mode is enabled, and the balanced respiration current generators are used in place of the BioZ transmit channel. |

## Interrupt1 Enable 1 (0xC0)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | A_FULL_EN1 | PPG_FRAME_RDY_EN1 | FIFO_DATA_RDY_EN1 | ALC_OVF_EN1 | EXP_OVF_EN1 | PPG_THRESH2_HILO_EN1 | PPG_THRESH1_HILO_EN1 | – |
| Reset | 0 | 0 | 0 | 0 | 0 | 0 | 0 | – |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | – |

### A_FULL_EN1

Enables the A_FULL[7](0x00) status bit to trigger the INT1 output pin.

| A_FULL_EN1 | Decode |
|---|---|
| 0 | The A_FULL status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when A_FULL is set to 1 |

### PPG_FRAME_RDY_EN1

Enables the FRAME_RDY[6](0x00) status bit to trigger the INT1 output pin.

| PPG_FRAME_RDY_EN1 | Decode |
|---|---|
| 0 | The FRAME_RDY status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when FRAME_RDY is set to 1. |

### FIFO_DATA_RDY_EN1

Enables the FIFO_DATA_RDY[5](0x00) status bit to trigger the INT1 output pin.

| FIFO_DATA_RDY_EN1 | Decode |
|---|---|
| 0 | The FIFO_DATA_RDY status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when FIFO_DATA_RDY is set to 1. |

### ALC_OVF_EN1

Enables the ALC_OVF[4](0x00) status bit to trigger the INT1 output pin.

| ALC_OVF_EN1 | Decode |
|---|---|
| 0 | The ALC_OVF status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when ALC_OVF is set to 1. |

### EXP_OVF_EN1

Enables the EXP_OVF[3](0x00) status bit to trigger the INT1 output pin.

| EXP_OVF_EN1 | Decode |
|---|---|
| 0 | The EXP_OVF status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when EXP_OVF is set to 1. |

### PPG_THRESH2_HILO_EN1

Enables the PPG_THRESH2_HILO[2](0x00) status bit to trigger the INT1 output pin.

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| PPG_THRESH2_HILO_EN1 | Decode |
|---|---|
| 0 | The PPG_THRESH2_HILO status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when PPG_THRESH2_HILO is set to 1. |

### PPG_THRESH1_HILO_EN1

Enables the PPG_THRESH1_HILO[1](0x00) status bit to trigger the INT1 output pin.

| PPG_THRESH1_HILO_EN1 | Decode |
|---|---|
| 0 | The PPG_THRESH1_HILO status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when PPG_THRESH1_HILO is set to 1. |

## Interrupt1 Enable 2 (0xC1)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | INVALID_PPG_CFG_EN1 | – | LED6_COMPB_EN1 | LED5_COMPB_EN1 | LED4_COMPB_EN1 | LED3_COMPB_EN1 | LED2_COMPB_EN1 | LED1_COMPB_EN1 |
| Reset | 0 | – | 0 | 0 | 0 | 0 | 0 | 0 |
| Access Type | Write, Read | – | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read |

### INVALID_PPG_CFG_EN1

Enables the INVALID_PPG_CFG[7](0x01) status bit to trigger the INT1 output pin.

| INVALID_PPG_CFG_EN1 | Decode |
|---|---|
| 0 | The INVALID_PPG_CFG status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when INVALID_PPG_CFG is set to 1. |

### LED6_COMPB_EN1

Enables the LED6_COMPB[5](0x01) status bit to trigger the INT1 output pin.

| LED6_COMPB_EN1 | Decode |
|---|---|
| 0 | The LED6_COMPB status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when LED6_COMPB is set to 1. |

### LED5_COMPB_EN1

Enables the LED5_COMPB[4](0x01) status bit to trigger the INT1 output pin.

| LED5_COMPB_EN1 | Decode |
|---|---|
| 0 | The LED5_COMPB status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when LED5_COMPB is set to 1. |

### LED4_COMPB_EN1

Enables the LED4_COMPB[3](0x01) status bit to trigger the INT1 output pin.

| LED4_COMPB_EN1 | Decode |
|---|---|
| 0 | The LED4_COMPB status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when LED4_COMPB is set to 1. |

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### LED3_COMPB_EN1

Enables the LED3_COMPB[2](0x01) status bit to trigger the INT1 output pin.

| LED3_COMPB_EN1 | Decode |
|---|---|
| 0 | The LED3_COMPB status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when LED3_COMPB is set to 1. |

### LED2_COMPB_EN1

Enables the LED2_COMPB[1](0x01) status bit to trigger the INT1 output pin.

| LED2_COMPB_EN1 | Decode |
|---|---|
| 0 | The LED2_COMPB status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when LED2_COMPB is set to 1. |

### LED1_COMPB_EN1

Enables the LED1_COMPB[0](0x01) status bit to trigger the INT1 output pin.

| LED1_COMPB_EN1 | Decode |
|---|---|
| 0 | The LED1_COMPB status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when LED1_COMPB is set to 1. |

## Interrupt1 Enable 3 (0xC2)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | – | – | FREQ_UNLOCK_EN1 | FREQ_LOCK_EN1 | PHASE_UNLOCK_EN1 | PHASE_LOCK_EN1 | – |
| Reset | – | – | – | | | 0 | 0 | – |
| Access Type | – | – | – | Write, Read | Write, Read | Write, Read | Write, Read | – |

### FREQ_UNLOCK_EN1

Enables the FREQ_UNLOCK[4](0x02) status bit to trigger the INT1 output pin.

| FREQ_UNLOCK_EN1 | Decode |
|---|---|
| 0 | The FREQ_UNLOCK status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when FREQ_UNLOCK is set to 1. |

### FREQ_LOCK_EN1

Enables the FREQ_LOCK[3](0x02) status bit to trigger the INT1 output pin.

| FREQ_LOCK_EN1 | Decode |
|---|---|
| 0 | The FREQ_LOCK status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when FREQ_LOCK is set to 1. |

### PHASE_UNLOCK_EN1

Enables PHASE_UNLOCK[2](0x02) to trigger the INT1 output pin. This interrupt is only asserted if PLL_EN[0](0x18) is set to 1.

| PHASE_UNLOCK_EN1 | Decode |
|---|---|
| 0 | The PHASE_UNLOCK status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when PHASE_UNLOCK is set to 1. |

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### PHASE_LOCK_EN1

Enables the PHASE_LOCK[1](0x02) status bit to trigger the INT1 output pin.

| PHASE_LOCK_EN1 | Decode |
|---|---|
| 0 | The PHASE_LOCK status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when PHASE_LOCK is set to 1. |

## Interrupt1 Enable 4 (0xC3)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | ECG_LON_EN1 | – | ECG_FAST_REC_EN1 | RLD_OOR_EN1 | ECG_LOFF_PH_EN1 | ECG_LOFF_PL_EN1 | ECG_LOFF_NH_EN1 | ECG_LOFF_NL_EN1 |
| Reset | 0 | – | 0 | 0 | 0 | 0 | 0 | 0 |
| Access Type | Write, Read | – | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read |

### ECG_LON_EN1

Enables ECG_LON[7](0x03) to trigger the INT1 output pin.

| ECG_LON_EN1 | Decode |
|---|---|
| 0 | The ECG_LON status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when ECG_LON is set to 1. |

### ECG_FAST_REC_EN1

Enables the ECG_FAST_REC[5](0x03) status bit to trigger the INT1 output pin.

| ECG_FAST_REC_EN1 | Decode |
|---|---|
| 0 | The ECG_FAST_REC status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when ECG_FAST_REC is set to 1. |

### RLD_OOR_EN1

Enables the RLD_OOR[4](0x03) status bit to trigger the INT1 output pin.

| RLD_OOR_EN1 | Decode |
|---|---|
| 0 | The RLD_OOR status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when RLD_OOR is set to 1. |

### ECG_LOFF_PH_EN1

Enables the ECG_LOFF_PH[3](0x03) status bit to trigger the INT1 output pin.

| ECG_LOFF_PH_EN1 | Decode |
|---|---|
| 0 | The ECG_LOFF_PH status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when ECG_LOFF_PH is set to 1. |

### ECG_LOFF_PL_EN1

Enables the ECG_LOFF_PL[2](0x03) status bit to trigger the INT1 output pin.

| ECG_LOFF_PL_EN1 | Decode |
|---|---|
| 0 | The ECG_LOFF_PL status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when ECG_LOFF_PL is set to 1. |

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### ECG_LOFF_NH_EN1

Enables the ECG_LOFF_NH[1](0x03) status bit to trigger the INT1 output pin.

| ECG_LOFF_NH_EN1 | Decode |
|---|---|
| 0 | The ECG_LOFF_NH status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when ECG_LOFF_NH is set to 1. |

### ECG_LOFF_NL_EN1

Enables the ECG_LOFF_NL[0](0x03) status bit to trigger the INT1 output pin.

| ECG_LOFF_NL_EN1 | Decode |
|---|---|
| 0 | The ECG_LOFF_NL status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when ECG_LOFF_NL is set to 1. |

## Interrupt1 Enable 5 (0xC4)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | BIOZ_LON_EN1 | BIOZ_OVER_EN1 | BIOZ_UNDR_EN1 | BIOZ_DRVP_OFF_EN1 | BIOZ_LOFF_PH_EN1 | BIOZ_LOFF_PL_EN1 | BIOZ_LOFF_NH_EN1 | BIOZ_LOFF_NL_EN1 |
| Reset | 0 | 0 | 0 | 0b0 | 0 | 0 | 0 | 0 |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read |

### BIOZ_LON_EN1

Enables the BIOZ_LON[7](0x04) status bit to trigger the INT1 output pin.

| BIOZ_LON_EN1 | Decode |
|---|---|
| 0 | The BIOZ_LON status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when BIOZ_LON is set to 1. |

### BIOZ_OVER_EN1

Enables the BIOZ_OVER[6](0x04) status bit to trigger the INT1 output pin.

| BIOZ_OVER_EN1 | Decode |
|---|---|
| 0 | The BIOZ_OVER status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when BIOZ_OVER is set to 1. |

### BIOZ_UNDR_EN1

Enables the BIOZ_UNDR[5](0x04) status bit to trigger the INT1 output pin.

| BIOZ_UNDR_EN1 | Decode |
|---|---|
| 0 | The BIOZ_UNDR status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when BIOZ_UNDR is set to 1. |

### BIOZ_DRVP_OFF_EN1

Enables RLD_OOR[3](0x04) to trigger the INT1 output pin.

| Value | Decode |
|---|---|
| 0 | RLD_OOR does not impact the INT1 output pin. |
| 1 | INT1 is triggered when RLD_OOR is set to 1. |

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### BIOZ_LOFF_PH_EN1

Enables the BIOZ_LOFF_PH[3](0x04) status bit to trigger the INT1 output pin.

| BIOZ_LOFF_PH_EN1 | Decode |
|---|---|
| 0 | The BIOZ_LOFF_PH status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when BIOZ_LOFF_PH is set to 1. |

### BIOZ_LOFF_PL_EN1

Enables BIOZ_LOFF_PL[3](0x04) to trigger the INT1 output pin.

| BIOZ_LOFF_PL_EN1 | Decode |
|---|---|
| 0 | The BIOZ_LOFF_PL status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when BIOZ_LOFF_PL is set to 1. |

### BIOZ_LOFF_NH_EN1

Enables the BIOZ_LOFF_NH[1](0x04) status bit to trigger the INT1 output pin.

| BIOZ_LOFF_NH_EN1 | Decode |
|---|---|
| 0 | The BIOZ_LOFF_NH status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when BIOZ_LOFF_NH is set to 1. |

### BIOZ_LOFF_NL_EN1

Enables the BIOZ_LOFF_NL[0](0x04) status bit to trigger the INT1 output pin.

| BIOZ_LOFF_NL_EN1 | Decode |
|---|---|
| 0 | The BIOZ_LOFF_NL status bit does not impact the INT1 output pin. |
| 1 | INT1 is triggered when BIOZ_LOFF_NL is set to 1. |

## Interrupt2 Enable 1 (0xC5)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | A_FULL_EN2 | PPG_FRAME_RDY_EN2 | FIFO_DATA_RDY_EN2 | ALC_OVF_EN2 | EXP_OVF_EN2 | PPG_THRESH2_HILO_EN2 | PPG_THRESH1_HILO_EN2 | – |
| Reset | 0 | 0 | 0 | 0 | 0 | 0 | 0 | – |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | – |

### A_FULL_EN2

Enables the A_FULL[7](0x00) status bit to trigger the INT2 output pin.

| A_FULL_EN2 | Decode |
|---|---|
| 0 | The A_FULL status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when A_FULL is set to 1 |

### PPG_FRAME_RDY_EN2

Enables the FRAME_RDY[6](0x00) status bit to trigger the INT2 output pin.

| PPG_FRAME_RDY_EN2 | Decode |
|---|---|
| 0 | The FRAME_RDY status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when FRAME_RDY is set to 1. |

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### FIFO_DATA_RDY_EN2

Enables the FIFO_DATA_RDY[5](0x00) status bit to trigger the INT2 output pin.

| FIFO_DATA_RDY_EN2 | Decode |
|---|---|
| 0 | The FIFO_DATA_RDY status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when FIFO_DATA_RDY is set to 1. |

### ALC_OVF_EN2

Enables the ALC_OVF[4](0x00) status bit to trigger the INT2 output pin.

| ALC_OVF_EN2 | Decode |
|---|---|
| 0 | The ALC_OVF status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when ALC_OVF is set to 1. |

### EXP_OVF_EN2

Enables the EXP_OVF[3](0x00) status bit to trigger the INT2 output pin.

| EXP_OVF_EN2 | Decode |
|---|---|
| 0 | EXP_OVF does not impact the INT2 output pin. |
| 1 | INT2 is triggered when EXP_OVF is set to 1. |

### PPG_THRESH2_HILO_EN2

Enables the PPG_THRESH2_HILO[2](0x00) status bit to trigger the INT2 output pin.

| PPG_THRESH2_HILO_EN2 | Decode |
|---|---|
| 0 | The PPG_THRESH2_HILO status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when PPG_THRESH2_HILO is set to 1. |

### PPG_THRESH1_HILO_EN2

Enables the PPG_THRESH1_HILO[1](0x00) status bit to trigger the INT2 output pin.

| PPG_THRESH1_HILO_EN2 | Decode |
|---|---|
| 0 | The PPG_THRESH1_HILO status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when PPG_THRESH1_HILO is set to 1. |

## Interrupt2 Enable 2 (0xC6)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | INVALID_PPG_CFG_EN2 | – | LED6_COMPB_EN2 | LED5_COMPB_EN2 | LED4_COMPB_EN2 | LED3_COMPB_EN2 | LED2_COMPB_EN2 | LED1_COMPB_EN2 |
| Reset | 0 | – | 0 | 0 | 0 | 0 | 0 | 0 |
| Access Type | Write, Read | – | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read |

### INVALID_PPG_CFG_EN2

Enables the INVALID_PPG_CFG[7](0x01) status bit to trigger the INT2 output pin.

| INVALID_PPG_CFG_EN2 | Decode |
|---|---|
| 0 | The INVALID_PPG_CFG status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when INVALID_PPG_CFG is set to 1. |

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### LED6_COMPB_EN2

Enables the LED6_COMPB[5](0x01) status bit to trigger the INT2 output pin.

| LED6_COMPB_EN2 | Decode |
|---|---|
| 0 | The LED6_COMPB status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when LED6_COMPB is set to 1. |

### LED5_COMPB_EN2

Enables the LED5_COMPB[4](0x01) status bit to trigger the INT2 output pin.

| LED5_COMPB_EN2 | Decode |
|---|---|
| 0 | The LED5_COMPB status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when LED5_COMPB is set to 1. |

### LED4_COMPB_EN2

Enables the LED4_COMPB[3](0x01) status bit to trigger the INT2 output pin.

| LED4_COMPB_EN2 | Decode |
|---|---|
| 0 | The LED4_COMPB status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when LED4_COMPB is set to 1. |

### LED3_COMPB_EN2

Enables the LED3_COMPB[2](0x01) status bit to trigger the INT2 output pin.

| LED3_COMPB_EN2 | Decode |
|---|---|
| 0 | The LED3_COMPB status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when LED3_COMPB is set to 1. |

### LED2_COMPB_EN2

Enables the LED2_COMPB[1](0x01) status bit to trigger the INT2 output pin.

| LED2_COMPB_EN2 | Decode |
|---|---|
| 0 | The LED2_COMPB status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when LED2_COMPB is set to 1. |

### LED1_COMPB_EN2

Enables the LED1_COMPB[0](0x01) status bit to trigger the INT2 output pin.

| LED1_COMPB_EN2 | Decode |
|---|---|
| 0 | The LED1_COMPB status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when LED1_COMPB is set to 1. |

## Interrupt2 Enable 3 (0xC7)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | – | – | – | FREQ_UNLOCK_EN2 | FREQ_LOCK_EN2 | PHASE_UNLOCK_EN2 | PHASE_LOCK_EN2 | – |
| Reset | – | – | – | | | 0 | 0 | – |
| Access Type | – | – | – | Write, Read | Write, Read | Write, Read | Write, Read | – |

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### FREQ_UNLOCK_EN2

Enables the FREQ_UNLOCK[4](0x02) status bit to trigger the INT2 output pin.

| FREQ_UNLOCK_EN2 | Decode |
|---|---|
| 0 | The FREQ_UNLOCK status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when FREQ_UNLOCK is set to 1. |

### FREQ_LOCK_EN2

Enables the FREQ_LOCK[3](0x02) status bit to trigger the INT2 output pin.

| FREQ_LOCK_EN2 | Decode |
|---|---|
| 0 | The FREQ_LOCK status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when FREQ_LOCK is set to 1. |

### PHASE_UNLOCK_EN2

Enables the PHASE_UNLOCK[2](0x02) status bit to trigger the INT2 output pin. This interrupt is only asserted if PLL_EN[0](0x18) is set to 1.

| PHASE_UNLOCK_EN2 | Decode |
|---|---|
| 0 | The PHASE_UNLOCK status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when PHASE_UNLOCK is set to 1. |

### PHASE_LOCK_EN2

Enables the PHASE_LOCK[1](0x02) status bit to trigger the INT2 output pin.

| PHASE_LOCK_EN2 | Decode |
|---|---|
| 0 | The PHASE_LOCK status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when PHASE_LOCK is set to 1. |

## Interrupt2 Enable 4 (0xC8)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | ECG_LON_EN2 | – | ECG_FAST_REC_EN2 | RLD_OOR_EN2 | ECG_LOFF_PH_EN2 | ECG_LOFF_PL_EN2 | ECG_LOFF_NH_EN2 | ECG_LOFF_NL_EN2 |
| Reset | 0 | – | 0 | 0 | 0 | 0 | 0 | 0 |
| Access Type | Write, Read | – | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read |

### ECG_LON_EN2

Enables ECG_LON[7](0x05) to trigger the INT2 output pin.

| ECG_LON_EN2 | Decode |
|---|---|
| 0 | The ECG_LON status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when ECG_LON is set to 1. |

### ECG_FAST_REC_EN2

Enables the ECG_FAST_REC[5](0x03) status bit to trigger the INT2 output pin.

| ECG_FAST_REC_EN2 | Decode |
|---|---|
| 0 | The ECG_FAST_REC status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when ECG_FAST_REC is set to 1. |

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### RLD_OOR_EN2

Enables the RLD_OOR[4](0x03) status bit to trigger the INT2 output pin.

| RLD_OOR_EN2 | Decode |
|---|---|
| 0 | The RLD_OOR status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when RLD_OOR is set to 1. |

### ECG_LOFF_PH_EN2

Enables ECG_LOFF_PH[3](0x03) to trigger the INT2 output pin.

| ECG_LOFF_PH_EN2 | Decode |
|---|---|
| 0 | The ECG_LOFF_PH status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when ECG_LOFF_PH is set to 1. |

### ECG_LOFF_PL_EN2

Enables ECG_LOFF_PL[2](0x03) to trigger the INT2 output pin.

| ECG_LOFF_PL_EN2 | Decode |
|---|---|
| 0 | The ECG_LOFF_PL status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when ECG_LOFF_PL is set to 1. |

### ECG_LOFF_NH_EN2

Enables the ECG_LOFF_NH[1](0x03) status bit to trigger the INT2 output pin.

| ECG_LOFF_NH_EN2 | Decode |
|---|---|
| 0 | The ECG_LOFF_NH status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when ECG_LOFF_NH is set to 1. |

### ECG_LOFF_NL_EN2

Enables the ECG_LOFF_NL[0](0x03) status bit to trigger the INT2 output pin.

| ECG_LOFF_NL_EN2 | Decode |
|---|---|
| 0 | The ECG_LOFF_NL status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when ECG_LOFF_NL is set to 1. |

## Interrupt2 Enable 5 (0xC9)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | BIOZ_LON_EN2 | BIOZ_OVER_EN2 | BIOZ_UNDR_EN2 | BIOZ_DRVP_OFF_EN2 | BIOZ_LOFF_PH_EN2 | BIOZ_LOFF_PL_EN2 | BIOZ_LOFF_NH_EN2 | BIOZ_LOFF_NL_EN2 |
| Reset | 0 | 0 | 0 | 0b0 | 0 | 0 | 0 | 0 |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read |

### BIOZ_LON_EN2

Enables the BIOZ_LON[7](0x04) status bit to trigger the INT2 output pin.

| BIOZ_LON_EN2 | Decode |
|---|---|
| 0 | The BIOZ_LON status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when BIOZ_LON is set to 1. |

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### BIOZ_OVER_EN2

Enables the BIOZ_OVER[6](0x04) status bit to trigger the INT2 output pin.

| BIOZ_OVER_EN2 | Decode |
|---|---|
| 0 | The BIOZ_OVER status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when BIOZ_OVER is set to 1. |

### BIOZ_UNDR_EN2

Enables the BIOZ_UNDR[5](0x04) status bit to trigger the INT2 output pin.

| BIOZ_UNDR_EN2 | Decode |
|---|---|
| 0 | The BIOZ_UNDR status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when BIOZ_UNDR is set to 1. |

### BIOZ_DRVP_OFF_EN2

Enables DRVN_OOR [4](0x04) to trigger the INT2 output pin.

| Value | Decode |
|---|---|
| 0 | The DRVP_OOR does not impact the INT2 output pin. |
| 1 | INT2 is triggered when DRVN_OOR is set to 1. |

### BIOZ_LOFF_PH_EN2

Enables the BIOZ_LOFF_PH[3](0x05) status bit to trigger the INT2 output pin.

| BIOZ_LOFF_PH_EN2 | Decode |
|---|---|
| 0 | The BIOZ_LOFF_PH status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when BIOZ_LOFF_PH is set to 1. |

### BIOZ_LOFF_PL_EN2

Enables the BIOZ_LOFF_PL[2](0x04) status bit to trigger the INT2 output pin.

| BIOZ_LOFF_PL_EN2 | Decode |
|---|---|
| 0 | The BIOZ_LOFF_PL status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when BIOZ_LOFF_PL is set to 1. |

### BIOZ_LOFF_NH_EN2

Enables the BIOZ_LOFF_NH[1](0x04) status bit to trigger the INT2 output pin.

| BIOZ_LOFF_NH_EN2 | Decode |
|---|---|
| 0 | The BIOZ_LOFF_NH status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when BIOZ_LOFF_NH is set to 1. |

### BIOZ_LOFF_NL_EN2

Enables the BIOZ_LOFF_NL[0](0x04) status bit to trigger the INT2 output pin.

| BIOZ_LOFF_NL_EN2 | Decode |
|---|---|
| 0 | The BIOZ_LOFF_NL status bit does not impact the INT2 output pin. |
| 1 | INT2 is triggered when BIOZ_LOFF_NL is set to 1. |

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## Part ID (0xFF)

| BIT | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| Field | PART_ID[7:0] | | | | | | | |
| Reset | 0x43 | | | | | | | |
| Access Type | Read Only | | | | | | | |

### PART_ID

This register stores the Part identifier for the chip.

| PART_ID[7:0] | Part Number | Description |
|---|---|---|
| 0x43 | MAX86178 | ECG + BioZ + Quad PD, Dual-Channel PPG with six LED Drivers |

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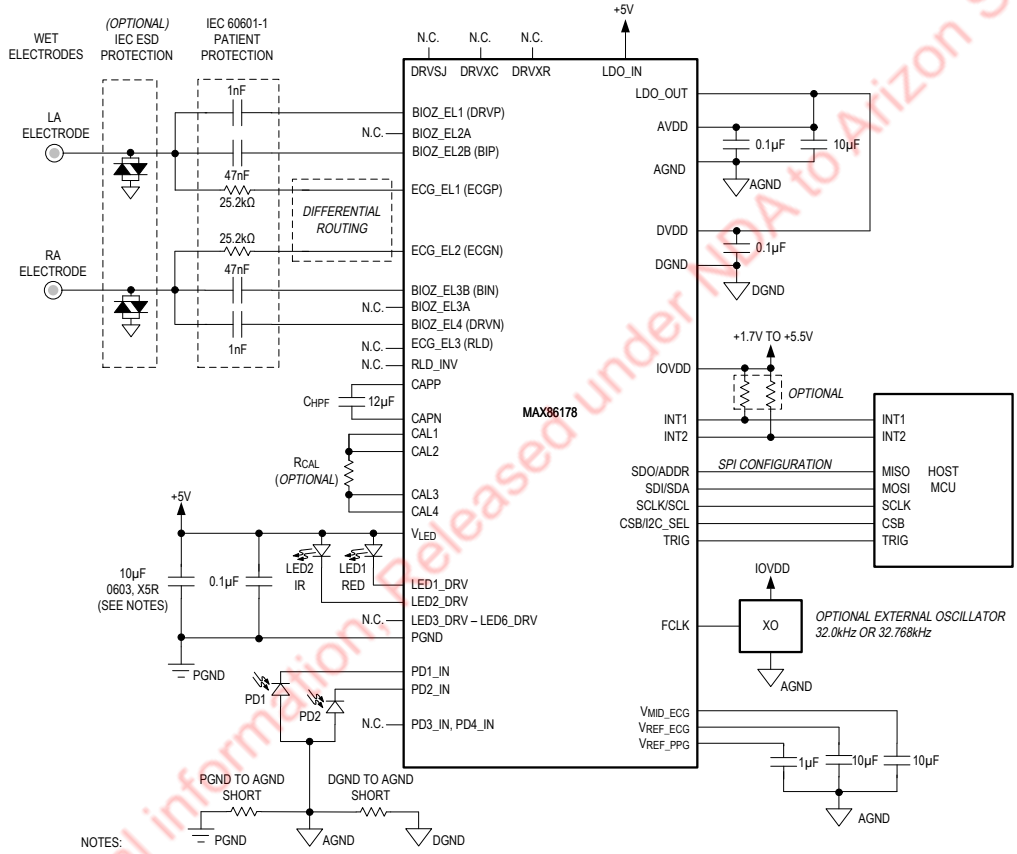
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## Typical Application Circuits

### Typical Application Circuit 1

**Chest Patch with 2 Wet Electrodes Supporting Continuous Respiration, Continuous ECG, and Continuous $SpO_2$**




NOTES:

- TWO ELECTRODE APPLICATIONS REQUIRE "ECG LEAD BIAS" TO BE ENABLED.
- WHEN ECG IS RUNNING IN PARALLEL WITH RESPIRATION MODE, THE DRVN AND DRVP CAPS ARE MINIMIZED TO 1NF IN ORDER TO MAINTAIN HIGH ECG INPUT IMPEDANCE.
- OPTIONAL THIRD ECG ELECTRODE FOR RLD (NOT SHOWN) CAN IMPROVE PERFORMANCE FOR HIGHER ELECTRODE IMPEDANCES.
- IEC 60601-1 PROTECTION COMPONENTS PROTECT THE PATIENT AGAINST DC CURRENTS IN A SINGLE-FAULT CONDITION SUCH AS BIOZ_ELX OR ECG_ELX PINS BEING SHORTED TO AVDD AND AGND.
- CHOOSE ECG PATIENT PROTECTION RESISTORS THAT ARE 1% TOLERANCE OR BETTER TO MAXIMIZE CMRR.
- AGND IS PLACED BOTH ADJACENT AND COPLANAR AROUND PD_IN LINES, SHIELDING THEM FROM OTHER RETURN OR CURRENT-CARRYING GROUNDS.
- THE 10µF CAPACITOR ON THE $V_{LED}$ LINE IS RECOMMENDED DUE TO ITS HIGH EFFECTIVE CAPACITANCE, WHICH IS GREATER THAN 4µF AT THE $V_{LED}$ VOLTAGE LEVEL (TYP +5V).
  - WHEN CHOOSING LESS THAN 0603-SIZED CAPACITORS, CONSIDER THE DERATING AT THE APPLIED VOLTAGE LEVEL TO ENSURE ENOUGH EFFECTIVE CAPACITANCE TO MAINTAIN HIGH SNR PERFORMANCE.
  - SMALLER BODY CAPACITORS DEGRADE SIGNIFICANTLY WHEN THE APPLIED VOLTAGES ($V_{LED}$) APPROACH THEIR VOLTAGE RATING.
- $C_{HPF}$ SETS THE HPF CORNER FREQUENCY. FOR WET ELECTRODES, 12µF SELECTS <0.05Hz. PLACE AS CLOSE AS POSSIBLE TO PINS AND SHIELD WITH AGND.
- $R_{FB}$ IS OPTIONAL. THIS OPTIONAL EXTERNAL FEEDBACK RESISTOR SETS THE GAIN OF THE RLD AMPLIFIER. INTERNAL REGISTER SETTINGS ALLOW FOR 12V/V, 24V/V, 48V/V, AND 97V/V GAIN.
- THE DRVXC CAPACITOR AC COUPLES THE DRVP BIOZ STIMULUS CURRENT, PREVENTING DC CURRENT FROM FLOWING INTO THE ELECTRODES.
- $R_{EXT}$ IS OPTIONAL AND SETS A CUSTOM BIOZ CURRENT STIMULUS MAGNITUDE. INTERNAL SETTINGS ARE AVAILABLE FROM 16nA$_{RMS}$ TO 1.28mA$_{RMS}$.
- FOR IEC 60601-2-47 COMPLIANCE, FCLK MUST BE SUPPLIED BY A LOW DRIFT 32.0kHz OR 32.768kHz OSCILLATOR.
- INT1 AND INT2 INTERRUPTS HAVE THE OPTION TO BE CONFIGURED AS ACTIVE DRIVE (ACTIVE-HIGH OR ACTIVE-LOW), ELIMINATING THE NEED FOR EXTERNAL PULLUPS.
- MULTIPLE PHOTODIODES CAN BE CONNECTED IN PARALLEL TO EACH PDM_IN LINE, TAKING NOTE TO ADJUST PD-CAPACITANCE BIAS REGISTER AS NEEDED.
- THE LDO SHOULD NOT BE USED TO POWER ANY OTHER DEVICE OTHER THEN THE MAX86178

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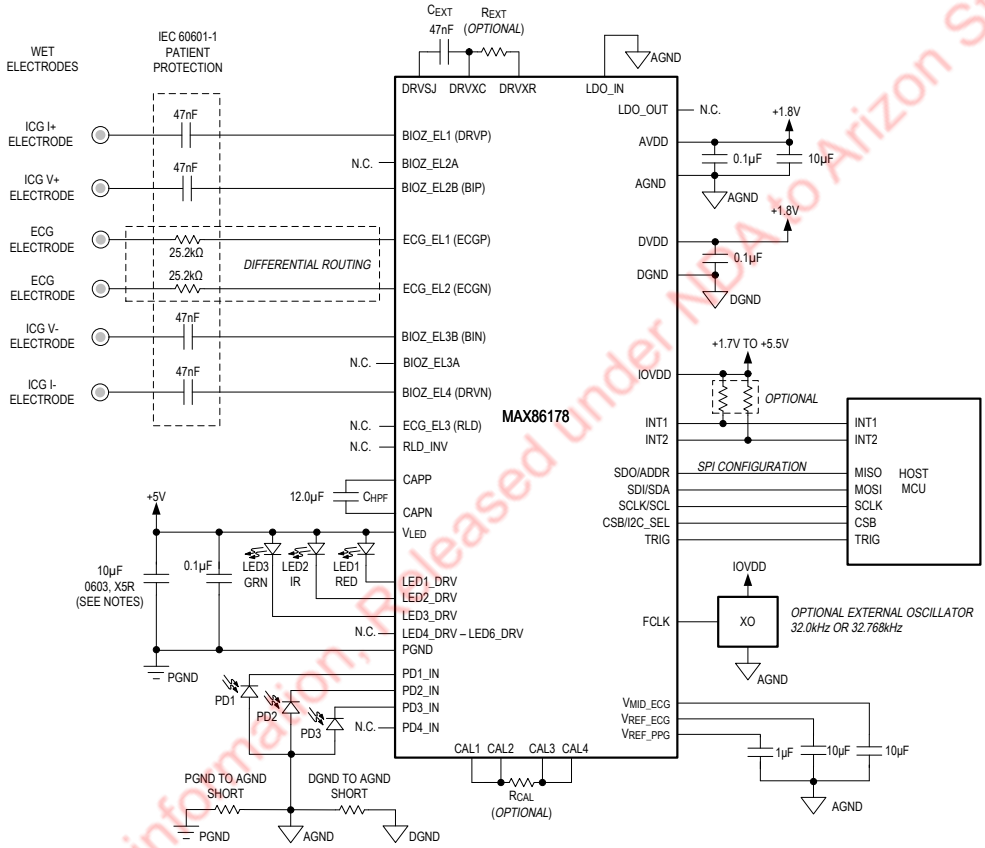
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## Typical Application Circuits (continued)

### Typical Application Circuit 2

**Wearable Device with 6 Wet Electrodes Supporting Continuous ECG, Continuous ICG, and Continuous $SpO_2$**




NOTES:

- ICG TYPICALLY REQUIRES A DIFFERENT ELECTRODE PLACEMENT WHICH MAY NOT SUPPORT OPTIMAL ECG, THEREFORE ECG AND BIOZ ELECTRODES CANNOT BE SHARED.
- CHOOSE ECG PATIENT PROTECTION RESISTORS THAT ARE 1% TOLERANCE OR BETTER TO MAXIMIZE CMRR.
- IEC 60601-1 PROTECTION COMPONENTS PROTECT THE PATIENT AGAINST DC CURRENTS IN A SINGLE-FAULT CONDITION SUCH AS BIOZ_ELX OR ECG_ELX PINS BEING SHORTED TO AVDD AND AGND.
- AGND IS PLACED BOTH ADJACENT AND COPLANAR AROUND PD_IN LINES, SHIELDING THEM FROM OTHER RETURN OR CURRENT-CARRYING GROUNDS.
- THE 10µF CAPACITOR ON THE $V_{LED}$ LINE IS RECOMMENDED DUE TO ITS HIGH EFFECTIVE CAPACITANCE, WHICH IS GREATER THAN 4µF AT THE $V_{LED}$ VOLTAGE LEVEL (TYP +5V).
  - WHEN CHOOSING LESS THAN 0603-SIZED CAPACITORS, CONSIDER THE DERATING AT THE APPLIED VOLTAGE LEVEL TO ENSURE ENOUGH EFFECTIVE CAPACITANCE TO MAINTAIN HIGH SNR PERFORMANCE.
  - SMALLER BODY CAPS DEGRADE SIGNIFICANTLY WHEN THE APPLIED VOLTAGES ($V_{LED}$) APPROACH THEIR VOLTAGE RATING.
- $C_{HPF}$ SETS THE HPF CORNER FREQUENCY. FOR TYPICAL USE CASES, 1.2µF SHOULD BE SELECTED FOR <0.67Hz. PLACE AS CLOSE AS POSSIBLE TO PINS AND SHIELD WITH AGND.
- $R_{FB}$ IS OPTIONAL. THIS OPTIONAL EXTERNAL FEEDBACK RESISTOR SETS THE GAIN OF THE RLD AMPLIFIER. INTERNAL REGISTER SETTINGS ALLOW FOR 12V/V, 24V/V, 48V/V, AND 97V/V GAIN.
- THE DRVXC CAPACITOR AC COUPLES THE DRVP BIOZ STIMULUS CURRENT, PREVENTING DC CURRENT FROM FLOWING INTO THE ELECTRODES.
- $R_{EXT}$ IS OPTIONAL AND SETS A CUSTOM BIOZ CURRENT STIMULUS MAGNITUDE. INTERNAL SETTINGS ARE AVAILABLE FROM 16nA$_{RMS}$ TO 1.28mA$_{RMS}$.
- FOR IEC 60601-2-47 COMPLIANCE, FCLK MUST BE SUPPLIED BY A LOW DRIFT 32.0kHz OR 32.768kHz OSCILLATOR.
- INT1 AND INT2 INTERRUPTS HAVE THE OPTION TO BE CONFIGURED AS ACTIVE DRIVE (ACTIVE HIGH OR ACTIVE LOW), ELIMINATING THE NEED FOR EXTERNAL PULLUPS.
- MULTIPLE PHOTODIODES CAN BE CONNECTED IN PARALLEL TO EACH PDM_IN LINE, TAKING NOTE TO ADJUST PD-CAPACITANCE BIAS REGISTER AS NEEDED.
- OPTIONAL ECG AND BIOZ ESD PROTECTION NOT SHOWN.
- THE LDO SHOULD NOT BE USED TO POWER ANY OTHER DEVICE OTHER THEN THE MAX86178

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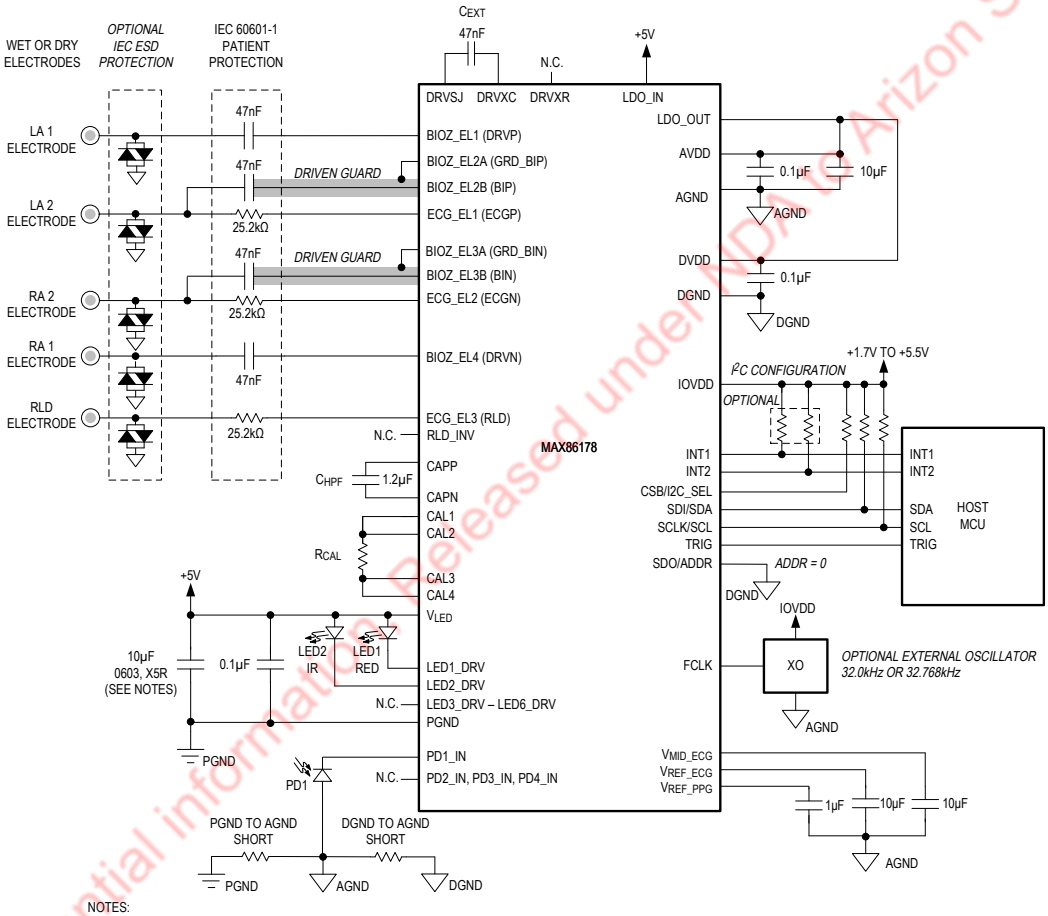
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## Typical Application Circuits (continued)

### Typical Application Circuit 3

**Wearable Device with Wet or Dry Electrodes Supporting Continuous $SpO_2$ and Either Continuous ECG and Continuous Respiration, or Intermittent BIA**




NOTES:

- IEC 60601-1 PROTECTION COMPONENTS PROTECT THE PATIENT AGAINST DC CURRENTS IN A SINGLE-FAULT CONDITION SUCH AS BIOZ_ELX OR ECG_ELX PINS BEING SHORTED TO AVDD AND AGND.
- GUARD TRACES REMOVE THE EFFECT OF THE PARASITIC PCB CAPACITANCES BY DRIVING A GUARD SIGNAL; THESE GUARD SIGNALS ARE ROUTED TO SURROUND BIP AND BIN NETS WITH BOTH CO-PLANAR AND ADJACENT SHIELDS.
- CHOOSE ECG PATIENT PROTECTION RESISTORS THAT ARE 1% TOLERANCE OR BETTER TO MAXIMIZE CMRR.
- AGND IS PLACED BOTH ADJACENT AND COPLANAR AROUND PD_IN LINES, SHIELDING THEM FROM OTHER RETURN OR CURRENT-CARRYING GROUNDS.
- THE 10µF CAPACITOR ON THE $V_{LED}$ LINE IS RECOMMENDED DUE TO ITS HIGH EFFECTIVE CAPACITANCE, WHICH IS GREATER THAN 4µF AT THE $V_{LED}$ VOLTAGE LEVEL (TYP +5V).
  - WHEN CHOOSING LESS THAN 0603-SIZED CAPACITORS, CONSIDER THE DERATING AT THE APPLIED VOLTAGE LEVEL TO ENSURE ENOUGH EFFECTIVE CAPACITANCE TO MAINTAIN HIGH SNR PERFORMANCE.
  - SMALLER BODY CAPACITORS DEGRADE SIGNIFICANTLY WHEN THE APPLIED VOLTAGES ($V_{LED}$) APPROACH THEIR VOLTAGE RATING.
- $C_{HPF}$ SETS THE HPF CORNER FREQUENCY. FOR TYPICAL USE CASES, 1.2µF SHOULD BE SELECTED FOR <0.67Hz. PLACE AS CLOSE AS POSSIBLE TO PINS AND SHIELD WITH AGND.
- FOR IEC 60601-2-47 TIMING ACCURACY COMPLIANCE, FCLK MUST BE SUPPLIED BY A LOW DRIFT 32.0kHz OR 32.768kHz OSCILLATOR.
- INT1 AND INT2 INTERRUPTS HAVE THE OPTION TO BE CONFIGURED AS ACTIVE DRIVE (ACTIVE-HIGH OR ACTIVE-LOW), ELIMINATING THE NEED FOR EXTERNAL PULLUPS.
- MULTIPLE PHOTODIODES CAN BE CONNECTED IN PARALLEL TO EACH PDm_IN LINE, TAKING NOTE TO ADJUST THE PD-CAPACITANCE BIAS REGISTER AS NEEDED.
- THE LDO SHOULD NOT BE USED TO POWER ANY OTHER DEVICE OTHER THEN THE MAX86178

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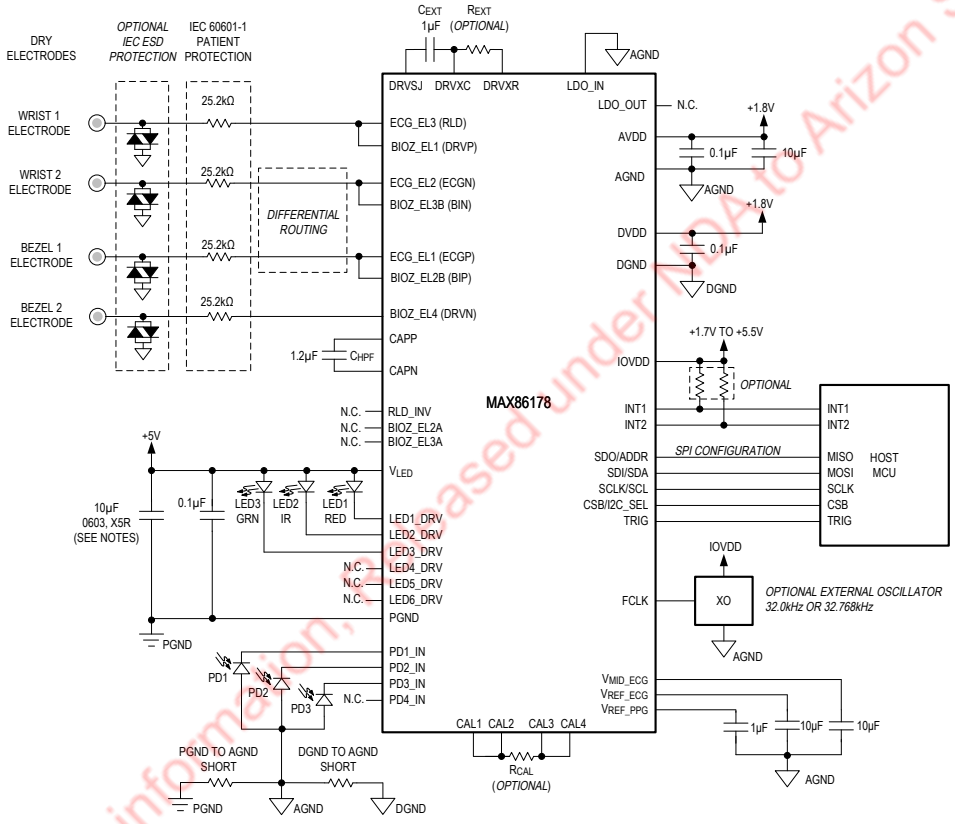
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## Typical Application Circuits (continued)

### Typical Application Circuit 4

**Wrist Device Supporting** Continuous HRM and $SpO_2$, with 4 Dry Electrodes Supporting On-Demand ECG or Continuous GSR/EDA




NOTES:

- EL1 AND EL3B ARE GROUPED TOGETHER ON THE WRIST TO ENABLE 2-ELECTRODE GSR/EDA ON THE WRIST. EL2B AND EL4 ARE GROUPED TOGETHER ON THE BEZEL TO ENABLE 2-ELECTRODE GSR/EDA ON THE FINGERS/PALM.
- THE DRVXC CAPACITOR AC COUPLES THE DRVP BIOZ STIMULUS CURRENT, PREVENTING DC CURRENT FROM FLOWING INTO THE ELECTRODES.
- DUE TO ITS LOWER IMPEDANCE, CHOOSING A ONE MICROFARAD CAP ON CEXT SUPPORTS THE GSR/EDA USE-CASE REQUIRING VERY LOW STIMULUS FREQUENCIES.
- REXT IS OPTIONAL AND SETS A CUSTOM BIOZ CURRENT STIMULUS MAGNITUDE. INTERNAL SETTINGS ARE AVAILABLE FROM 16nARMS TO 1.28mARMS.
- IEC 60601-1 PROTECTION COMPONENTS PROTECT THE PATIENT AGAINST DC CURRENTS IN A SINGLE-FAULT CONDITION SUCH AS BIOZ_ELX OR ECG_ELX PINS BEING SHORTED TO AVDD AND AGND.
- CHOOSE ECG PATIENT PROTECTION RESISTORS THAT ARE 1% TOLERANCE OR BETTER TO MAXIMIZE CMRR.
- AGND IS PLACED BOTH ADJACENT AND COPLANAR AROUND PD_IN LINES, SHIELDING THEM FROM OTHER RETURN OR CURRENT-CARRYING GROUNDS.
- THE 10µF CAPACITOR ON THE VLED LINE IS RECOMMENDED DUE TO ITS HIGH EFFECTIVE CAPACITANCE, WHICH IS GREATER THAN 4µF AT THE VLED VOLTAGE LEVEL (TYP +5V).
  - WHEN CHOOSING LESS THAN 0603-SIZED CAPACITORS, CONSIDER THE DERATING AT THE APPLIED VOLTAGE LEVEL TO ENSURE ENOUGH EFFECTIVE CAPACITANCE TO MAINTAIN HIGH SNR PERFORMANCE.
  - SMALLER BODY CAPS DEGRADE SIGNIFICANTLY WHEN THE APPLIED VOLTAGES (VLED) APPROACH THEIR VOLTAGE RATING.
- CHPF SETS THE HPF CORNER FREQUENCY. FOR TYPICAL USE CASES, 1.2µF SHOULD BE SELECTED FOR <0.67Hz. PLACE AS CLOSE AS POSSIBLE TO PINS AND SHIELD WITH AGND.
- AN OPTIONAL EXTERNAL FEEDBACK RESISTOR (NOT SHOWN) SETS THE GAIN OF THE RLD AMPLIFIER. INTERNAL REGISTER SETTINGS ALLOW FOR 12V/V, 24V/V, 48V/V, AND 97V/V GAIN.
- FOR IEC 60601-2-47 COMPLIANCE, FCLK MUST BE SUPPLIED BY A LOW DRIFT 32.0kHz OR 32.768kHz OSCILLATOR.
- INT1 AND INT2 INTERRUPTS HAVE THE OPTION TO BE CONFIGURED AS ACTIVE DRIVE (ACTIVE HIGH OR ACTIVE LOW), ELIMINATING THE NEED FOR EXTERNAL PULLUPS.
- MULTIPLE PHOTODIODES CAN BE CONNECTED IN PARALLEL TO EACH PDM_IN LINE, TAKING NOTE TO ADJUST PD-CAPACITANCE BIAS REGISTER AS NEEDED.
- THE LDO SHOULD NOT BE USED TO POWER ANY OTHER DEVICE OTHER THEN THE MAX86178

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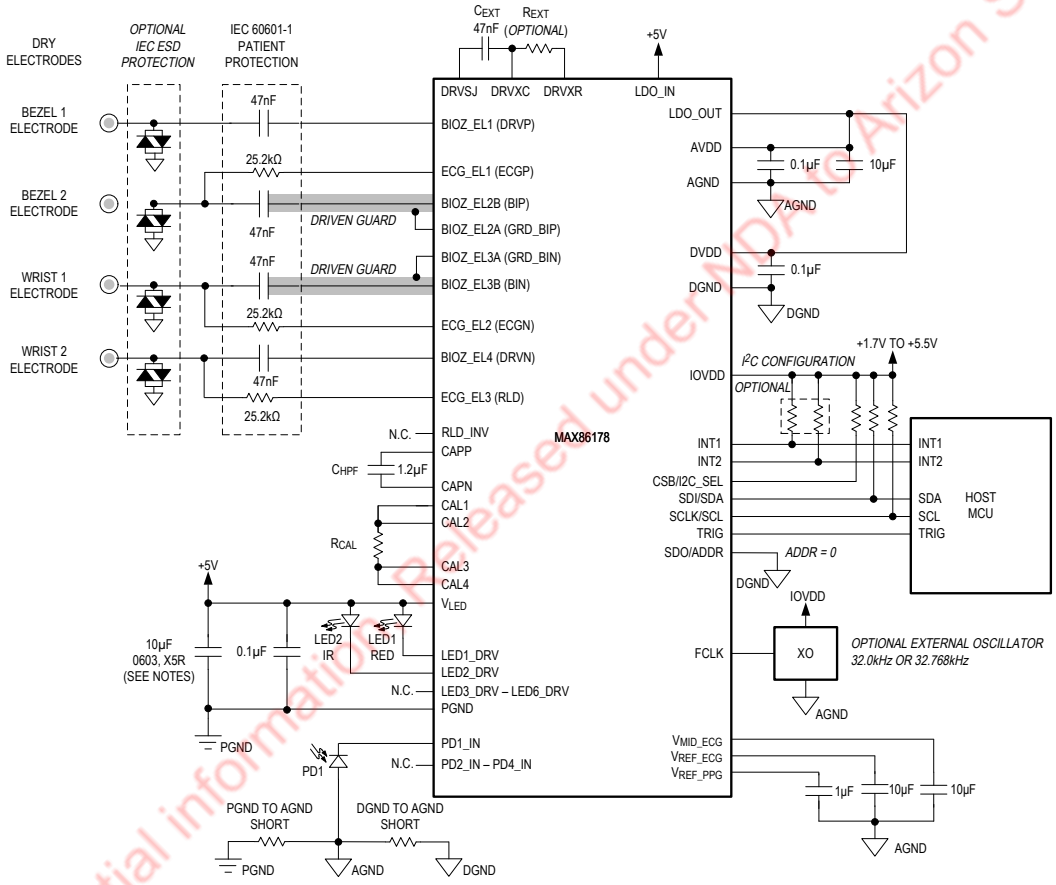
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## Typical Application Circuits (continued)

### Typical Application Circuit 5

**Wrist Device Supporting Continuous HRM and $SpO_2$, with 4 Dry Electrodes Supporting Either On-Demand BIA or On-Demand ECG**




NOTES:

- WHEN RUNNING ON-DEMAND BIA, THE ECG CIRCUIT SHOULD BE DISABLED TO MAINTAIN ABSOLUTE ACCURACY NECESSARY FOR THIS APPLICATION.
- GUARD TRACES REMOVE THE EFFECT OF THE PARASITIC PCB CAPACITANCES BY DRIVING A GUARD SIGNAL; THESE GUARD SIGNALS ARE ROUTED TO SURROUND BIP AND BIN NETS WITH BOTH CO-PLANAR AND ADJACENT SHIELDS.
- $R_{EXT}$ IS OPTIONAL AND SETS A CUSTOM BIOZ CURRENT STIMULUS MAGNITUDE. INTERNAL SETTINGS ARE AVAILABLE FROM 16nA$_{RMS}$ TO 1.28mA$_{RMS}$.
- IEC 60601-1 PROTECTION COMPONENTS PROTECT THE PATIENT AGAINST DC CURRENTS IN A SINGLE-FAULT CONDITION SUCH AS BIOZ_ELX OR ECG_ELX PINS BEING SHORTED TO AVDD AND AGND.
- CHOOSE ECG PATIENT PROTECTION RESISTORS THAT ARE 1% TOLERANCE OR BETTER TO MAXIMIZE CMRR.
- AGND IS PLACED BOTH ADJACENT AND COPLANAR AROUND PD_IN LINES, SHIELDING THEM FROM OTHER RETURN OR CURRENT-CARRYING GROUNDS.
- THE 10µF CAPACITOR ON THE $V_{LED}$ LINE IS RECOMMENDED DUE TO ITS HIGH EFFECTIVE CAPACITANCE, WHICH IS GREATER THAN 4µF AT THE $V_{LED}$ VOLTAGE LEVEL (TYP +5V).
  - WHEN CHOOSING LESS THAN 0603-SIZED CAPACITORS, CONSIDER THE DERATING AT THE APPLIED VOLTAGE LEVEL TO ENSURE ENOUGH EFFECTIVE CAPACITANCE TO MAINTAIN HIGH SNR PERFORMANCE.
  - SMALLER BODY CAPACITORS DEGRADE SIGNIFICANTLY WHEN THE APPLIED VOLTAGES ($V_{LED}$) APPROACH THEIR VOLTAGE RATING.
- $C_{HPF}$ SETS THE HPF CORNER FREQUENCY. FOR TYPICAL USE CASES, 1.2µF SHOULD BE SELECTED FOR <0.67Hz. PLACE AS CLOSE AS POSSIBLE TO PINS AND SHIELD WITH AGND.
- FOR IEC 60601-2-47 TIMING ACCURACY COMPLIANCE, FCLK MUST BE SUPPLIED BY A LOW DRIFT 32.0kHz OR 32.768kHz OSCILLATOR.
- INT1 AND INT2 INTERRUPTS HAVE THE OPTION TO BE CONFIGURED AS ACTIVE DRIVE (ACTIVE-HIGH OR ACTIVE-LOW), ELIMINATING THE NEED FOR EXTERNAL PULLUPS.
- MULTIPLE PHOTODIODES CAN BE CONNECTED IN PARALLEL TO EACH PDm_IN LINE, TAKING NOTE TO ADJUST THE PD-CAPACITANCE BIAS REGISTER AS NEEDED.
- THE LDO SHOULD NOT BE USED TO POWER ANY OTHER DEVICE OTHER THEN THE MAX86178

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## Ordering Information

| PART NUMBER | TEMP. RANGE | PIN-PACKAGE |
|---|---|---|
| MAX86178ENJ+ | -40ºC to +85ºC | 49 WLP |
| MAX86178ENJ+T | -40ºC to +85ºC | 49 WLP |

*+Denotes lead(Pb)-free/RoHS compliance.*

*T = Tape-and-reel.*

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## Revision History

| REVISION NUMBER | REVISION DATE | DESCRIPTION | PAGES CHANGED |
|---|---|---|---|
| 0 | 5/21 | Release for market intro | — |
| 1 | 8/21 | Updated IOVDD to DGND Absolute Maximum Rating, IOVDD Supply Voltage Max in Electrical Characteristics, Timing Subsystem, ECG EMI Filtering, BioZ EMI Filtering, BIOZ_KDIV, ECG_LOFF_FREQ, BIOZ Mux Configuration 1, Typical Application Circuits | 9, 41, 73, 97, 125, 143, 182, 195-196, 215-217 |
| 2 | 10/21 | Updated the Benefits and Features section and Applications section | 1 |
| 3 | 6/22 | Updated Maximum Free-Running Frame Rate and Biopotential-Channel Gain Error in Electrical Characteristics, Typical Operating Characteristics 40 and 80, Table 14, Register 0x28 Decode Table, Typical Application Circuits | 13,18, 30, 34, 75, 150, 217-219 |
| 4 | 8/23 | Updated the Benefits and Features, Updated the Common-Mode Input Range parameter name and Notes in Electrical Characteristics, Added the missing description about the unused pins in the Pin Description. Updated the Pin Functional Configuration(0x13), ECG Lead Bias Configuration 1(0x90), RLD Configuration 1(0x92), Status 3(0x02), BioZ Configuration 1(0xA0), System Sync(0x10), Threshold Hyst(0x71), and Status 1(0x00) registers in Register Map. Updated the Timing Subsystem, ECG Sample Rate and AC Lead-Off Frequency, FIFO Description, PPG, ECG, BioZ, PLL Synchronization, and Digital Interface sections in Detailed Description. Updated the FIFO Configuration examples 2 and 3 in the FIFO description. Updated Figure 1, Figure 28, Figure 44, Figure 58, Figure 59, Figure 60, and Figure 61. | 1, 10, 15, 24, 26, 37, 38, 40, 41, 45, 46, 47, 65, 70, 73, 85, 87, 97, 109, 110, 118, 119,128, 129, 137, 140, 174, 183, 184 and 187 |

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